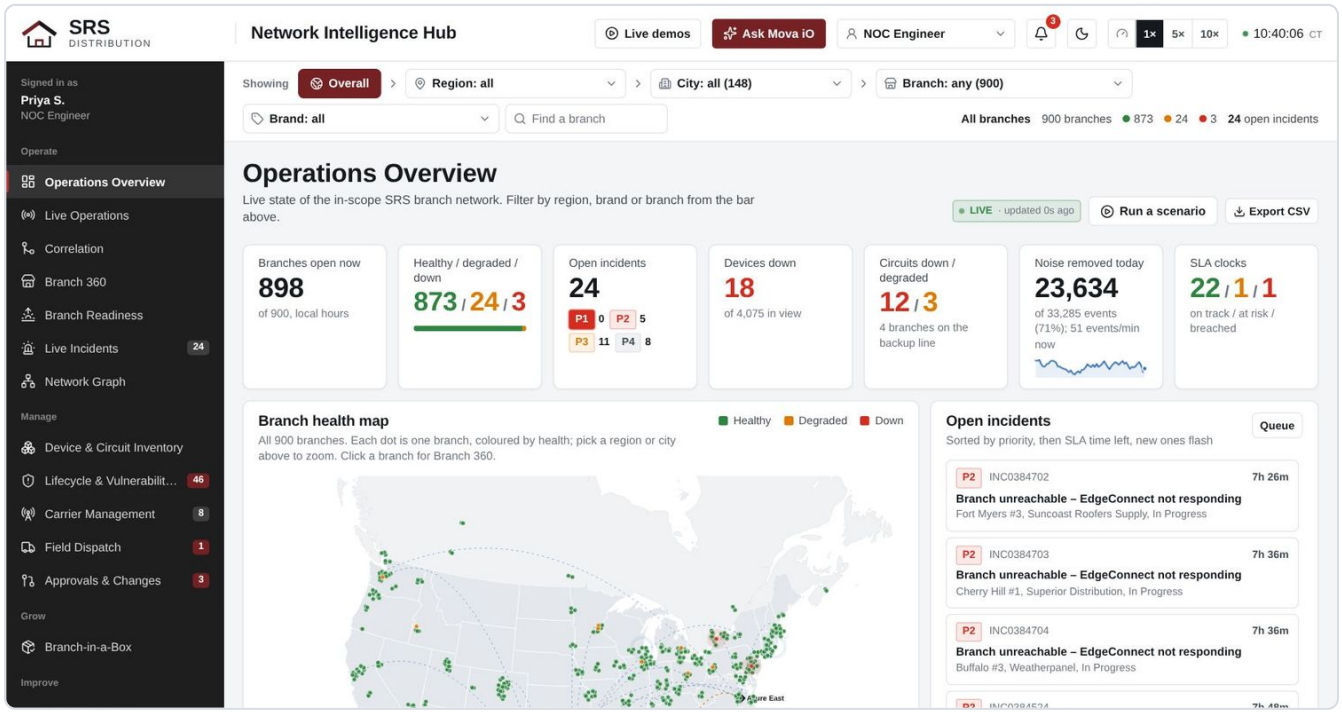
SRS DISTRIBUTION

SRS Network Intelligence Hub

The complete guide to the console: every screen, every visual, every demo



Managed Network Services · Movate for SRS Distribution

Version 2.0 · October 2026

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How to use this guide

This guide explains every screen of the SRS Network Intelligence Hub: each tile, chart, table, tab and button, what it measures, how to read it and what to do about it. It is written for anyone who uses or presents the console, from a NOC engineer on shift to a branch manager or an executive sponsor.

The guide is in four parts:

1. **Getting started:** what the console is, how the screen is laid out, the common way every screen is read, and where the data comes from.
2. **Screen by screen:** one chapter for each of the 20 screens, in the order of the left menu. Every visual has a picture and the same five points: what it shows, its parameters, how to read it, good and bad, and what to do.
3. **The live demos:** all 33 demos, what each one proves, its control mode and where it comes from in the proposal documents.
4. **Presenting and reference:** a suggested walkthrough, answers to common questions and a glossary.

The console runs on realistic sample data that moves every two seconds, so numbers in the pictures will differ slightly from what you see on screen. The structure, the thresholds and the rules are the same.

Part I. Getting started

1. The console at a glance

The Network Intelligence Hub is the working console for SRS Distribution's managed network: 900 branches across the United States and Canada, 4,075 managed devices and 1,800 WAN and internet circuits. It brings the network tools together on one screen and shows what Mova iO, the AI layer, is doing about each problem.

Estate	Count	Notes
Branches	900	Nine regions, from Southeast to Canada; each branch keeps its local brand name
SD-WAN appliances	933	HPE Aruba EdgeConnect EC-XS; larger branches have an HA pair. End of sale 31 Jan 2026, supported until 31 Jan 2031; new units are EdgeConnect 10104
Switches	1,856	Aruba 2530 (728, end of support in 2027), CX 6100, 6200M and 6300; 1,268 are standalone (outside Aruba Central)
Access points	1,286	Indoor AP-635 and outdoor AP-565 and AP-585 for yards; 812 are standalone
Circuits	1,800	A primary and a backup path per branch, from AT&T, Lumen, Comcast, Spectrum, Verizon, Cox and others

The tools behind the console:

Tool	Role in the console
HPE Aruba EdgeConnect Orchestrator	SD-WAN paths, tunnels, loss, latency and jitter for every branch
Aruba Central	Switch, access point and client health; firmware; config groups. It manages 1,062 switches and APs; the other 2,080 are standalone and monitored by OpsRamp only
HPE OpsRamp	Monitoring and event management: SNMP, syslog, config backups, correlation
New Relic	Synthetic checks of the payment and ERP paths from each branch
ServiceNow	Incidents, changes, problems, CMDB and the service catalog
Mova iO	AI agents that correlate, diagnose, run approved runbooks, verify and record
Carrier portals	Carrier cases, loopback tests, maintenance calendars and outage maps

Scope is the network only: SD-WAN, circuits, switching, Wi-Fi, network services such as DHCP, configuration, firmware and the network paths to the data centre and cloud. Applications, end-user devices and building systems are outside it.

2. The screen layout

Every screen has the same frame: a header across the top, the menu on the left, a filter bar under the header and the working area in the middle.

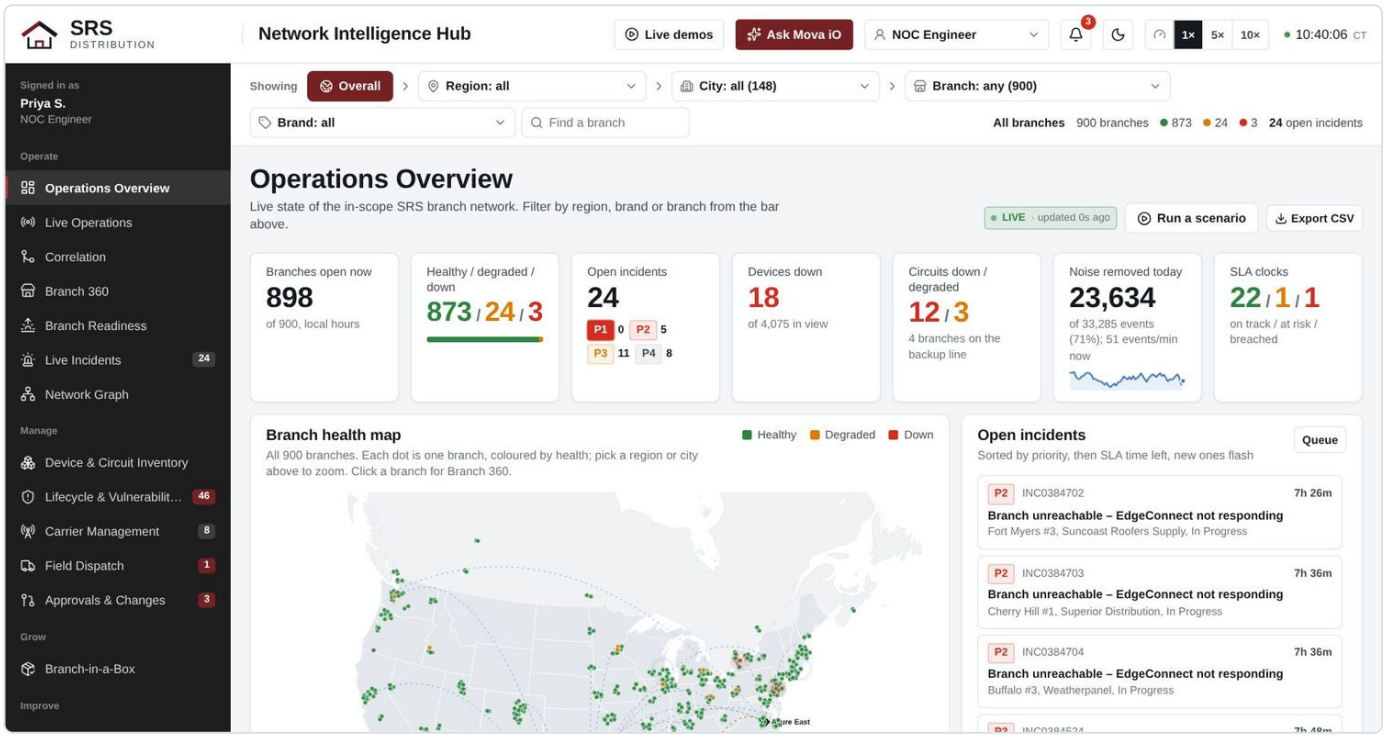


Figure 1. The Operations Overview with the header, left menu and filter bar

Header

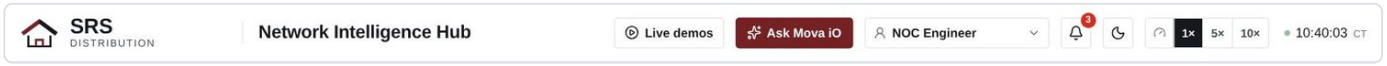


Figure 2. Header

Element	What it does
SRS mark and "Network Intelligence Hub"	The console name.
Tool sync (wide screens only)	OpsRamp, Aruba Central, ServiceNow and Mova iO, each with a green dot and the seconds since the last sync. A dot that is not green means that feed is late.
Live demos	Opens the demo library (chapter 24). On narrower screens it shows as a play icon.
Ask Mova iO	Opens the question panel on the right.
Role selector	Switches the viewing role. The role sets the landing page, the suggested questions and some defaults.
Bell	Notifications. A red number shows how many are new. Short pop-up messages (for example a new P1, or a change of role) appear at the top right under the header; each has a close button and clears itself after a few seconds.

Element	What it does
Moon or sun	Switches between light and dark mode.
Speed (1×, 5×, 10×)	How fast live data refreshes: every 2 seconds at 1×, every 0.4 seconds at 5× and every 0.2 seconds at 10×. Charts, numbers and streams all move at the chosen speed. Use 5× or 10× to show a lot of activity in a short meeting.
Clock	Current time in Central Time (CT), which the whole console uses.

Left menu

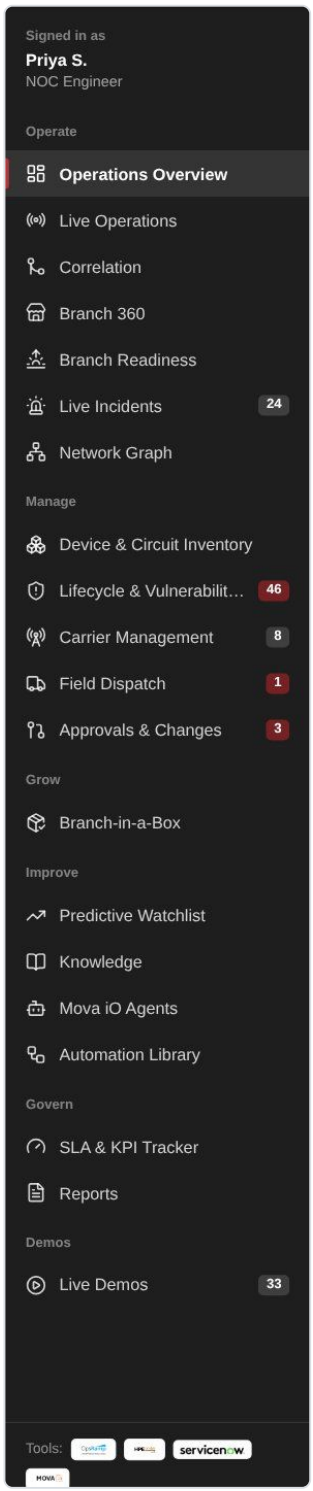


Figure 3. Left menu

The menu groups the 20 screens by what you are doing:

Group	Screens
Operate	Operations Overview, Live Operations, Correlation, Branch 360, Branch Readiness, Live Incidents, Network Graph

Group	Screens
Manage	Device & Circuit Inventory, Lifecycle & Vulnerabilities, Carrier Management, Field Dispatch, Approvals & Changes
Grow	Branch-in-a-Box
Improve	Predictive Watchlist, Knowledge, Mova iO Agents, Automation Library
Govern	SLA & KPI Tracker, Reports
Demos	Live Demos, at the bottom of the menu

The console opens on the Operations Overview. A number next to a screen is a live count: open incidents, open carrier cases, demos available. A maroon number means something is waiting for a person: devices with a critical vulnerability, unassigned work orders or approvals pending. "Signed in as" at the top shows the person and role in use. **Collapse** at the bottom shrinks the menu to icons.

Filter bar



Figure 4. Filter bar

The filter bar sets which branches every screen shows. It works left to right:

1. **Overall** resets to all 900 branches.
2. **Region** narrows to one of nine regions; the count in brackets is its branches.
3. **City** lists only the cities in the chosen region.
4. **Branch** lists the branches in that city. Picking one on the Overview or Branch 360 opens that branch's Branch 360.
5. **Brand** narrows to one of the SRS family brands.
6. **Find a branch** searches by name, brand, city, state or branch ID.

The right end of the bar always summarises the current view: its name, the number of branches, how many are green, amber and red, and the open incidents among them. The filter stays in place as you move between screens.

Ask Mova iO

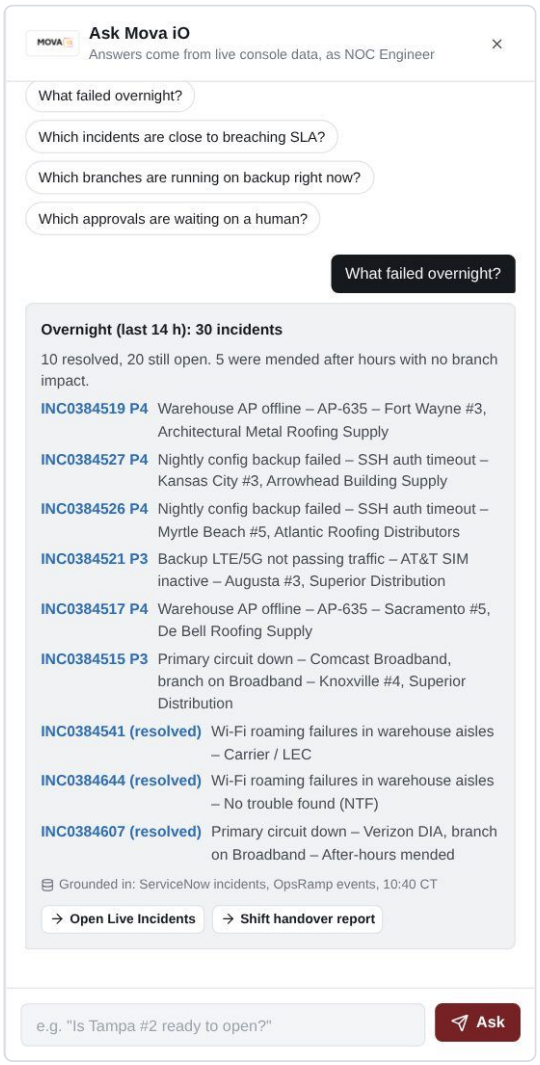


Figure 5. Ask Mova iO panel

The ask panel answers questions from the live console data. Each role gets four suggested questions, and you can type your own, such as "Is Tampa #2 ready to open?". Every answer shows its lines, names you can click to open the incident or branch, the source it was grounded in with the time, and buttons that jump to the right screen.

Notifications

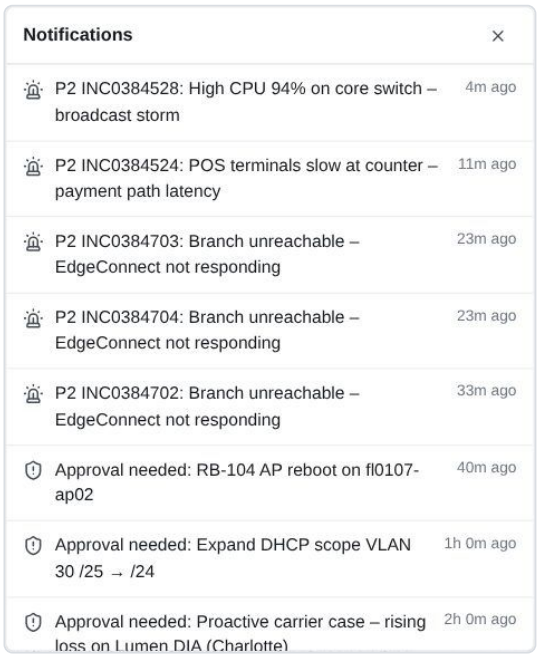


Figure 6. Notifications

The bell collects things that need a person: approvals waiting, new P1 or P2 incidents, breach risks and demo results. Click an entry to go to it.

Roles

The role selector changes the landing page and the suggested questions. The data and the rules do not change.

Role	Person in the sample	Lands on	Typical questions
NOC Engineer	Priya S.	Live Incidents	What failed overnight? Which incidents are close to breaching?
Network Engineer L3	Marcus L.	Approvals & Changes	Which carriers breached this week? Where is config drift?
StoreOps Lead	Jenna H.	Branch Readiness	Which branches are not ready to open?
Branch Manager	Rick D.	Operations Overview, for one branch	Is my branch ready to open? When is the technician arriving?
Field Technician	Carlos M.	Field Dispatch	What are my jobs today? Which spares are in my depot?
M&A Integration Lead	Alicia R.	Branch-in-a-Box	Status of this weekend's cutovers? Which projects are red?
Security Lead	Omar N.	Approvals & Changes, firmware tab	Which devices are below firmware policy? Who approved changes today?

The Branch Manager role locks the console to one branch and uses plain language for it.

Live updates

The console refreshes every two seconds at normal speed, faster when the header speed control is set to 5× or 10×. A green **LIVE** label means the card is updating; click it on a screen to pause updates while you explain something, and click again to resume. New rows slide in at the top of streams, changed numbers flash briefly, and "updated 1s ago" shows the age of the last reading. Live panels at the top of many screens show the stream for that screen, with a small chart and the latest events. Branch-in-a-Box is a project board, so it does not refresh live: its numbers change when a project moves forward, not every few seconds.

3. How to read any screen

The same conventions apply on every screen.

Colours

Colour	Means	Used for
Green	Healthy, passed, on track	Branches up, checks passed, SLA clocks with time left
Amber	Degraded, warning, at risk	A branch on its backup path, a check that passed with a warning, a clock past 75%
Red	Down, failed, breached	A branch with no path, a failed check, a breached SLA
Blue	Information and data series	Chart lines and bars that are measurements, not judgements
Maroon (SRS brand)	Actions and navigation	Buttons, the active menu item, badges that need a person

Maroon is never used for health, so a red item always means a problem.

Severity on alerts and events follows the ITOM convention: **Critical** (red), **Major** (orange), **Minor** (yellow) and **Info** (blue).

Priorities and SLA clocks

Priority	When it is used	Respond	Resolve
P1	Several branches down or a critical service lost	15 min	4 h
P2	One branch with both paths down, or a significant impact at one branch	15 min	8 h
P3	One path down while the branch trades on its backup, or degradation with a workaround	30 min	24 h
P4	Minor issues and planned work	30 min	96 h

An SLA clock shows the time left to resolve. It turns amber when more than 75% of the time is used and red when it is breached. "On track / at risk / breached" triples are always read in that colour order.

Numbers and small charts

- A triple such as **872 / 25 / 3** is healthy / degraded / down, in green, amber and red.
- A sparkline is the recent history of the number next to it; the last point is now.
- Where a note says "square-root scale", small values are drawn larger than their share so they stay visible. Read the number, not the bar length.
- Percentages for availability use two or three decimals because the targets are 99.9% and above.

Time

All times are in Central Time (CT) unless a card says otherwise. Branch opening and closing times use each branch's local time, which matters on Branch Readiness, where branches open in waves across five time zones.

Confidence and the gate rule

Every Mova iO decision carries **evidence** (why it thinks so), a **confidence** from 0 to 100%, a **rollback** (how to undo it) and a **verification** (how it checks the fix worked). A person must approve before anything changes when any of these is true:

- confidence is below 80%;
- the action changes a device at a branch that is open for trading;
- a freeze window covers the device or region.

Some actions are never autonomous, whatever the confidence: firmware and OS upgrades, routing and security policy changes, QoS changes, device reloads, rollbacks on hub devices and any action on a device without a confirmed alternate path.

The three control modes

Mode	Catalog tier	What happens	A person's role
Autonomous	Act	Mova iO acts on its own inside a policy SRS and Movate set and audit	Over the loop: reviews outcomes and the policy
Human-on-the-loop	Act and notify	Mova iO acts and tells a named person at once; the person can stop it during a stop window (5 minutes in production, 10 seconds in demos)	On the loop: watches and can stop
Human-in-the-loop	Stage for approval	Mova iO prepares the plan, evidence and rollback, then waits	In the loop: nothing changes until they approve

New actions start at the Stage tier. They move up only on evidence: a record of clean runs, proposed by the Catalog Steward and approved by SRS.

Network terms in plain words

Every reading in the console is a network measurement. This section explains each term once: what it is, what is normal at SRS, and what it means for the branch and its customers. The glossary in chapter 28 repeats the short definitions.

How a branch connects

Term	What it is	At SRS	Why the business cares
LAN (branch network)	The switches, access points and cabling inside the building	1 to 6 switches and 1 to 3 APs per branch	Counters, office PCs, phones and warehouse handhelds all plug into it
WAN	The lines that connect the branch to the data centre, Azure and the internet	Two per branch	ERP, pricing, card payments and calls all cross the WAN

Term	What it is	At SRS	Why the business cares
Line, circuit	One WAN connection bought from a carrier	1,800 in total	A line down means the branch relies on its other line
Primary line, backup line	The main line the branch normally uses, and the second line kept for failures	Every branch has both	With one line left there is no spare if it also fails
MPLS	A private line inside a carrier's network, steady but expensive	22% of primaries (AT&T, Lumen, Bell)	Very steady latency; slow to install
DIA (dedicated internet access)	A business internet line with guaranteed speed	30% of primaries	Steady, with a carrier repair commitment
Broadband	A cable or fibre internet line shared with other customers	48% of primaries, many backups	Cheaper and fast, but busier at peak times
LTE/5G	A cellular line from a router with a SIM	Backup at 55% of branches	Works when wires are cut, but adds 25 to 40 ms and is metered
Carrier, LEC	The company that sells the line; the local exchange carrier that owns the last mile to the building	AT&T, Lumen, Comcast, Spectrum, Verizon, Cox, Bell, Rogers	The carrier case and its repair time decide how long a line stays down
Demarc, NTE	The point where the carrier's equipment ends and the branch's begins	One per line	Proving a fault is on the carrier's side of the demarc gets it fixed faster
SD-WAN	Software that uses both lines together and picks the best line for each application	On every branch	Payments and ERP keep working when one line is poor
EdgeConnect	The HPE Aruba SD-WAN appliance at each branch (model EC-XS; EdgeConnect 10104 for new and replacement units)	933 at 900 branches	It measures both lines every second and moves traffic between them
HA pair	Two EdgeConnects at one branch, one taking over if the other fails	33 larger branches	A hardware fault does not take the branch down
Orchestrator	The central system that configures every EdgeConnect and collects its statistics	One for the estate	The source of the latency, jitter, loss and tunnel readings
Tunnel	An encrypted path over one line from the branch EdgeConnect to the data centre	Two per appliance, 1,866 in total	A tunnel down means one line cannot carry business traffic
Overlay, business intent policy	Rules that say which line each kind of traffic should use (payments, ERP, voice, guest)	Set centrally in Orchestrator	Critical traffic gets the best line; guest Wi-Fi never competes with payments
Steering, failover, failback	Moving traffic to the other line; doing it automatically when a line fails; moving it back once the main line is stable	Failover takes about 4 seconds	Customers at the counter do not notice a failover
Bandwidth	The capacity of a line, in megabits per second (Mb/s)	20 to 1,000 Mb/s primaries, 150 Mb/s LTE	A line with too little bandwidth fills up at peak times
WAN use	The share of a line's bandwidth in use right now	Usually 10 to 60%	Above 85% traffic queues, which raises latency and loss for everyone

Line quality and what the business feels

Term	What it is	Normal at SRS	Warning	What the business feels
Latency (to the data centre)	Time for a request to reach the McKinney data centre and come back, in milliseconds	7 to 15 ms in Texas, 35 to 50 ms on the coasts, about 90 ms in Hawaii; LTE adds 25 to 40 ms; 32 ms on average	Normal plus 25 ms; critical at normal plus 60 ms	ERP, pricing and order screens respond slowly
Jitter	How much the delay varies from one packet to the next	0.5 to 3 ms on wired lines, 5 to 12 ms on LTE	Above 20 ms; critical above 30 ms	Choppy calls; card terminals time out more often
Packet loss	The share of data that never arrives and must be sent again	Under 0.2%	Above 1%; critical above 2%	Card payments retry, calls break up, ERP sessions drop
Degraded line	A line that works but with high latency, jitter or loss	A handful at any time	Any degraded primary at an open branch	The branch trades, but slowly, until traffic moves to the other line
Payment path round trip	A synthetic test payment from the counter VLAN (PCI segment) to the payment processor and back	100 to 200 ms	Above 400 ms	Card payments are slow at the counter
ERP login	A synthetic login to the ERP from the branch	About 1 second	Above 2 seconds	Staff wait on every order and stock check
Voice call quality (MOS)	A score from 1 to 4.4 for how a call sounds, worked out from jitter, loss and delay	4.2 to 4.4	Below 3.6	Customers and branches struggle to hear each other
Wi-Fi experience	Aruba Central's score out of 100 for the Wi-Fi inside the branch: signal, connection success and roaming	90 to 97	Below 80; critical below 65	Handhelds and laptops in the warehouse and office struggle. It does not depend on the WAN: Wi-Fi can be good while a WAN line is poor
p95	The 95th percentile: 95% of measurements are at or below this value	–	–	Shows what most users get, without being skewed by one bad sample
Synthetic probe	A scripted test run every minute from each EdgeConnect: payment path, ERP login, DNS, DHCP	5,400 checks every 10 minutes	A failed probe	Finds problems before a customer does

Inside the branch

Term	What it is	Why the business cares
Switch, port, uplink	The box that connects wired devices; each socket is a port; the uplink connects one switch to another or to the EdgeConnect	An uplink fault cuts off everything behind it
PoE (Power over Ethernet)	Power delivered to APs, phones and cameras through the network cable	A switch out of PoE budget switches devices off

Term	What it is	Why the business cares
CRC errors	Corrupted frames on a link, usually a failing cable or optic	Slow or dropping connections; often worse over days
Optic, SFP	The plug-in module for a fibre link	A cheap part that is a common cause of uplink faults
VLAN	A separate network segment on the same switches: 10 corporate, 20 payments (PCI), 30 warehouse Wi-Fi, 40 voice, 50 cameras, 60 guest	Keeps payments, staff and guests apart
PCI segment	VLAN 20, the isolated network for card terminals, with no routes to other VLANs	Required for card payment compliance
Access point (AP), SSID	The Wi-Fi radio; the network name it broadcasts, such as SRS-SCAN for warehouse handhelds	Handhelds, laptops and yard devices depend on it
Roaming	A handheld moving from one AP to the next as staff walk the aisles	Failures stop scanning and picking mid-task
Channel utilisation	How busy an AP's radio channel is	Above about 70% Wi-Fi slows for everyone on that AP
DHCP, scope	The service that gives each device an address; the pool of addresses for a VLAN	A full scope means new devices cannot connect
Broadcast storm, loop	Traffic copied endlessly round a cabling loop, often from an unmanaged desk switch	The whole branch network slows or stops
BPDU guard	A switch setting that shuts a port when another switch is plugged into it	Stops loops before they become storms
OOB (out-of-band) console	A cellular console path to devices when the WAN is down	Engineers can still reach a branch that has lost both lines

Monitoring, events and incidents

Term	What it is	At SRS
Telemetry, poll, trap, syslog	Readings collected from devices; a regular query (SNMP poll); a message a device sends when something happens (trap); a device's log lines (syslog)	OpsRamp polls every device every minute
Event	Any single message or threshold crossing from a tool	About 50 a minute across the estate
Noise removal	Dropping duplicates, flaps and information-only events before they become alerts	About 70% of events (KPI at least 50%)
Duplicate, flap, hold-down	The same event repeated; a line or port that goes down and up quickly; the wait (for example 30 to 40 seconds) before acting, in case it recovers	Flaps that recover in the hold-down never become tickets
Alert	An event worth acting on, left after noise removal	Most clear on their own
Auto-cleared	An alert that recovered by itself and was checked by ping and SNMP	About 70% of alerts
Digest	A daily summary of warnings that do not need a ticket	Read by the morning shift

Term	What it is	At SRS
Correlation, correlation group	Grouping related alerts by topology, time, change, carrier and similarity into one incident	One incident per branch per event
Incident	A ServiceNow record of a problem, with a priority and SLA clocks	20 to 40 a day
Confidence	How sure Mova iO is about a diagnosis or action, 0 to 100%	Below 80% a person decides
Root cause	The underlying fault behind an incident, for example a carrier fault upstream of the demarc	Recorded in the RCA
Problem record	A ServiceNow record for the cause behind repeat incidents	Opened after three incidents in 30 days
Repeat offender	A circuit with three or more tickets in 90 days	Gets a carrier review
NTF (no trouble found)	A carrier closing a ticket without finding a fault	23% of tickets in the last six months

Devices, firmware and lifecycle

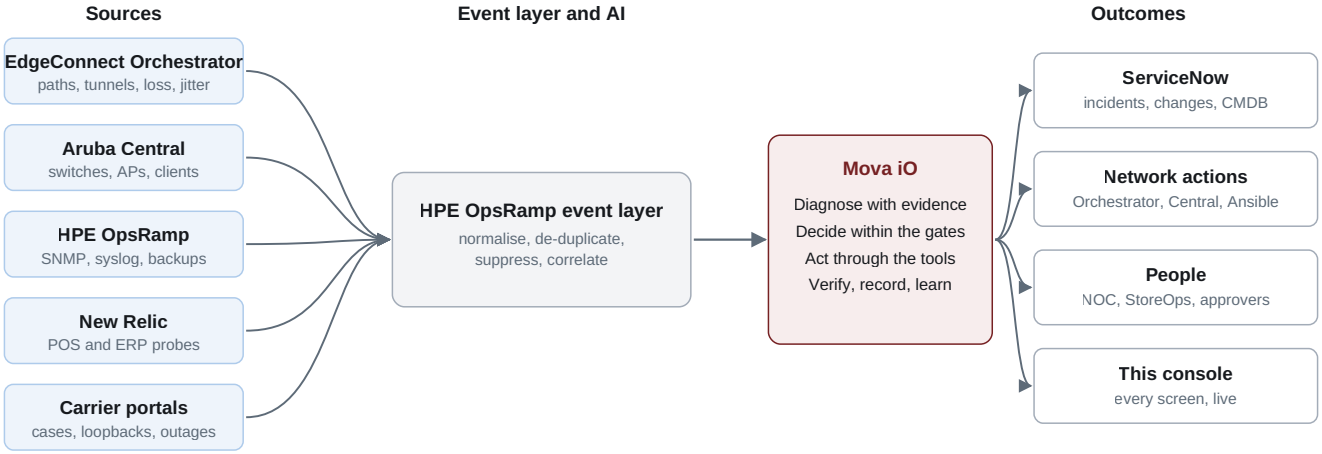
Term	What it is	At SRS
Aruba Central, standalone	HPE Aruba's cloud management for switches and APs; a standalone device is managed one by one and only monitored by OpsRamp	1,062 in Central, 2,080 standalone
Firmware, release, N, N-1, N-2	The device software; the target release (N), one behind (N-1) and two behind (N-2)	Policy: EdgeConnects at N or N-1, others no older than N-2
Config backup	A nightly copy of each device's configuration	02:00 CT every night
Golden config, drift, Network-as-Code	The approved configuration kept as files in a repository; any difference on a device is drift; changes go through reviewed merge requests	Drift is checked every night
Advisory, CVE, CVSS	A vendor security bulletin; the public ID of a vulnerability; its severity score from 0 to 10 (9.0 and above is critical)	Each advisory is matched to the firmware every device runs
Patch, canary, ring, wave	A fixed release; a small first group upgraded and checked before the rest; the groups that follow; one batch of upgrades after branch close	Canary first, then waves by region
End of sale	The last day the vendor sells a model; it is still supported	EC-XS: 31 Jan 2026. Aruba 2530: 31 Jan 2022 (48-port), 31 Mar 2022 (24-port)
End of support	The last day of vendor support and security fixes; after it the device should be replaced	EC-XS: 31 Jan 2031. Aruba 2530: 31 Jan 2027 (48-port), 31 Mar 2027 (24-port). Other models: not announced
Limited lifetime warranty	HPE's warranty for Aruba switches and APs, which runs until five years after the model's end of sale	Switches and APs show "Lifetime"
Support contract	The yearly HPE support agreement that covers the EdgeConnects, which have a 1-year warranty	Renewed every year

Service measures and automation

Term	What it is	At SRS
SLA, respond, resolve	The agreed time to start working on an incident and to fix it	P1: respond in 15 minutes, resolve in 4 hours
MSL, ESL, KPI	Minimum service level (below it a service credit applies); expected service level (the target); key performance indicator	All from Annexure B
Availability	The share of time a service works	Branches online: target 99.95%
MTTR	Mean time to restore service	Tracked per priority
Error budget, burn rate	The failure a target allows over 30 days (99.9% allows about 43 minutes); how fast it is being used compared with an even spend	Burning fast holds non-essential changes
Runbook	An approved, tested automation for one kind of fix, numbered RB-1xx	10 in the library
Act, Act and notify, Stage	The three catalog tiers: acts on its own; acts and tells a person at once; prepares and waits for approval	New runbooks start at Stage
Gate rule	The rule that sends a decision to a person: confidence below 80%, a branch open for trading, or a freeze	Applies to every action
Verification, rollback	The checks that prove a fix worked; the prepared way to undo it	Every action has both
Mova iO agent	One of Movate's 13 AI agents, each with one job: noise reduction, correlation, diagnosis, runbooks, carrier cases and others	Every action lands in the audit log

4. Where the data comes from

Five monitoring sources feed one event layer. Mova iO correlates the events, decides what to do within the gate rules, acts through the tools and writes the record to ServiceNow.



Data flow: five sources feed one event layer; Mova iO decides within the gate rules and acts through the tools; every step is recorded in ServiceNow and the audit log.

Source	What it sends	How often
EdgeConnect Orchestrator	Path and tunnel health per appliance: loss, latency, jitter, failover	Stats every 30 s, alarms live
Aruba Central	Switch, AP and client health, PoE, radios, roaming, firmware	Every minute, events live
HPE OpsRamp	SNMP and syslog from every device, config backups	Polls every minute, traps live
New Relic	Synthetic payment-path and ERP checks from each branch	Every minute
Carrier portals	Ticket status, loopback results, maintenance windows, outage maps	On change

Typical ticket triggers are a device unreachable for three polls, an SD-WAN tunnel down for more than two minutes, packet loss above 2% for ten minutes, sustained latency or jitter above the path's limit, recurring interface errors and a failed synthetic check.

Part II. Screen by screen

5. Operations Overview

The Operations Overview answers "is the network all right, and if not, where?" in one screen. It covers every branch in the current filter. The header buttons are the **LIVE** label (click to pause), **Run a scenario**, which replays one of five scripted events into the console, and **Export**, which downloads the branch health list as a CSV file.

Headline tiles



Figure 7. Headline tiles

Tile	What it counts	Good	Look closer when	Click
Branches open now	Branches inside their local trading hours, out of the branches in view	Follows the clock: low at night, near 900 in the day	Not applicable	–
Healthy / degraded / down	Branches by health: green has both lines and its devices up; amber is trading but impaired, for example on its backup line or with an unstable main line; red has no working line and cannot trade	Green above 97%, red at 0	Any red, or amber rising	–
Open incidents	All open incidents in view, with the split by priority. "+N new" flashes when incidents arrive	No P1	Any P1 (tile turns red), or P2 count rising	Opens Live Incidents
Devices down	EdgeConnects, switches and APs not answering, out of all devices in view. Every device at a branch that is down counts, because none of them can be reached	Single digits	A jump, which usually means a branch is down	–
Circuits down / degraded	WAN lines (circuits) down (red) or degraded (amber), primary and backup together, and how many branches are trading on their backup line. A branch that is down counts both its lines	Low and steady	Several on one carrier or in one area	Opens Carrier Management
Noise removed today	Events removed before they became alerts: duplicates, flaps that recovered within the hold-down and information-only messages. Shown with all events since midnight, the share removed and events per minute now	Above 50% (KPI); usually near 70%	The share drops, so engineers see more noise	–
SLA clocks	Open incidents on track, at risk (more than 75% of resolve time used) and breached	No breached clocks	Any amber or red	Opens SLA & KPI Tracker

Branch health map

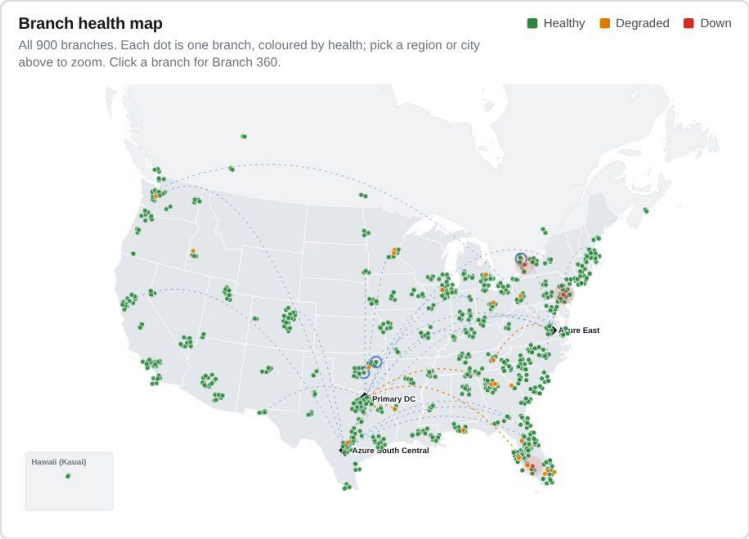


Figure 8. Branch health map

What it shows. All 900 branches on a map of the United States and Canada, one dot per branch, coloured by health. Dashed moving lines are live traffic from the busiest branches to the three hosting points: the Primary DC, Azure South Central and Azure East (black diamonds).

Parameter	Meaning
Dot colour	Green healthy, amber degraded, red down. Red dots pulse
Faded dots	Branches outside the current filter, kept for context
Flow line colour	Blue normal traffic, amber from a degraded branch, red from a branch that is down
Zoom	Picking a region or city in the filter bar zooms the map to it

How to read it. Look for clusters, not single dots. A cluster of amber or red in one metro usually means a carrier or weather event, which the Correlation screen will show as one major incident. **What to do:** hover a dot for the branch name and state; click it to open Branch 360.

Open incidents

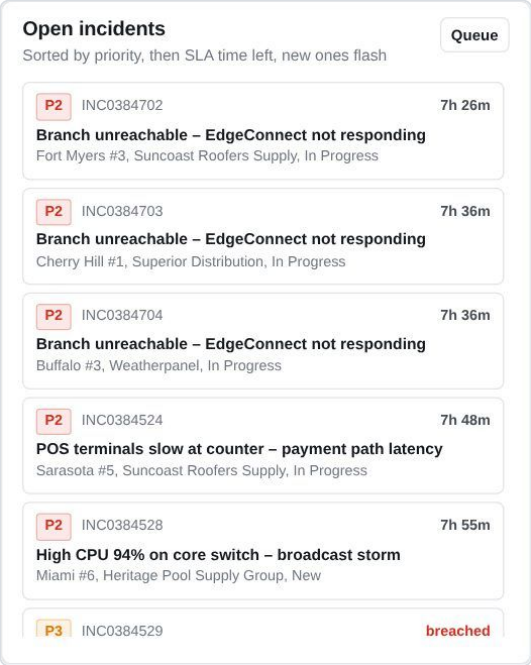


Figure 9. Open incidents

The 14 most urgent open incidents, sorted by priority and then by SLA time left. Each row shows the priority, incident number, MAJOR or NEW flags, time left to resolve (amber past 75% used, red "breached"), the title and the branch, brand and state. New incidents flash. **Queue** opens Live Incidents; clicking a row opens that incident.

Service levels and error budgets

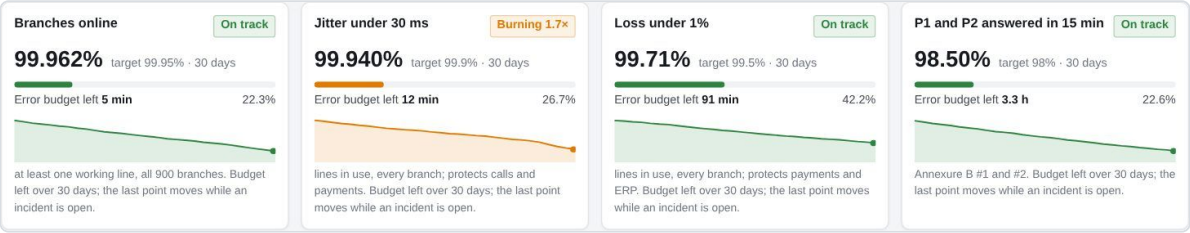


Figure 10. Error budgets

Four service-level objectives over 30 days: **Branches online** (at least one working line), **Jitter under 30 ms** on the lines in use (protects calls and card payments), **Loss under 1%** on the lines in use (protects payments and ERP) and **P1/P2 answered in 15 minutes**.

Parameter	Meaning
Large figure	Achieved level over the last 30 days, against the target shown beside it
Badge	**On track**, or **Burning N×** when the budget is being used faster than an even spend (more than 1.5 times)
Bar and percentage	Share of the 30-day error budget still left
Error budget left	The same budget as time: minutes or hours of failure still allowed this month
Line	Budget left over the 30 days; the last point moves while an incident is open

How to read it. The error budget is the failure the target allows. A 99.9% target over 30 days allows about 43 minutes. Green means plenty left; amber means burning fast; red means less than 20% left. **What it means for change:** when a budget burns fast, non-essential changes in that area wait (see the error budget demo in Part III).

Incident outlook, next 12 hours



Figure 11. Incident outlook

What it shows. New incidents per hour for the last 24 hours (solid) and the forecast for the next 12 hours (dashed), with a shaded band for the 80% likely range. Above the chart: expected new incidents, the busiest hour and the weather risk.

How to read it. The forecast uses the hourly pattern, the weather outlook and open changes. Evening hours are busier because most incidents are worked after branches close. If the actual line runs above the band, something unusual is happening. **What to do:** plan the evening shift and surge cover from the busiest hour.

Prevention in flight

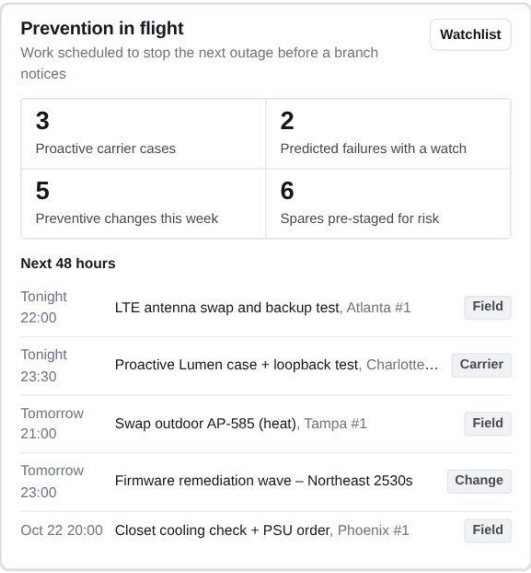


Figure 12. Prevention in flight

Work already booked to stop the next outage before a branch notices: proactive carrier cases, predicted failures under watch, preventive changes this week and spares pre-staged for risk, then a list of the next 48 hours with time, task, branch and owner (Field, Carrier or Change). **Watchlist** opens the Predictive Watchlist.

Event flow

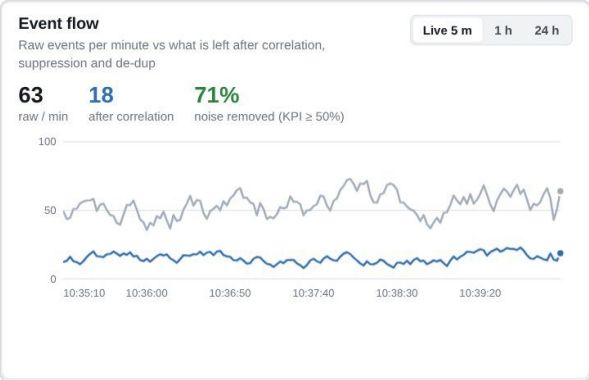


Figure 13. Event flow

Parameter	Meaning
Raw / min	Events per minute arriving from all monitoring tools (grey line)
After correlation	What is left once duplicates, flaps and children of an open incident are removed (blue line)
Noise removed	1 minus after/raw. The KPI asks for at least 50%; the console usually runs near 70%
Tabs	Live 5 minutes, 1 hour, 24 hours

How to read it. The gap between the lines is the work Mova iO saves. A sudden rise in raw events with a flat blue line is a storm being contained. A rising blue line means more real events, which should match new incidents.

WAN quality – live

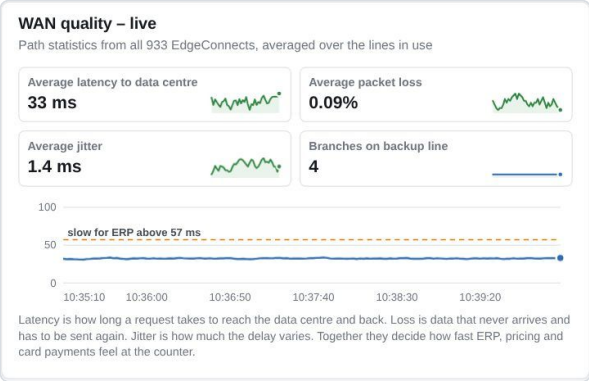


Figure 14. WAN quality

The quality of the lines every branch is using right now, averaged across all 933 EdgeConnects from Orchestrator path statistics and refreshed with the live data:

Tile	What it is	Normal at SRS	Look closer when	What the business feels
Average latency to data centre	Time for a request to travel from the branch to the McKinney data centre and back	About 32 ms	More than 10 ms above normal	ERP, pricing and order screens respond more slowly

Tile	What it is	Normal at SRS	Look closer when	What the business feels
Average packet loss	Share of data that never arrives and must be sent again	Under 0.2%	Above 0.5% on average	Card payments retry, calls break up
Average jitter	How much the delay varies from one packet to the next	1 to 3 ms	Above 10 ms on average	Voice becomes choppy; payment terminals time out more often
Branches on backup line	Branches whose main line is down, or whose traffic was moved off it	Under 15	Above 20, or a cluster in one area	Trading continues, but LTE is slower and there is no spare line

The chart tracks average latency with a dashed line at normal plus 25 ms, the point where staff start to notice ERP screens slowing. A rise across many branches at once usually means a carrier or regional problem; one branch alone shows on Top branches and Branch 360.

Live event stream

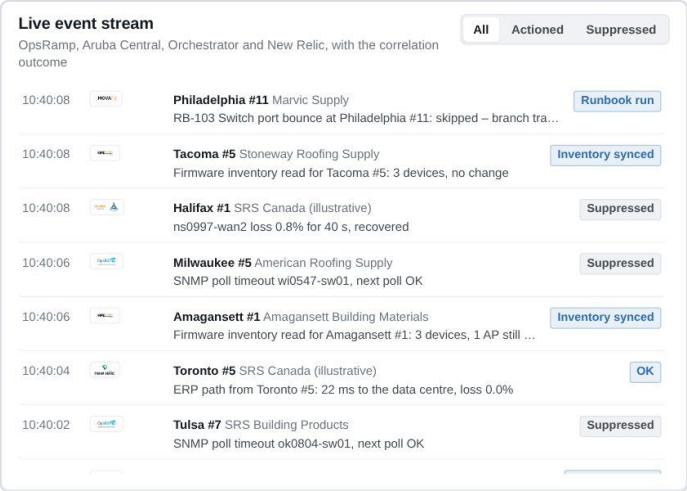


Figure 15. Live event stream

Every event in view as it arrives: time, source tool, branch and brand, the message and the outcome. Outcome badges: **OK** (a probe passed), **Suppressed** or **Duplicate** (grey), **Watching** (amber), **Auto-resolved** or **Recovered** (green), **New incident** (red), and others such as **Accepted** (a knowledge article accepted by an engineer). Tabs filter to **All**, **Actioned** (events that led to something) or **Suppressed**. Click a row to open its incident, or the branch if there is none.

Top branches

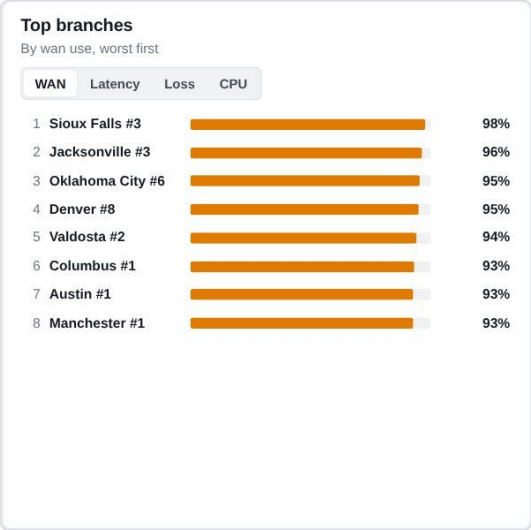
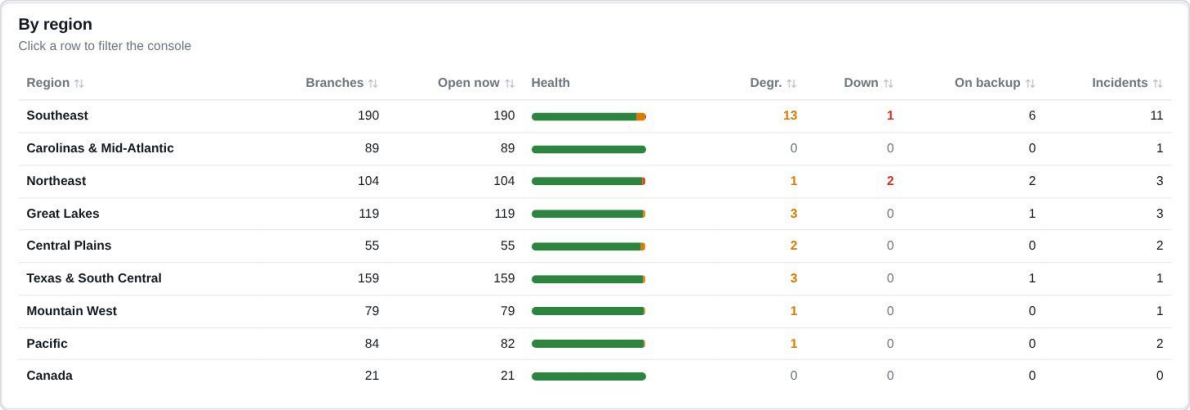


Figure 16. Top branches

The eight worst branches on one measure, chosen with the tabs: **WAN** use (%, amber from 80%), **Latency** (ms, amber from 80 ms), **Loss** (%, amber from 1%) or EdgeConnect **CPU** (%, amber from 75%). Use it to find the branch that is about to become an incident. Click a row for Branch 360.

By region



Region	Branches	Open now	Health	Degr.	Down	On backup	Incidents
Southeast	190	190		13	1	6	11
Carolinas & Mid-Atlantic	89	89		0	0	0	1
Northeast	104	104		1	2	2	3
Great Lakes	119	119		3	0	1	3
Central Plains	55	55		2	0	0	2
Texas & South Central	159	159		3	0	1	1
Mountain West	79	79		1	0	0	1
Pacific	84	82		1	0	0	2
Canada	21	21		0	0	0	0

Figure 17. By region

One row per region: branches, open now, a health bar (green, amber, red), degraded, down, on backup and incidents. Click a row to filter the whole console to that region.

Alerts per minute – live



Figure 18. Alerts per minute

Alerts per minute after correlation for the last 30 minutes, stacked by severity: Critical (red), Major (orange) and Minor (yellow). A tall red bar is the moment to look at the event stream.

Circuits down or degraded



Figure 19. Circuits down or degraded

Every circuit not fully up: branch, primary or secondary, carrier and type, and Down (red dot) or Degraded (amber dot). A branch listed twice has lost both paths. **Carriers** opens Carrier Management; a row opens the branch.

Ticket mix – last 6 months

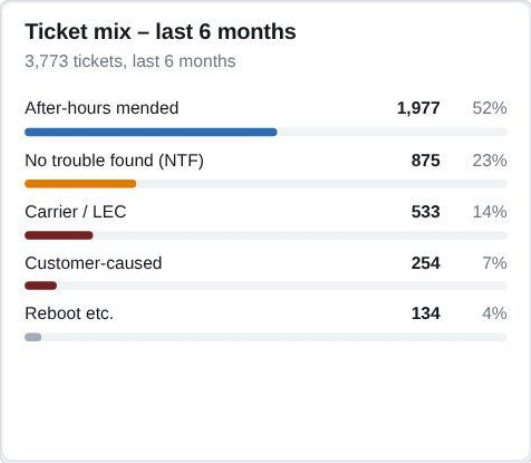


Figure 20. Ticket mix

The 3,773 network tickets of the last six months by type: after-hours mended (52%), no trouble found (23%), carrier or LEC (14%), customer-caused (7%) and reboots (4%). This is the baseline the automation targets: after-hours fixes and no-trouble-found tickets are where Mova iO saves the most.

Unified timeline

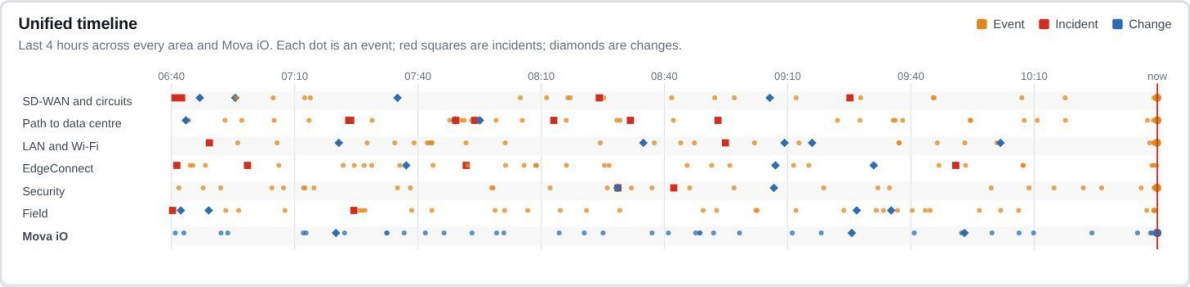


Figure 21. Unified timeline

The last four hours on one axis, one lane per area (SD-WAN and circuits, path to data centre, LAN and Wi-Fi, EdgeConnect, security, field) plus a lane for Mova iO. Dots are events (larger when new), red squares are incidents and blue diamonds are changes; the red line is now. Use it to see whether an incident followed a change, or whether one area's events lead another's.

6. Live Operations

Live Operations is the "watch it happen" screen: the network as it runs, line by line, with alerts and automation as they arrive, for every branch in the current filter. Every tile says what its number means for trading. Use it on a wall screen or when something is unfolding.

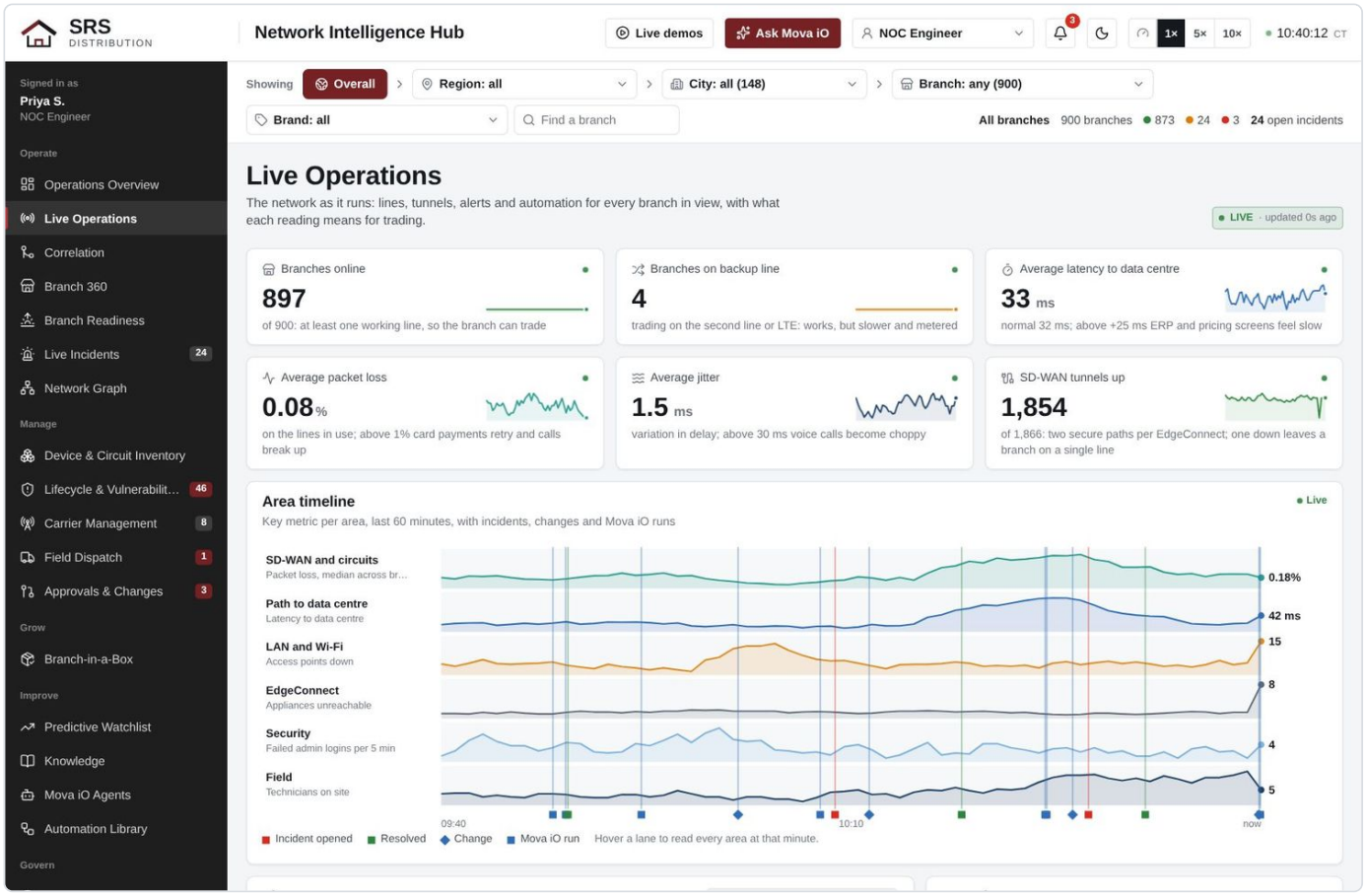


Figure 22. Live Operations

Live tiles

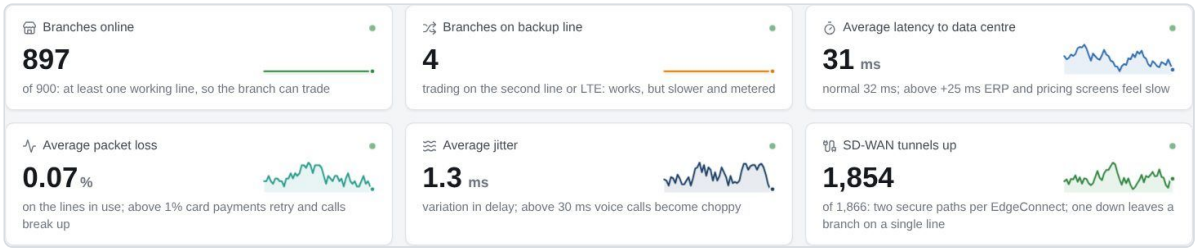


Figure 23. Live tiles

Tile	What it is	Normal at SRS	What it means for the business
Branches online	Branches with at least one working WAN line	897 to 900 of 900	A branch online can take orders and card payments. A branch that drops off cannot trade until a line returns

Tile	What it is	Normal at SRS	What it means for the business
Branches on backup line	Branches whose main line is down, or whose traffic was moved off it, running on their second line (broadband or LTE/5G)	Under 15	Trading continues; LTE is slower and metered, and there is no spare line if it fails too
Average latency to data centre	Round trip from each branch to the McKinney data centre on the line in use, averaged. The subtitle shows the normal value	About 32 ms: 7 to 15 ms in Texas, 35 to 50 ms on the coasts, about 90 ms in Hawaii	Above normal plus 25 ms, ERP, pricing and order entry feel slow
Average packet loss	Data lost on the lines in use, averaged	Under 0.2%	Above 1% card payments retry and calls break up; above 2% ERP sessions drop
Average jitter	Variation in delay on the lines in use, averaged	1 to 3 ms	Above 30 ms voice calls become choppy and payment terminals time out more
SD-WAN tunnels up	Encrypted paths from each EdgeConnect to the data centre, one per line, out of 1,866	1,866 minus the lines that are down	Each tunnel down leaves a branch on a single line

Each tile has a sparkline of its recent history; the green dot shows it is live. The numbers follow the header speed control (1×, 5× or 10×).

Area timeline

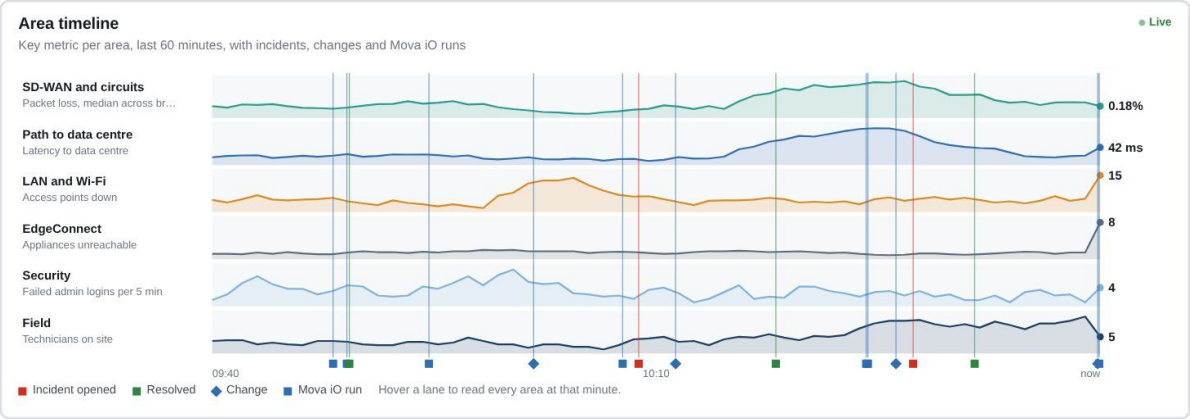


Figure 24. Area timeline

What it shows. One lane per area for the last 60 minutes, each with its key metric and the value now at the right.

Lane	Metric
SD-WAN and circuits	Packet loss, median across branches (%)
Path to data centre	Latency to the data centre (ms)
LAN and Wi-Fi	Access points down
EdgeConnect	Appliances unreachable
Security	Failed admin logins per 5 minutes

Lane	Metric
Field	Technicians on site

Markers along the bottom: red squares for incidents opened, green for resolved, blue diamonds for changes and dark blue for Mova iO runs, each with a thin line up through the lanes. **How to read it.** Each lane is scaled to its own range, so read the shape, not the height between lanes. A bump in one lane followed by bumps in others shows cause and effect, for example a loss rise on SD-WAN followed by higher latency on the path to the data centre. Hover anywhere to read every lane at that minute.

Live event stream

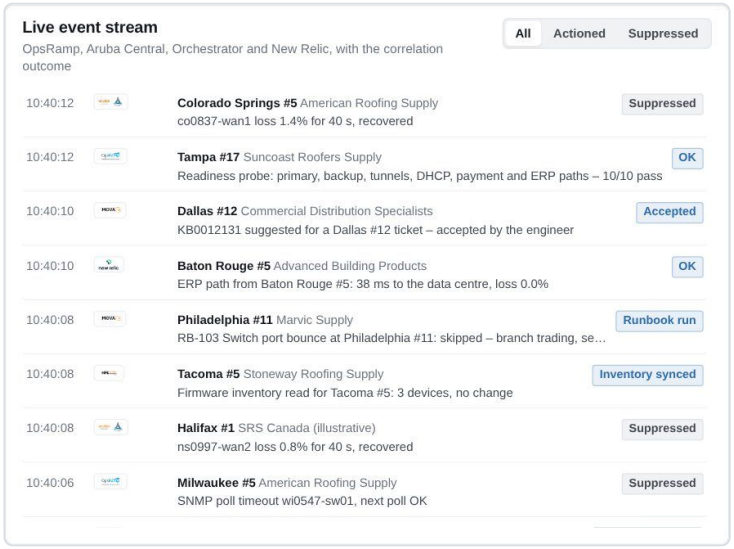


Figure 25. Live event stream

The same stream as on the Overview (chapter 5): every event with its source, branch, message and outcome badge, filtered by **All**, **Actioned** or **Suppressed**.

Mova iO agents and Needs attention

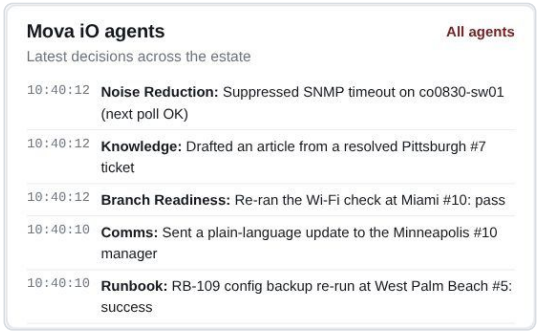


Figure 26. Latest Mova iO decisions

The last eight decisions by any agent, with time and agent name; **All agents** opens the Mova iO Agents screen.

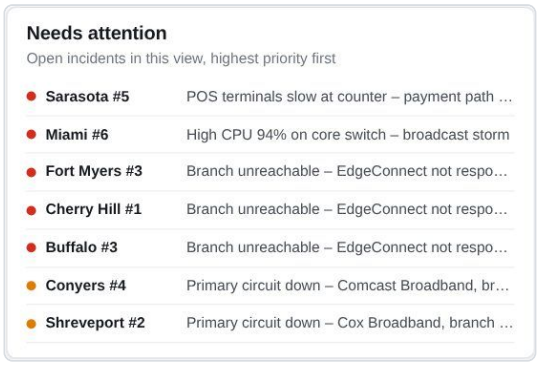


Figure 27. Needs attention

Open incidents in view, highest priority first: a red dot for P1 and P2, amber for P3 and P4, then the branch and title. Click to open the incident.

Alerts by area and hour

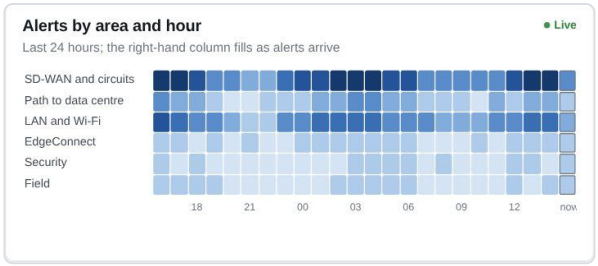


Figure 28. Alerts by area and hour

A heat map of alerts by area (rows) and hour for the last 24 hours (columns). Darker blue means more alerts; the outlined right-hand column is the current hour and fills as alerts arrive. Read it for patterns: SD-WAN alerts usually peak in the evening, when carriers do maintenance and branches are closed.

Alerts per minute – live

The same severity bars as on the Overview: Critical, Major and Minor after correlation, last 30 minutes.

Tools

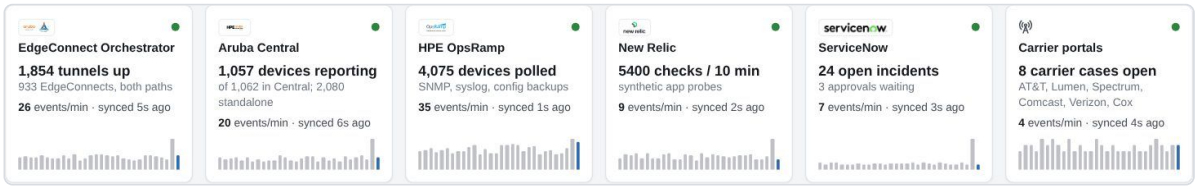


Figure 29. Tool status strip

One tile per source system: EdgeConnect Orchestrator, Aruba Central, HPE OpsRamp, New Relic, ServiceNow and the carrier portals.

Parameter	Meaning
Headline	What the tool holds now: tunnels up, devices reporting, devices polled, synthetic checks every 10 minutes, open incidents and approvals, open carrier cases

Parameter	Meaning
Events/ min	Events arriving from that tool this minute
Synced	Seconds since the last sync
Bars	Events per minute for the last half hour; the blue bar is now

A tool whose bars drop to nothing, or whose sync time grows, is a monitoring gap, not a quiet network.

Branch heartbeat

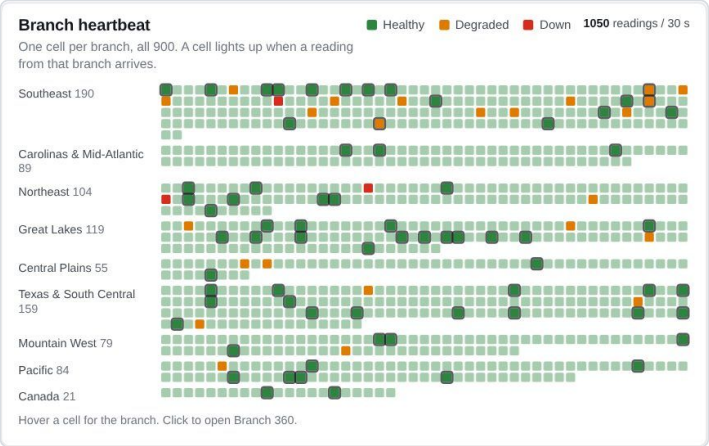


Figure 30. Branch heartbeat

What it shows. One small square per branch, all 900, grouped by region. The colour is health; a square lights up with an outline when a reading from that branch arrives. The counter shows readings per 30 seconds.

How to read it. A healthy estate twinkles evenly. A square that stays dim for a long time is a branch that has stopped reporting, which matters even if it is still green. Hover for the branch; click for Branch 360.

Branch health score distribution

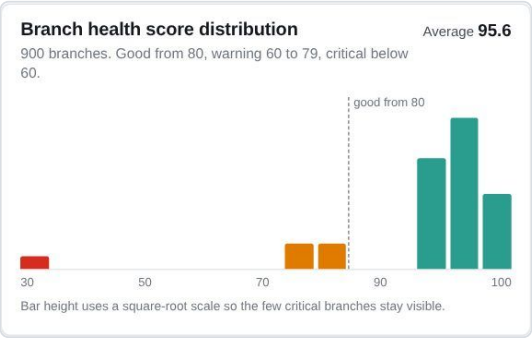


Figure 31. Health score distribution

A histogram of the branch health score, from 30 to 100, for every branch in view, with the average at top right. Bars from 80 up are green-blue (good), 60 to 79 amber (warning) and below 60 red (critical). The dashed line marks "good from 80". Height uses a square-root scale so the few critical branches stay visible. The score combines path health, device health, incidents and recent events; Branch 360 shows how one branch's score is built.

Top branches

The eight worst branches by WAN use, latency, loss or EdgeConnect CPU, as on the Overview.

7. Correlation

Correlation shows how tens of thousands of events become a handful of incidents: one ServiceNow incident per branch per event. Every screen uses the same pipeline:

1. **Events** arrive from Orchestrator, Aruba Central, OpsRamp and New Relic: traps, syslog lines and threshold crossings, about 50 a minute across the estate.
2. **Noise removal** drops duplicates, flaps that recover within the hold-down and information-only messages, usually about 70% of events.
3. What is left are **alerts**. Most clear on their own (**auto-cleared**), warnings go to the morning **digest**, and the rest are **correlated** into incidents.
4. **Incidents** go to ServiceNow, one per branch per event, each with all its alerts as evidence.

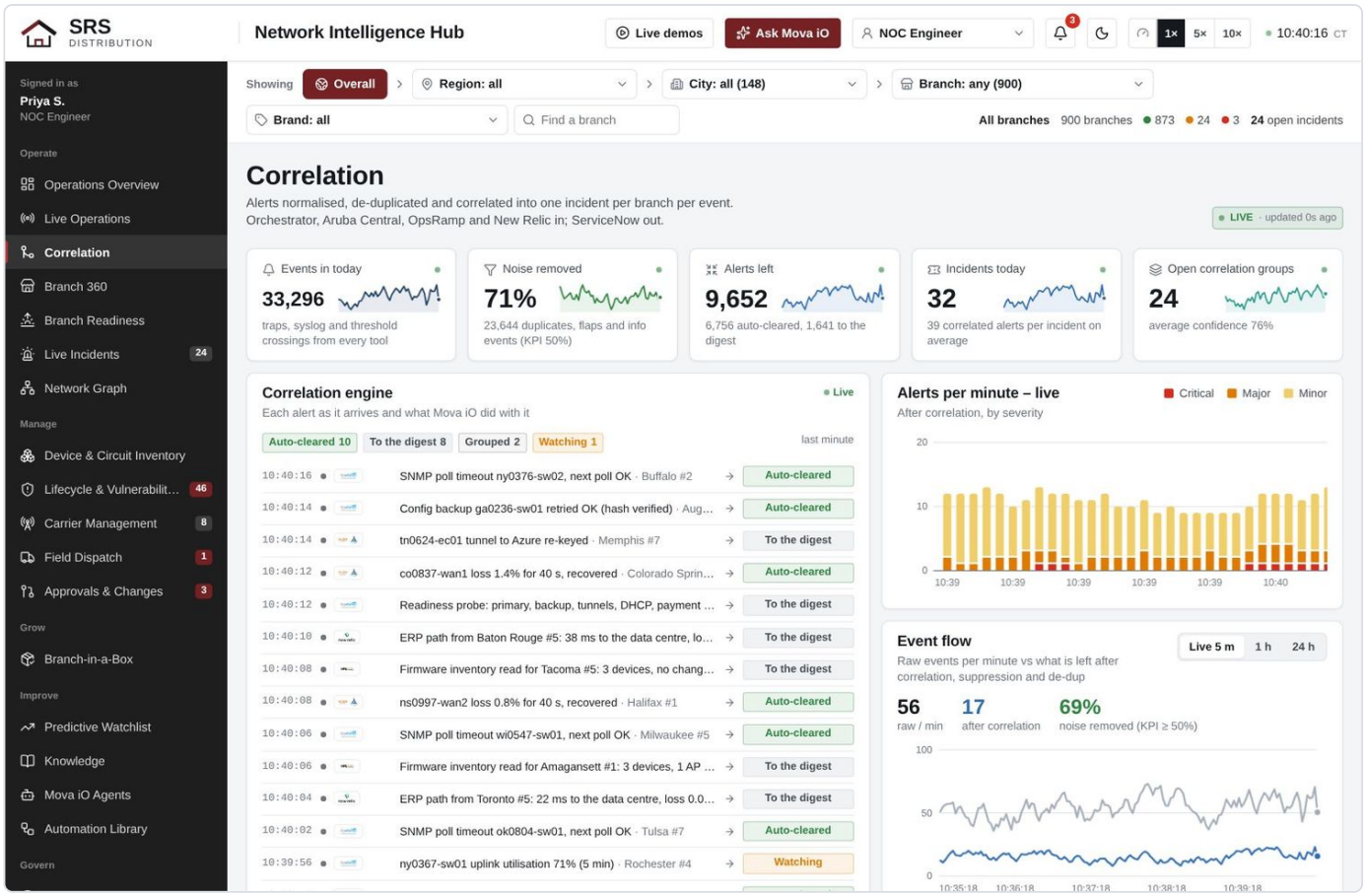


Figure 32. Correlation

Live tiles



Figure 33. Correlation tiles

Tile	Meaning
Events in today	All events received since midnight from every tool
Noise removed	Share of today's events removed before they became alerts, with the count (KPI at least 50%)
Alerts left	Alerts after noise removal, with how many cleared on their own and how many went to the digest
Incidents today	Incidents created today, and the average number of correlated alerts in each
Open correlation groups	Open incidents, each a group of correlated alerts, with their average confidence

The same numbers appear on the Overview (**Noise removed today**) and in the funnel below, so the screens always agree.

Correlation engine

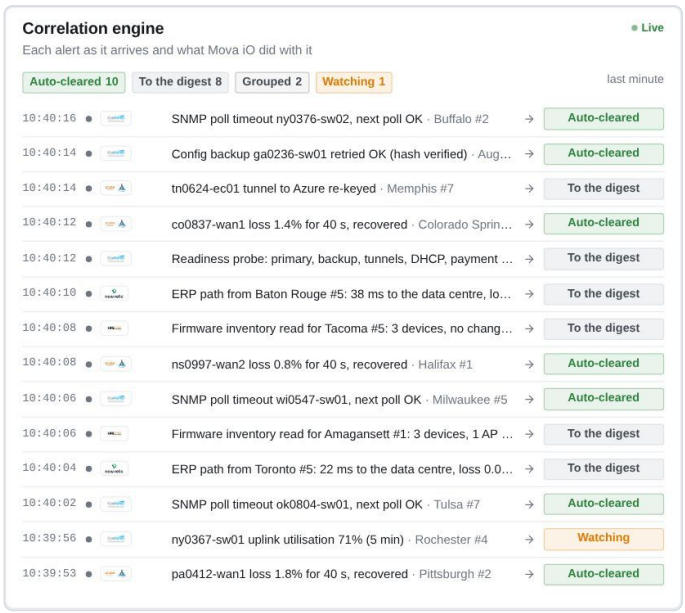


Figure 34. Correlation engine

What it shows. Each alert as it arrives (time, severity dot, source tool, message and branch) and, after the arrow, what Mova iO did with it. The counters above count the last minute.

Outcome	Meaning
Auto-cleared	A transient: checked by ping and SNMP and cleared after the hold-down (for example a 40-second flap)
To the digest	An informational warning, bundled into the daily digest instead of a ticket
Grouped (into nnnn)	Linked to an open incident as one more piece of its evidence
Watching	Held for a re-check before deciding

Click a row to open the incident it was grouped into, or the branch.

How alerts were grouped



Figure 35. How alerts were grouped

One lane per open incident for the last 90 minutes, labelled with the last four digits of the incident and its title. Each dot is an alert grouped into it, coloured by source: Orchestrator, OpsRamp, Central or New Relic. A lane with many colours is an incident confirmed by several tools, which is why its confidence is high. A lane with one dot is a single-signal incident; check it is not a noisy device. Click a lane to open the incident.

From events to incidents



Figure 36. From events to incidents

A funnel of today's counts since 00:00 CT:

Stage	Meaning
Events received	Every trap, syslog line and threshold crossing from the four tools
Alerts	Left after noise removal
Auto-cleared	Alerts that recovered by themselves: checked by ping and SNMP after the hold-down
Rolled to digest	Warnings bundled into the morning digest instead of tickets
Correlated	Alerts grouped by topology, time, change and carrier into incidents
Incidents	ServiceNow incidents, one per branch per event

Bar length uses a square-root scale so the small stages stay visible; read the numbers. The two ratios to watch are events to alerts (noise removed) and correlated alerts per incident.

Alert flow, today

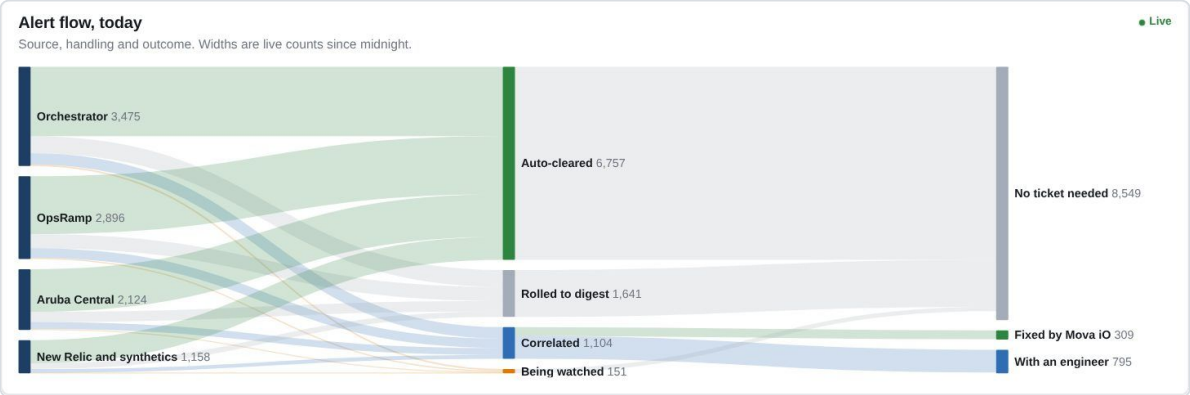


Figure 37. Alert flow

The same day as a flow from left to right. Left: source tools with their alert counts. Middle: how alerts were handled (auto-cleared, rolled to digest, correlated, being watched). Right: the outcome (no ticket needed, fixed by Mova iO, with an engineer). Band widths are live counts since midnight. Use it to explain in one picture why engineers see only a small share of what the tools raise.

Alerts by area and hour

The 24-hour heat map from Live Operations (chapter 6).

Area-to-area correlations

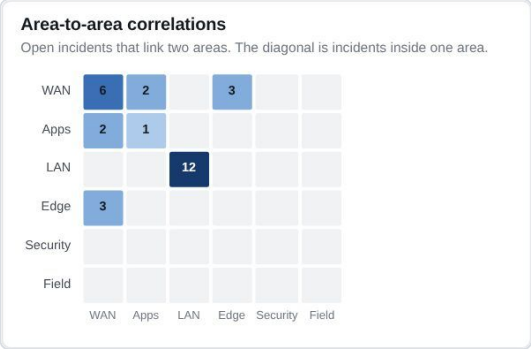


Figure 38. Area-to-area correlations

A grid of open incidents by the area where they started (rows) and the area they affect (columns): WAN, Apps (the path to payments and ERP), LAN, Edge (EdgeConnect), Security and Field. The diagonal counts incidents that stay inside one area; cells off the diagonal count incidents that link two areas, such as a WAN problem slowing the POS path. Darker cells hold more incidents.

Correlation methods

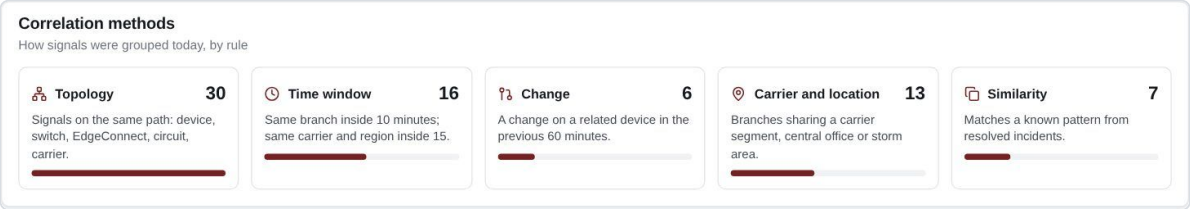


Figure 39. Correlation methods

How today's signals were grouped, by rule, with a count and a bar for each:

Method	Rule
Topology	Signals on the same path: device, switch, EdgeConnect, circuit, carrier
Time window	Same branch inside 10 minutes; same carrier and region inside 15
Change	A change on a related device in the previous 60 minutes
Carrier and location	Branches sharing a carrier segment, central office or storm area
Similarity	Matches a known pattern from resolved incidents

Topology does most of the work. A rising share for "Change" means changes are causing incidents and the change process needs attention.

8. Branch 360

Branch 360 is everything about one branch on one screen: its live metrics, health, risk, who is working on it, its circuits, applications, devices, vulnerabilities and history. Open it by clicking any branch anywhere in the console, by picking a branch in the filter bar, or from **Branch 360** at the end of a demo.

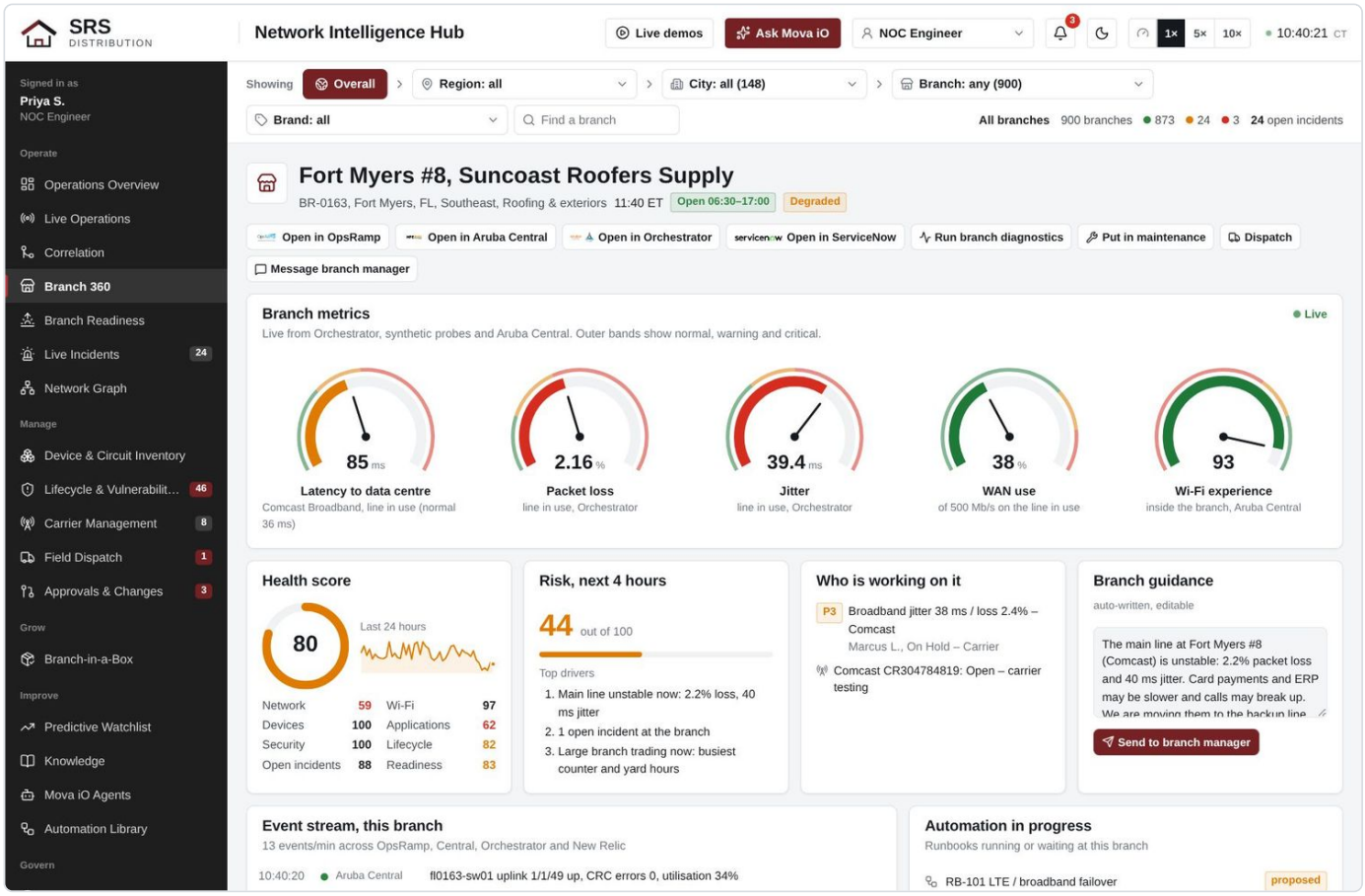


Figure 40. Branch 360

Header and actions

The header gives the branch name and brand, branch ID, city, state, region and line of business, the local time, the opening hours (green when open now) and badges such as **Degraded** or **On backup path**.

Button	What it does
Open in OpsRamp, Aruba Central, Orchestrator, ServiceNow	Opens the same branch in that tool
Run branch diagnostics	Runs the read-only checks: reachability, both lines, tunnels, payment and ERP probes, switch and AP health
Put in maintenance	Holds alerts for planned work so they do not page anyone; recorded in the audit log
Dispatch	Creates a field work order for the branch
Message branch manager	Sends the guidance text below to the branch manager

Branch metrics



Figure 41. Branch metrics

Five live dials. The outer band of each dial shows the normal (green), warning (amber) and critical (red) ranges; the needle and inner arc show the value now. The first four describe the WAN line the branch is using at this moment; the fifth describes Wi-Fi inside the building.

Dial	What it is	Source	Warning	Critical	What the business feels
Latency to data centre (ms)	Round trip from this branch to the McKinney data centre on the line in use. The label shows the line's normal value, which depends on distance and line type	Orchestrator	normal plus 25 ms	normal plus 60 ms	ERP, pricing and order screens slow down
Packet loss (%)	Data lost on the line in use	Orchestrator	above 1%	above 2%	Card payments retry, calls break up, ERP sessions can drop
Jitter (ms)	How much the delay varies on the line in use	Orchestrator	above 20 ms	above 30 ms	Choppy calls; payment terminals time out more often
WAN use (%)	Share of the line's bandwidth in use (for example of 500 Mb/s)	Orchestrator	above 70%	above 85%	A full line queues traffic, which raises latency and loss for everyone
Wi-Fi experience (score out of 100)	Aruba Central's score for the Wi-Fi inside the branch: signal, connection success and roaming for every client	Aruba Central	below 80	below 65	Handhelds and laptops in the office, counter and warehouse struggle

Why the readings move together. When a line is congested or faulty, latency, jitter and loss rise at the same time; the dials are drawn from that one line, so they always tell the same story. Wi-Fi is measured inside the building and does not depend on the WAN, so a branch can show a poor WAN line and good Wi-Fi at the same time. That is normal: staff can connect, but anything that crosses the WAN, such as ERP and payments, is affected.

Normal values. A branch in Texas normally sees 7 to 15 ms to the data centre, one on the east or west coast 35 to 50 ms, Hawaii about 90 ms; an LTE/5G backup adds 25 to 40 ms. Jitter is normally 0.5 to 3 ms on wired lines and 5 to 12 ms on LTE; loss is normally under 0.2%.

Health score, risk, owner and guidance



Figure 42. Health score

Health score is a number from 0 to 100 with its last 24 hours, built from eight parts: Network, Wi-Fi, Devices, Applications, Security, Lifecycle, Open incidents and Readiness. The ring is green from 85, amber from 70 and red below 70; each part is scored the same way, with amber and red parts highlighted. The lowest parts explain a low score; for example Network drops when the main line is unstable, Applications when card payments, ERP or calls are affected, Security when a device has an unpatched critical vulnerability, Lifecycle when devices are past support.



Figure 43. Risk, next 4 hours

Risk, next 4 hours is the chance of a business-impacting problem soon, out of 100, with the top drivers listed in order, for example the main line unstable now, trading on the backup line with no spare, a backup line not passing traffic, a repeat-offender circuit, open incidents, unpatched critical vulnerabilities, a regional storm, or a large branch at its busiest time. Below 25 is low (green), 25 to 49 medium (amber), 50 and above high (red).

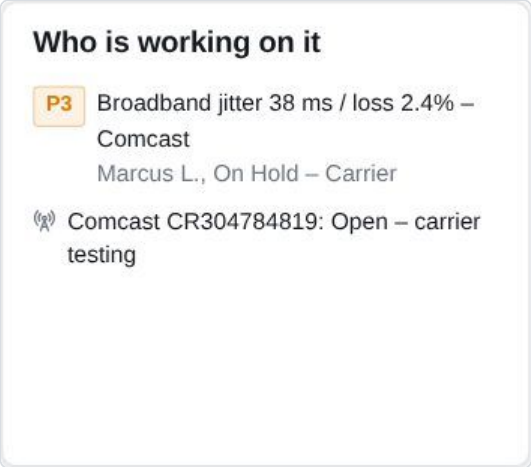


Figure 44. Who is working on it

Who is working on it lists each open incident with its priority, title, assignee and state, and any carrier case with its number and status. Nothing listed means nothing is open.

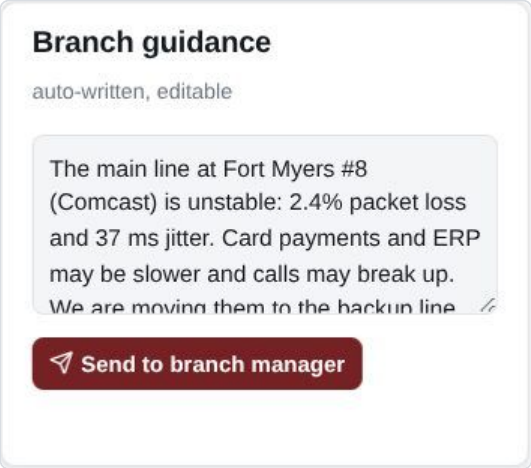


Figure 45. Branch guidance

Branch guidance is a plain-language note for the branch manager, written by Mova iO from the branch's state and matching the dials: down on both lines, running on its backup line, the main line unstable (with the loss and jitter, what staff may notice and that traffic is being moved), or the incidents being worked. It is editable; **Send to branch manager** sends it and records it.

Event stream and automation

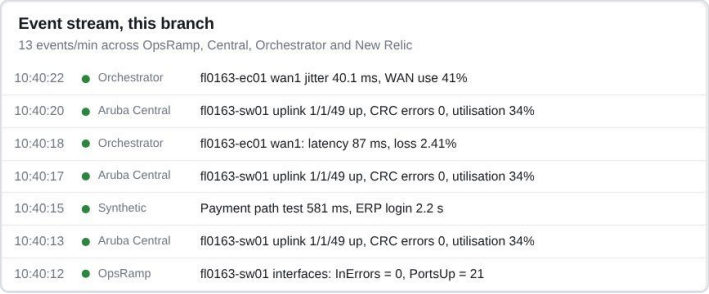


Figure 46. Event stream for this branch

Every reading and event from this branch as it arrives, with time, source and message, and the events per minute in the subtitle. Use it to watch a fix land in real time.

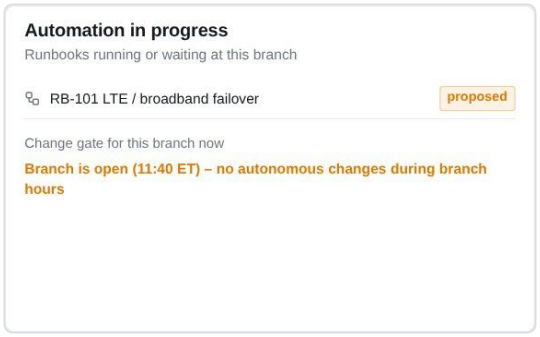


Figure 47. Automation in progress

Runbooks running, proposed or waiting at this branch, with their state, and the **change gate for this branch now**: whether autonomous changes are allowed at this moment, for example "Branch is open – no autonomous changes during branch hours".

Connectivity, applications and network today

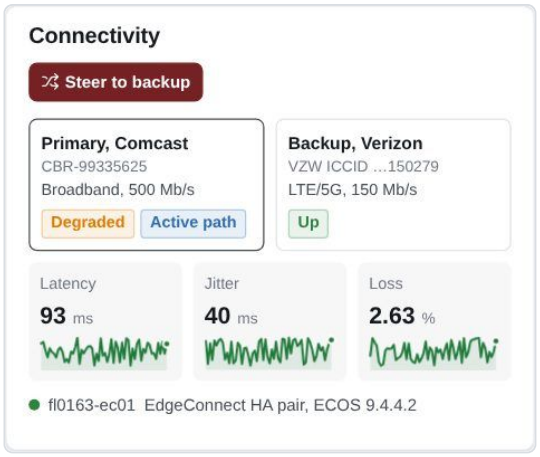


Figure 48. Connectivity

Connectivity shows both circuits side by side: primary and backup, carrier, circuit ID, type and bandwidth, and status (Up, Degraded or Down). The active path has a dark border, with its live latency, jitter and loss. Below are the EdgeConnect name, whether it is a single unit or an HA pair, and its ECOS release. **Steer to backup** moves card payments, ERP and voice to the backup line (an SD-WAN override in Orchestrator); at an open branch it follows the gate rules.

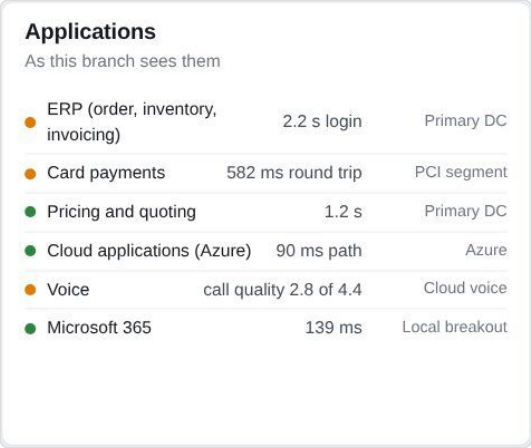


Figure 49. Applications

Applications shows what the business feels, worked out from the line in use: ERP login time (normally about 1 s), card payment round trip from the counter to the payment processor (normally 100 to 200 ms; slow above 400 ms), pricing and quoting, the path to Azure, voice call quality on the 1 to 4.4 scale (good above 4.0, poor below 3.6) and Microsoft 365, each with where it is hosted. An amber dot means that service is slower than it should be at this branch.

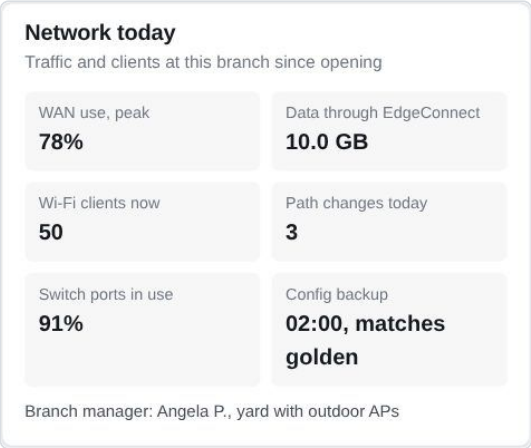


Figure 50. Network today

Network today gives the day since opening: peak WAN use, data through the EdgeConnect, Wi-Fi clients now, path changes, switch ports in use, the last config backup and whether it matches the golden config, and the branch manager's name.

LAN and Wi-Fi

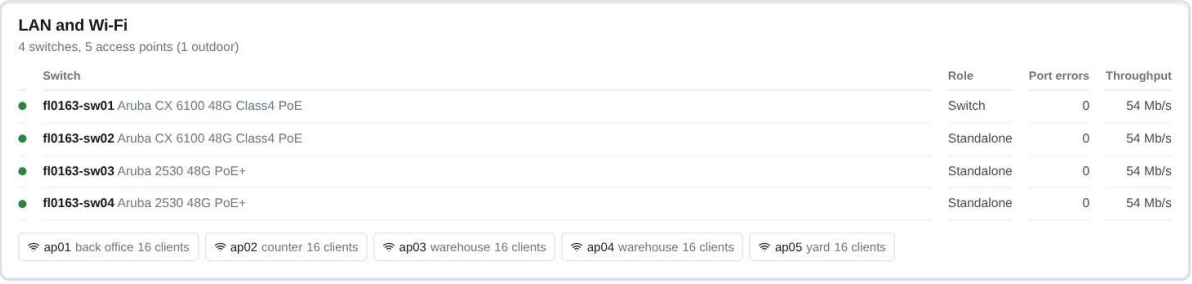


Figure 51. LAN and Wi-Fi

Every switch with model, role (Switch or Standalone), port errors and throughput, then every access point with its zone (back office, counter, warehouse, yard) and clients. Port errors above zero, or an AP with no clients during trading hours, are worth a look.

Dependencies

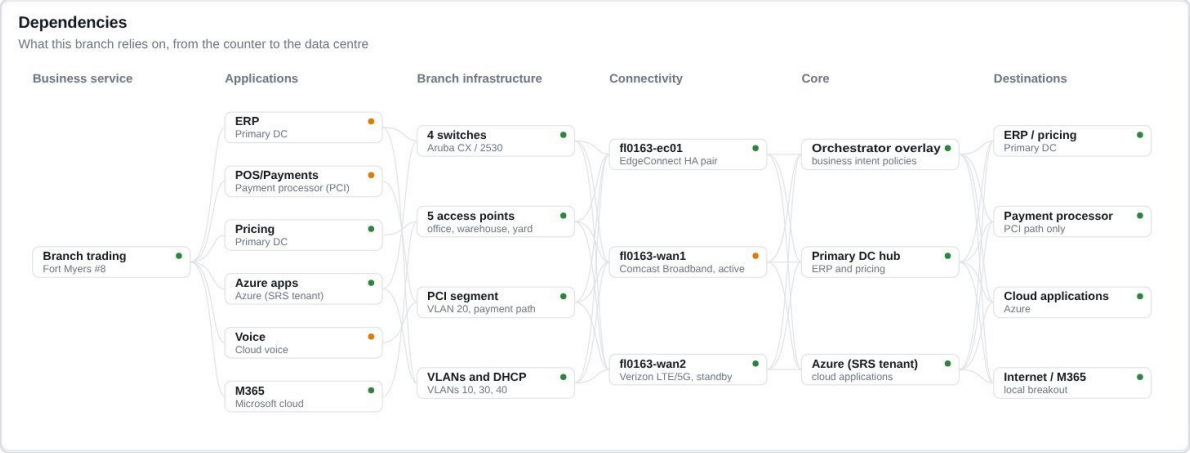
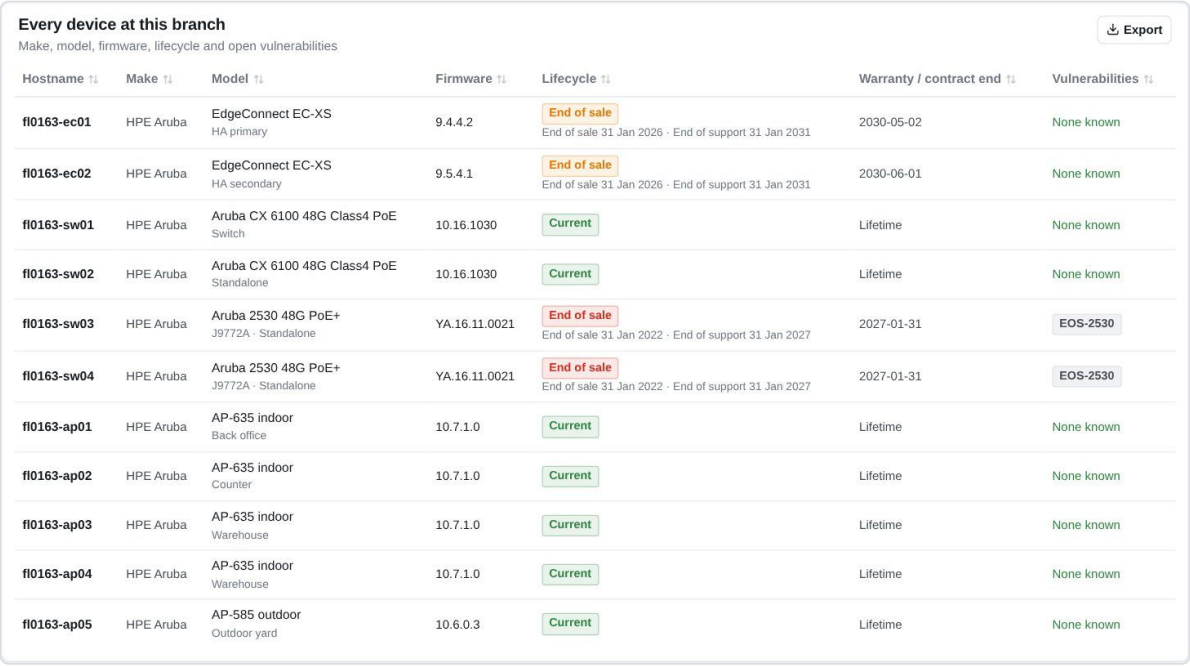


Figure 52. Dependencies

What it shows. Everything the branch relies on, from left to right: the business service (branch trading), the applications, the branch infrastructure (switches, APs, PCI segment, client VLANs and DHCP), connectivity (EdgeConnect, both circuits, the Orchestrator overlay), the core and the destinations (Primary DC, payment processor, Azure, cloud voice, Microsoft cloud). Each box has a status dot.

How to read it. Follow a red or amber dot to the right to see what it affects, or to the left to see what it depends on. A degraded circuit with healthy applications to its right means the backup path is doing its job.

Tab: Devices and lifecycle



Hostname	Make	Model	Firmware	Lifecycle	Warranty / contract end	Vulnerabilities
fl0163-ec01	HPE Aruba	EdgeConnect EC-XS HA primary	9.4.4.2	End of sale End of sale 31 Jan 2026 · End of support 31 Jan 2031	2030-05-02	None known
fl0163-ec02	HPE Aruba	EdgeConnect EC-XS HA secondary	9.5.4.1	End of sale End of sale 31 Jan 2026 · End of support 31 Jan 2031	2030-06-01	None known
fl0163-sw01	HPE Aruba	Aruba CX 6100 48G Class4 PoE Switch	10.16.1030	Current	Lifetime	None known
fl0163-sw02	HPE Aruba	Aruba CX 6100 48G Class4 PoE Standalone	10.16.1030	Current	Lifetime	None known
fl0163-sw03	HPE Aruba	Aruba 2530 48G PoE+ J9772A · Standalone	YA.16.11.0021	End of sale End of sale 31 Jan 2022 · End of support 31 Jan 2027	2027-01-31	EOS-2530
fl0163-sw04	HPE Aruba	Aruba 2530 48G PoE+ J9772A · Standalone	YA.16.11.0021	End of sale End of sale 31 Jan 2022 · End of support 31 Jan 2027	2027-01-31	EOS-2530
fl0163-ap01	HPE Aruba	AP-635 indoor Back office	10.7.1.0	Current	Lifetime	None known
fl0163-ap02	HPE Aruba	AP-635 indoor Counter	10.7.1.0	Current	Lifetime	None known
fl0163-ap03	HPE Aruba	AP-635 indoor Warehouse	10.7.1.0	Current	Lifetime	None known
fl0163-ap04	HPE Aruba	AP-635 indoor Warehouse	10.7.1.0	Current	Lifetime	None known
fl0163-ap05	HPE Aruba	AP-585 outdoor Outdoor yard	10.6.0.3	Current	Lifetime	None known

Figure 53. Every device at this branch

Every device at this branch lists each device: hostname, make, model and role, firmware, lifecycle (Current, End of sale or End of support, with both dates under the badge), warranty or support contract end (Lifetime for switches and APs) and open vulnerabilities with their CVSS score. **Export** downloads the list.

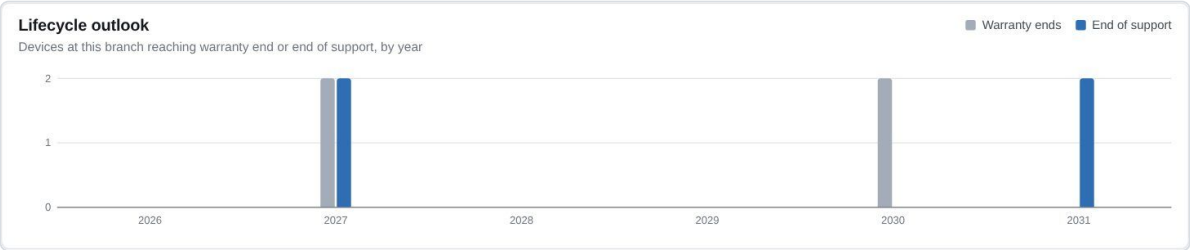


Figure 54. Lifecycle outlook

Lifecycle outlook counts the branch's devices whose warranty or support contract ends, or whose end of support falls, in each year, to plan a refresh.

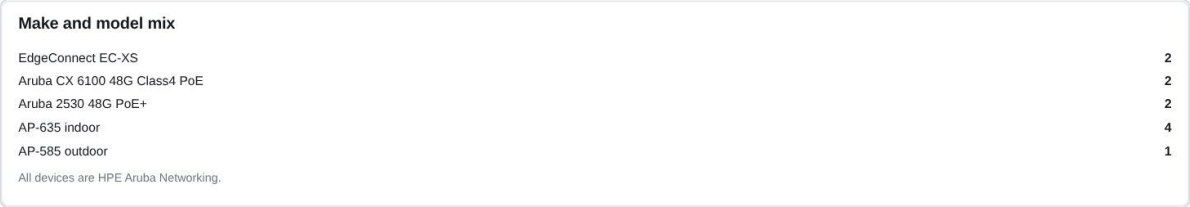


Figure 55. Make and model mix

Make and model mix counts devices by model.

Tab: Vulnerabilities



Figure 56. Vulnerabilities at this branch

Each finding matched from HPE Aruba Networking security bulletins against the running firmware of this branch's devices: severity and CVSS score, title, CVE and bulletin number with date, attack vector, the affected devices here, the fix and the mitigation available now. **Schedule fix after close** creates the change for after trading hours; **Apply mitigation** applies the interim protection (for example restricting management access) and follows the gate rules. Lifecycle risks, such as switches past end of sale, are listed the same way.

Tab: Open work and history

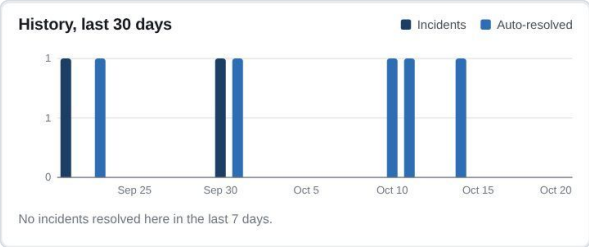


Figure 57. History, last 30 days

Incidents per day for 30 days, with those that were auto-resolved.

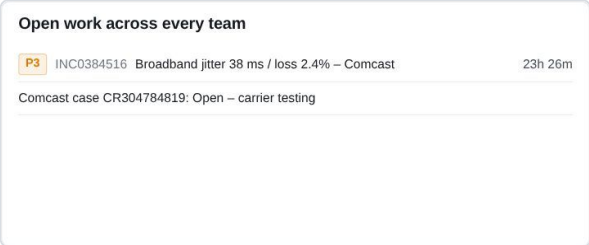


Figure 58. Open work across every team

Everything open for this branch in one list: incidents with time left, carrier cases, work orders and changes.

9. Branch Readiness

Branch Readiness answers "will every branch open on time with a working network?". At 05:30 local time, Mova iO runs ten network checks at every branch. Problems are fixed overnight where the rules allow, or given an owner before the doors open. **Run check now** re-runs the checks for the branches in view; **Export** downloads the results.

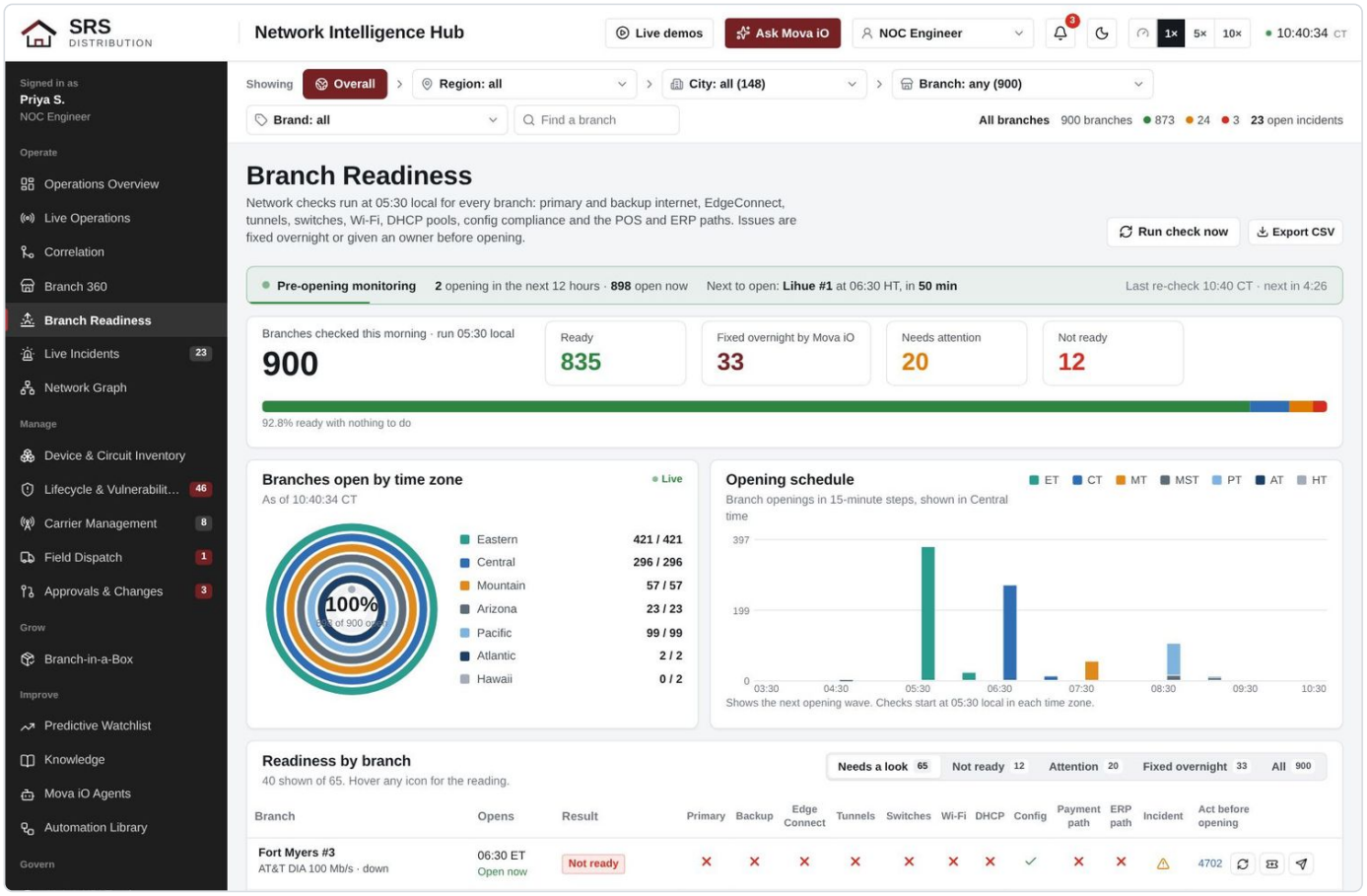


Figure 59. Branch Readiness

This morning's result



Figure 60. Readiness summary

The bar splits every branch checked this morning into four results:

Result	Meaning	Colour
Ready	All ten checks passed	Green
Fixed overnight by Mova iO	A check failed overnight and an approved runbook fixed it before 05:30, with no engineer woken	Blue

Result	Meaning	Colour
Needs attention	The branch can trade, but something is degraded (for example the backup line is down while the primary is clean)	Amber
Not ready	A check that stops normal trading failed (for example no working line, or the payment path over its limit)	Red

The headline "ready with nothing to do" is Ready plus Fixed overnight as a share of all branches.

The ten checks

Check	Passes when	Source
Primary	The primary internet line is up and clean	Orchestrator
Backup	The backup line is up; LTE backups pass their weekly failover test	Orchestrator
EdgeConnect	The EdgeConnect is reachable, its HA state is healthy and its firmware is in policy	Orchestrator
Tunnels	Overlay tunnels to the data centre and Azure are up on the active path	Orchestrator
Switches	All switches are up, uplinks clean, no port errors	Aruba Central
Wi-Fi	All access points are up in office, warehouse and yard	Aruba Central
DHCP	Address pools are below 85% on every branch VLAN	OpsRamp
Config	Last night's config backup was taken and there is no drift from the golden config	OpsRamp
Payment path	A test payment from the counter VLAN (PCI segment) reaches the payment processor and back in under 400 ms	Synthetic probe
ERP path	The path to the data centre is within its latency and loss limits	Synthetic probe

A failed check on the primary line, the EdgeConnect, the tunnels or the payment path makes the branch **Not ready** and raises a P2; other failures raise a P3. A backup line down while the primary is clean is a warning, not a failure.

Branches open by time zone and the opening schedule



Figure 61. Branches open by time zone

How many branches are open now in each time zone (Eastern, Central, Mountain, Arizona, Pacific, Atlantic, Hawaii), with the total share open. Read it with the clock: at 06:00 CT the Eastern wave is open and the Pacific wave is still hours away.

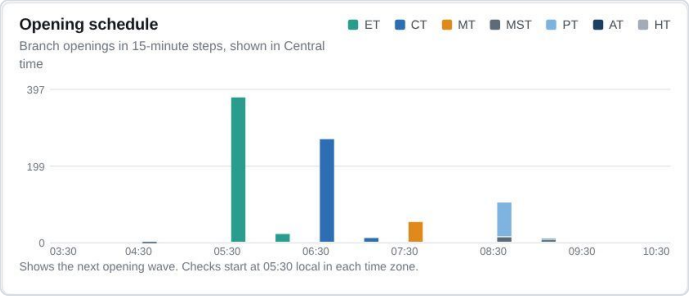


Figure 62. Opening schedule

Branch openings in 15-minute steps, shown in Central Time and coloured by time zone. Checks start at 05:30 local in each zone, so the NOC works the problems in waves: east first, west last.

Readiness by branch

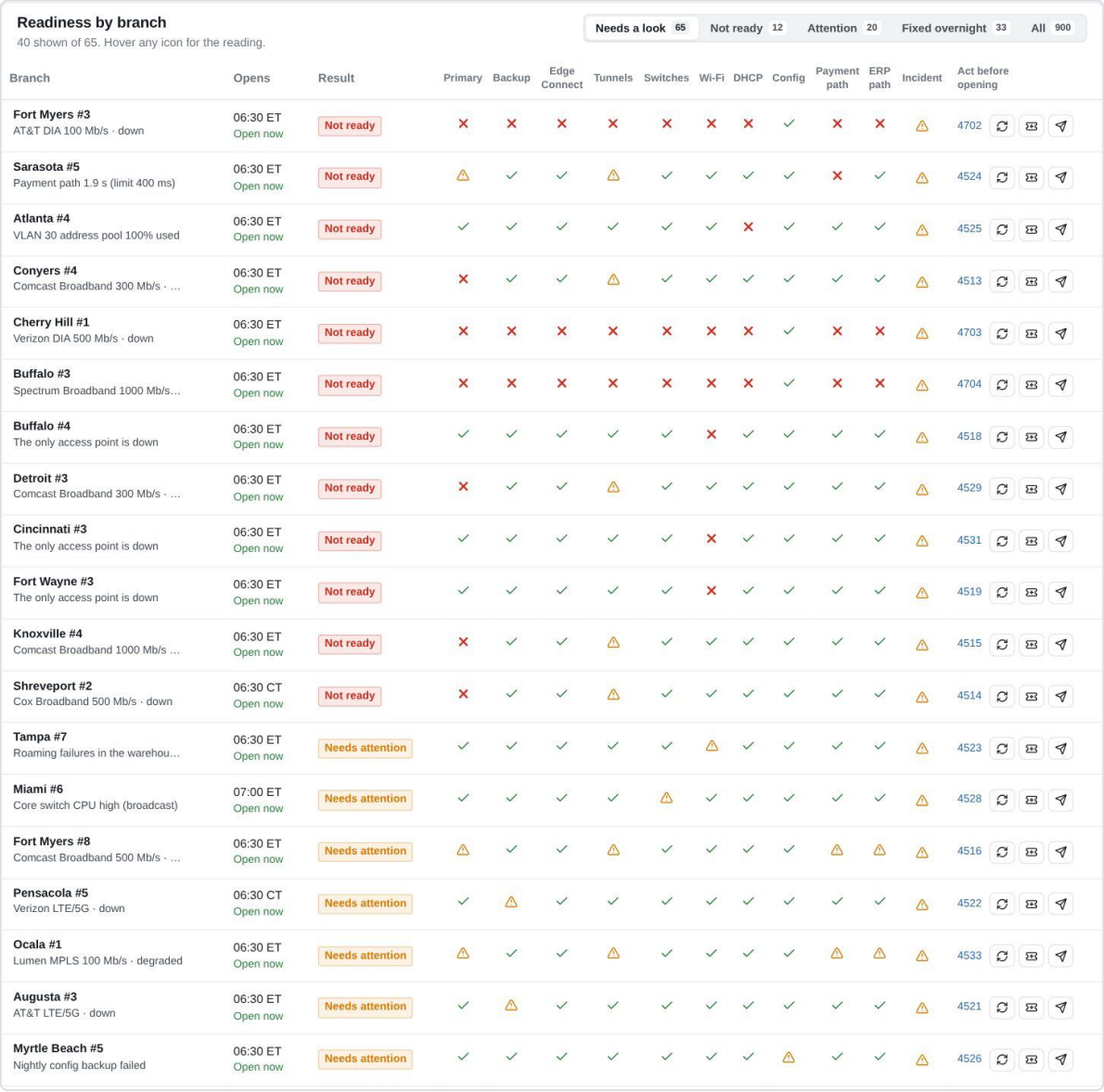


Figure 63. Readiness by branch

What it shows. One row per branch with a problem, worst first. The tabs choose the rows: **Needs a look** (everything not plainly ready), **Not ready**, **Attention**, **Fixed overnight** and **All**.

Column	Meaning
Branch	Name, with the first failing reading under it
Opens	Local opening time and "Open now" once trading
Result	Ready, Fixed overnight, Needs attention or Not ready
Ten check columns	A green tick, amber warning, red cross or blue wrench (fixed overnight). Hover an icon for the reading
Incident	The incident raised for the failure, if any
Act before opening	**Re-run** the checks or **Raise** an incident; "No action" when ready

Show 40 more loads further rows. Click a branch name for Branch 360.

Pre-opening checks



Figure 64. Pre-opening checks

The live stream of readiness readings: probes of the primary and backup lines, tunnels, DHCP and the payment and ERP paths, with the branch and the result. The subtitle shows readings per minute.

Fixed overnight

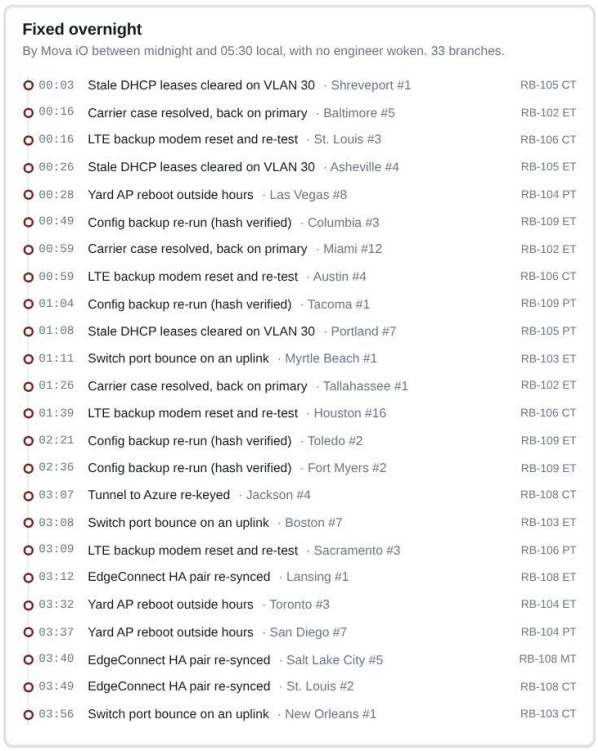


Figure 65. Fixed overnight

What Mova iO fixed between midnight and 05:30 local, with no engineer woken: time, what was fixed, branch, the runbook used and the time zone. Typical fixes are an LTE modem reset and re-test (RB-106), a yard AP reboot outside hours (RB-104), a config backup re-run (RB-109), stale DHCP leases cleared (RB-105), a tunnel to Azure re-keyed (RB-108) and a return to the primary after a carrier fix (RB-102). These count as Ready at opening.

Ready at opening, 14 days

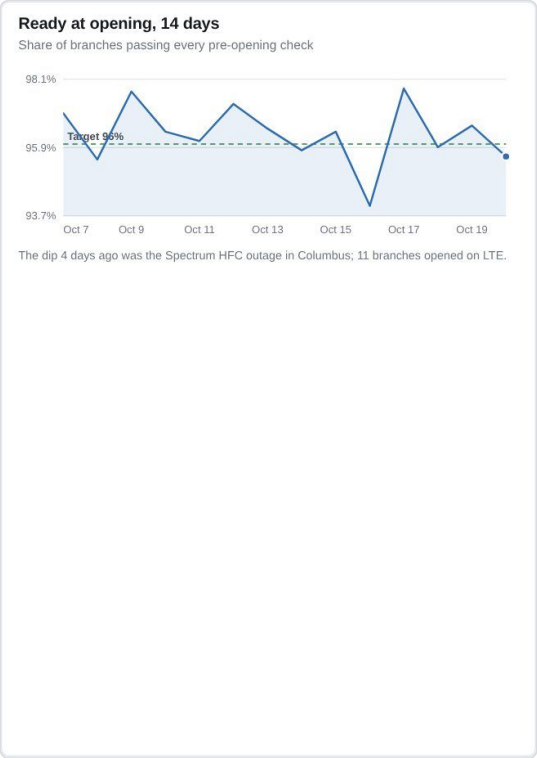


Figure 66. Ready at opening, 14 days

The share of branches that passed every check at opening, each day for 14 days, against the 96% target line. The note explains any dip, for example a carrier outage that sent a group of branches to LTE.

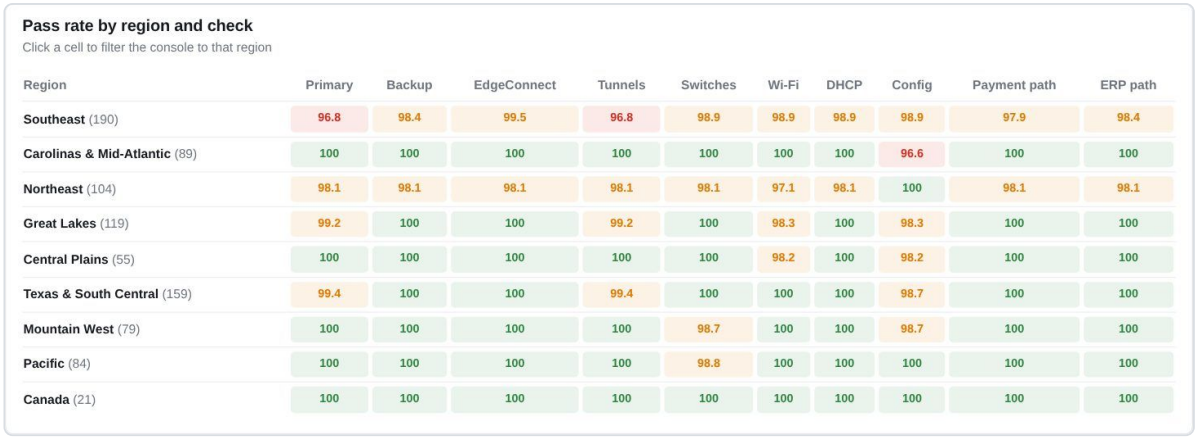
Tomorrow's opening risk



Figure 67. Tomorrow's opening risk

Branches the model expects to fail a check tomorrow morning, with the probability, the reason and the preventive step to take tonight. **Schedule** books that step. This is where tomorrow's "Not ready" branches are prevented.

Pass rate by region and check



Region	Primary	Backup	EdgeConnect	Tunnels	Switches	Wi-Fi	DHCP	Config	Payment path	ERP path
Southeast (190)	96.8	98.4	99.5	96.8	98.9	98.9	98.9	98.9	97.9	98.4
Carolinas & Mid-Atlantic (89)	100	100	100	100	100	100	100	96.6	100	100
Northeast (104)	98.1	98.1	98.1	98.1	98.1	97.1	98.1	100	98.1	98.1
Great Lakes (119)	99.2	100	100	99.2	100	98.3	100	98.3	100	100
Central Plains (55)	100	100	100	100	100	98.2	100	98.2	100	100
Texas & South Central (159)	99.4	100	100	99.4	100	100	100	98.7	100	100
Mountain West (79)	100	100	100	100	98.7	100	100	98.7	100	100
Pacific (84)	100	100	100	100	98.8	100	100	100	100	100
Canada (21)	100	100	100	100	100	100	100	100	100	100

Figure 68. Pass rate by region and check

The share of branches passing each check, by region. Green cells are 100%, amber 97% or more, red below 97%. A red column across many regions points at a common cause, such as a carrier or a template; a red row points at one region. Click a cell to filter the console to that region.

10. Live Incidents

Live Incidents is the NOC queue: every open incident with its SLA clocks, and the full record of each one. It is the NOC Engineer's landing page.

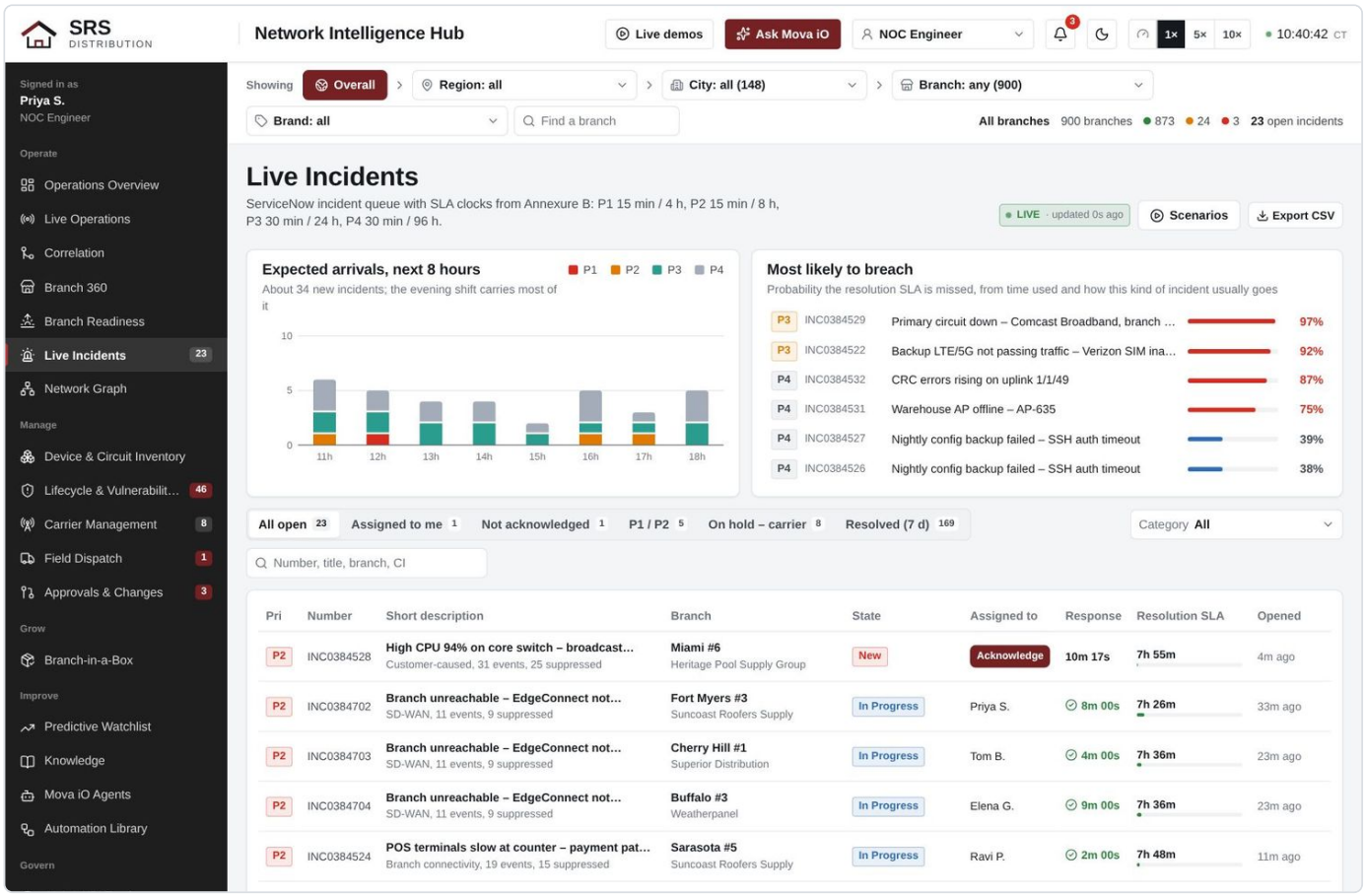


Figure 69. Live Incidents

Expected arrivals and breach risk



Figure 70. Expected arrivals, next 8 hours

New incidents expected per hour for the next eight hours, stacked by priority (P1 red, P2 orange, P3 blue, P4 grey), with the total in the subtitle. Use it to staff the next shift.

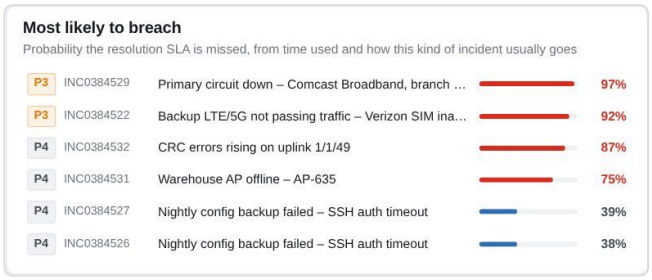


Figure 71. Most likely to breach

Open incidents ranked by the chance their resolution SLA is missed, worked out from the time already used and how this kind of incident usually goes. Anything above 70% needs action now: escalate, chase the carrier or dispatch.

The queue

Pri	Number	Short description	Branch	State	Assigned to	Response	Resolution SLA	Opened
P2	INC0384528	**High CPU 94% on core switch – broadcast…** Customer-caused, 31 events, 25 suppressed	**Miami #6** Heritage Pool Supply Group	New	Acknowledge	10m 15s	7h 55m	4m ago
P2	INC0384702	**Branch unreachable – EdgeConnect not…** SD-WAN, 11 events, 9 suppressed	**Fort Myers #3** Suncoast Roofers Supply	In Progress	Priya S.	8m 00s	7h 26m	33m ago
P2	INC0384703	**Branch unreachable – EdgeConnect not…** SD-WAN, 11 events, 9 suppressed	**Cherry Hill #1** Superior Distribution	In Progress	Tom B.	4m 00s	7h 36m	23m ago
P2	INC0384704	**Branch unreachable – EdgeConnect not…** SD-WAN, 11 events, 9 suppressed	**Buffalo #3** Weatherpanel	In Progress	Elena G.	9m 00s	7h 36m	23m ago
P2	INC0384524	**POS terminals slow at counter – payment pat…** Branch connectivity, 19 events, 15 suppressed	**Sarasota #5** Suncoast Roofers Supply	In Progress	Ravi P.	2m 00s	7h 48m	11m ago
P3	INC0384529	**Primary circuit down – Comcast Broadband,…** Carrier / LEC, 17 events, 14 suppressed	**Detroit #3** Wimsatt Building Materials	On Hold – Carrier	Neha R.	3m 00s	-1h 00m	1d ago
P3	INC0384522	**Backup LTE/5G not passing traffic – Verizon…** Carrier / LEC, 1 events, 1 suppressed	**Pensacola #5** Suncoast Roofers Supply	On Hold – Carrier	Tom B.	2m 00s	2h 19m	21h 40m ago
P3	INC0384521	**Backup LTE/5G not passing traffic – AT&T SI…** Carrier / LEC, 1 events, 1 suppressed	**Augusta #3** Superior Distribution	On Hold – Carrier	Dana W.	3m 00s	17h 19m	6h 40m ago
P3	INC0384515	**Primary circuit down – Comcast Broadband,…** Carrier / LEC, 17 events, 14 suppressed	**Knoxville #4** Superior Distribution	On Hold – Carrier	Ravi P.	9m 00s	19h 39m	4h 20m ago
P3	INC0384530	**Wi-Fi roaming failures in warehouse aisles** Wireless, 6 events, 5 suppressed	**Minneapolis #10** American Roofing Supply	In Progress	Dana W.	7m 00s	21h 39m	2h 20m ago
P3	INC0384513	**Primary circuit down – Comcast Broadband,…** Carrier / LEC, 17 events, 14 suppressed	**Conyers #4** SRS Distribution Inc.	On Hold – Carrier	Neha R.	3m 00s	22h 27m	1h 32m ago
P3	INC0384525	**DHCP scope exhausted on VLAN 30…** Customer-caused, 4 events, 3 suppressed	**Atlanta #4** Heritage Landscape Supply Group	In Progress	Sanjay K.	5m 00s	22h 44m	1h 15m ago
P3	INC0384523	**Wi-Fi roaming failures in warehouse aisles** Wireless, 6 events, 5 suppressed	**Tampa #7** Suncoast Roofers Supply	In Progress	Dana W.	8m 00s	23h 07m	52m ago
P3	INC0384514	**Primary circuit down – Cox Broadband, branc…** Carrier / LEC, 17 events, 14 suppressed	**Shreveport #2** SRS Distribution Inc.	On Hold – Carrier	Ravi P.	5m 00s	23h 18m	41m ago
P3	INC0384516	**Broadband jitter 38 ms / loss 2.4% – Comcast** Carrier / LEC, 24 events, 19 suppressed	**Fort Myers #8** Suncoast Roofers Supply	On Hold – Carrier	Marcus L.	4m 00s	23h 26m	33m ago
P3	INC0384533	**MPLS jitter 38 ms / loss 2.4% – Lumen** Carrier / LEC, 24 events, 19 suppressed	**Ocala #1** Sunniland	On Hold – Carrier	Dana W.	6m 00s	23h 50m	9m ago
P4	INC0384532	**CRC errors rising on uplink 1/1/49** Switching, 2 events, 2 suppressed	**Phoenix #12** Heritage Pool Supply Group	In Progress	Ravi P.	8m 00s	1d 16h	2d ago
P4	INC0384531	**Warehouse AP offline – AP-635** Wireless, 3 events, 2 suppressed	**Cincinnati #3** Marsh Building Products	In Progress	Dana W.	6m 00s	1d 23h	2d ago
P4	INC0384519	**Warehouse AP offline – AP-635** Wireless, 3 events, 2 suppressed	**Fort Wayne #3** Architectural Metal Roofing Supply	In Progress	Dana W.	8m 00s	3d 13h	10h 10m ago
P4	INC0384527	**Nightly config backup failed – SSH auth timeout** Config management, 0 events	**Kansas City #3** Arrowhead Building Supply	In Progress	Tom B.	5m 00s	3d 15h	8h 0m ago

1–20 of 23 Page 1 / 2

Figure 72. Incident queue

The tabs filter the queue: **All open**, **Assigned to me**, **Not acknowledged**, **P1 / P2**, **On hold – carrier** and **Resolved (7 d)**.

Column	Meaning
Pri	P1 to P4
Number	The ServiceNow incident number
Short description	Title, then category, the number of events correlated into it and how many were suppressed
Branch	Branch and brand
State	New, In Progress, On Hold – Carrier, Resolved
Assigned to	The engineer or agent, or an **Acknowledge** button
Response	Time left to acknowledge (15 or 30 minutes by priority), or the time taken
Resolution SLA	Time left to resolve; amber past 75% used; negative and red once breached; "Met" when resolved in time
Opened	How long ago

Scenarios replays a scripted event into the queue; **Export CSV** downloads it. Click a row to open the incident.

When incidents happen

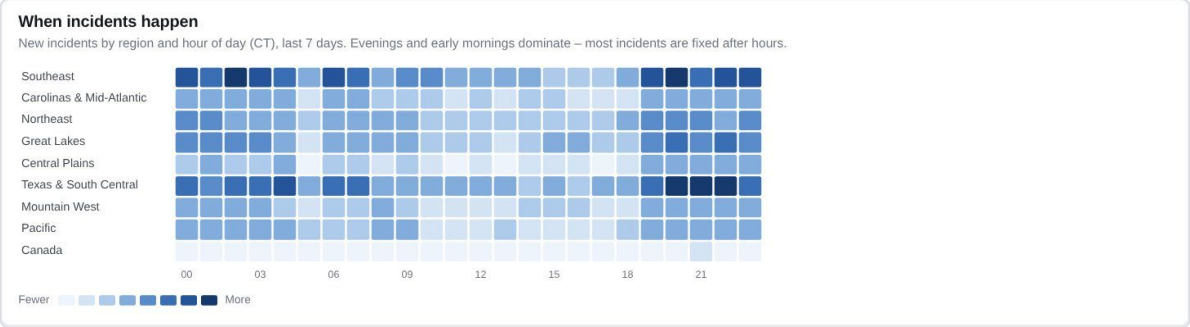


Figure 73. When incidents happen

New incidents by region (rows) and hour of day in CT (columns) over the last seven days; darker means more. Evenings and early mornings dominate because most fixes are made after branches close and carriers work at night.

The incident record

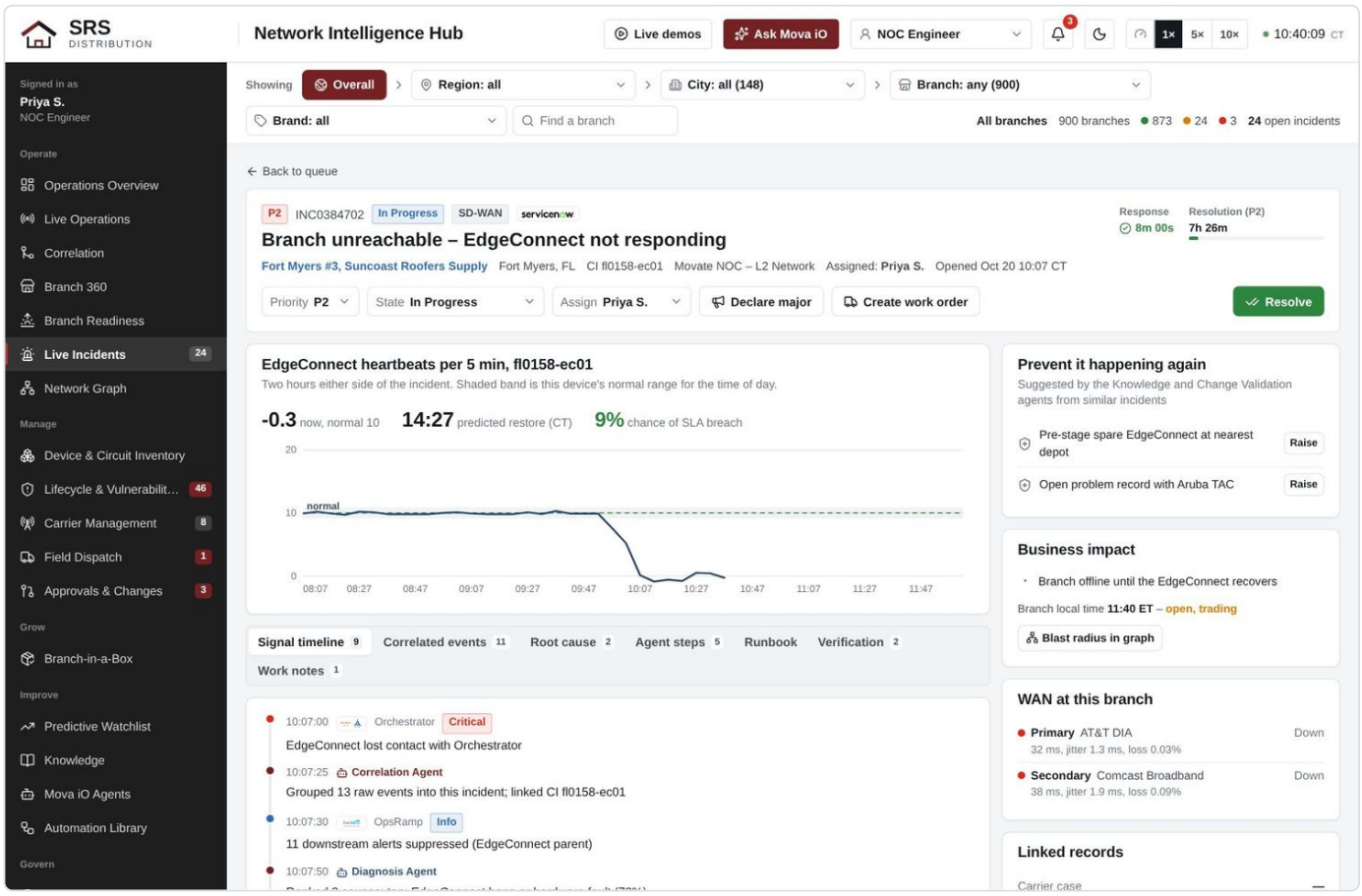


Figure 74. Incident detail

The top of the record gives the priority, number, state, category, the ServiceNow link, the title, branch, city, CI, assignment group, assignee and opening time, with the response and resolution clocks on the right.

Action	What it does
Acknowledge & take	Stops the response clock and assigns the incident to you
Priority, State, Assign	Change them; every change is in the audit log
Declare major	Makes it a major incident with a bridge and leadership updates
Create work order	Opens a field visit with skills and parts
Resolve	Closes with a close code once verification passes

Under the actions:

- **Device chart:** the key metric of the CI for two hours either side of the incident, with the device's normal band for that time of day, the value now, the predicted restore time and the chance of an SLA breach.
- **Prevent it happening again:** fixes suggested by the Knowledge and Change Validation agents from similar incidents; **Raise** creates the change or task.
- **Business impact:** what the branch feels and whether it is open and trading; **Blast radius in graph** opens the Network Graph on this CI.

- **WAN at this branch:** both circuits with latency, jitter, loss and status.
- **Linked records:** carrier case, work order, change and problem.

The record tabs

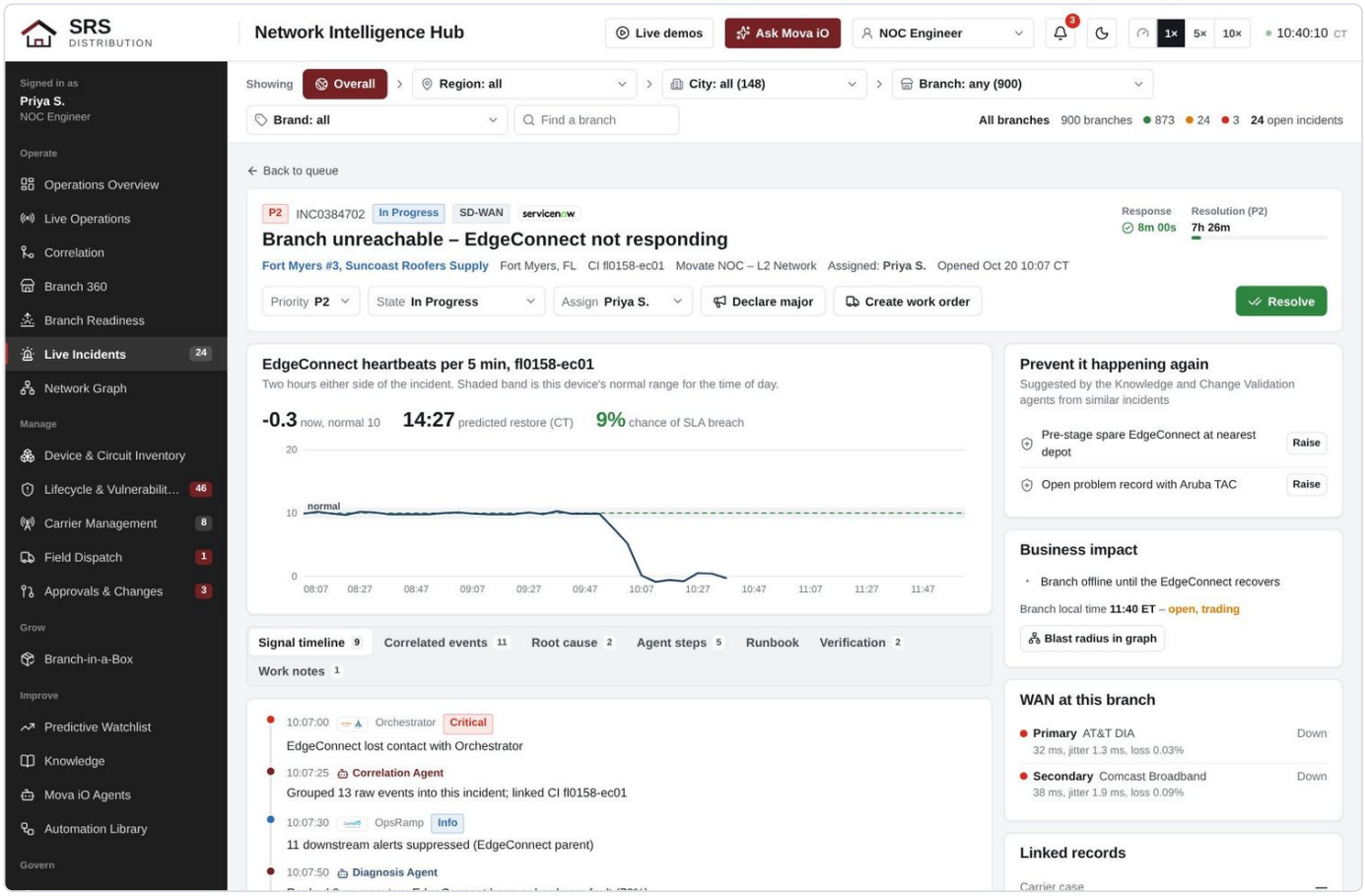


Figure 75. Signal timeline

Tab	What it holds
Signal timeline	Every signal, agent step and note in time order, with source and severity
Correlated events	All raw events grouped into this incident and which were suppressed
Root cause	Ranked causes, each with its confidence and the evidence for it
Agent steps	What each Mova iO agent did, in order
Runbook	The recommended runbook, its steps, autonomy level, gate result, and **Run**, **Send to approvals queue** or **Rollback**
Verification	The checks that must pass before the incident can close, with values
Work notes	Notes from engineers and agents; **Post** adds one

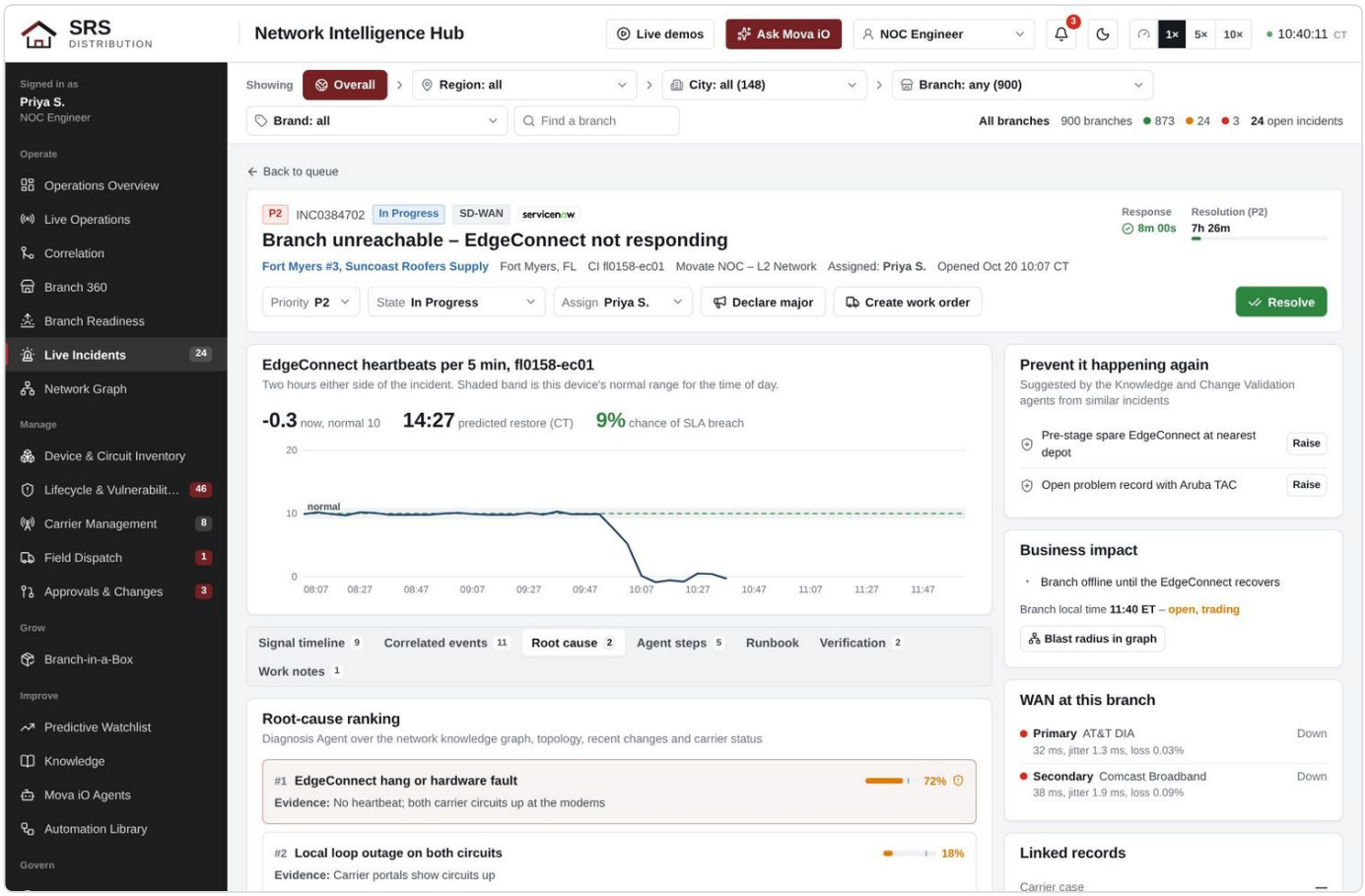


Figure 76. Root cause tab

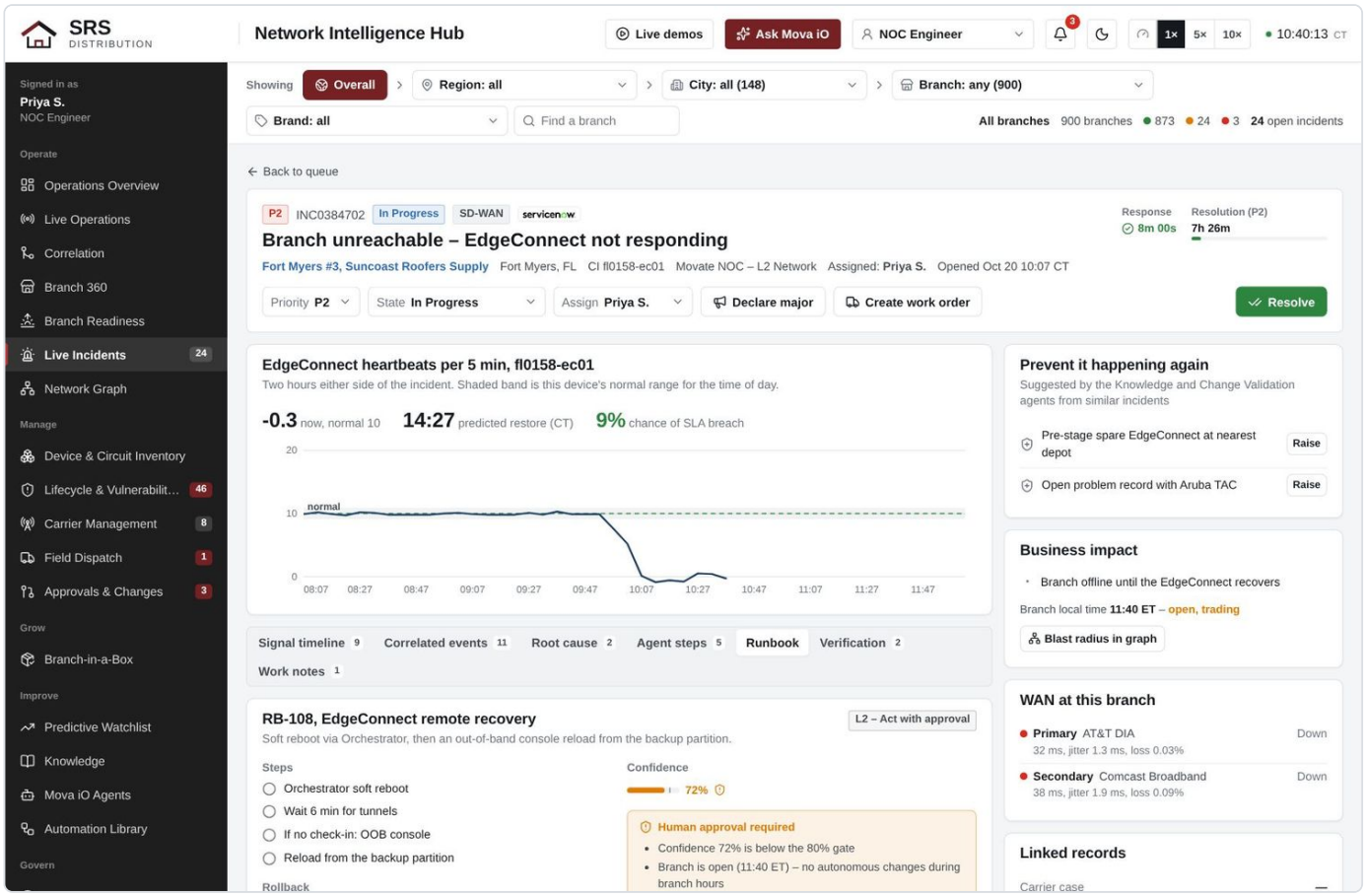


Figure 77. Runbook tab

How to read the root cause. The top cause carries the confidence that drives the gate rule: 80% or more allows the runbook to run within its tier, below 80% sends it to a person. The lower causes show what was considered and why it is less likely.

11. Network Graph

The Network Graph shows how each branch reaches the business: its LAN and EdgeConnect, both WAN lines, the carrier networks, the hosting points and the applications. It is built from OpsRamp, Orchestrator and the ServiceNow CMDB. Click anything to see what depends on it.

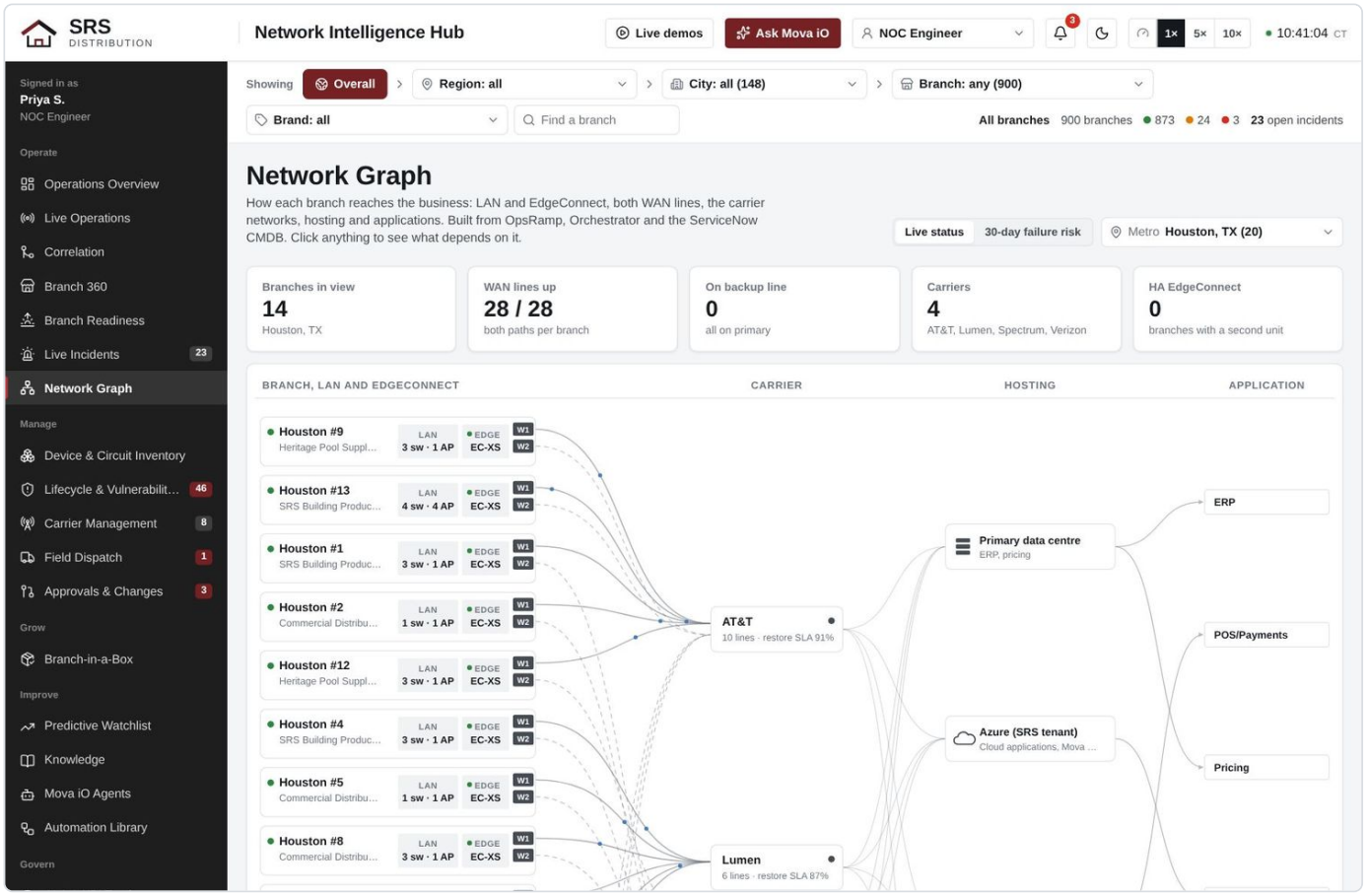


Figure 78. Network Graph

Controls

Control	What it does
Live status / 30-day failure risk	Colours the graph by health now, or by each element's predicted chance of failing in the next 30 days
Metro	Picks the metro area to draw, with its branch count. The graph draws up to the first 14 branches in that metro; the filter bar narrows further

Tiles

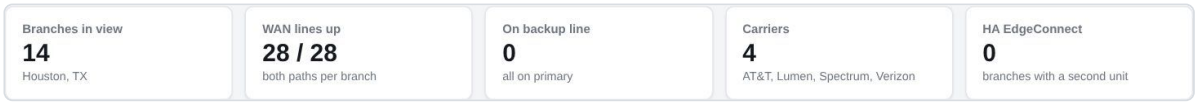


Figure 79. Graph tiles

Tile	Meaning
Branches in view	Branches drawn, and the metro
WAN lines up	Lines up out of all lines drawn (two per branch)
On backup line	Branches running on their second line now
Carriers	Carriers serving these branches
HA EdgeConnect	Branches with a second EdgeConnect unit

Reading the graph

The graph runs left to right in four columns.

Column	Nodes	What to look at
Branch, LAN and EdgeConnect	One card per branch: name, brand, the LAN (switches and APs), the EdgeConnect model, and the two WAN ports W1 and W2	The dot by the name is the branch health; W1 and W2 turn amber or red when a line is degraded or down
Carrier	One node per carrier, with the number of lines and its restore SLA over six months	A carrier with many lines and a low restore SLA is a concentration risk
Hosting	Primary data centre (ERP, pricing) and Azure (SRS tenant: cloud applications and Mova iO)	–
Application	ERP, POS and payments, pricing and the other business services	–

Solid lines are primary circuits, dashed lines are backups. Moving dots on a line show traffic. Branches are sorted by carrier so lines do not cross more than they must, and each carrier sits next to the branches it serves.

What depends on it



Figure 80. What depends on it

Click a carrier to see which branches would go down (both lines on that carrier) and which would fail over (only one line on it). Click a branch's LAN, EdgeConnect or a W1 or W2 port to see what depends on that element. This is the **blast radius**: the set of branches and services affected if that one element fails. It is the same blast radius Mova iO checks before any change.

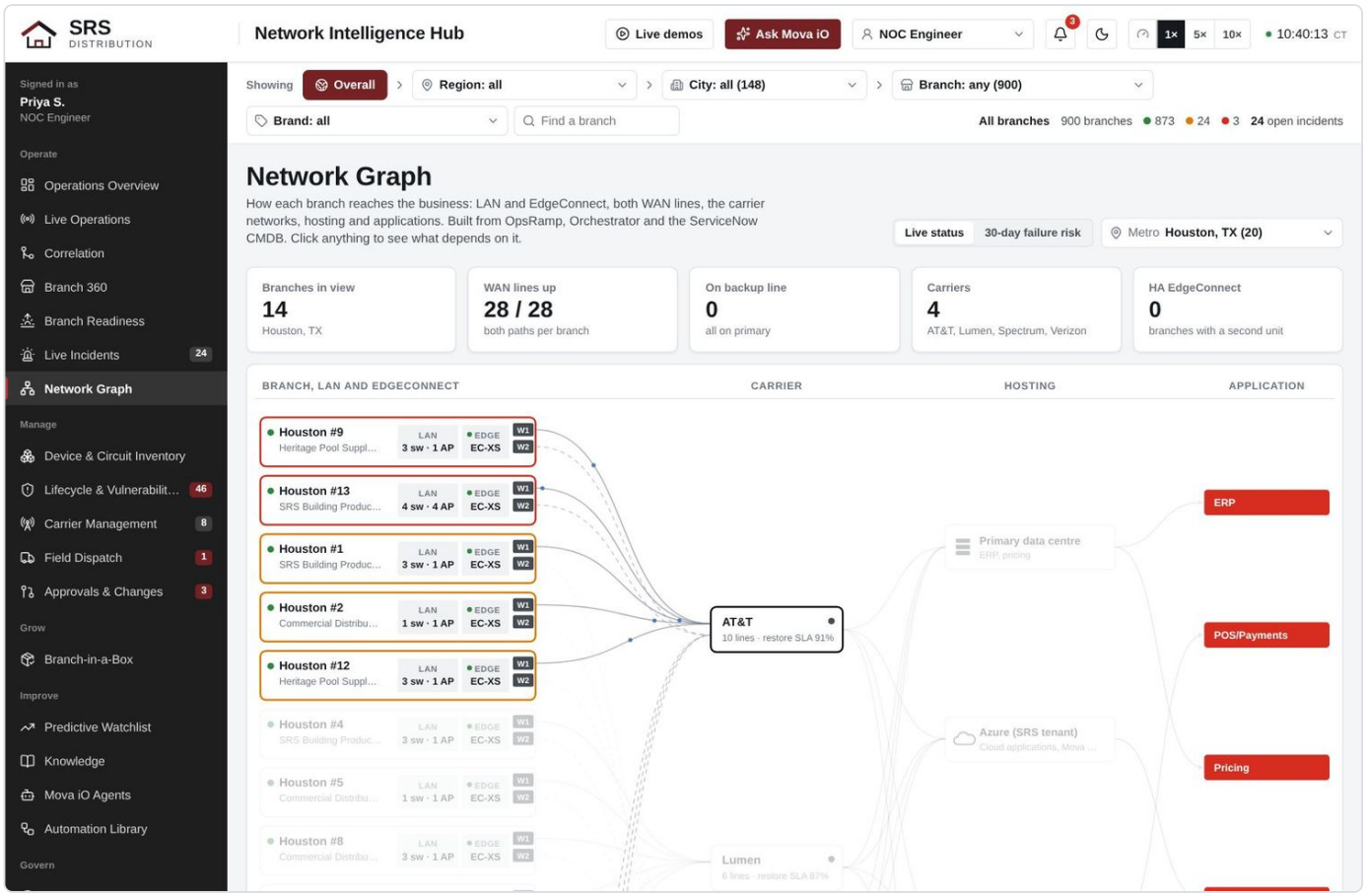


Figure 81. Blast radius after clicking a carrier

Line detail

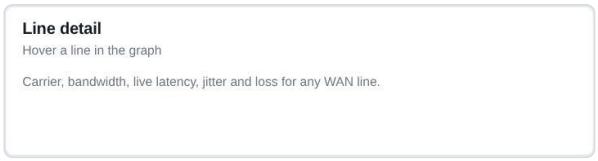


Figure 82. Line detail

Hover any line for its carrier, bandwidth and live latency, jitter and loss.

30-day failure risk

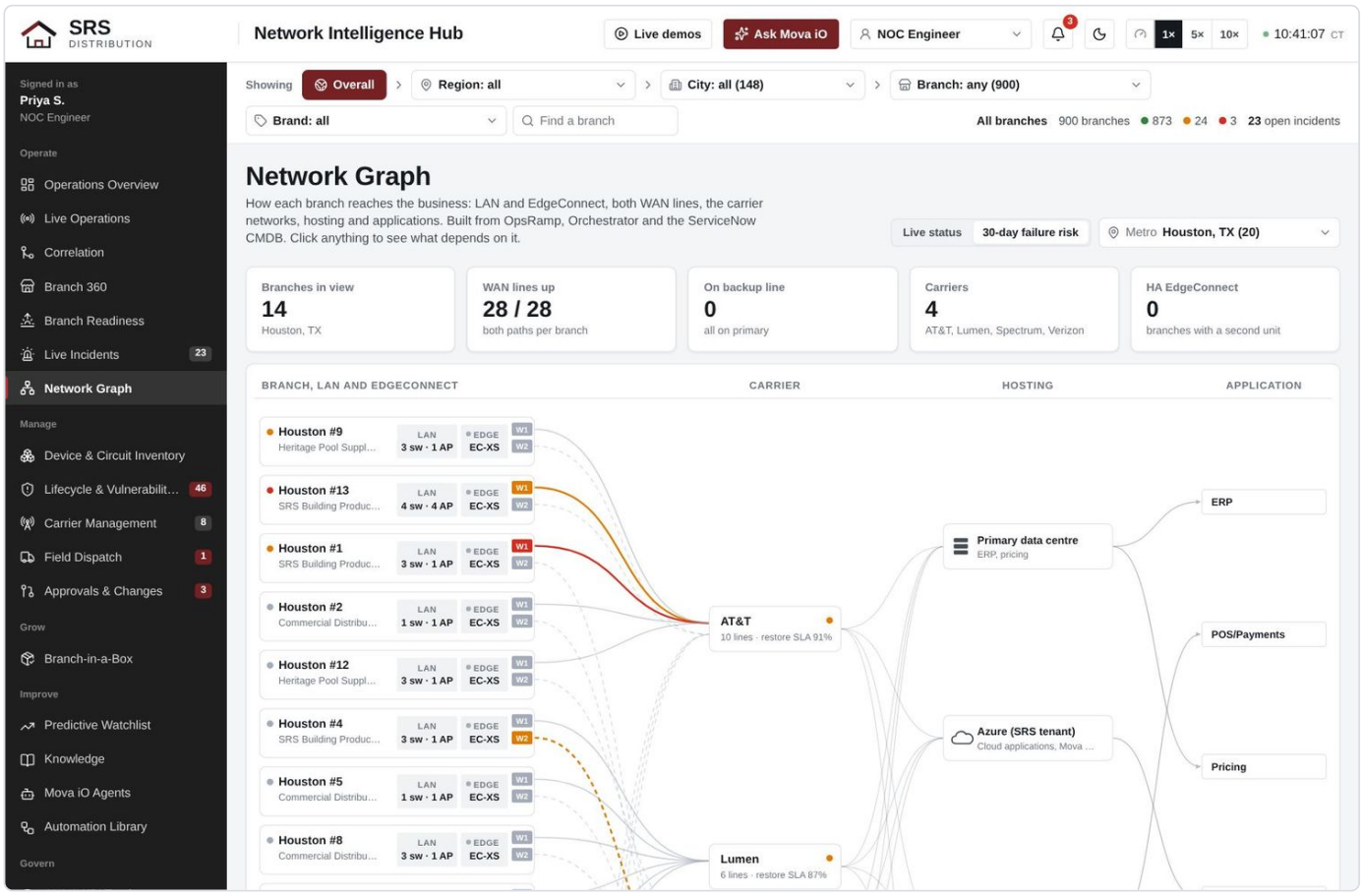


Figure 83. 30-day failure risk

The same graph coloured by predicted risk instead of health. Use it before a change window or a busy season: the red elements are where to raise a proactive case, pre-stage a spare or test a failover first.

Network activity – live

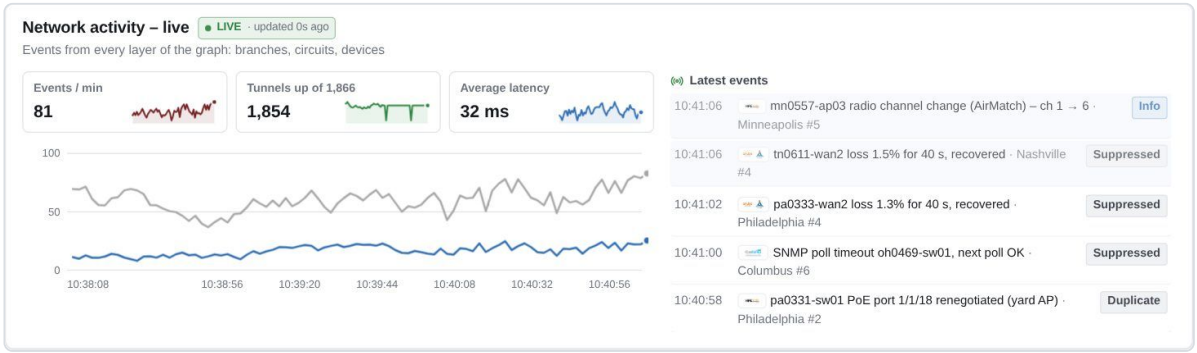


Figure 84. Network activity

The live panel for this screen: events per minute from every layer of the graph, tunnels up out of 1,866 and the average latency to the data centre, with a chart and the latest events. Every screen in Manage, Grow and Improve has a live panel like this one at the top, showing the stream that matters for that screen.

12. Device & Circuit Inventory

The inventory is the CMDB view of every managed device and circuit: model, firmware, config backup, warranty and circuit details. It has three tabs: **Devices**, **Circuits** and **Reconciliation**.

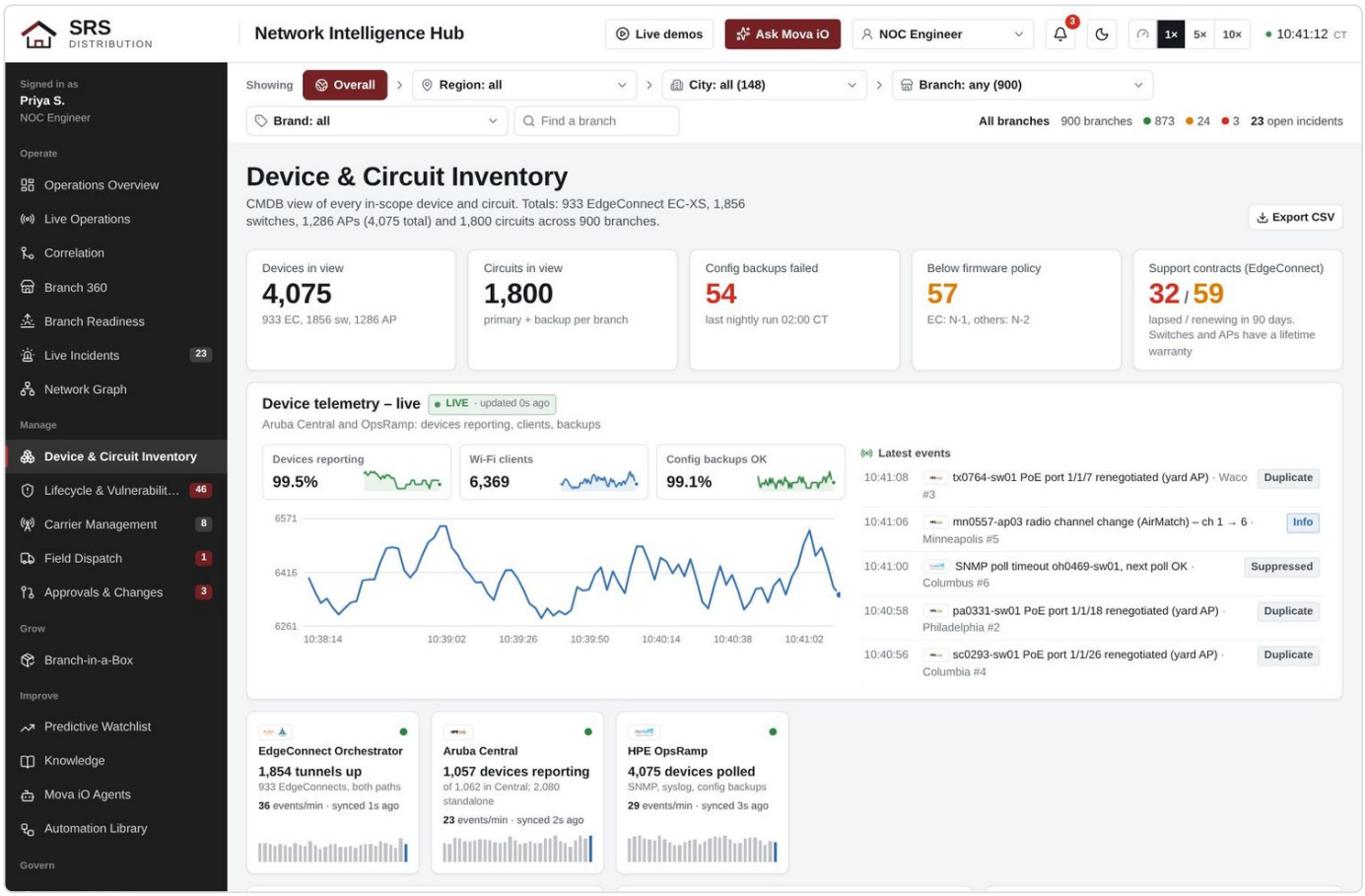


Figure 85. Device & Circuit Inventory

Tiles

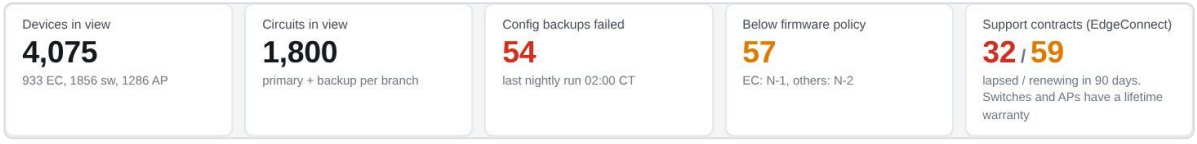


Figure 86. Inventory tiles

Tile	Meaning	Look closer when
Devices in view	All managed devices, split into EdgeConnects, switches and APs	The count does not match the contract (see Reconciliation)
Circuits in view	Primary and backup lines, two per branch	–
Config backups failed	Devices whose last nightly backup (02:00 CT) failed	More than about 2% of devices with backups

Tile	Meaning	Look closer when
Below firmware policy	Devices further behind than the policy allows: EdgeConnects more than one release behind (N-1), others more than two (N-2)	Rising after a wave
Support contracts (EdgeConnect)	EdgeConnect support contracts that have lapsed, and those renewing within 90 days. Switches and APs carry HPE's limited lifetime warranty, which runs until 5 years after the model's end of sale, so they have no yearly date	Any lapsed contract

Device telemetry – live

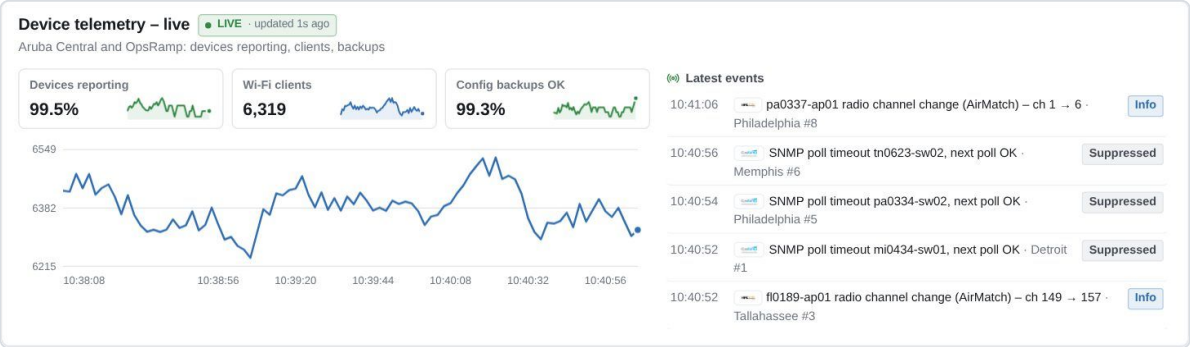


Figure 87. Device telemetry

Devices reporting (%), Wi-Fi clients connected and config backups OK (%), with a chart of clients and the latest device events.

Firmware levels by platform

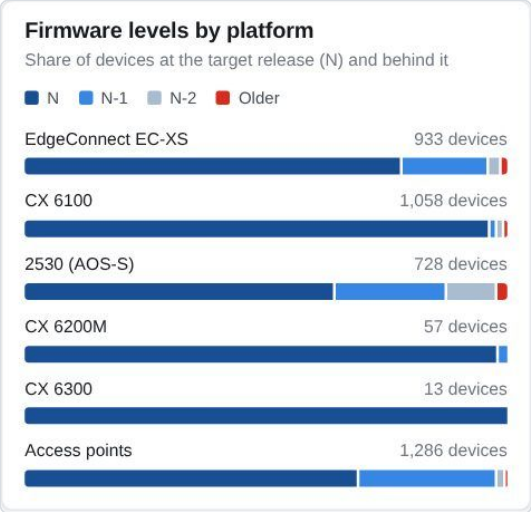


Figure 88. Firmware levels by platform

For each platform (EdgeConnect EC-XS, CX 6100, 2530, CX 6200M, CX 6300 and access points), a bar split into the share of devices on the target release (N), one behind (N-1), two behind (N-2) and older. More of the bar in the darkest shade (N) is better. A large "Older" share on one platform means it missed waves; on the 2530 it reflects the end-of-sale model.

Warranty and support contracts ending, next 8 quarters

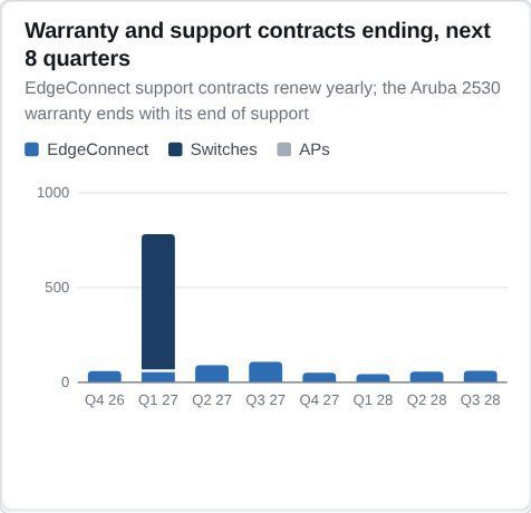


Figure 89. Warranty and support contracts

Dates ending in each of the next eight quarters, stacked by EdgeConnect, switches and APs. EdgeConnects have a 1-year hardware warranty and are covered by an HPE support contract renewed every year, so their bars are contract renewals. The Aruba 2530 warranty ends with its end of support in 2027. Current switches and APs have a lifetime warranty and do not appear.

Config backup success, 14 days

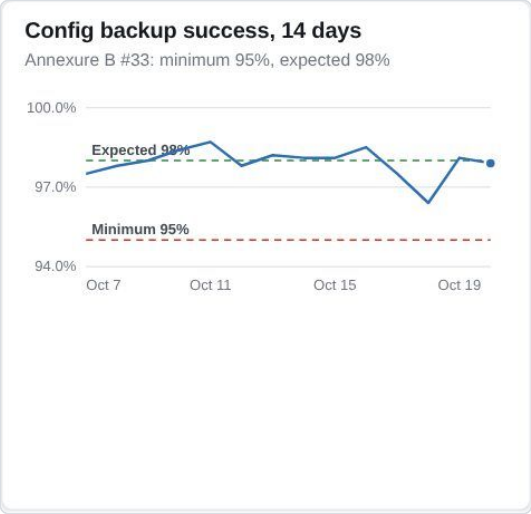


Figure 90. Config backup success

Daily backup success for 14 days against the two lines from service level #33: minimum 95% and expected 98%. Below 95% in a month triggers a service credit.

Hardware likely to fail in 30 days

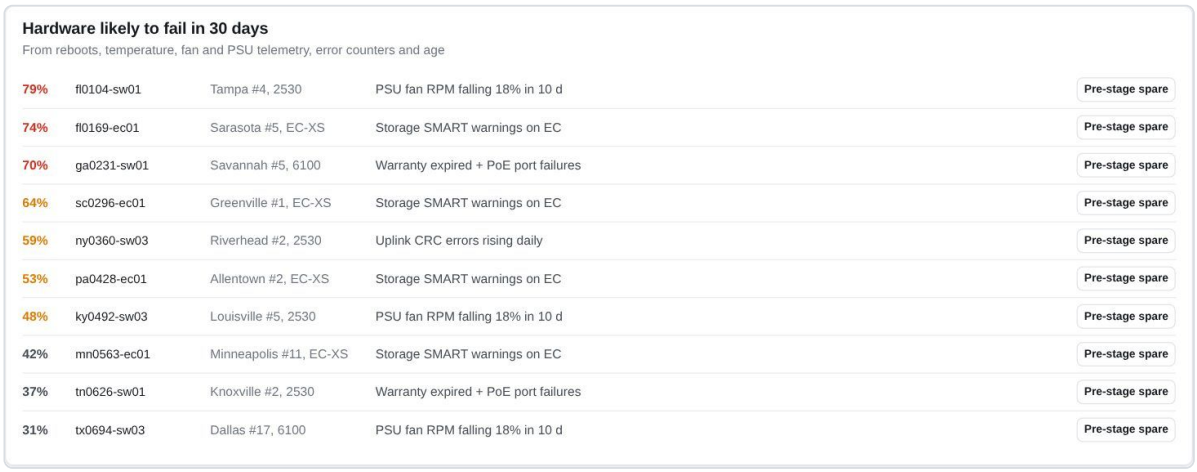
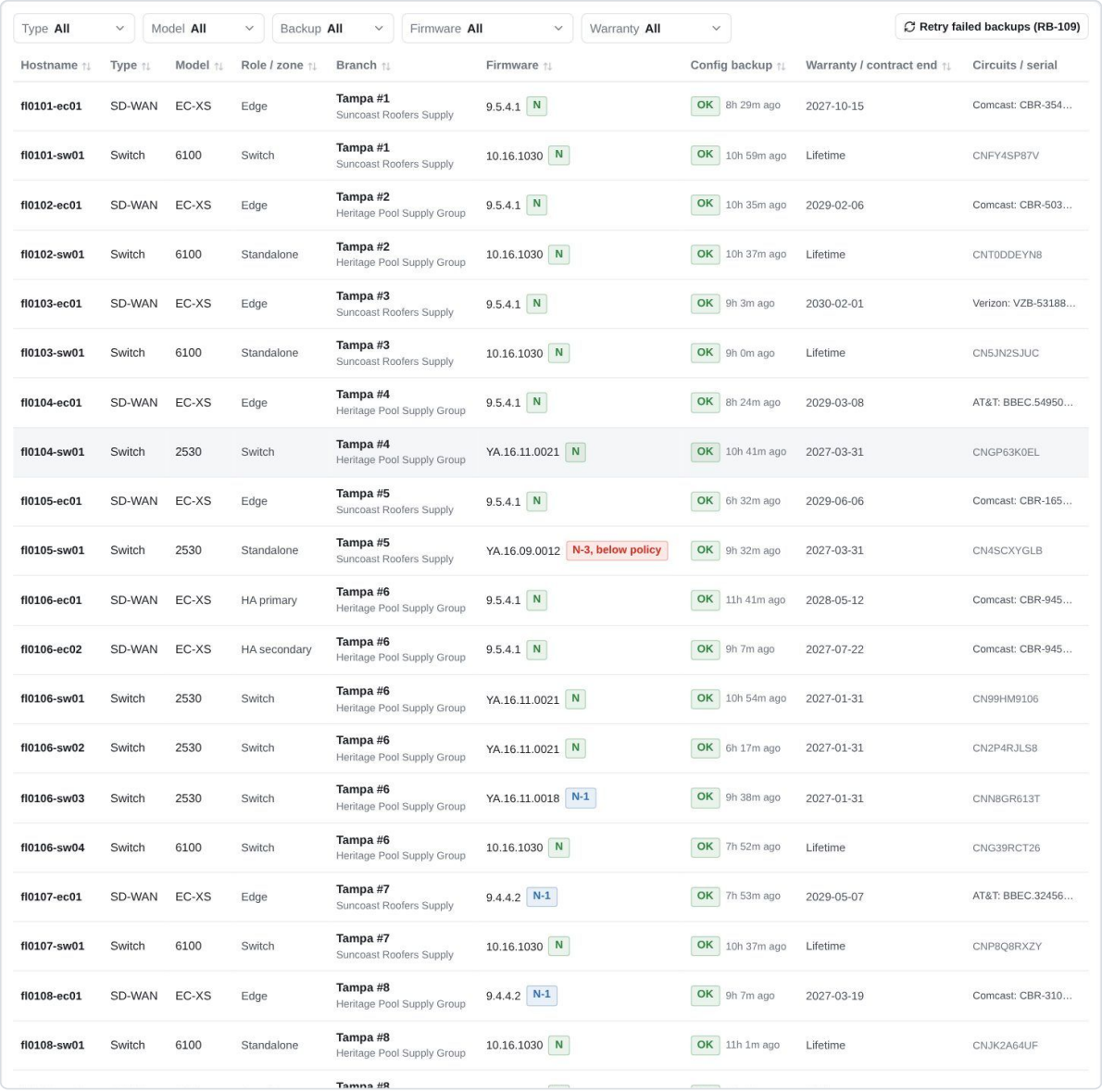


Figure 91. Hardware likely to fail

Devices ranked by their chance of failing in 30 days, worked out from reboots, temperature, fan and power-supply telemetry, error counters and age. Each row shows the device, branch and model, the signal behind the prediction, and **Pre-stage spare**, which reserves a spare at the nearest depot.

Devices tab



Hostname	Type	Model	Role / zone	Branch	Firmware	Config backup	Warranty / contract end	Circuits / serial
fl0101-ec01	SD-WAN	EC-XS	Edge	Tampa #1 Suncoast Roofers Supply	9.5.4.1 N	OK 8h 29m ago	2027-10-15	Comcast: CBR-354…
fl0101-sw01	Switch	6100	Switch	Tampa #1 Suncoast Roofers Supply	10.16.1030 N	OK 10h 59m ago	Lifetime	CNFY4SP87V
fl0102-ec01	SD-WAN	EC-XS	Edge	Tampa #2 Heritage Pool Supply Group	9.5.4.1 N	OK 10h 35m ago	2029-02-06	Comcast: CBR-503…
fl0102-sw01	Switch	6100	Standalone	Tampa #2 Heritage Pool Supply Group	10.16.1030 N	OK 10h 37m ago	Lifetime	CNT0DDEYN8
fl0103-ec01	SD-WAN	EC-XS	Edge	Tampa #3 Suncoast Roofers Supply	9.5.4.1 N	OK 9h 3m ago	2030-02-01	Verizon: VZB-53188…
fl0103-sw01	Switch	6100	Standalone	Tampa #3 Suncoast Roofers Supply	10.16.1030 N	OK 9h 0m ago	Lifetime	CN5JN2SJUC
fl0104-ec01	SD-WAN	EC-XS	Edge	Tampa #4 Heritage Pool Supply Group	9.5.4.1 N	OK 8h 24m ago	2029-03-08	AT&T: BBEC.54950…
fl0104-sw01	Switch	2530	Switch	Tampa #4 Heritage Pool Supply Group	YA.16.11.0021 N	OK 10h 41m ago	2027-03-31	CNGP63K0EL
fl0105-ec01	SD-WAN	EC-XS	Edge	Tampa #5 Suncoast Roofers Supply	9.5.4.1 N	OK 6h 32m ago	2029-06-06	Comcast: CBR-165…
fl0105-sw01	Switch	2530	Standalone	Tampa #5 Suncoast Roofers Supply	YA.16.09.0012 N-3, below policy	OK 9h 32m ago	2027-03-31	CN4SCXYGLB
fl0106-ec01	SD-WAN	EC-XS	HA primary	Tampa #6 Heritage Pool Supply Group	9.5.4.1 N	OK 11h 41m ago	2028-05-12	Comcast: CBR-945…
fl0106-ec02	SD-WAN	EC-XS	HA secondary	Tampa #6 Heritage Pool Supply Group	9.5.4.1 N	OK 9h 7m ago	2027-07-22	Comcast: CBR-945…
fl0106-sw01	Switch	2530	Switch	Tampa #6 Heritage Pool Supply Group	YA.16.11.0021 N	OK 10h 54m ago	2027-01-31	CN99HM9106
fl0106-sw02	Switch	2530	Switch	Tampa #6 Heritage Pool Supply Group	YA.16.11.0021 N	OK 6h 17m ago	2027-01-31	CN2P4RJLS8
fl0106-sw03	Switch	2530	Switch	Tampa #6 Heritage Pool Supply Group	YA.16.11.0018 N-1	OK 9h 38m ago	2027-01-31	CNN8GR613T
fl0106-sw04	Switch	6100	Switch	Tampa #6 Heritage Pool Supply Group	10.16.1030 N	OK 7h 52m ago	Lifetime	CNG39RCT26
fl0107-ec01	SD-WAN	EC-XS	Edge	Tampa #7 Suncoast Roofers Supply	9.4.4.2 N-1	OK 7h 53m ago	2029-05-07	AT&T: BBEC.32456…
fl0107-sw01	Switch	6100	Switch	Tampa #7 Suncoast Roofers Supply	10.16.1030 N	OK 10h 37m ago	Lifetime	CNP8Q8RXZY
fl0108-ec01	SD-WAN	EC-XS	Edge	Tampa #8 Heritage Pool Supply Group	9.4.4.2 N-1	OK 9h 7m ago	2027-03-19	Comcast: CBR-310…
fl0108-sw01	Switch	6100	Standalone	Tampa #8 Heritage Pool Supply Group	10.16.1030 N	OK 11h 1m ago	Lifetime	CNJK2A64UF

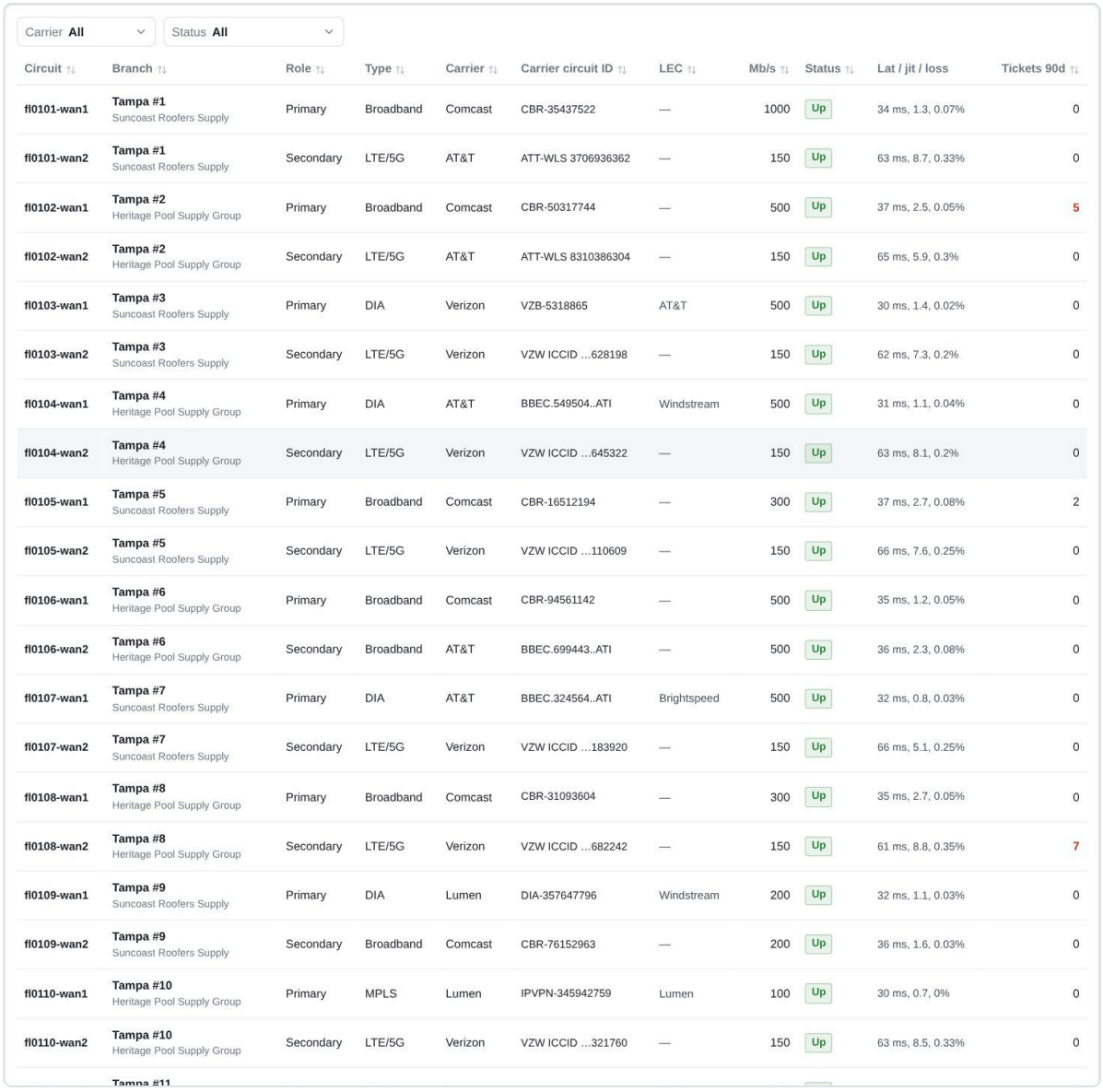
Figure 92. Device table

Filters at the top: type, model, backup state (OK, Failed, Stale, N/A), firmware (below policy) and warranty or contract (ending within 90 days, lapsed). **Retry failed backups (RB-109)** re-runs every failed backup in view.

Column	Meaning
Hostname	Device name, built from state, branch number and device type
Type, model, role or zone	For example SD-WAN EC-XS Edge, Switch CX 6100, AP-635 warehouse
Branch	Branch and brand
Firmware	Running release and its level: N, N-1, N-2 or older
Config backup	OK, Failed or Stale, with the age of the last good backup

Column	Meaning
Warranty / contract end	Date for EdgeConnect support contracts and the Aruba 2530; Lifetime for current switches and APs. Amber within 90 days, red when lapsed
Circuits or serial	Circuits for an EdgeConnect, serial number for others

Circuits tab

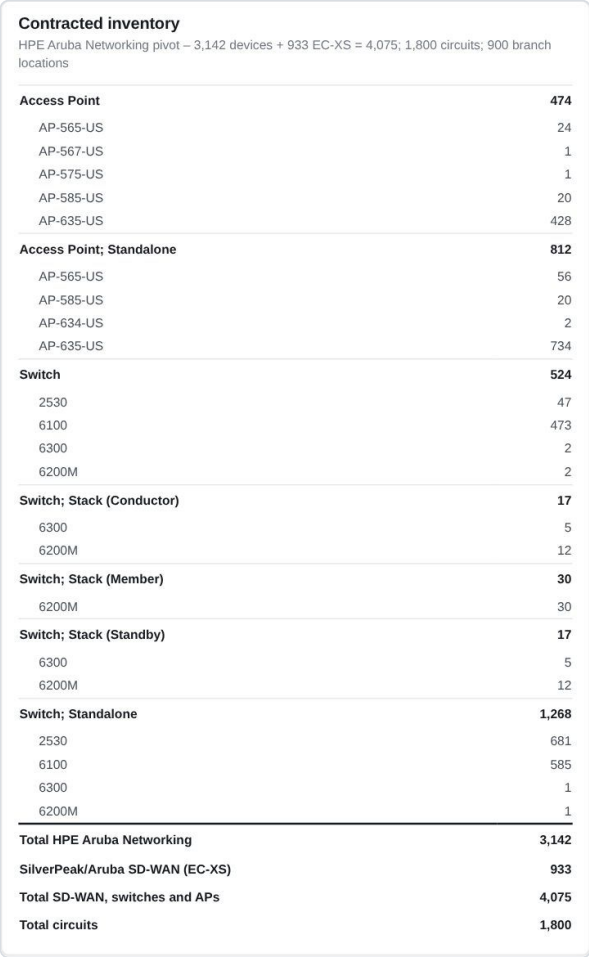


Circuit	Branch	Role	Type	Carrier	Carrier circuit ID	LEC	Mb/s	Status	Lat / jit / loss	Tickets 90d
fl0101-wan1	Tampa #1 Suncoast Roofers Supply	Primary	Broadband	Comcast	CBR-35437522	—	1000	Up	34 ms, 1.3, 0.07%	0
fl0101-wan2	Tampa #1 Suncoast Roofers Supply	Secondary	LTE/5G	AT&T	ATT-WLS 3706936362	—	150	Up	63 ms, 8.7, 0.33%	0
fl0102-wan1	Tampa #2 Heritage Pool Supply Group	Primary	Broadband	Comcast	CBR-50317744	—	500	Up	37 ms, 2.5, 0.05%	5
fl0102-wan2	Tampa #2 Heritage Pool Supply Group	Secondary	LTE/5G	AT&T	ATT-WLS 8310386304	—	150	Up	65 ms, 5.9, 0.3%	0
fl0103-wan1	Tampa #3 Suncoast Roofers Supply	Primary	DIA	Verizon	VZB-5318865	AT&T	500	Up	30 ms, 1.4, 0.02%	0
fl0103-wan2	Tampa #3 Suncoast Roofers Supply	Secondary	LTE/5G	Verizon	VZW ICCID …628198	—	150	Up	62 ms, 7.3, 0.2%	0
fl0104-wan1	Tampa #4 Heritage Pool Supply Group	Primary	DIA	AT&T	BBEC.549504..ATI	Windstream	500	Up	31 ms, 1.1, 0.04%	0
fl0104-wan2	Tampa #4 Heritage Pool Supply Group	Secondary	LTE/5G	Verizon	VZW ICCID …645322	—	150	Up	63 ms, 8.1, 0.2%	0
fl0105-wan1	Tampa #5 Suncoast Roofers Supply	Primary	Broadband	Comcast	CBR-16512194	—	300	Up	37 ms, 2.7, 0.08%	2
fl0105-wan2	Tampa #5 Suncoast Roofers Supply	Secondary	LTE/5G	Verizon	VZW ICCID …110609	—	150	Up	66 ms, 7.6, 0.25%	0
fl0106-wan1	Tampa #6 Heritage Pool Supply Group	Primary	Broadband	Comcast	CBR-94561142	—	500	Up	35 ms, 1.2, 0.05%	0
fl0106-wan2	Tampa #6 Heritage Pool Supply Group	Secondary	Broadband	AT&T	BBEC.699443..ATI	—	500	Up	36 ms, 2.3, 0.08%	0
fl0107-wan1	Tampa #7 Suncoast Roofers Supply	Primary	DIA	AT&T	BBEC.324564..ATI	Brightspeed	500	Up	32 ms, 0.8, 0.03%	0
fl0107-wan2	Tampa #7 Suncoast Roofers Supply	Secondary	LTE/5G	Verizon	VZW ICCID …183920	—	150	Up	66 ms, 5.1, 0.25%	0
fl0108-wan1	Tampa #8 Heritage Pool Supply Group	Primary	Broadband	Comcast	CBR-31093604	—	300	Up	35 ms, 2.7, 0.05%	0
fl0108-wan2	Tampa #8 Heritage Pool Supply Group	Secondary	LTE/5G	Verizon	VZW ICCID …682242	—	150	Up	61 ms, 8.8, 0.35%	7
fl0109-wan1	Tampa #9 Suncoast Roofers Supply	Primary	DIA	Lumen	DIA-357647796	Windstream	200	Up	32 ms, 1.1, 0.03%	0
fl0109-wan2	Tampa #9 Suncoast Roofers Supply	Secondary	Broadband	Comcast	CBR-76152963	—	200	Up	36 ms, 1.6, 0.03%	0
fl0110-wan1	Tampa #10 Heritage Pool Supply Group	Primary	MPLS	Lumen	IPVPN-345942759	Lumen	100	Up	30 ms, 0.7, 0%	0
fl0110-wan2	Tampa #10 Heritage Pool Supply Group	Secondary	LTE/5G	Verizon	VZW ICCID …321760	—	150	Up	63 ms, 8.5, 0.33%	0
	Tampa #11									

Figure 93. Circuit table

Filters: carrier, status and **Repeat offenders** (circuits with repeated tickets). Columns: circuit, branch, role (primary or secondary), type (DIA, broadband, MPLS, LTE/5G), carrier, carrier circuit ID, LEC, bandwidth, status, live latency, jitter and loss, and tickets in 90 days.

Reconciliation tab



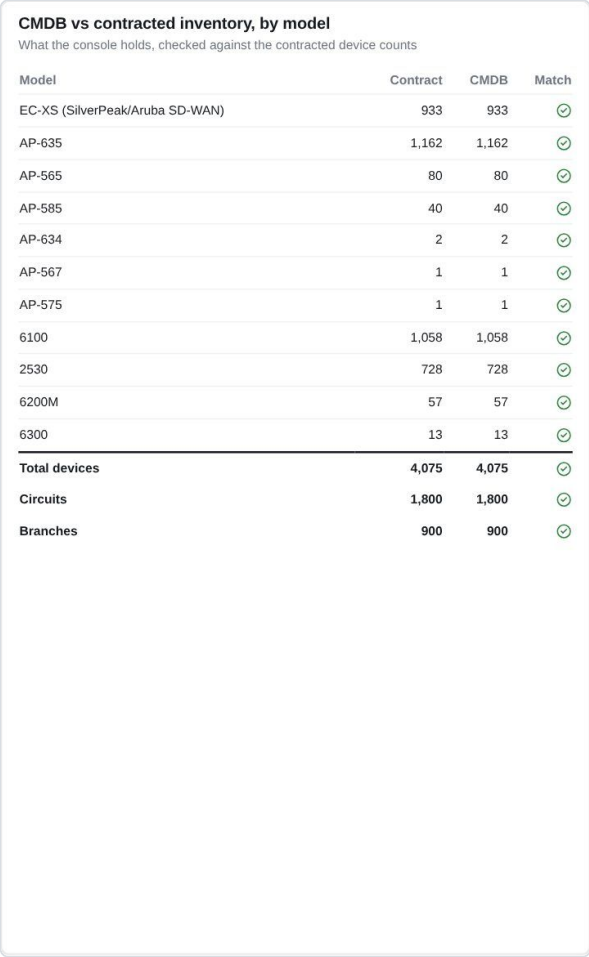
Contracted inventory

HPE Aruba Networking pivot – 3,142 devices + 933 EC-XS = 4,075; 1,800 circuits; 900 branch locations

Access Point	**474**
AP-565-US	24
AP-567-US	1
AP-575-US	1
AP-585-US	20
AP-635-US	428
Access Point; Standalone	**812**
AP-565-US	56
AP-585-US	20
AP-634-US	2
AP-635-US	734
Switch	**524**
2530	47
6100	473
6300	2
6200M	2
Switch; Stack (Conductor)	**17**
6300	5
6200M	12
Switch; Stack (Member)	**30**
6200M	30
Switch; Stack (Standby)	**17**
6300	5
6200M	12
Switch; Standalone	**1,268**
2530	681
6100	585
6300	1
6200M	1
Total HPE Aruba Networking	**3,142**
SilverPeak/Aruba SD-WAN (EC-XS)	**933**
Total SD-WAN, switches and APs	**4,075**
Total circuits	**1,800**

Figure 94. Contracted inventory

Contracted inventory is the device list from the contract, by role and model: 3,142 HPE Aruba Networking devices plus 933 EC-XS for 4,075 in total, 1,800 circuits and 900 branch locations. It shows how many access points and switches are managed in Aruba Central and how many are standalone: 1,268 switches and 812 APs, 2,080 devices or 51% of the estate, are standalone. Standalone devices are monitored by OpsRamp (SNMP and syslog) but cannot be configured or upgraded from Central; demo 25 brings them under Central. The device panel shows **Managed by** for each device.



CMDB vs contracted inventory, by model

What the console holds, checked against the contracted device counts

Model	Contract	CMDB	Match
EC-XS (SilverPeak/Aruba SD-WAN)	933	933	✓
AP-635	1,162	1,162	✓
AP-565	80	80	✓
AP-585	40	40	✓
AP-634	2	2	✓
AP-567	1	1	✓
AP-575	1	1	✓
6100	1,058	1,058	✓
2530	728	728	✓
6200M	57	57	✓
6300	13	13	✓
Total devices	**4,075**	**4,075**	✓
Circuits	**1,800**	**1,800**	✓
Branches	**900**	**900**	✓

Figure 95. CMDB versus contract

CMDB vs contracted inventory, by model checks what the console holds against the contracted counts for every model, circuits and branches. A tick means they match; a mismatch is a CMDB clean-up task.

13. Lifecycle & Vulnerabilities

This screen matches every device's running firmware against HPE Aruba Networking security bulletins and lifecycle notices, and plans the fixes. It has four tabs: **Advisories**, **Models and lifecycle**, **Exposure by region** and **Most exposed branches**.

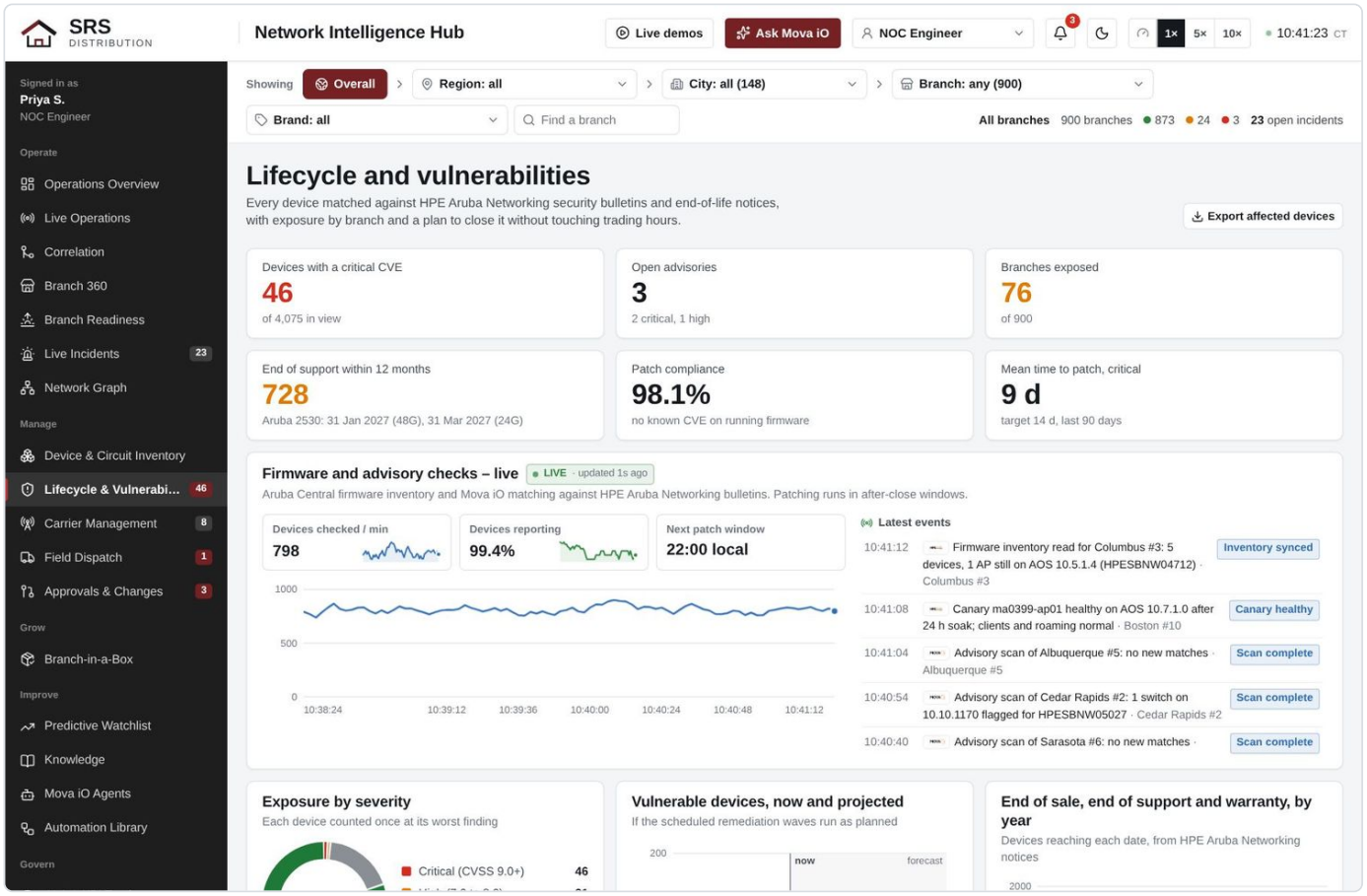


Figure 96. Lifecycle & Vulnerabilities

Tiles

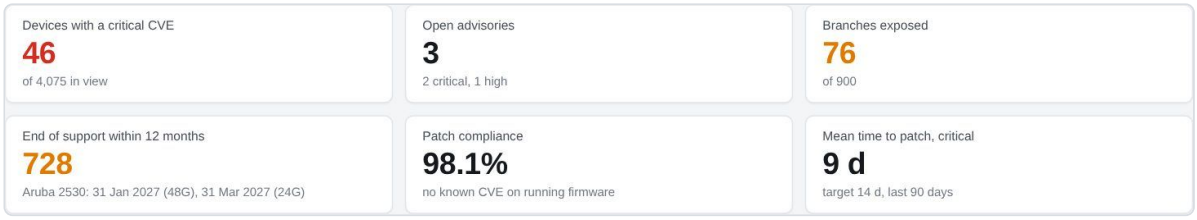


Figure 97. Vulnerability tiles

Tile	Meaning	Target
Devices with a critical CVE	Devices whose running firmware has a CVSS 9.0+ vulnerability	Falling with each wave
Open advisories	Advisories that affect at least one device, by severity	–

Tile	Meaning	Target
Branches exposed	Branches with at least one affected device	–
End of support within 12 months	Devices whose end of support falls in the next 12 months: the 728 Aruba 2530 switches (31 Jan 2027 for the 48-port, 31 Mar 2027 for the 24-port)	Falling with the refresh
Patch compliance	Share of devices with no known CVE on their running firmware	Rising
Mean time to patch, critical	Average days from advisory to patched, last 90 days	14 days

Firmware and advisory checks – live

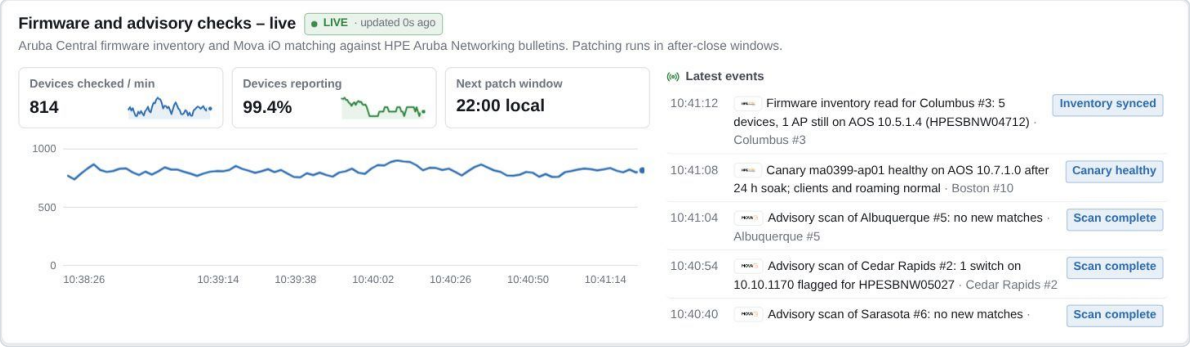


Figure 98. Firmware and advisory checks

Devices checked per minute, devices reporting and the next patch window (22:00 local, after close), with the latest inventory and matching events.

Exposure by severity

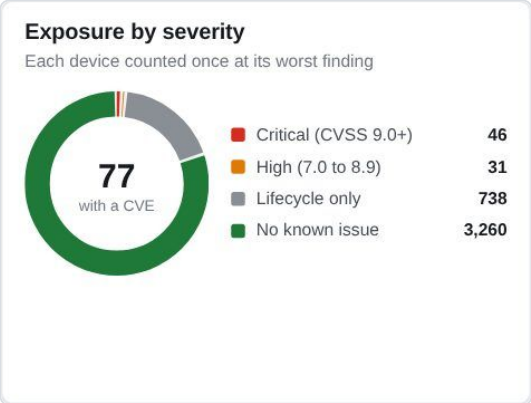


Figure 99. Exposure by severity

Every device counted once at its worst finding: critical (CVSS 9.0+), high (7.0 to 8.9), lifecycle only (no CVE, but near end of support or on an older software release) and no known issue.

Vulnerable devices, now and projected

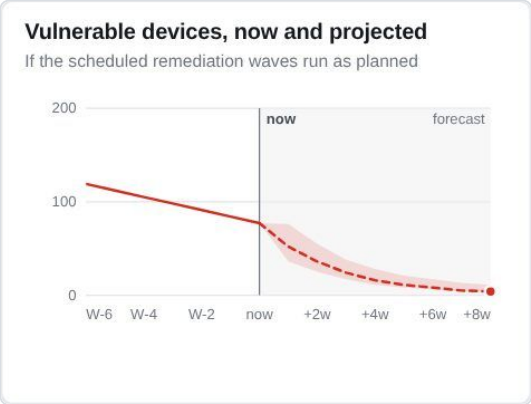


Figure 100. Vulnerable devices

Vulnerable devices for the last six weeks and, after "now", the forecast for six weeks if the scheduled remediation waves run as planned. A forecast that does not fall means waves are missing.

End of sale, end of support and warranty, by year

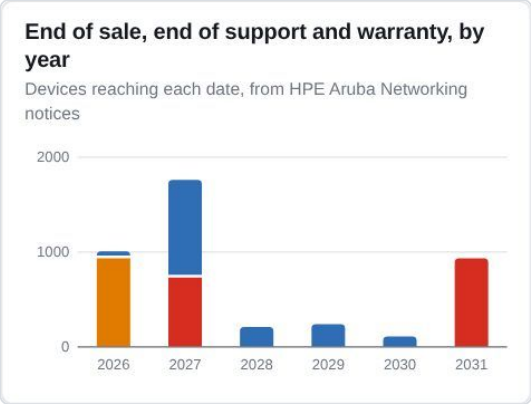


Figure 101. End of sale, end of support and warranty

Devices reaching each date in each year, from HPE Aruba Networking notices: **end of sale** (the last day the model can be ordered), **end of support** (the last day of vendor support and security fixes) and warranty or contract end. The 2026 end-of-sale bar is the 933 EdgeConnect EC-XS; the 2027 end-of-support bar is the 728 Aruba 2530 switches; the 2031 bar is the EC-XS end of support.

Advisory cards

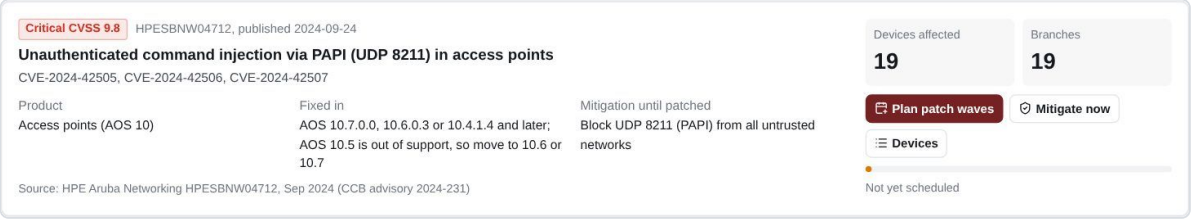


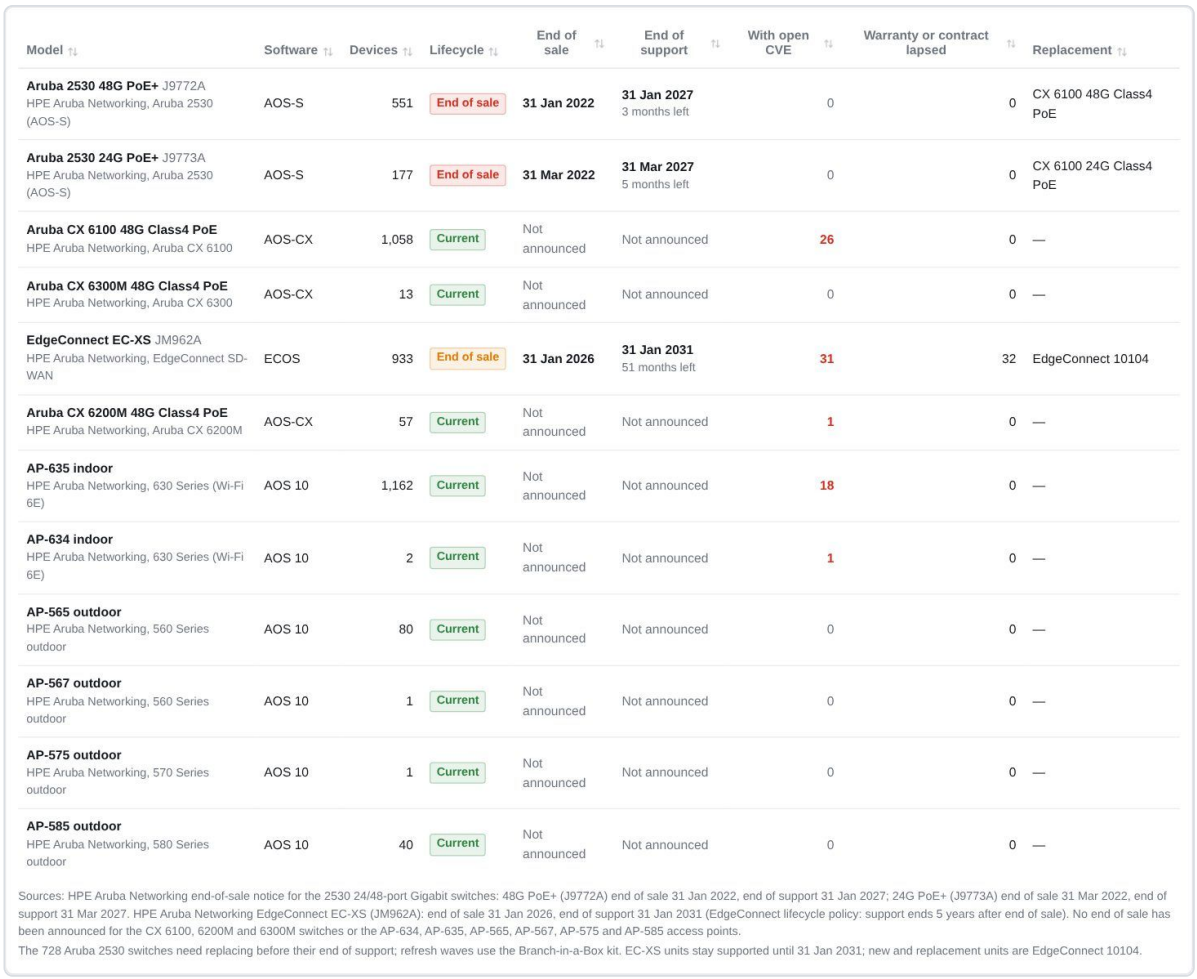
Figure 102. Advisory card

One card per advisory:

Field	Meaning
Severity and CVSS	Critical, High or Lifecycle, with the score
Bulletin and date	HPE Aruba Networking bulletin number and publication date
Title and CVEs	What the flaw is and its CVE numbers
Product	Which platform and software is affected
Fixed in	The release that fixes it
Mitigation until patched	What can be done now without a reboot
Devices and branches affected	Counts in the current view
Status	Not yet scheduled, waves scheduled, mitigated or patched

Plan patch waves creates region-by-region changes after branch close, canary first. **Mitigate now** applies the policy mitigation on every branch; it is a change with no client impact, so it can run sooner. **Devices** lists the affected devices. Lifecycle cards offer **Plan refresh** instead, using the Branch-in-a-Box standard kit.

Models and lifecycle tab



Model	Software	Devices	Lifecycle	End of sale	End of support	With open CVE	Warranty or contract lapsed	Replacement
Aruba 2530 48G PoE+ J9772A HPE Aruba Networking, Aruba 2530 (AOS-S)	AOS-S	551	End of sale	**31 Jan 2022**	**31 Jan 2027** 3 months left	0	0	CX 6100 48G Class4 PoE
Aruba 2530 24G PoE+ J9773A HPE Aruba Networking, Aruba 2530 (AOS-S)	AOS-S	177	End of sale	**31 Mar 2022**	**31 Mar 2027** 5 months left	0	0	CX 6100 24G Class4 PoE
Aruba CX 6100 48G Class4 PoE HPE Aruba Networking, Aruba CX 6100	AOS-CX	1,058	Current	Not announced	Not announced	26	0	—
Aruba CX 6300M 48G Class4 PoE HPE Aruba Networking, Aruba CX 6300	AOS-CX	13	Current	Not announced	Not announced	0	0	—
EdgeConnect EC-XS JM962A HPE Aruba Networking, EdgeConnect SD-WAN	ECOS	933	End of sale	**31 Jan 2026**	**31 Jan 2031** 51 months left	31	32	EdgeConnect 10104
Aruba CX 6200M 48G Class4 PoE HPE Aruba Networking, Aruba CX 6200M	AOS-CX	57	Current	Not announced	Not announced	1	0	—
AP-635 indoor HPE Aruba Networking, 630 Series (Wi-Fi 6E)	AOS 10	1,162	Current	Not announced	Not announced	18	0	—
AP-634 indoor HPE Aruba Networking, 630 Series (Wi-Fi 6E)	AOS 10	2	Current	Not announced	Not announced	1	0	—
AP-565 outdoor HPE Aruba Networking, 560 Series outdoor	AOS 10	80	Current	Not announced	Not announced	0	0	—
AP-567 outdoor HPE Aruba Networking, 560 Series outdoor	AOS 10	1	Current	Not announced	Not announced	0	0	—
AP-575 outdoor HPE Aruba Networking, 570 Series outdoor	AOS 10	1	Current	Not announced	Not announced	0	0	—
AP-585 outdoor HPE Aruba Networking, 580 Series outdoor	AOS 10	40	Current	Not announced	Not announced	0	0	—

Sources: HPE Aruba Networking end-of-sale notice for the 2530 24/48-port Gigabit switches: 48G PoE+ (J9772A) end of sale 31 Jan 2022, end of support 31 Jan 2027; 24G PoE+ (J9773A) end of sale 31 Mar 2022, end of support 31 Mar 2027. HPE Aruba Networking EdgeConnect EC-XS (JM962A): end of sale 31 Jan 2026, end of support 31 Jan 2031 (EdgeConnect lifecycle policy: support ends 5 years after end of sale). No end of sale has been announced for the CX 6100, 6200M and 6300M switches or the AP-634, AP-635, AP-565, AP-567, AP-575 and AP-585 access points.

The 728 Aruba 2530 switches need replacing before their end of support; refresh waves use the Branch-in-a-Box kit. EC-XS units stay supported until 31 Jan 2031; new and replacement units are EdgeConnect 10104.

Figure 103. Models and lifecycle

One row per model, with the Aruba 2530 split into its 48-port and 24-port SKUs because their dates differ:

Column	Meaning
Lifecycle	**Current** (no end of sale announced) or **End of sale** (no longer sold, still supported). Red when end of support is within 12 months
End of sale	The last day HPE Aruba Networking sold the model, or "Not announced"
End of support	The last day of vendor support and security fixes, with the months left, or "Not announced"
With open CVE	Devices of this model with an open vulnerability
Warranty or contract lapsed	EdgeConnect support contracts not renewed; zero for lifetime-warranty models
Replacement	The model it is replaced with

The dates come from HPE Aruba Networking notices:

Model	End of sale	End of support	Replacement
Aruba 2530 48G PoE+ (J9772A)	31 Jan 2022	31 Jan 2027	CX 6100 48G Class4 PoE
Aruba 2530 24G PoE+ (J9773A)	31 Mar 2022	31 Mar 2027	CX 6100 24G Class4 PoE
EdgeConnect EC-XS (JM962A)	31 Jan 2026	31 Jan 2031	EdgeConnect 10104
CX 6100, CX 6200M, CX 6300M	Not announced	Not announced	–
AP-634, AP-635, AP-565, AP-567, AP-575, AP-585	Not announced	Not announced	–

HPE's lifecycle policy ends support five years after end of sale. The RFP gives the 2530 date as 1 Feb 2027, the day after the 48-port end of support. EC-XS units already installed stay supported until 2031; new and replacement units are EdgeConnect 10104.

Exposure by region tab

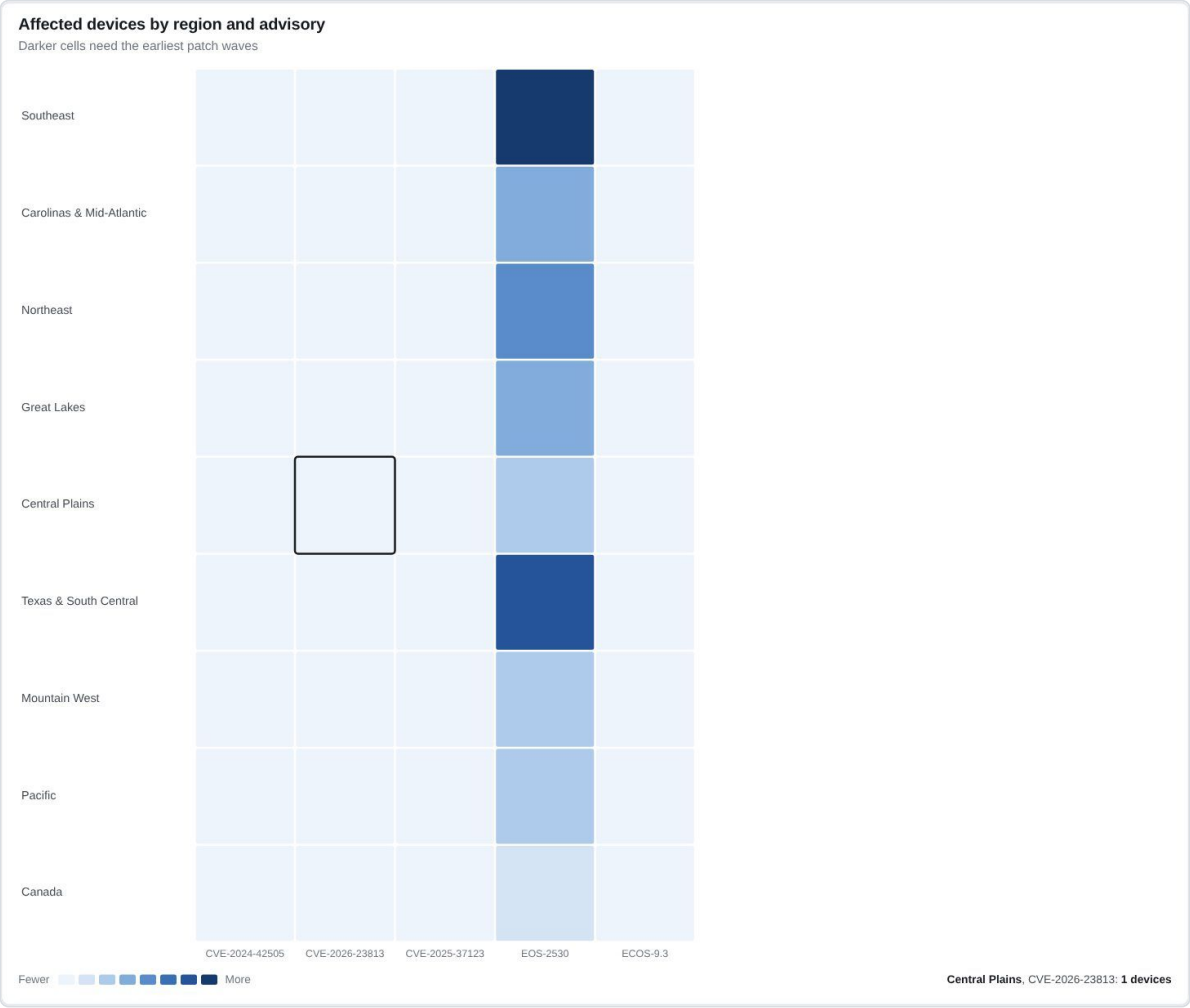


Figure 104. Affected devices by region and advisory

A heat map of affected devices by region (rows) and advisory (columns). Darker cells need the earliest patch waves. Hover a cell for the count.

Most exposed branches tab



Figure 105. Most exposed branches

Branches ranked by a score weighted by the severity of their open findings. Click a branch to open its Branch 360 vulnerability tab.

14. Carrier Management

Carrier Management tracks every carrier case from opening to close, scores the carriers and drives down "no trouble found" (NTF) closures. Its tabs are **Open cases**, **Closed (7 d)**, **Scorecard**, **Repeat offenders** and **NTF tracking**.

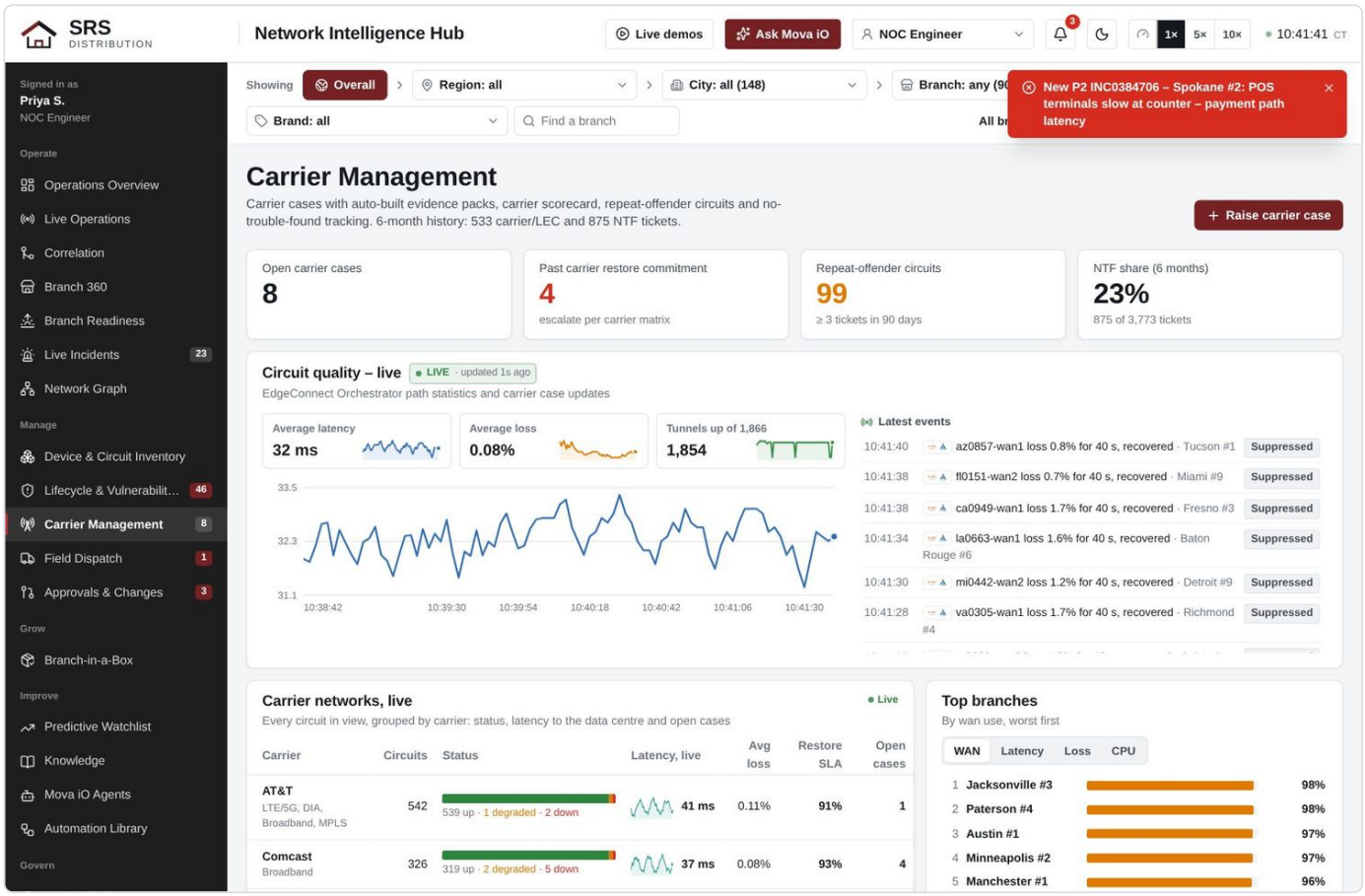


Figure 106. Carrier Management

Tiles



Figure 107. Carrier tiles

Tile	Meaning	Act when
Open carrier cases	Cases open with any carrier	–
Past carrier restore commitment	Open cases past the time the carrier committed to	Above zero: escalate per the carrier matrix
Repeat-offender circuits	Circuits with three or more tickets in 90 days	Each one should have a problem record
NTF share (6 months)	Tickets the carrier closed as no trouble found: 875 of 3,773	The aim is to bring this down with evidence

Circuit quality – live



Figure 108. Circuit quality

Average latency to the data centre, average loss and tunnels up, with a chart and the latest path events from Orchestrator and carrier case updates. Short loss bursts that recover are marked Suppressed: they are watched, not ticketed.

Carrier networks, live

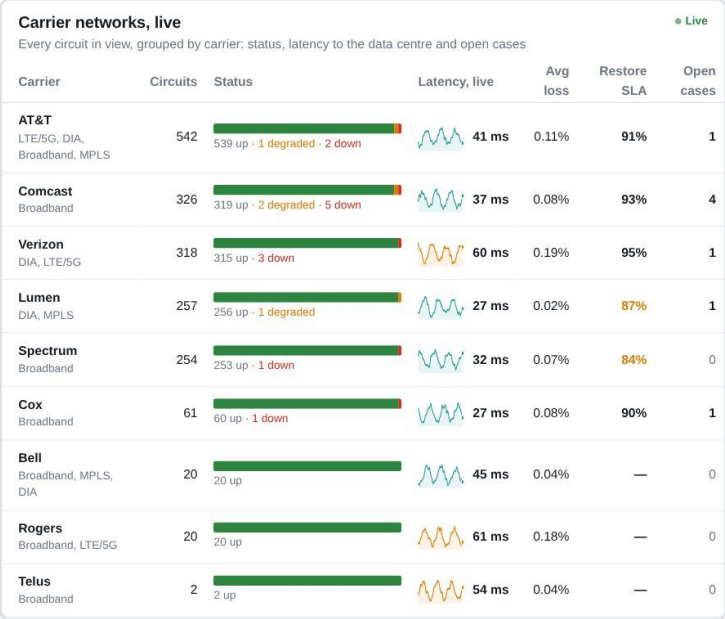


Figure 109. Carrier networks

One row per carrier for every circuit in view:

Column	Meaning
Carrier	Name and the circuit types it provides (DIA, broadband, MPLS, LTE/5G)
Circuits	How many lines in view
Status	A bar of up (green), degraded (amber) and down (red), with counts
Latency, live	Average latency to the data centre with its recent trend; amber above 45 ms
Avg loss	Average packet loss

Column	Meaning
Restore SLA	Share of carrier faults restored within the carrier's commitment over six months; amber below 90%
Open cases	Click to see them

Circuits likely to fail this week



Figure 110. Circuits likely to fail

Circuits the prediction model expects to fail, with the probability, the time window, the carrier, the signal behind it and **Raise case**. A proactive case raised while the branch is still on its primary gives the carrier time to fix it before anyone notices.

Carrier restore time, 6 months

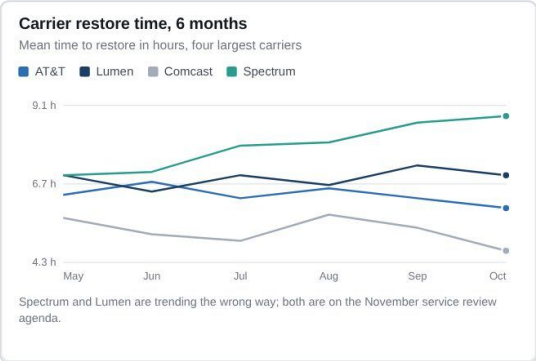
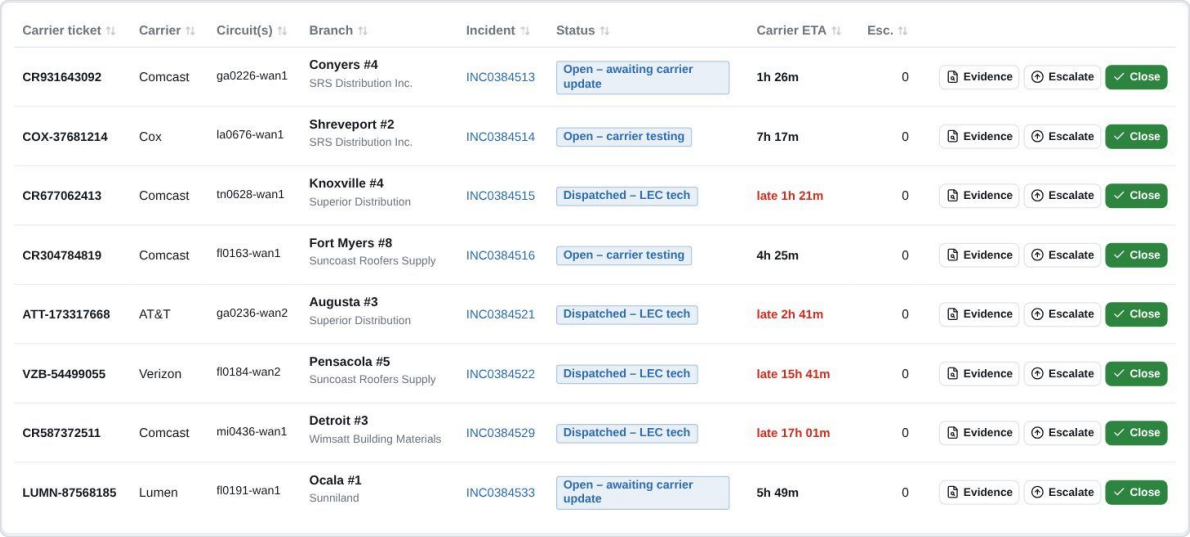


Figure 111. Carrier restore time

Mean time to restore in hours for the four largest carriers, month by month, with a note on which carriers are trending the wrong way and are on the next service review agenda.

Open cases



Carrier ticket	Carrier	Circuit(s)	Branch	Incident	Status	Carrier ETA	Esc.	
CR931643092	Comcast	ga0226-wan1	Conyers #4 SRS Distribution Inc.	INC0384513	Open – awaiting carrier update	1h 26m	0	Evidence Escalate Close
COX-37681214	Cox	la0676-wan1	Shreveport #2 SRS Distribution Inc.	INC0384514	Open – carrier testing	7h 17m	0	Evidence Escalate Close
CR677062413	Comcast	tn0628-wan1	Knoxville #4 Superior Distribution	INC0384515	Dispatched – LEC tech	late 1h 21m	0	Evidence Escalate Close
CR304784819	Comcast	fl0163-wan1	Fort Myers #8 Suncoast Roofers Supply	INC0384516	Open – carrier testing	4h 25m	0	Evidence Escalate Close
ATT-173317668	AT&T	ga0236-wan2	Augusta #3 Superior Distribution	INC0384521	Dispatched – LEC tech	late 2h 41m	0	Evidence Escalate Close
VZB-54499055	Verizon	fl0184-wan2	Pensacola #5 Suncoast Roofers Supply	INC0384522	Dispatched – LEC tech	late 15h 41m	0	Evidence Escalate Close
CR587372511	Comcast	mi0436-wan1	Detroit #3 Wimsatt Building Materials	INC0384529	Dispatched – LEC tech	late 17h 01m	0	Evidence Escalate Close
LUMN-87568185	Lumen	fl0191-wan1	Ocala #1 Sunniland	INC0384533	Open – awaiting carrier update	5h 49m	0	Evidence Escalate Close

Figure 112. Open carrier cases

Column	Meaning
Carrier ticket	The carrier's case number
Carrier, circuit(s), branch	Which line; a master case lists several circuits
Incident	The linked ServiceNow incident
Status	For example Open – carrier testing, Dispatched – LEC tech, Open – proactive
Carrier ETA	Time left to the carrier's restore commitment
Esc.	Escalation level reached

Evidence shows the evidence pack attached (loss and jitter graphs, MTR traces, loopback results, timestamps). **Escalate** moves the case up the carrier's escalation matrix. **Close** closes it with a close code. The **Closed (7 d)** tab lists recent cases with their close code: carrier fault fixed, or NTF.

Scorecard tab

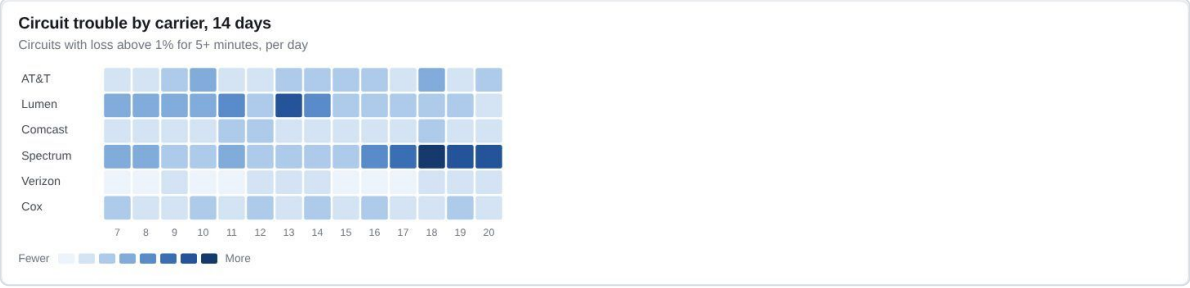


Figure 113. Circuit trouble by carrier

Circuit trouble by carrier, 14 days: circuits with loss above 1% for five minutes or more, per carrier per day. Darker cells mean more circuits in trouble; a dark run across days is a carrier-side problem.

Carrier scorecard – last 6 months

533 carrier/LEC and 875 no-trouble-found tickets, by carrier

Carrier	Circuits	Carrier faults	NTF	Faults / 100 circuits	MTTR (h)	Restore commit met	Repeat ckts
AT&T	542	148	231	27.3	6.1	91%	9
Lumen	257	131	214	51	7.4	87%	11
Comcast	326	97	168	29.8	5.2	93%	6
Spectrum	254	86	152	33.9	8.3	84%	10
Verizon	318	41	61	12.9	4.6	95%	3
Cox	61	22	37	36.1	6.8	90%	2
Canadian carriers	42	8	12	19	5.9	94%	0

Figure 114. Carrier scorecard

Carrier scorecard, last 6 months: circuits, carrier faults, NTF tickets, faults per 100 circuits (the fair comparison between large and small carriers), mean time to restore in hours, restore commitment met and repeat-offender circuits. Use it in the monthly carrier review.

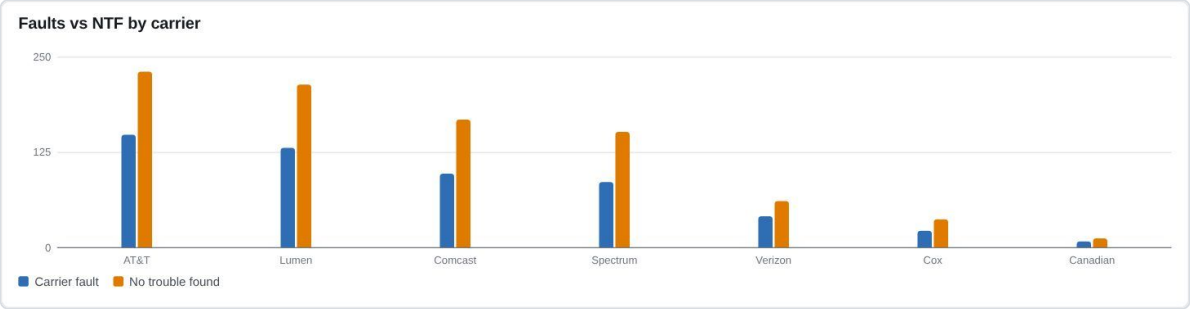


Figure 115. Faults versus NTF

Faults vs NTF by carrier: carrier faults and NTF tickets side by side. A carrier with more NTF than faults is closing tickets it should be fixing.

Repeat offenders tab

Circuit	Branch	Carrier	Type	LEC	Tickets 90d	NTF 90d	
fl0143-wan1	Miami #1 Suncoast Roofers Supply	AT&T	DIA	Frontier	8	1	Problem record Carrier review
il0514-wan1	Chicago #2 Prairie State Exteriors	Comcast	Broadband	—	8	2	Problem record Carrier review
fl0108-wan2	Tampa #8 Heritage Pool Supply Group	Verizon	LTE/5G	—	7	0	Problem record Carrier review
fl0170-wan2	Sarasota #6 Suncoast Roofers Supply	AT&T	LTE/5G	—	7	1	Problem record Carrier review
ny0376-wan2	Buffalo #2 Weatherpanel	Verizon	LTE/5G	—	7	0	Problem record Carrier review
oh0484-wan1	Cincinnati #4 Marsh Building Products	Spectrum	Broadband	—	7	1	Problem record Carrier review
mo0595-wan1	St. Louis #8 Arrowhead Building Supply	Spectrum	Broadband	—	7	2	Problem record Carrier review
tn0627-wan2	Knoxville #3 Superior Distribution	AT&T	Broadband	—	7	1	Problem record Carrier review
co0819-wan1	Denver #3 American Roofing Supply	Comcast	Broadband	—	7	3	Problem record Carrier review
az0841-wan1	Phoenix #1 American Roofing Supply	AT&T	DIA	Lumen	7	1	Problem record Carrier review
wa0919-wan1	Bellingham #2 CB Wholesale	Lumen	DIA	Brightspeed	7	0	Problem record Carrier review
on0981-wan1	Toronto #4 SRS Canada (illustrative)	Rogers	Broadband	—	7	0	Problem record Carrier review
ga0207-wan1	Atlanta #3 Superior Distribution	AT&T	DIA	Windstream	6	3	Problem record Carrier review
nc0242-wan1	Charlotte #2 Heritage Pool Supply Group	Lumen	MPLS	Frontier	6	2	Problem record Carrier review
fl0102-wan1	Tampa #2 Heritage Pool Supply Group	Comcast	Broadband	—	5	2	Problem record Carrier review
fl0152-wan2	Miami #10 Heritage Pool Supply Group	Comcast	Broadband	—	5	1	Problem record Carrier review
fl0179-wan2	West Palm Beach #8 Heritage Pool Supply Group	Comcast	Broadband	—	5	1	Problem record Carrier review
ga0214-wan2	Atlanta #10 Heritage Landscape Supply …	Verizon	LTE/5G	—	5	0	Problem record Carrier review
sc0288-wan2	Myrtle Beach #5 Atlantic Roofing Distributors	AT&T	LTE/5G	—	5	0	Problem record Carrier review
pa0332-wan1	Philadelphia #3 Marvic Supply	Lumen	DIA	AT&T	5	3	Problem record Carrier review
ct0386-wan1	Hartford #6 Metro Roofing Supplies	Comcast	Broadband	—	5	0	Problem record Carrier review

Figure 116. Repeat offenders

Circuits with three or more tickets in 90 days: circuit, branch, carrier, type, LEC, tickets and NTF in 90 days. **Problem record** opens a problem in ServiceNow; **Carrier review** asks the carrier's account team for a chronic-circuit review instead of another repair ticket.

NTF tracking tab

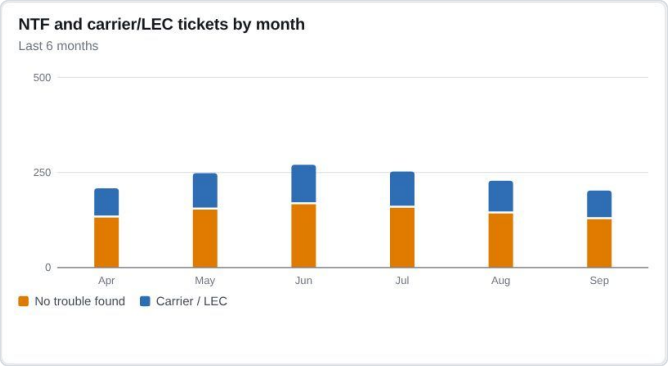


Figure 117. NTF and carrier tickets by month

NTF and carrier/LEC tickets per month for six months. The target is a falling NTF bar.

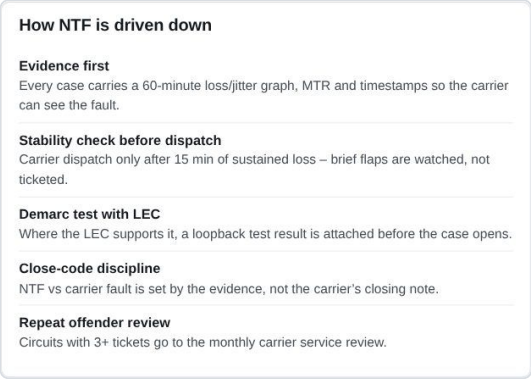


Figure 118. How NTF is driven down

The four rules that bring NTF down:

1. **Evidence first:** every case carries a 60-minute loss and jitter graph, an MTR trace and timestamps, so the carrier can see the fault.
2. **Stability check before dispatch:** the carrier is dispatched only after 15 minutes of sustained loss; brief flaps are watched, not ticketed.
3. **Demarc test with the LEC:** where the LEC supports it, a loopback result is attached before the case opens.
4. **Close-code discipline:** NTF or carrier fault is set by the evidence, not by the carrier's closing note.

15. Field Dispatch

Field Dispatch matches work orders to technicians by skill, depot spares and branch hours, and tracks every visit. Its tabs are **Work orders**, **Technicians** and **Depot spares**. It is the Field Technician's landing page. Remote fixes come first; a visit is booked only when the fix needs hands on site.

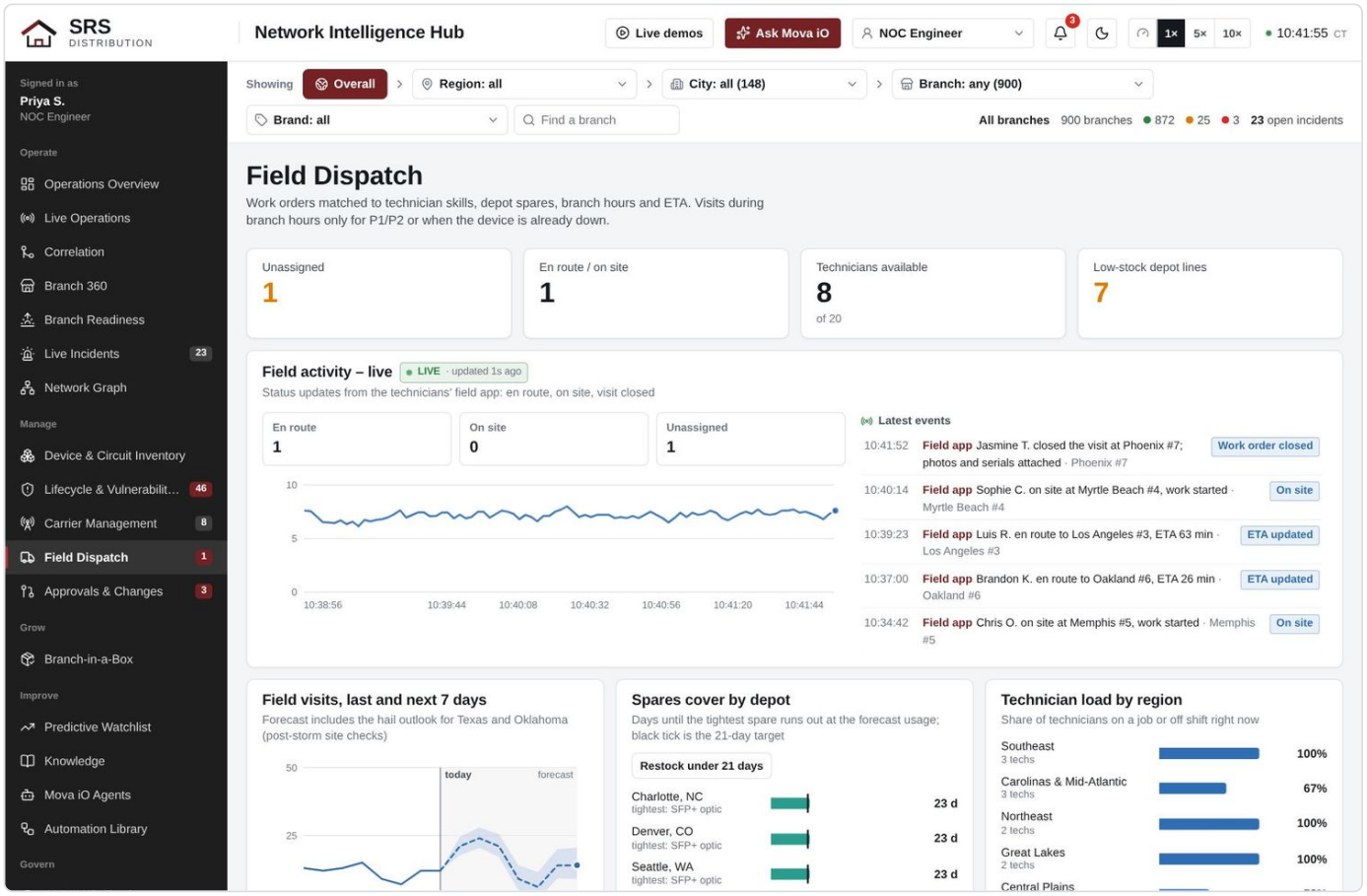


Figure 119. Field Dispatch

Tiles



Figure 120. Dispatch tiles

Tile	Meaning
Unassigned	Work orders with no technician yet (maroon badge in the menu)
En route / on site	Visits in progress
Technicians available	Technicians free now, out of the total
Low-stock depot lines	Spare lines below their minimum at any depot

Field activity – live

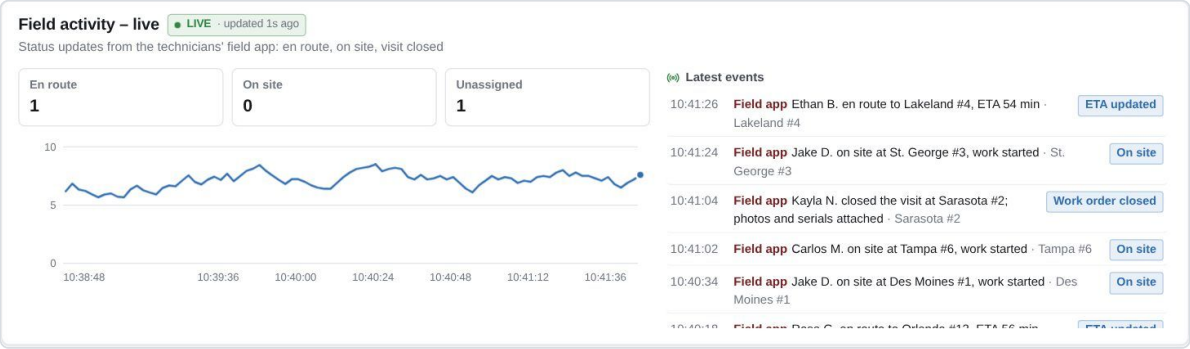


Figure 121. Field activity

Status updates from the technicians' field app: en route with ETA, on site, visit closed.

Forecasts and capacity

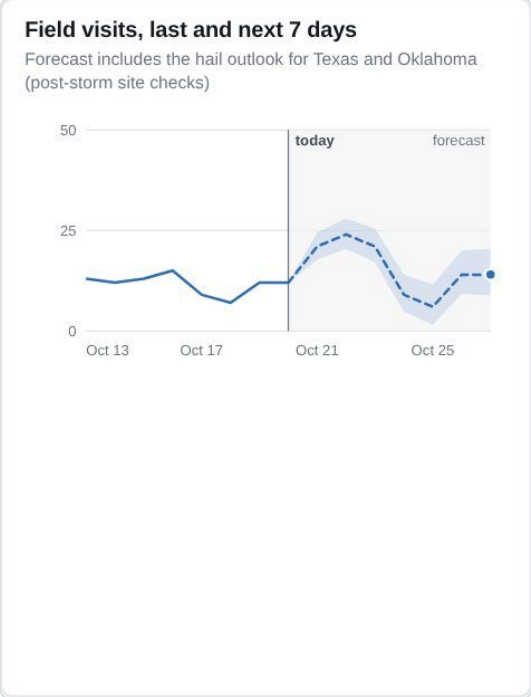


Figure 122. Field visits

Field visits, last and next 7 days: visits per day, with a forecast that includes the weather outlook, such as post-storm site checks after hail.



Figure 123. Spares cover by depot

Spares cover by depot: for each depot, the days until its tightest spare runs out at the forecast usage, and which part that is. The black tick is the 21-day target; bars under it need a restock, which **Restock under 21 days** raises.

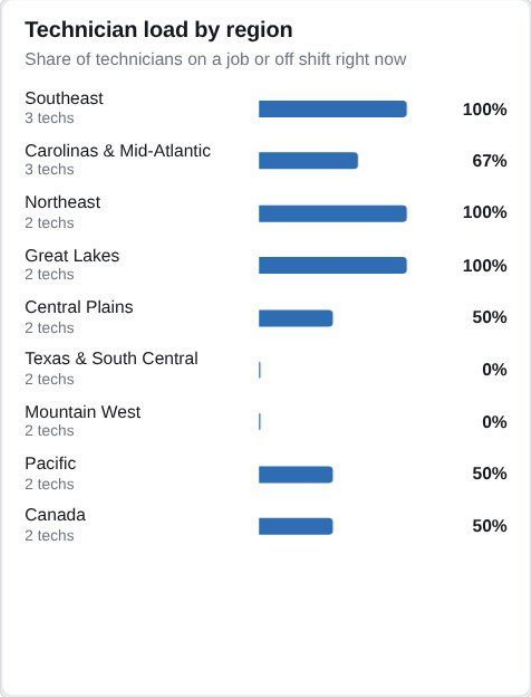


Figure 124. Technician load by region

Technician load by region: the share of technicians on a job or off shift now, per region, with the head count. A region at 100% has no one free for a new P1.

Work orders tab

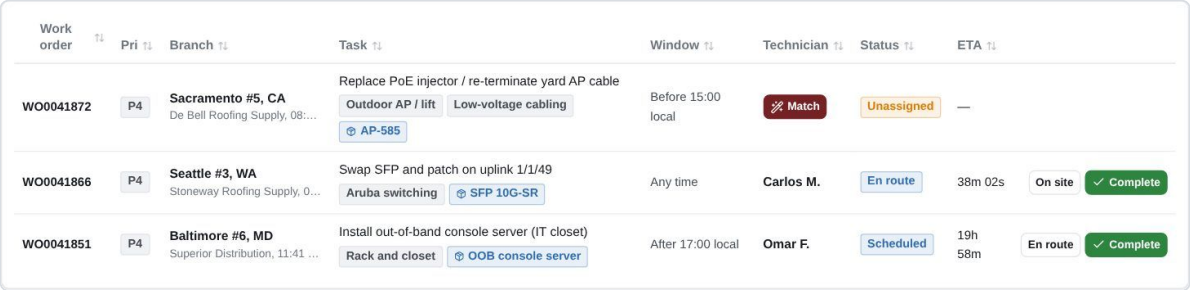


Figure 125. Work orders

Column	Meaning
Work order	The WO number
Pri	Priority of the linked incident
Branch	Branch, brand, local time and whether it is open
Task	What to do, the skills needed and the parts
Window	When the visit may happen, for example after close, before 15:00 local or "Now (P1)"
Technician	The assigned technician, or **Match** to pick the best fit
Status	Unassigned, Dispatched, En route, On site, Closed
ETA	Time to arrival

Match ranks technicians by skill fit, distance, current load and branch hours.

Technicians tab

Technician	Region	Base	Skills	Status	Jobs 30d	Rating
Luis R.	Southeast	Tampa	Outdoor AP / lift, Aruba wireless, EdgeConnect swap	On site	28	4.9
Brandon K.	Carolinas & Mid-Atlantic	Charlotte	Aruba switching, Outdoor AP / lift	On site	8	4.8
Jasmine T.	Northeast	Philadelphia	Aruba switching, LTE / cellular	On site	21	4.3
Carlos M.	Great Lakes	Detroit	Outdoor AP / lift, LTE / cellular, Aruba switching, EdgeConnect swap	On site	21	4.4
Derek H.	Central Plains	Des Moines	Low-voltage cabling, Aruba wireless	On site	25	4.4
Amanda P.	Texas & South Central	Lafayette	EdgeConnect swap, LTE / cellular, Aruba switching	Available	25	4.5
Tyrell J.	Mountain West	Colorado Springs	EdgeConnect swap, Aruba switching, Rack and closet	Available	6	4.6
Megan S.	Pacific	Yakima	Low-voltage cabling, Rack and closet, Aruba wireless, EdgeConnect swap	Available	14	4.7
Omar F.	Canada	Tampa	Low-voltage cabling, Rack and closet, EdgeConnect swap	Available	9	4.5
Hannah W.	Southeast	Orlando	Aruba switching, EdgeConnect swap, Aruba wireless, Outdoor AP / lift	On site	26	4.7
Jake D.	Carolinas & Mid-Atlantic	Greensboro	Low-voltage cabling, LTE / cellular	Available	28	4.6
Nina V.	Northeast	Amagansett	Low-voltage cabling, Aruba switching, Aruba wireless	On site	18	4.5
Chris O.	Great Lakes	Columbus	Aruba switching, Rack and closet, LTE / cellular, EdgeConnect swap	Off shift	14	4.7
Rosa G.	Central Plains	St. Louis	Aruba wireless, Aruba switching	Available	16	4.3
Ethan B.	Texas & South Central	Fort Worth	Rack and closet, EdgeConnect swap	Available	8	4.8
Kayla N.	Mountain West	Tucson	Aruba wireless, Low-voltage cabling, Aruba switching, Outdoor AP / lift, EdgeConnect swap	Available	13	4.5
Andre L.	Pacific	Oakland	Rack and closet, LTE / cellular, EdgeConnect swap	On site	17	4.8
Sophie C.	Canada	Orlando	Rack and closet, Aruba wireless, Low-voltage cabling	On site	10	4.6
Victor A.	Southeast	Miami	Aruba switching, LTE / cellular, Outdoor AP / lift, EdgeConnect swap	On site	17	4.5
Grace Y.	Carolinas & Mid-Atlantic	Greenville	Aruba switching, EdgeConnect swap, Low-voltage cabling	Off shift	13	4.9

Figure 126. Technicians

Every technician with region, base, skills (for example outdoor AP and lift, Aruba switching, LTE and cellular, EdgeConnect swap), status, jobs in 30 days and rating.

Depot spares tab

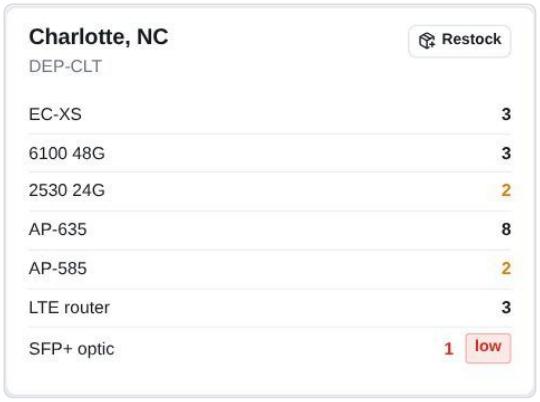


Figure 127. Depot card

One card per depot with its code and the count of each spare: EC-XS, CX 6100, 2530, AP-635, AP-585 or AP-565, LTE router and SFP+ optic. "low" marks a line below its minimum. **Restock** raises the order. Spares for a predicted failure are reserved here from the Predictive Watchlist and the inventory.

16. Approvals & Changes

Approvals & Changes is where people stay in control: the human-in-the-loop queue, the change calendar with freeze windows, Network-as-Code changes, drift from the golden configuration and firmware compliance. It is the landing page for the Network Engineer L3 and, on the firmware tab, the Security Lead.

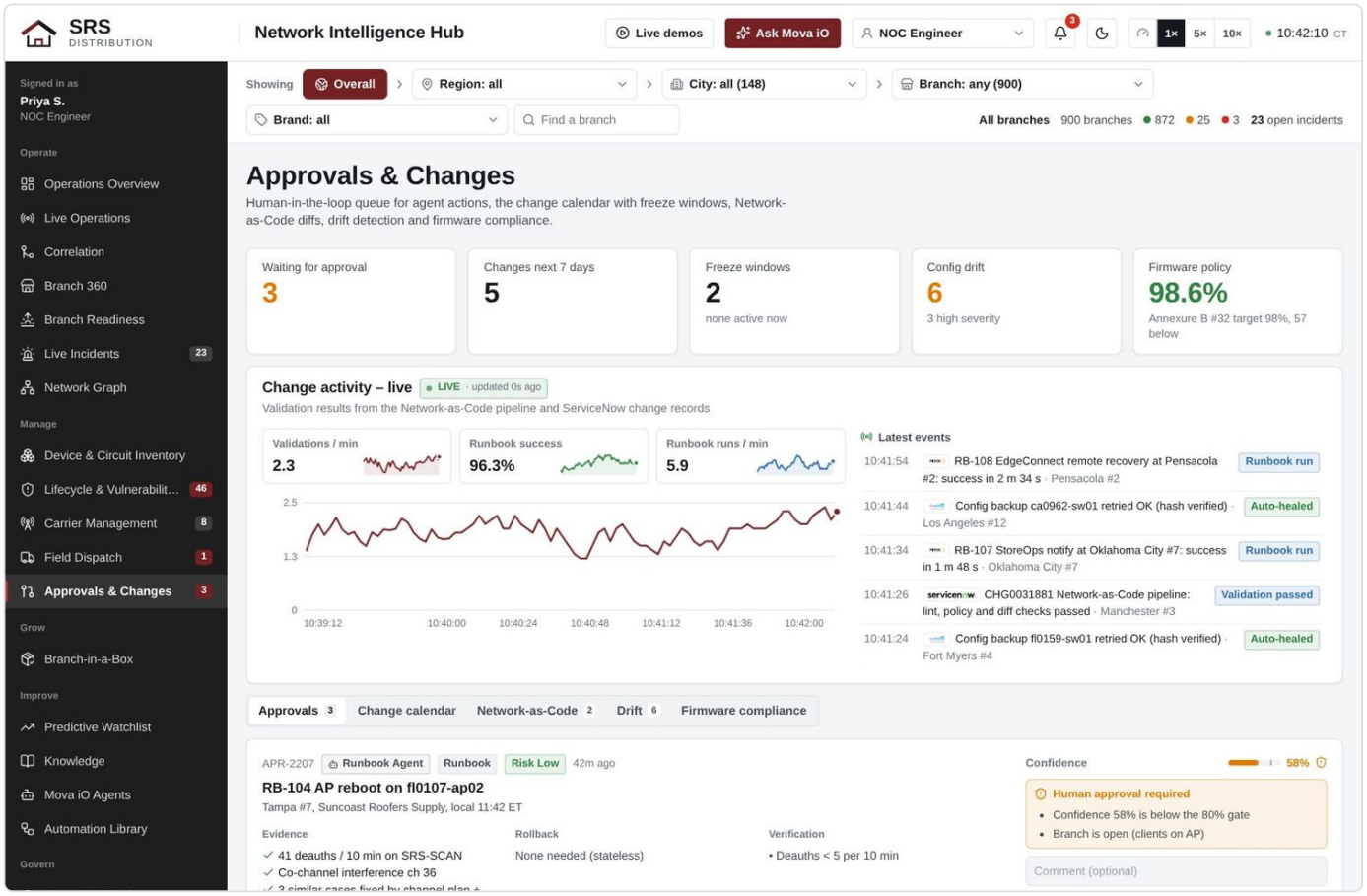


Figure 128. Approvals & Changes

Tiles

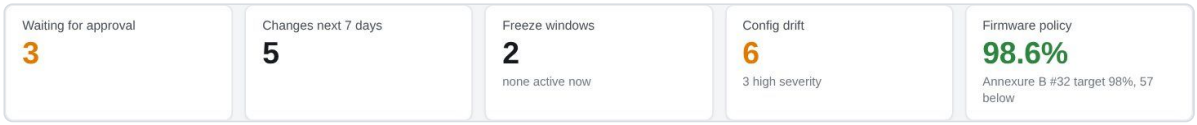


Figure 129. Change tiles

Tile	Meaning
Waiting for approval	Mova iO actions held for a person (maroon badge in the menu)
Changes next 7 days	Scheduled changes in the coming week
Freeze windows	Freeze windows defined, and whether one is active now
Config drift	Devices differing from the golden config, with how many are high severity
Firmware policy	Share of devices within the firmware policy, against service level #32 (98%), and how many are below it

Change activity – live

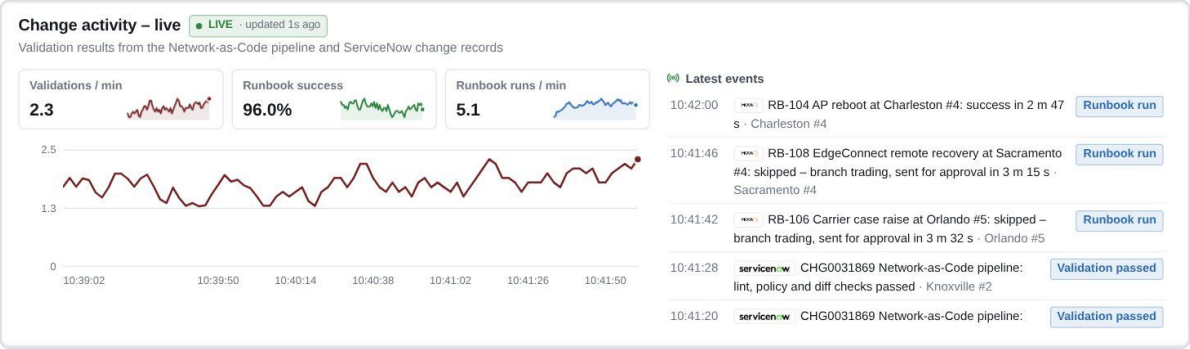


Figure 130. Change activity

Network-as-Code validations per minute, runbook success rate and runbook runs per minute, with the latest pipeline and runbook events, such as "skipped – branch trading, sent for approval".

Approvals tab



Figure 131. Approval card

Each approval is a card with everything needed to decide:

Part	Meaning
ID, agent, type, risk, age	Who is asking (for example Runbook Agent, Change Validation Agent, Prediction Agent), what kind of action, its risk and how long it has waited
Title and branch	The action and where, with the branch's local time
Evidence	Why the agent proposes it
Rollback	How it would be undone
Verification	How success will be checked
Confidence	The agent's confidence
Why a person is needed	The gate reason: confidence below 80%, branch open, freeze, or "predictions never act directly"

Approve runs the action within its runbook and records the approver. **Reject** keeps it as a recommendation and returns it to the engineer. The open incident link goes to the incident.

Change calendar tab

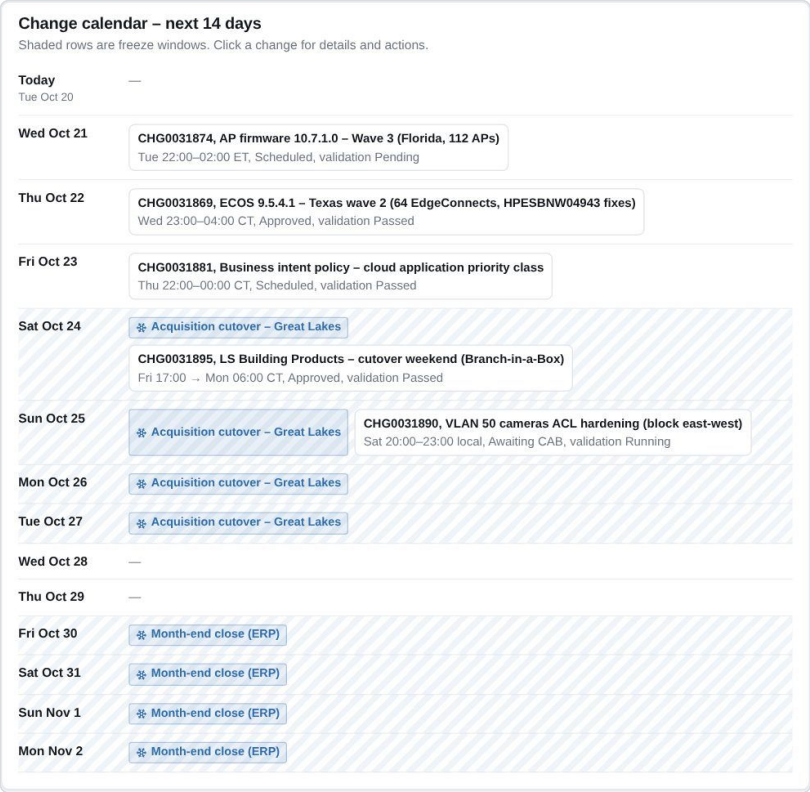


Figure 132. Change calendar

Change calendar, next 14 days: one row per day with each change, its window in local time, state (Scheduled, Approved, In progress, Paused) and validation result. Shaded rows are freeze windows. Click a change for details and actions, including **Pause** and **Rollback**.

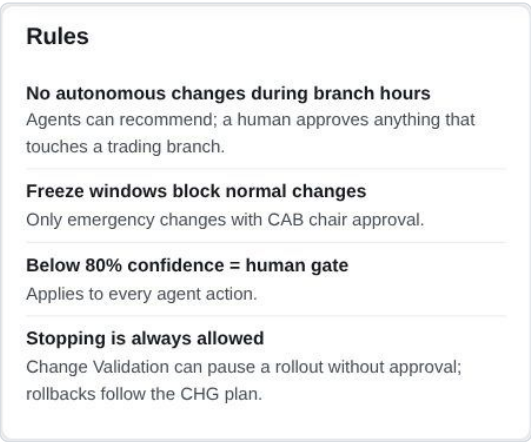


Figure 133. Rules

Rules lists the four change rules: no autonomous changes during branch hours; freeze windows block normal changes (only emergency changes with CAB chair approval); below 80% confidence means a person approves; stopping is always allowed (Change Validation can pause a rollout without approval, and rollbacks follow the change plan).

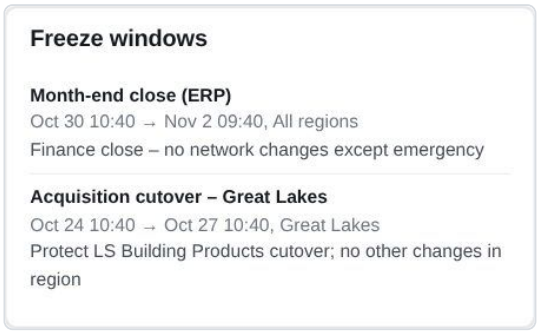


Figure 134. Freeze windows

Freeze windows lists each freeze with its dates, scope and reason, for example month-end ERP close (all regions) or an acquisition cutover in one region.

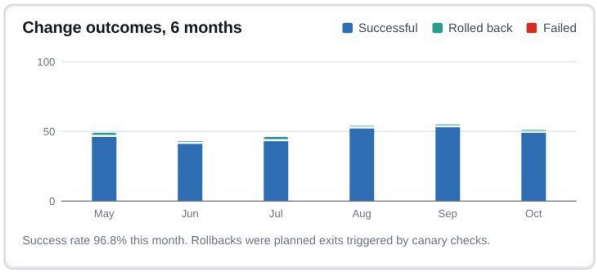


Figure 135. Change outcomes

Change outcomes, 6 months: changes per month that succeeded, were rolled back or failed, with this month's success rate. Rollbacks triggered by a canary are planned exits, not failures.

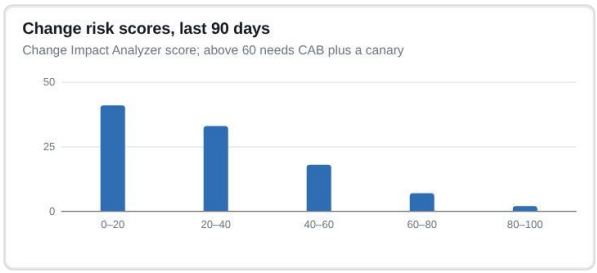


Figure 136. Change risk scores

Change risk scores, last 90 days: how many changes fell in each risk band of the Change Impact Analyzer score (0 to 100). Above 60 needs the Change Review Board and a canary.

Network-as-Code tab

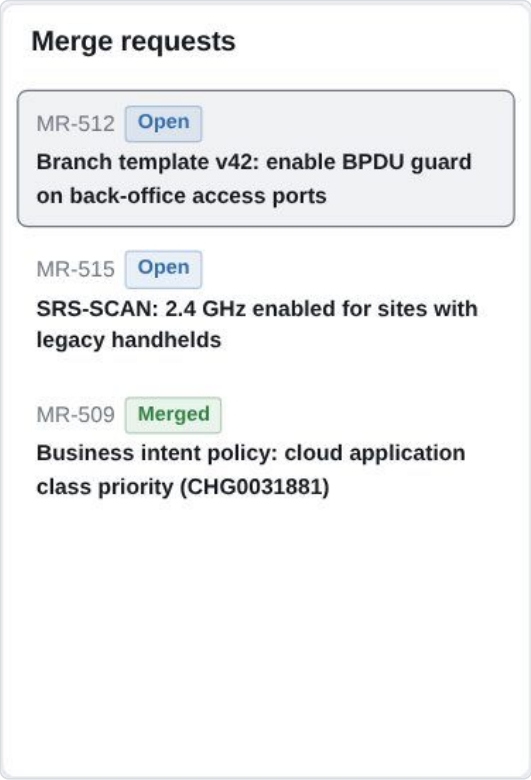


Figure 137. Merge requests

Merge requests lists proposed changes to the templates and site files in the Network-as-Code repository, open or merged.

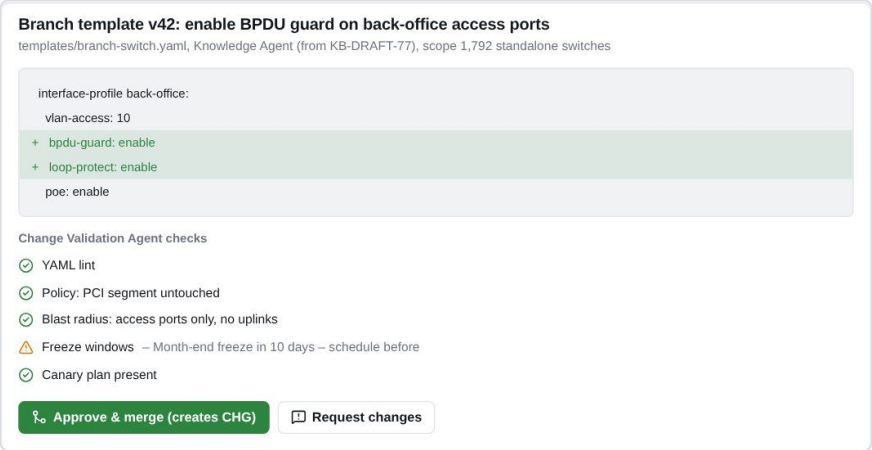


Figure 138. Merge request detail

The selected merge request shows the file, who proposed it and from what (for example the Knowledge Agent from a draft article), its scope, the diff (green lines added), and the Change Validation Agent's checks: YAML lint, policy (PCI segment untouched), blast radius, freeze windows and whether a canary plan is present. **Approve & merge** creates the change record; **Request changes** sends it back.

Drift tab

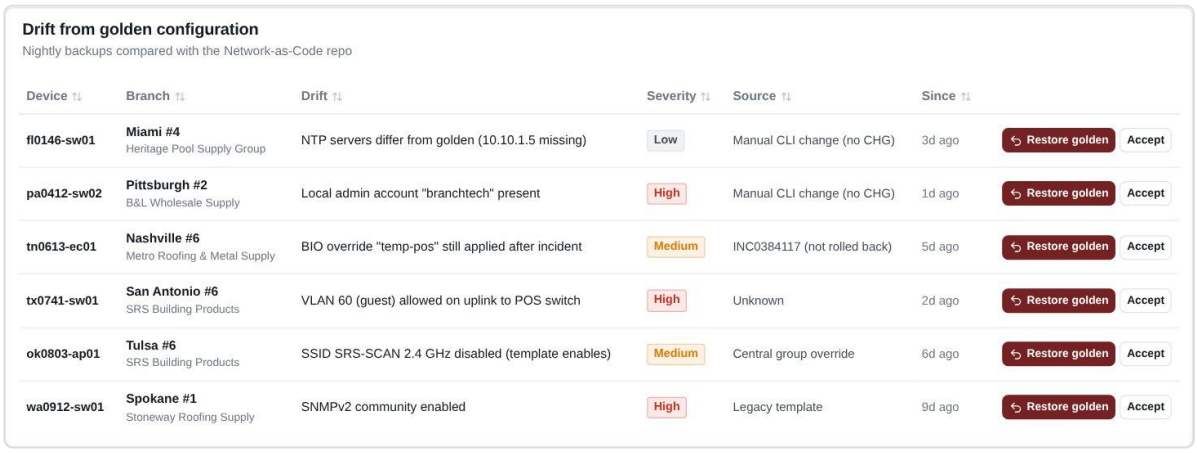


Figure 139. Drift from golden configuration

Nightly backups compared with the Network-as-Code repository. Each row shows the device, branch, the drift found, its severity, its source (for example a manual CLI change with no change record) and how long ago. **Restore golden** puts the approved config back (a change, so it follows the gate rules); **Accept** makes the drift the new approved state through a change.

Firmware compliance tab

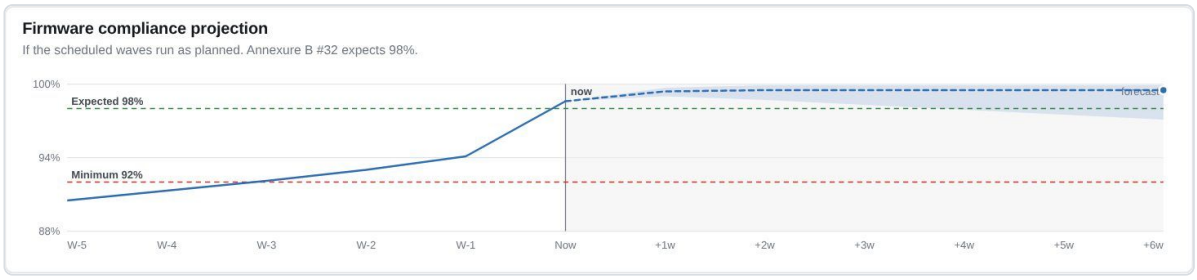


Figure 140. Firmware compliance projection

Firmware compliance projection: compliance for the last five weeks and the forecast for six weeks if the scheduled waves run, against the expected 98% and minimum 92% lines from service level #32.

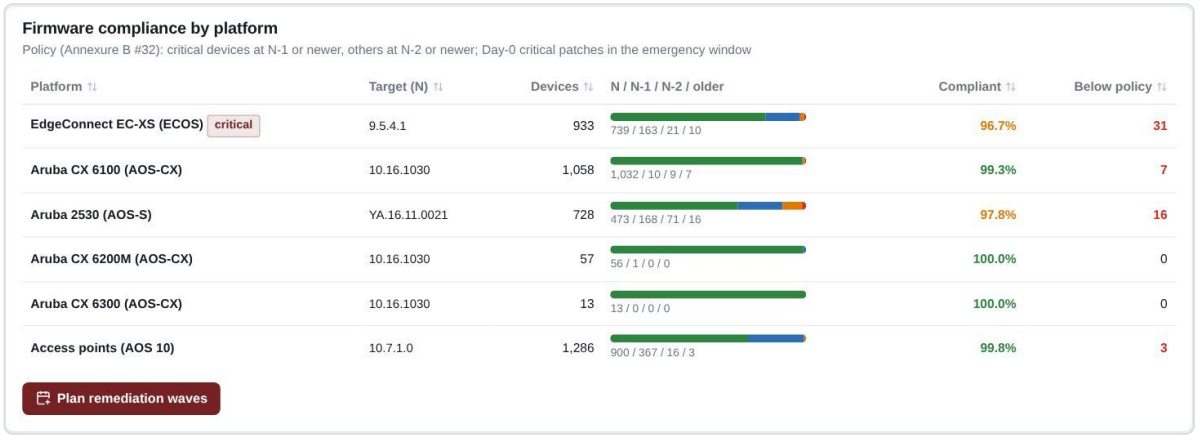


Figure 141. Firmware compliance by platform

Firmware compliance by platform: for each platform, the target release (N), device count, how many are on N, N-1, N-2 and older, the share compliant and how many are below policy. The policy: critical devices (EdgeConnects) at N-1 or newer, others at N-2 or newer, and Day-0 critical patches inside the emergency window.

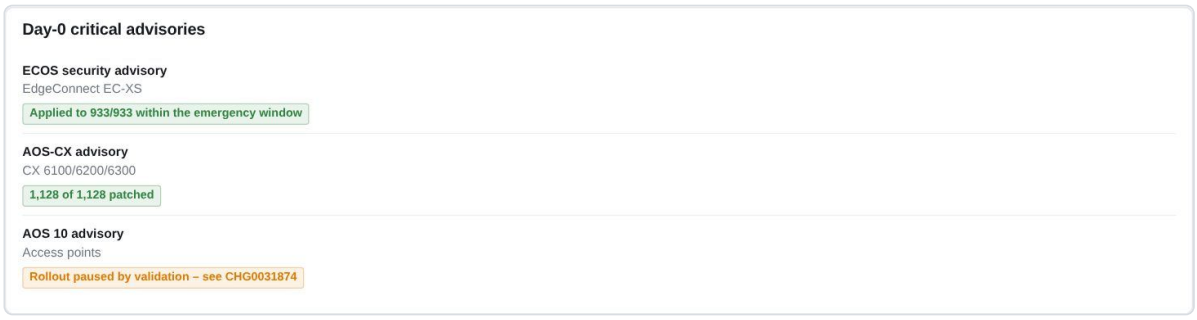


Figure 142. Day-0 critical advisories

Day-0 critical advisories: each critical advisory and its status: applied to every affected device within the emergency window, partly patched, or paused by validation with the change number.

17. Branch-in-a-Box

Branch-in-a-Box is the standard, repeatable kit and process that brings any new or acquired branch onto the SRS standard network, usually over one weekend, with the business trading again on Monday morning. Acquired branches keep their brand name. It is the M&A Integration Lead's landing page. It is a project board, not a live feed: nothing on it changes every few seconds; numbers move only when a gate passes, a stage completes or a project is created. Its tabs are **Portfolio**, **Two tracks**, **Kit & VLANs**, **Floor plan**, **Acquired-branch steps**, **Cutover weekend** and **Day-1 parity**.

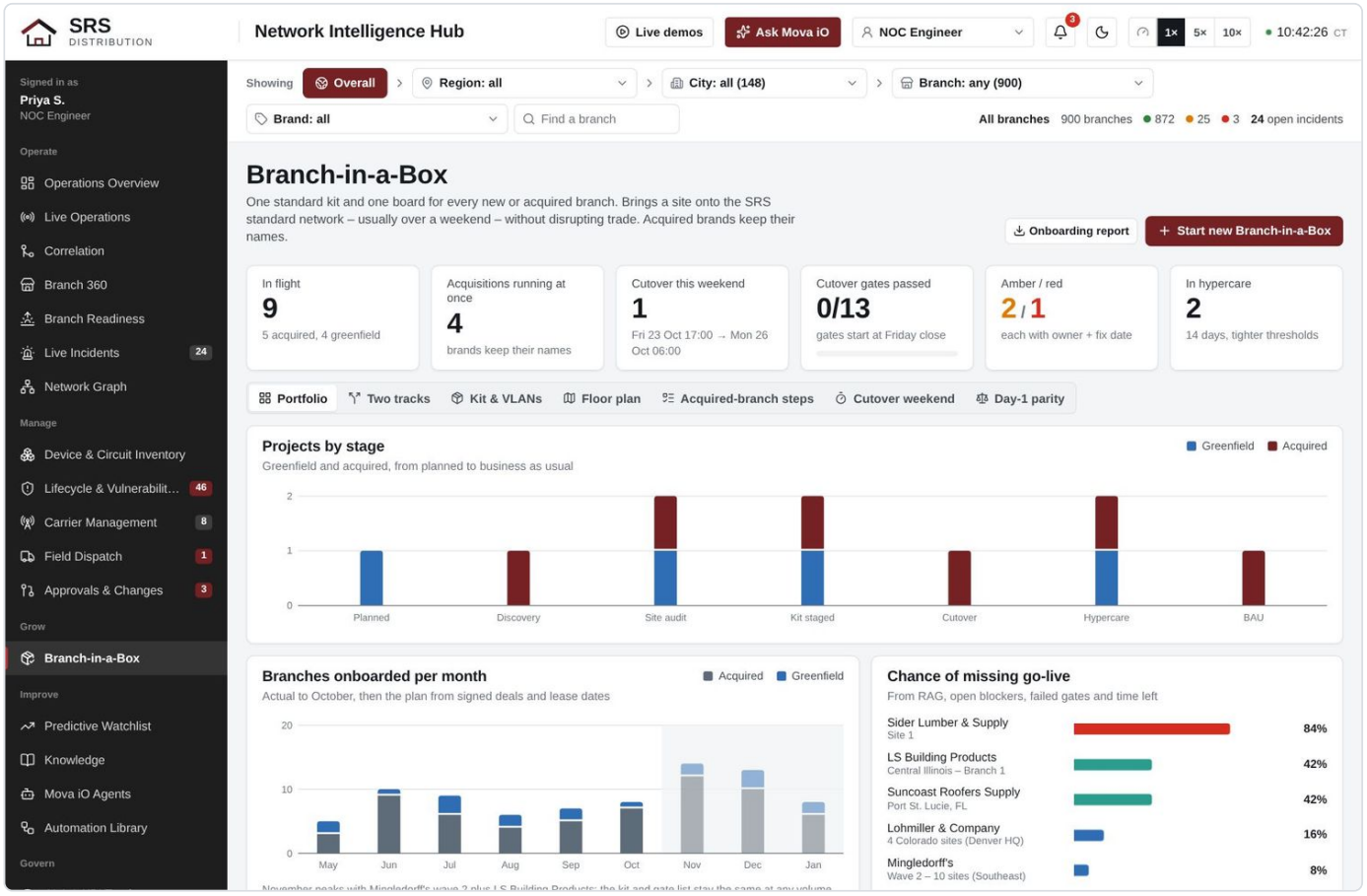


Figure 143. Branch-in-a-Box

Tiles

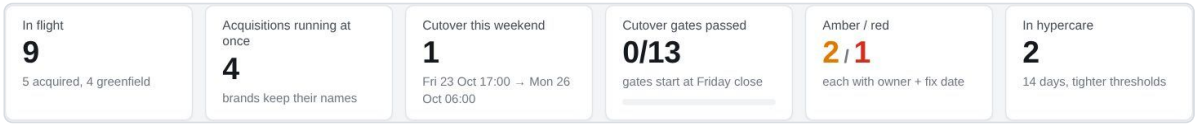


Figure 144. Branch-in-a-Box tiles

Tile	Meaning
In flight	Projects running, split into acquired and greenfield
Acquisitions running at once	Acquired brands being onboarded in parallel; brands keep their names

Tile	Meaning
Cutover this weekend	Projects whose cutover runs this coming weekend, with the dates (Friday 17:00 close to Monday 06:00)
Cutover gates passed	Gates passed out of 13 for that cutover; the gates start at Friday close, so the count is 0 during the week
Amber / red	Projects at risk, each with an owner and a fix date
In hypercare	Projects in their 14 days of tighter thresholds after go-live

Start new Branch-in-a-Box opens a wizard that creates a project from the template; **Onboarding report** downloads the report for the selected project.

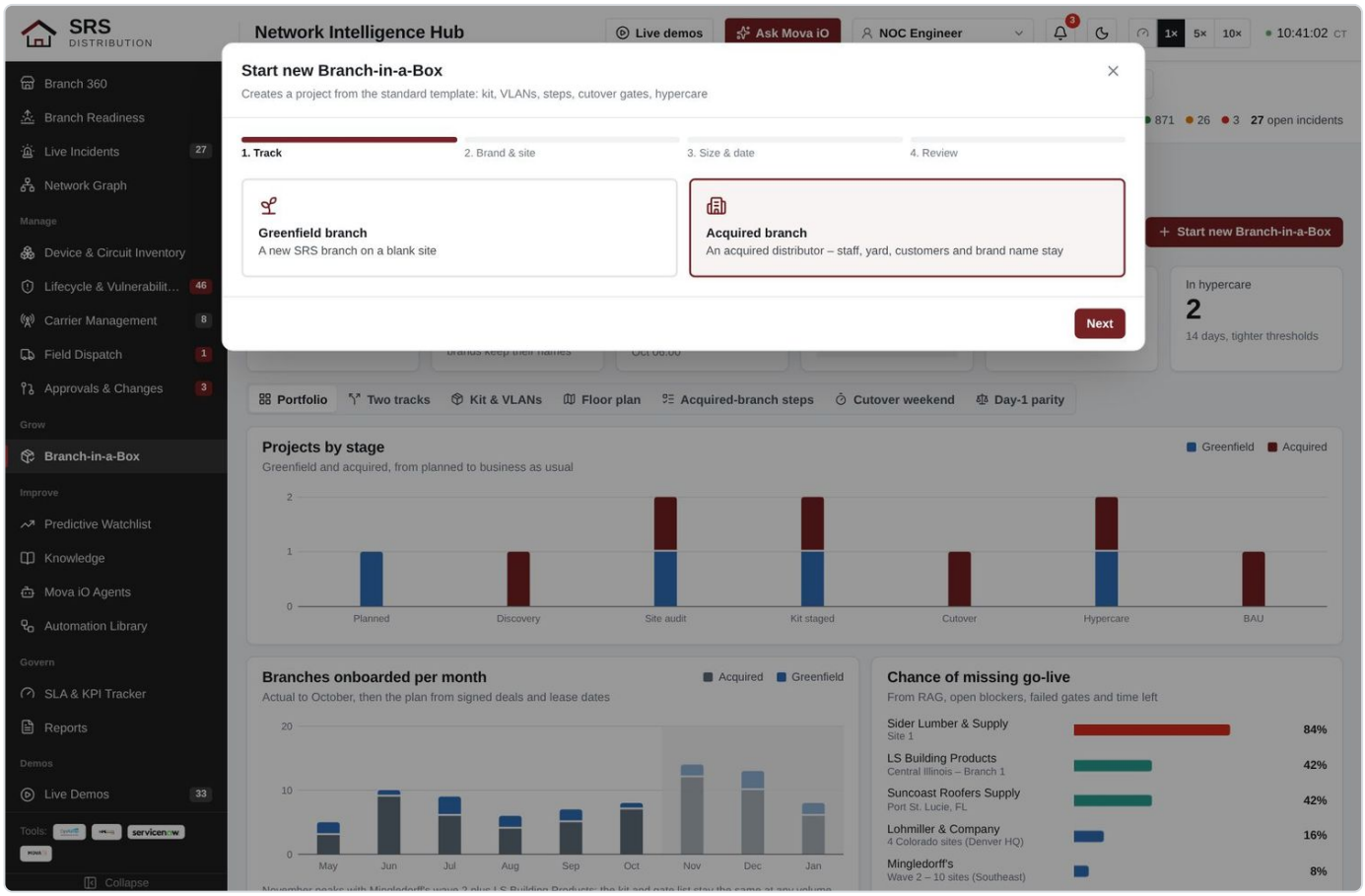


Figure 145. New project wizard

Portfolio tab

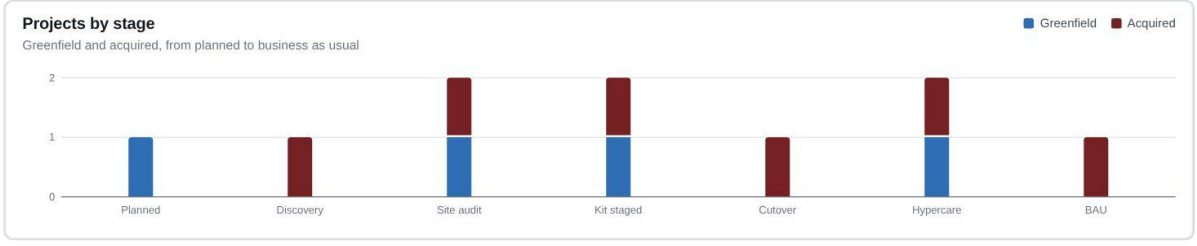


Figure 146. Projects by stage

Projects by stage: greenfield and acquired projects counted at each stage: Planned, Discovery, Site audit, Kit staged, Cutover, Hypercare, BAU.

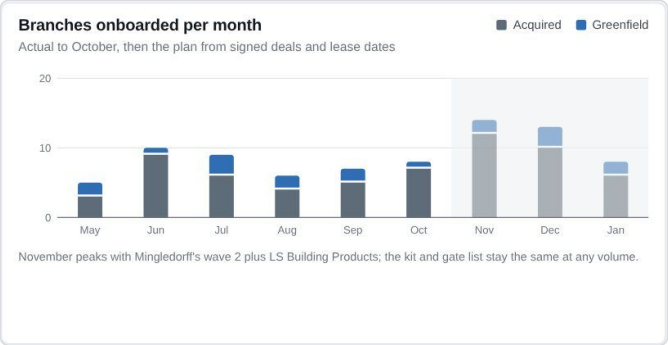


Figure 147. Branches onboarded per month

Branches onboarded per month: actual onboarding to date, then the plan from signed deals and lease dates, split into acquired and greenfield. The same kit and gate list apply at any volume.



Figure 148. Chance of missing go-live

Chance of missing go-live: each project's risk of missing its date, worked out from its RAG status, open blockers, failed gates and time left. Above 50% needs the integration lead's attention this week.

The project table on this tab lists every project with brand, site, track, stage, RAG, gates passed, days to go-live and blockers.

Two tracks tab

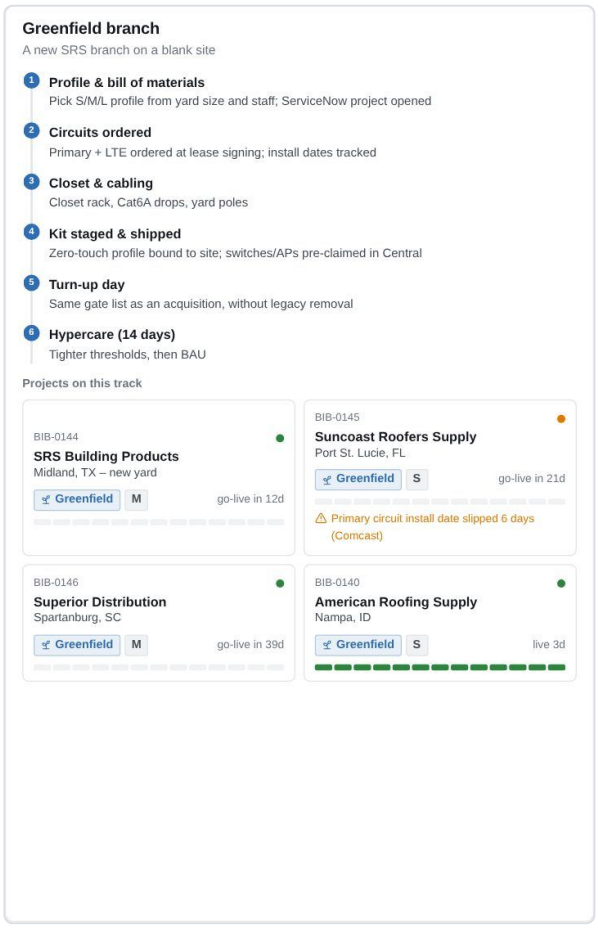


Figure 149. Greenfield track

Greenfield branch (a new SRS branch on a blank site): profile and bill of materials, circuits ordered at lease signing, closet and cabling, kit staged and shipped with zero-touch profiles, turn-up day using the same gates as an acquisition, and 14 days of hypercare.

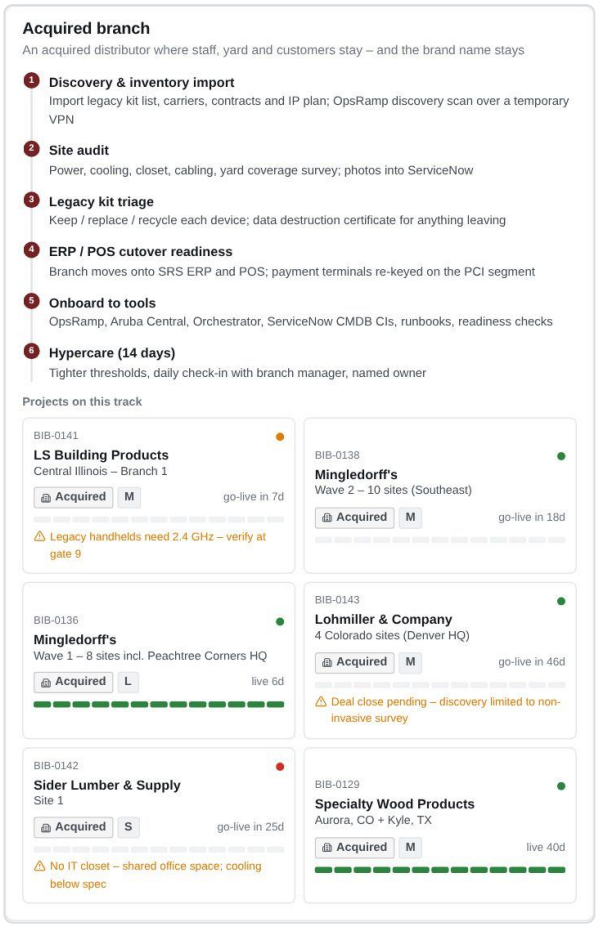


Figure 150. Acquired track

Acquired branch (staff, yard, customers and brand stay): discovery and inventory import, site audit, legacy kit triage (keep, replace or recycle, with a data destruction certificate), ERP and POS cutover readiness, onboarding to the tools, and hypercare. Each track lists its current projects.

Kit & VLANs tab

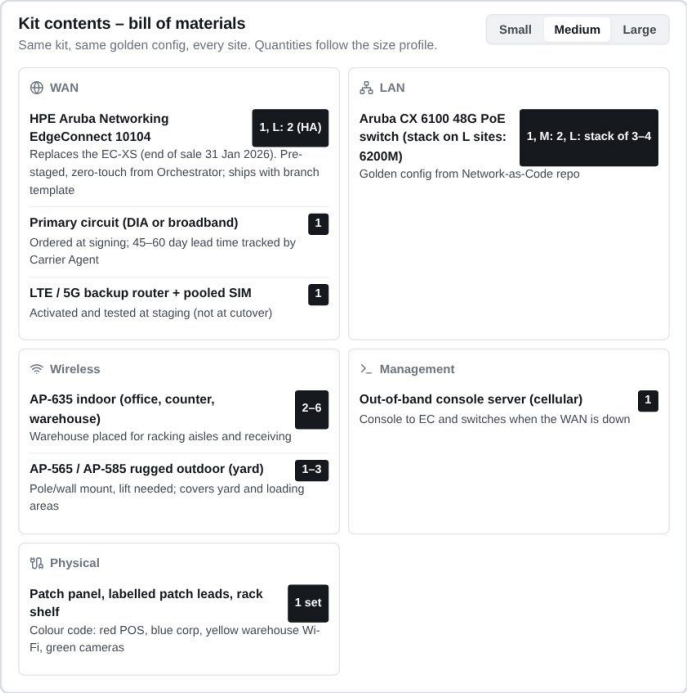


Figure 151. Kit contents

Kit contents, bill of materials: the same kit and golden config at every site, with quantities by size profile (Small, Medium, Large):

Group	Items
WAN	EdgeConnect 10104 (two as an HA pair on large sites), which replaces the EC-XS after its end of sale on 31 Jan 2026, pre-staged for zero-touch; a primary circuit ordered at signing; an LTE/5G backup router with a pooled SIM, tested at staging
LAN	Aruba CX 6100 48-port PoE switches (a 6200M stack on large sites) with the golden config
Wireless	AP-635 indoor for office, counter and warehouse; AP-565 or AP-585 rugged outdoor for the yard
Management	An out-of-band console server on cellular, for console access when the WAN is down
Physical	Patch panel, labelled leads and a colour code: red POS, blue corporate, yellow warehouse Wi-Fi, green cameras

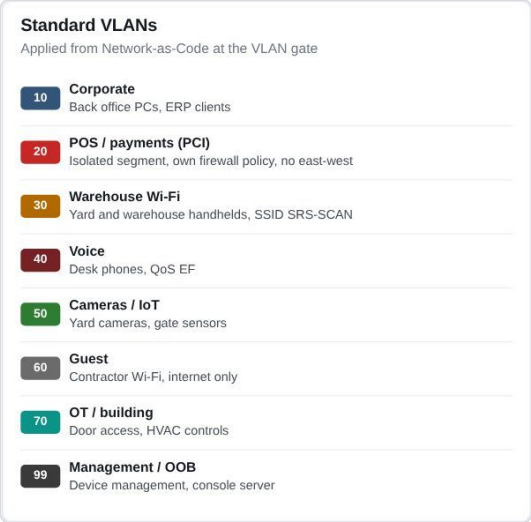


Figure 152. Standard VLANs

Standard VLANs, applied from Network-as-Code at the VLAN gate: 10 Corporate, 20 POS and payments (PCI), 30 Warehouse Wi-Fi (SSID SRS-SCAN), 40 Voice (QoS EF), 50 Cameras and IoT, 60 Guest (internet only), 70 OT and building, 99 Management and out-of-band.

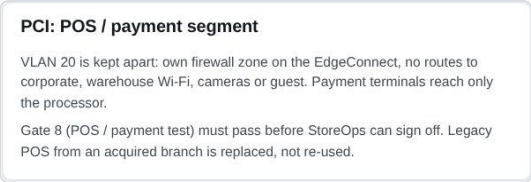


Figure 153. PCI segment

PCI: POS and payment segment. VLAN 20 has its own firewall zone on the EdgeConnect and no routes to corporate, warehouse Wi-Fi, cameras or guest; payment terminals reach only the processor. Gate 8 must pass before StoreOps can sign off, and legacy POS from an acquired branch is replaced, not re-used.

Floor plan tab



Figure 154. Standard floor plan

The standard medium branch layout: counter and showroom, back office, IT closet, warehouse, loading docks, outdoor yard and staging area, each with its drops and VLANs, and the AP positions (indoor and rugged outdoor). Click a zone to see its devices, drops, VLANs and what is tested there at cutover.

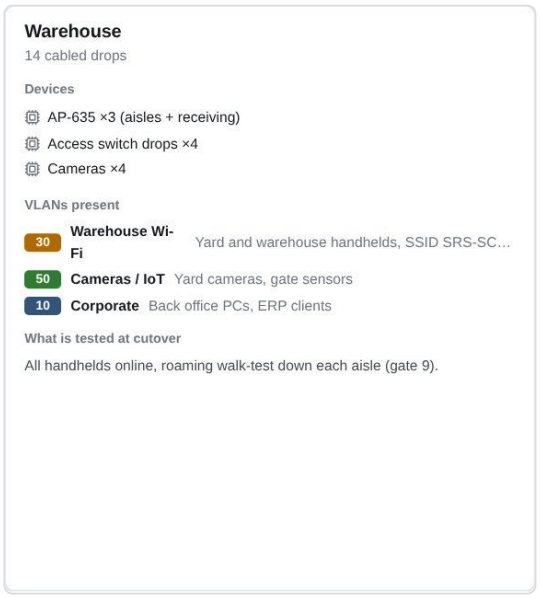


Figure 155. Zone detail

Acquired-branch steps tab

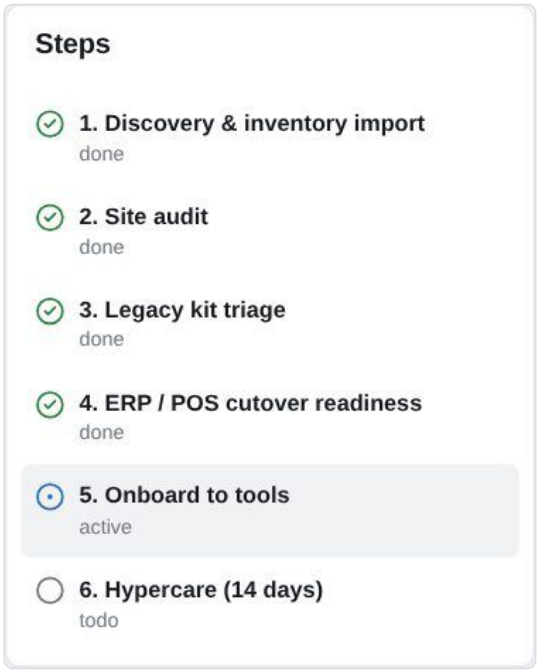


Figure 156. Acquired-branch steps

The six steps for the selected acquired project, each done, active or to do. The active step shows its checklist; for **Onboard to tools** that is OpsRamp discovery and monitors, Aruba Central claim and group, Orchestrator template, ServiceNow CIs and relationships, runbooks RB-101 to RB-110 enabled, readiness checks scheduled at 05:30 local and Mova iO agents scoped to the site. **Run onboarding automation** performs the checklist; **Complete step** moves on.

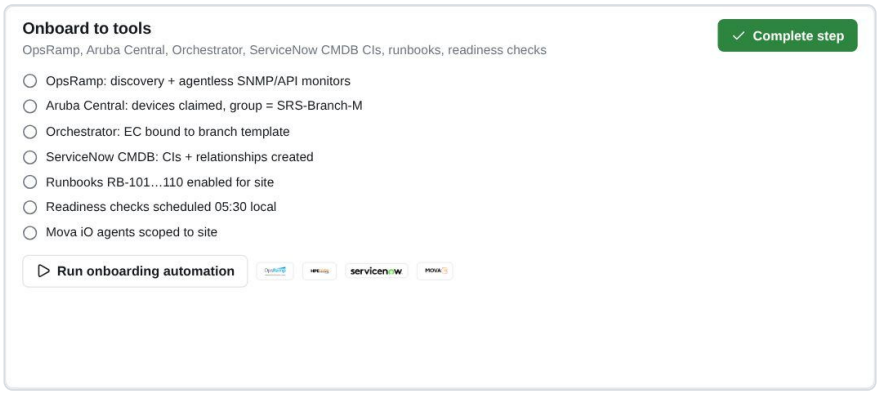


Figure 157. Onboard to tools

Cutover weekend tab

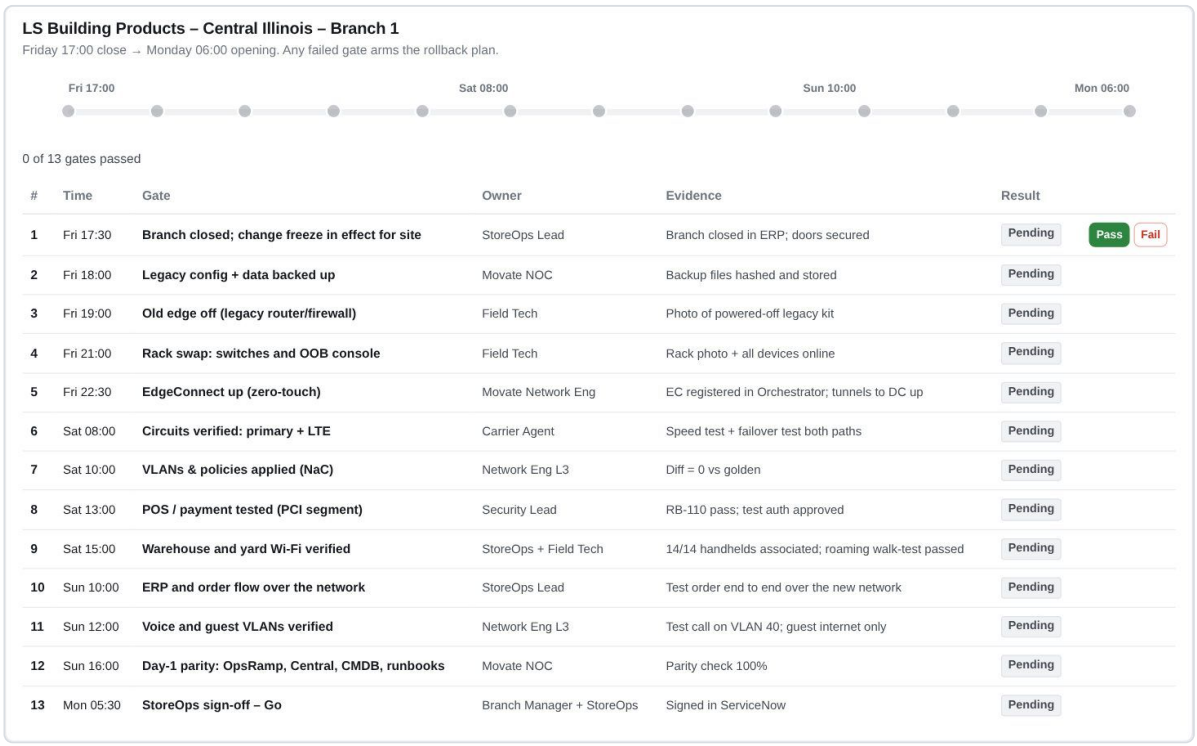


Figure 158. Cutover gate tracker

The gate tracker for the cutover weekend, from Friday 17:00 close to Monday 06:00 opening, with a timeline and the gates passed. Each of the 13 gates has a time, an owner, the evidence required and a result (Pending, Pass or Fail, with **Pass** and **Fail** buttons). Any failed gate arms the rollback plan.

#	Time	Gate	Owner
1	Fri 17:30	Branch closed; change freeze in effect for site	StoreOps Lead
2	Fri 18:00	Legacy config and data backed up	Movate NOC
3	Fri 19:00	Old edge off (legacy router or firewall)	Field Tech
4	Fri 21:00	Rack swap: switches and OOB console	Field Tech
5	Fri 22:30	EdgeConnect up (zero-touch)	Movate Network Eng

#	Time	Gate	Owner
6	Sat 08:00	Circuits verified: primary and LTE	Carrier Agent
7	Sat 10:00	VLANs and policies applied (Network-as-Code)	Network Eng L3
8	Sat 13:00	POS and payment tested (PCI segment)	Security Lead
9	Sat 15:00	Warehouse and yard Wi-Fi verified	StoreOps and Field Tech
10	Sun 10:00	ERP and order flow over the network	StoreOps Lead
11	Sun 12:00	Voice and guest VLANs verified	Network Eng L3
12	Sun 16:00	Day-1 parity: OpsRamp, Central, CMDB, runbooks	Movate NOC
13	Mon 05:30	StoreOps sign-off: Go	Branch Manager and StoreOps

Day-1 parity tab



Figure 159. Parity with standard branches

Day-1 parity checks that the new branch has the same telemetry, automation, runbooks and AI coverage as every other branch from its first day: monitored in OpsRamp, claimed in Central, bound in Orchestrator, in the CMDB, runbooks enabled, readiness checks scheduled and agents scoped. The percentage reaches 100% at gate 12.

18. Predictive Watchlist

The Predictive Watchlist holds every prediction Mova iO has made about the network: degrading circuits, access points and switches likely to fail, capacity running out and backup paths at risk. A prediction never changes anything by itself. It can only create a watch or ask for approval. The tabs filter by kind: **All**, **Degrading circuit**, **AP failure risk**, **Switch failure risk**, **Capacity forecast** and **Backup path risk**.

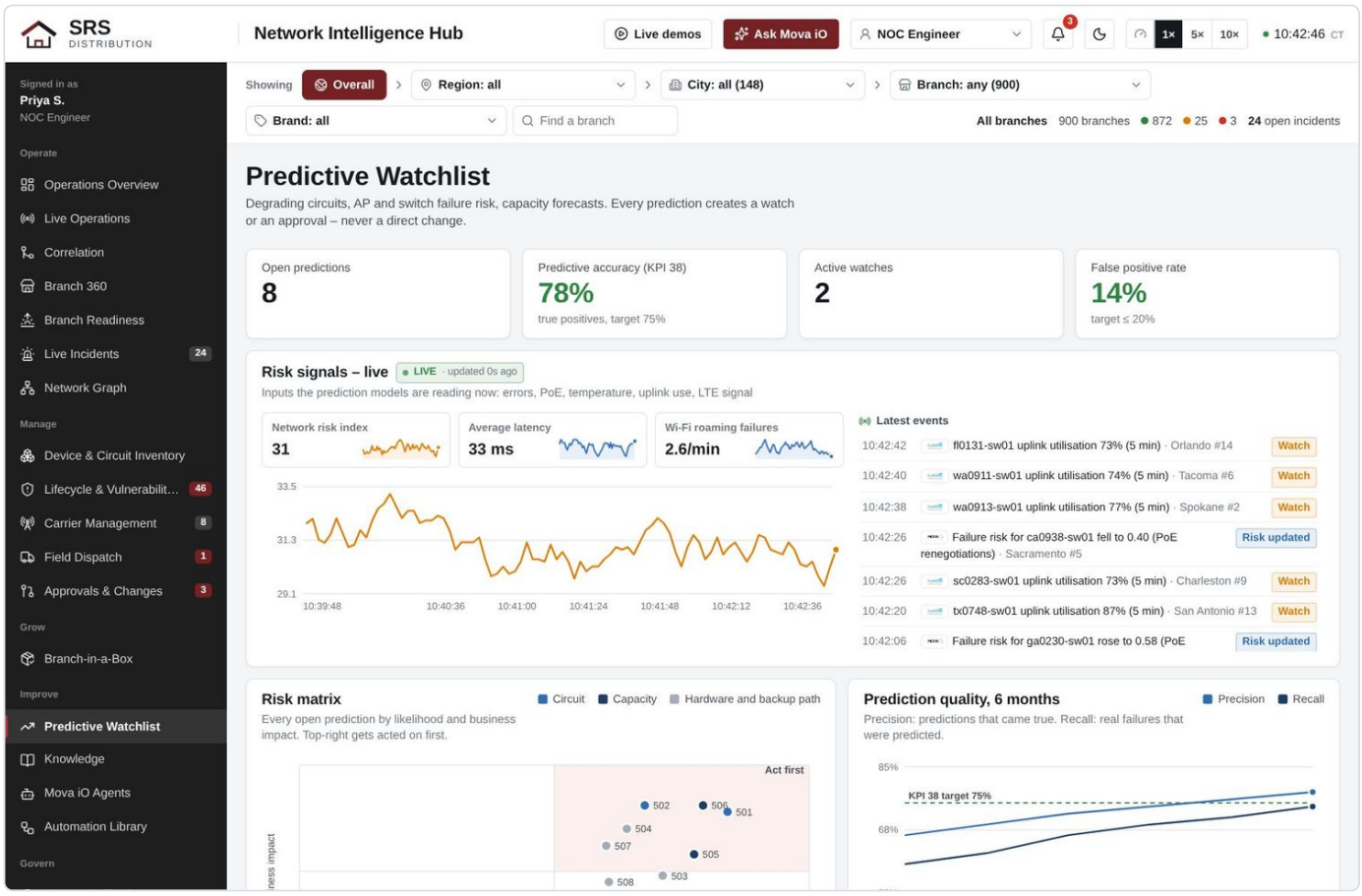


Figure 160. Predictive Watchlist

Tiles



Figure 161. Prediction tiles

Tile	Meaning	Target
Open predictions	Predictions not yet resolved or dismissed	–
Predictive accuracy (KPI 38)	Share of predictions that came true	75% or more
Active watches	Predictions being watched with an alert set	–
False positive rate	Share of predictions marked false	20% or less

Risk signals – live

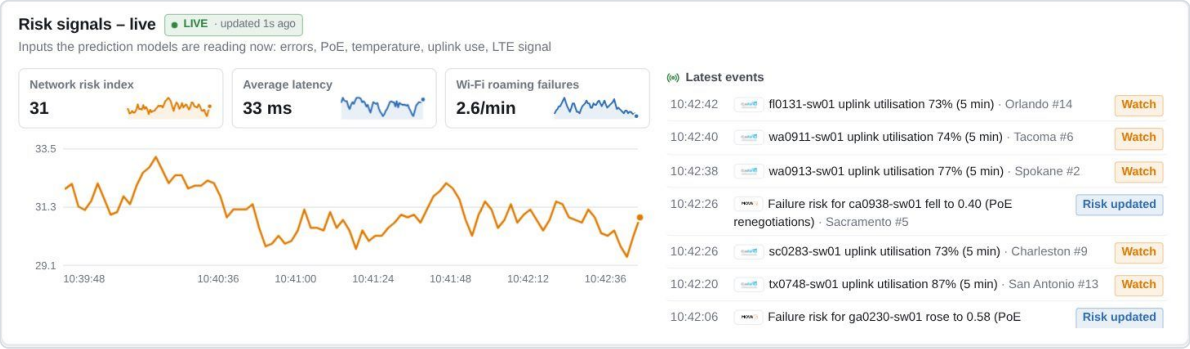


Figure 162. Risk signals

The inputs the models are reading now: a network risk index (the average risk score across branches), the average latency to the data centre and Wi-Fi roaming failures, with the latest risk events such as uplink use, PoE renegotiations, temperature and LTE signal.

Risk matrix



Figure 163. Risk matrix

Every open prediction placed by likelihood within its horizon (across) and business impact (up), labelled with its number and coloured by kind: circuit, capacity, or hardware and backup path. The top-right corner is **Act first**; the bottom-left is **Watch**.

Prediction quality, 6 months

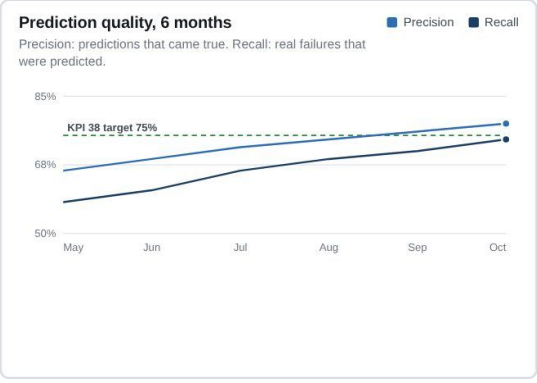


Figure 164. Prediction quality

Precision is the share of predictions that came true; **recall** is the share of real failures that were predicted. Both should stay above the 75% line of KPI 38. Precision falling means too many false alarms; recall falling means failures are being missed.

Preventive maintenance, next 9 days

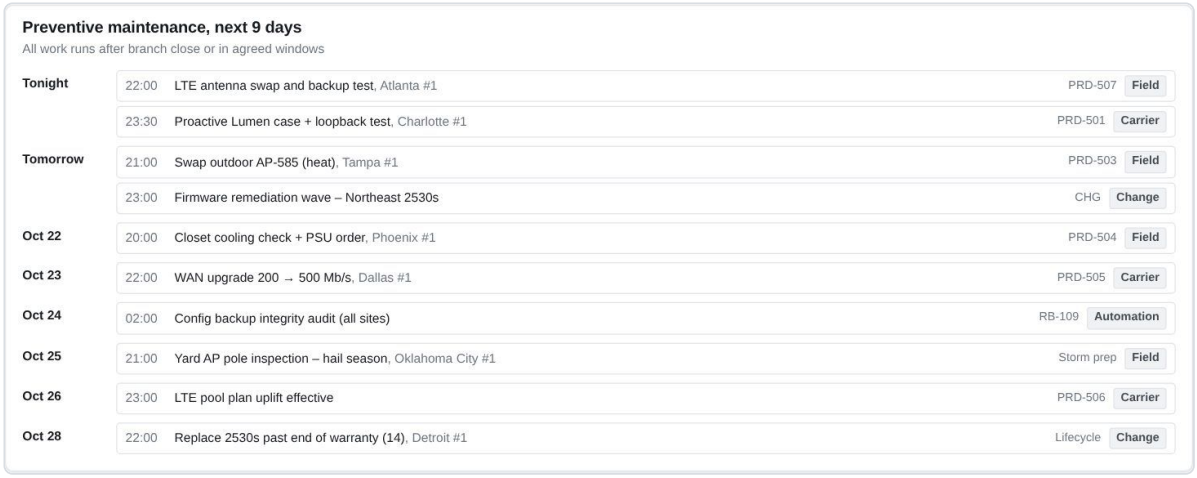


Figure 165. Preventive maintenance

Work booked from predictions, by day and time, with the branch, the prediction or change it came from and the owner (Field, Carrier or Change). All of it runs after branch close or in agreed windows.

Prediction cards



Figure 166. Prediction card

Part	Meaning
Number, kind, horizon, state	For example PRD-501, degrading circuit, 48 to 72 hours, New or Watching
Target	The circuit, device or pool, with its branch
Signal	What the model sees, such as loss rising from 0.1% to 0.9% over five days
Chart	The metric for the last nine days and the forecast, against its limit
Probability	Chance the failure happens within the horizon
Suggests	The preventive action, and whether it needs approval

Create watch sets an alert on the metric. **Request approval** sends the suggested action to Approvals & Changes. **False positive** dismisses the prediction and feeds back into the model's accuracy.

Capacity forecast



Figure 167. Capacity forecast

For the capacity prediction in view: p95 utilisation for ten weeks of actuals and four weeks of forecast, with the likely range and the 90% upgrade threshold. Where the forecast crosses 90% is when the upgrade must be in place.

19. Knowledge

Knowledge holds the articles engineers use, finds similar past incidents, reviews articles Mova iO drafts from resolved incidents and tracks problem records. Its tabs are **Search**, **Auto-generated drafts** and **Problem records**.

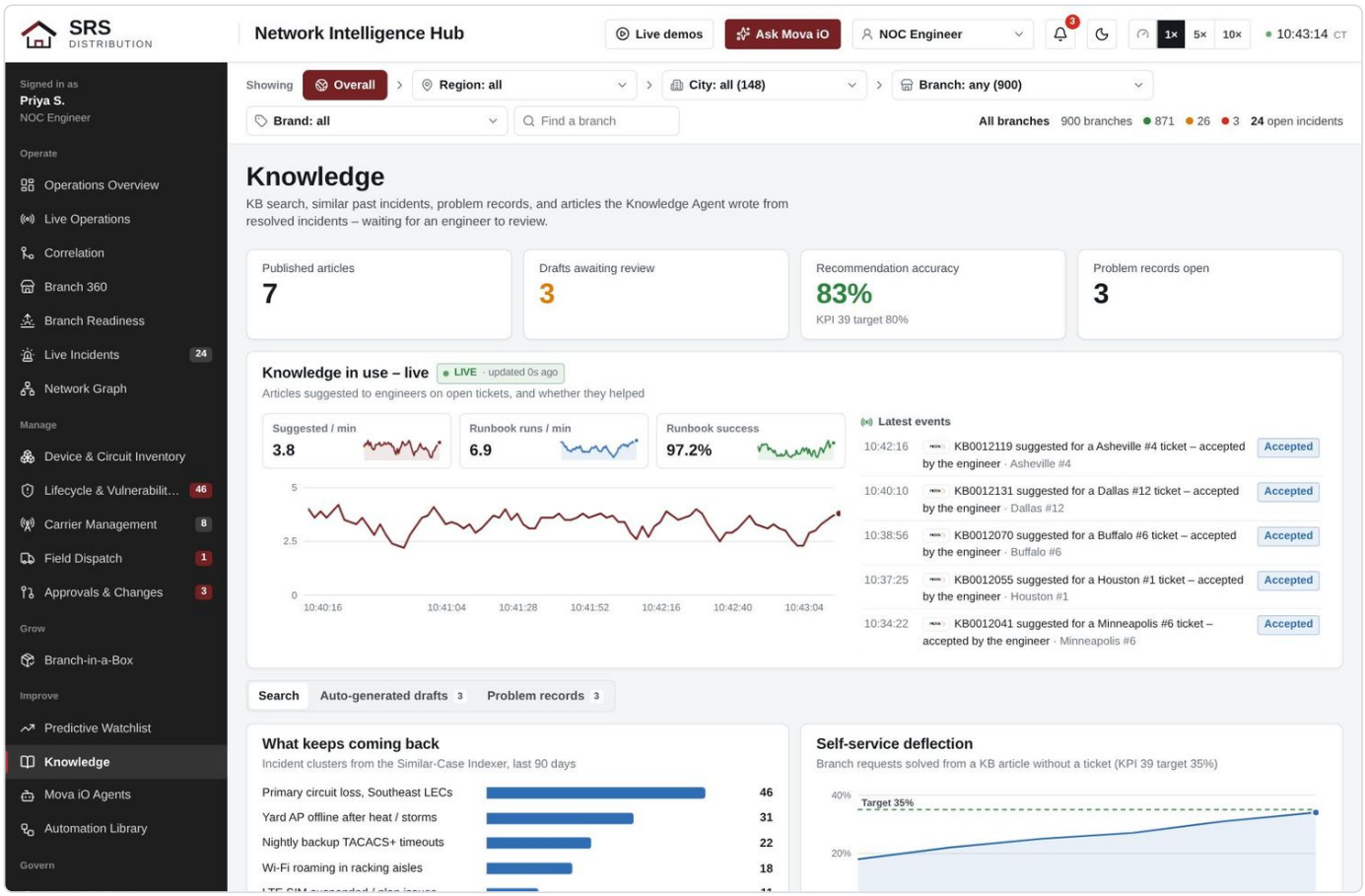


Figure 168. Knowledge

Tiles



Figure 169. Knowledge tiles

Tile	Meaning	Target
Published articles	Articles live in the knowledge base	–
Drafts awaiting review	Articles written by the Knowledge Agent, waiting for an engineer	–
Recommendation accuracy (KPI 39)	Share of suggested articles that engineers found relevant	80%
Problem records open	Open problems: the causes behind repeat incidents	–

Knowledge in use – live

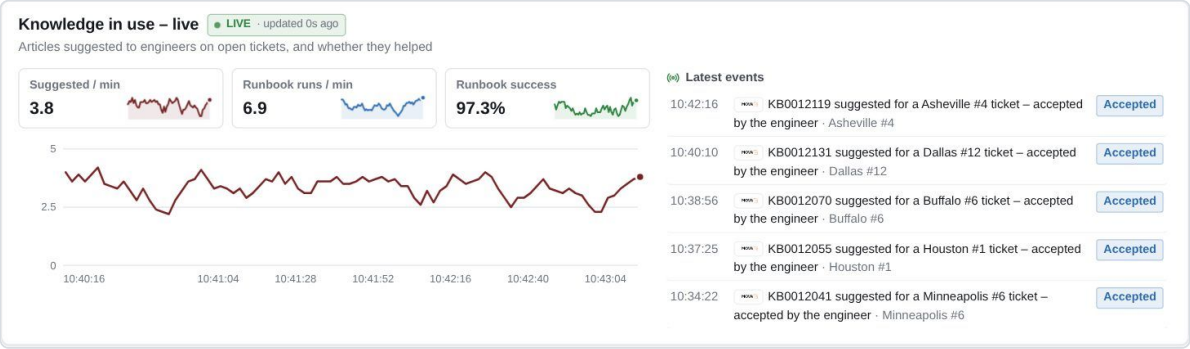


Figure 170. Knowledge in use

Articles suggested per minute on open tickets, runbook runs and success, and the latest suggestions with whether the engineer accepted the article or marked it not relevant.

What keeps coming back

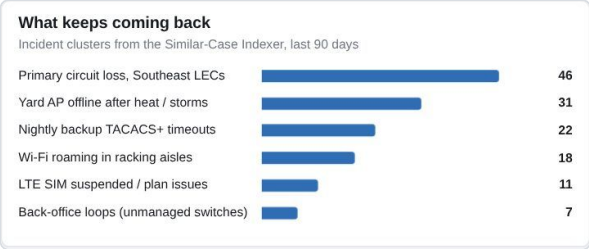


Figure 171. What keeps coming back

Incident clusters found by the Similar-Case Indexer over 90 days, longest bar first: for example primary circuit loss on Southeast LECs, yard APs offline after heat or storms, backup timeouts. Each cluster is a candidate for a problem record and a permanent fix.

Self-service deflection



Figure 172. Self-service deflection

Branch requests solved from a knowledge article without a ticket, by month, against the 35% target of KPI 39.

Most used articles

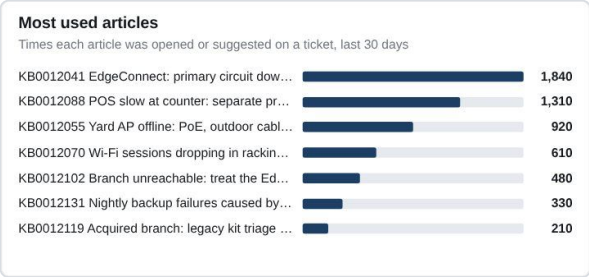


Figure 173. Most used articles

The seven articles opened or suggested most often on tickets in the last 30 days, with the count. The top articles are usually the runbook-backed ones for the commonest faults, such as a primary circuit down or a slow payment path.

Suggestions accepted by engineers



Figure 174. Suggestions accepted

Articles Mova iO suggested on tickets each week, split into those the engineer accepted and those marked not relevant. A rising green share means the recommendations are getting better (KPI 39 asks for 80% relevant).

Knowledge base and similar incidents

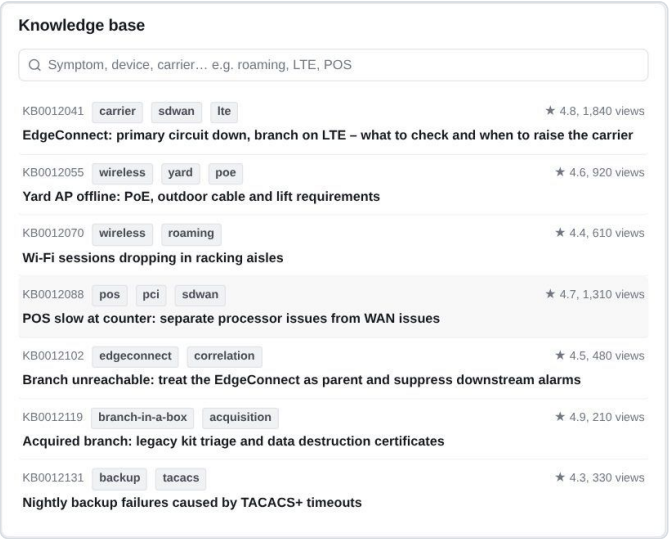


Figure 175. Knowledge base

Each article shows its number, tags, rating and views, and its title. Click to read it.

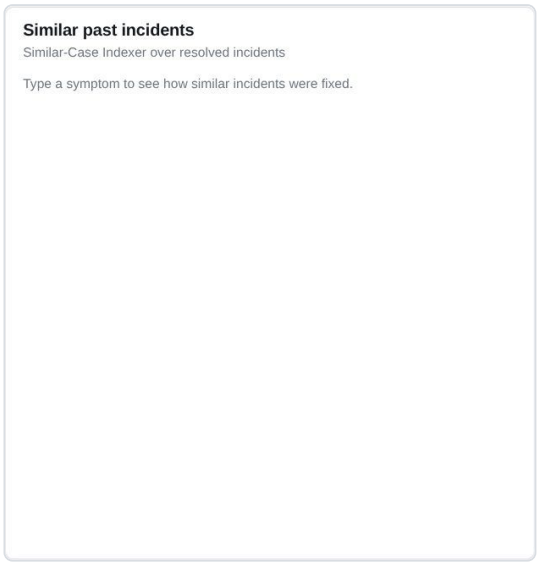


Figure 176. Similar past incidents

Type a symptom to see how similar incidents were fixed, ranked by similarity.

Auto-generated drafts tab



Figure 177. Draft article

Each draft shows the agent, the incident it came from, its age, the title, the symptom and fix, and the agent's confidence. **Approve & publish** makes it live, **Edit** changes it first, **Reject** discards it. No draft is published without an engineer.

Problem records tab



Problem	Title	Linked incidents	Owner	Opened	State
PRB0004211	Repeat Lumen DIA loss at Southeast branches on shared LEC	14	Network Eng L3	18d ago	Root cause analysis
PRB0004218	Outdoor AP-585 reboots in high heat (Sun Belt yards)	9	Wireless team	26d ago	Known error
PRB0004230	TACACS+ node timeouts during 02:00 backup window	22	Security Lead	9d ago	Fix in progress

Figure 178. Problem records

Each problem has a number, title, linked incidents, owner, age and state, moved along New, Root cause analysis, Known error, Fix in progress and Closed. A problem is opened when an incident repeats three or more times in 30 days, after every P1, or when a cluster appears in "What keeps coming back".

20. Mova iO Agents

This screen shows the 13 Mova iO agents: what each is doing now, how often it acts and how often it gets it right. Every action follows the gate rules and lands in the audit log. The header shows how many agents are working right now and the actions taken today.

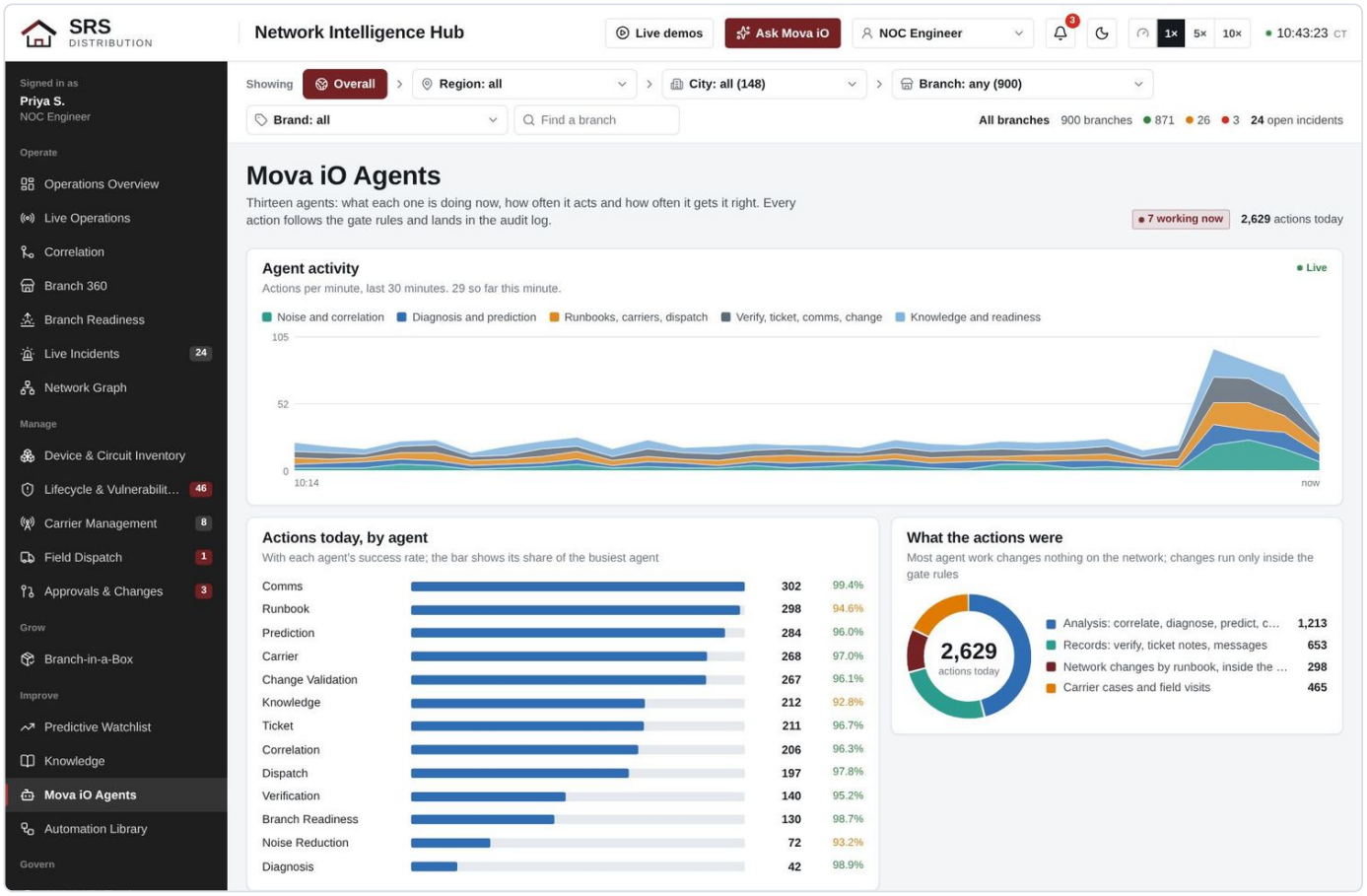


Figure 179. Mova iO Agents

Agent activity

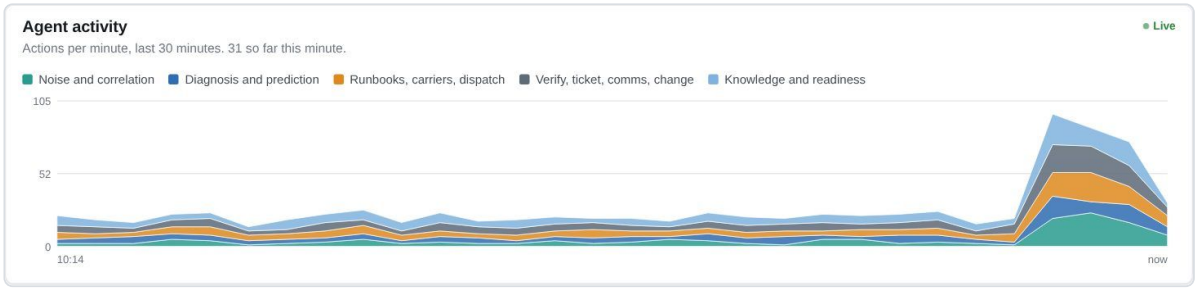


Figure 180. Agent activity

Actions per minute for the last 30 minutes, stacked by agent group: noise and correlation; diagnosis and prediction; runbooks, carriers and dispatch; verify, ticket, comms and change; knowledge and readiness. The subtitle gives the count so far this minute. A busy noise-and-correlation band with a thin runbook band is normal: most work is filtering.

Actions today, by agent

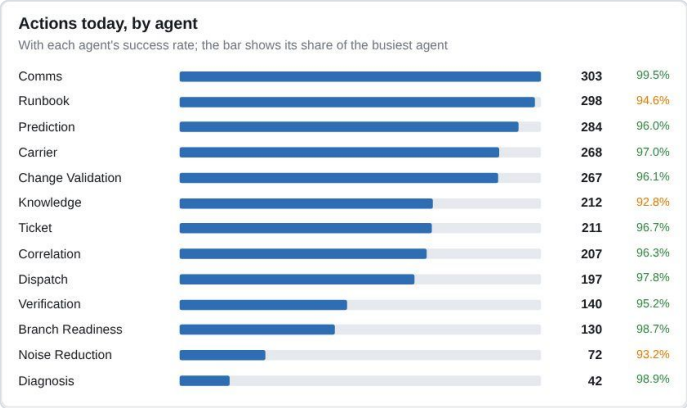


Figure 181. Actions today by agent

Every agent ranked by actions today, with its success rate on the right (green from 95%). Comms, runbooks and prediction are usually the busiest; diagnosis acts only when an incident needs a root cause.

What the actions were

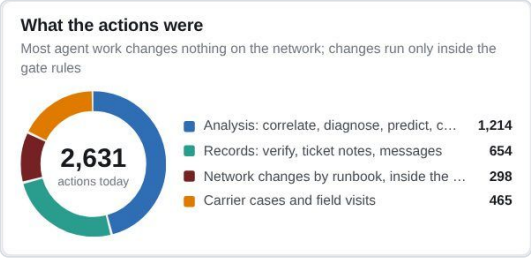


Figure 182. What the actions were

Today's actions by kind: analysis (correlate, diagnose, predict, check readiness and changes), records (verification, ticket notes, messages), network changes by runbook inside the gate rules, and carrier cases and field visits. Most agent work changes nothing on the network; the maroon slice is the only part that touches devices.

Agent cards

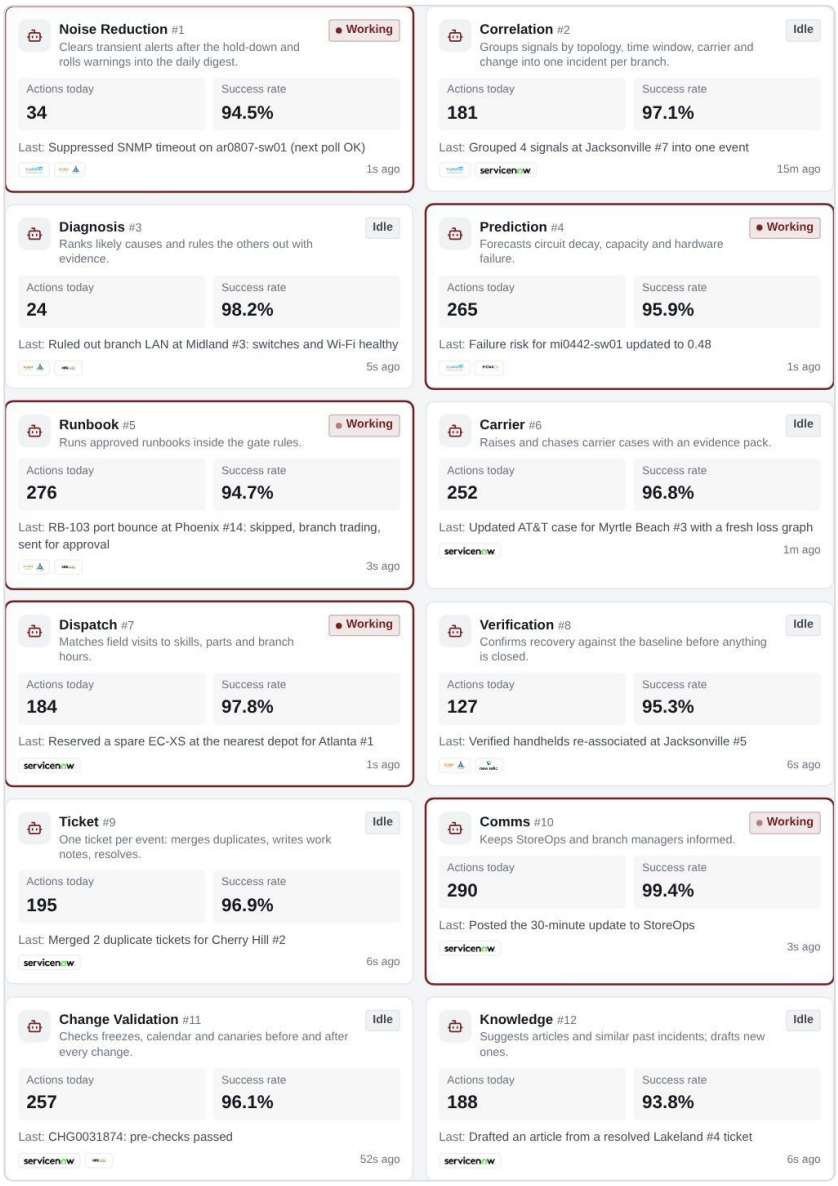


Figure 183. Agent cards

One card per agent: its name and number, what it does, **Working** (maroon, it acted in the last few seconds) or **Idle**, actions today, success rate, its last action, the tools it acts through and how long ago.

Agent	What it does
Noise Reduction	Clears transient alerts after the hold-down and rolls warnings into the daily digest
Correlation	Groups signals by topology, time window, carrier and change into one incident per branch
Diagnosis	Ranks likely causes and rules the others out with evidence
Prediction	Forecasts circuit decay, capacity and hardware failure
Runbook	Runs approved runbooks inside the gate rules
Carrier	Raises and chases carrier cases with an evidence pack

Agent	What it does
Dispatch	Matches field visits to skills, parts and branch hours
Verification	Confirms recovery against the baseline before anything is closed
Ticket	One ticket per event: merges duplicates, writes work notes, resolves
Comms	Keeps StoreOps and branch managers informed
Change Validation	Checks freezes, the calendar and canaries before and after every change
Knowledge	Suggests articles and similar past incidents; drafts new ones
Branch Readiness	Runs the pre-opening checks at 05:30 local for every branch

Agent activity stream



Figure 184. Agent activity stream

Every agent action as it happens, with the agent, time and what it did.

21. Automation Library

The Automation Library is the catalog of approved runbooks: what each does, how much it may do on its own, how often it runs and how often it succeeds.

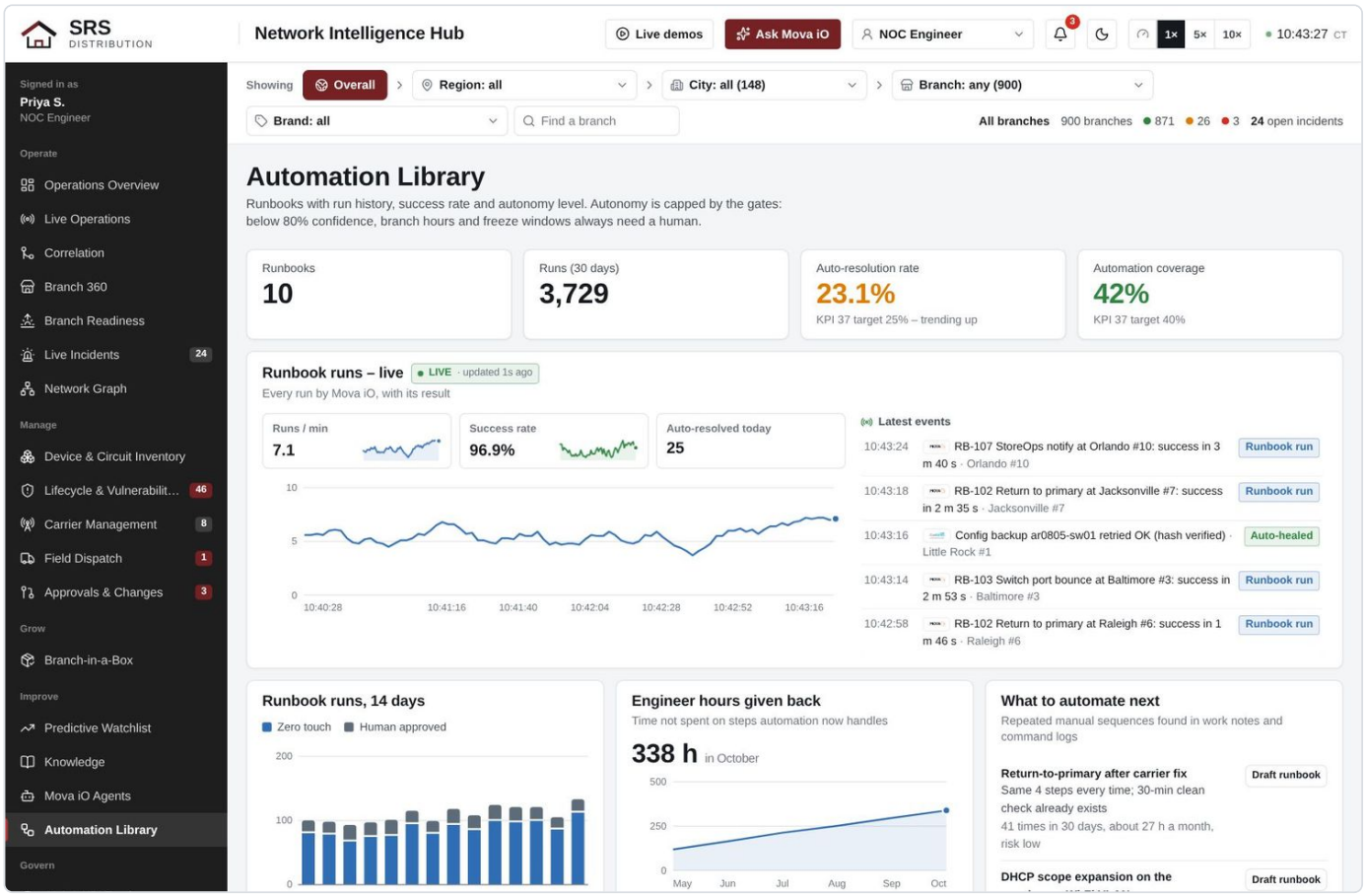


Figure 185. Automation Library

Tiles

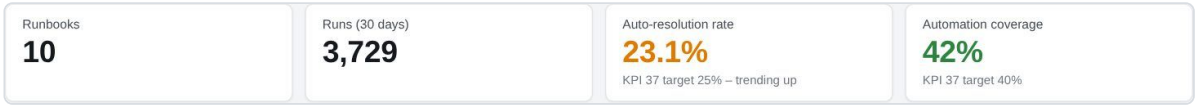


Figure 186. Automation tiles

Tile	Meaning	Target
Runbooks	Runbooks in the catalog	–
Runs (30 days)	All runs in the last 30 days	–
Auto-resolution rate (KPI 37)	Incidents closed by a runbook with verification and no human touch	25% in year 1, rising
Automation coverage (KPI 37)	Share of incident types with an approved runbook	40%

Runbook runs – live

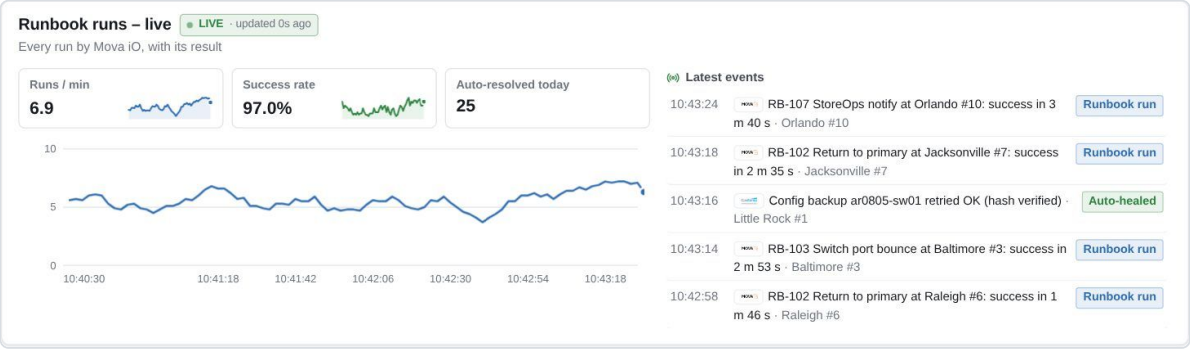


Figure 187. Runbook runs

Runs per minute, success rate and incidents auto-resolved today, with the latest runs and their results.

Runbook runs, 14 days and hours given back

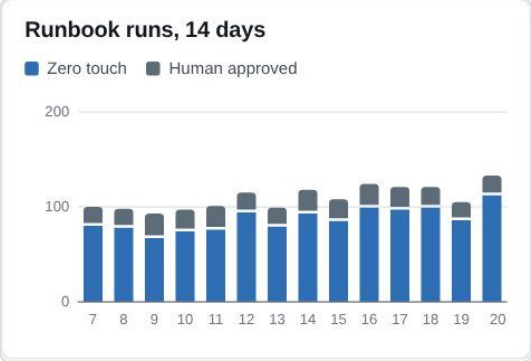


Figure 188. Runbook runs, 14 days

Runs per day split into **zero touch** (ran inside its tier with no approval) and **human approved**. Zero touch growing is autonomy growing.

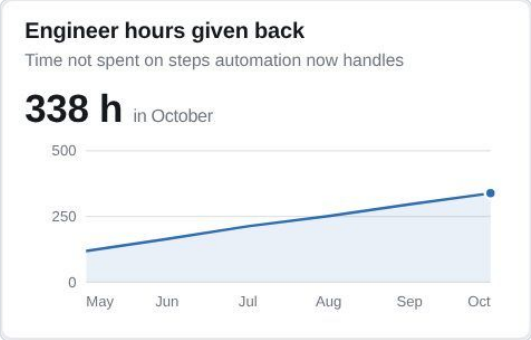


Figure 189. Engineer hours given back

Engineer hours not spent on steps automation now handles, this month and by month.

What to automate next

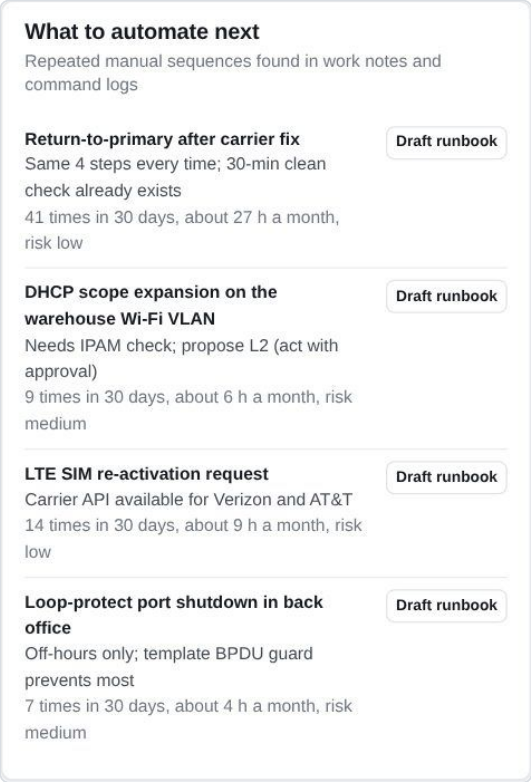
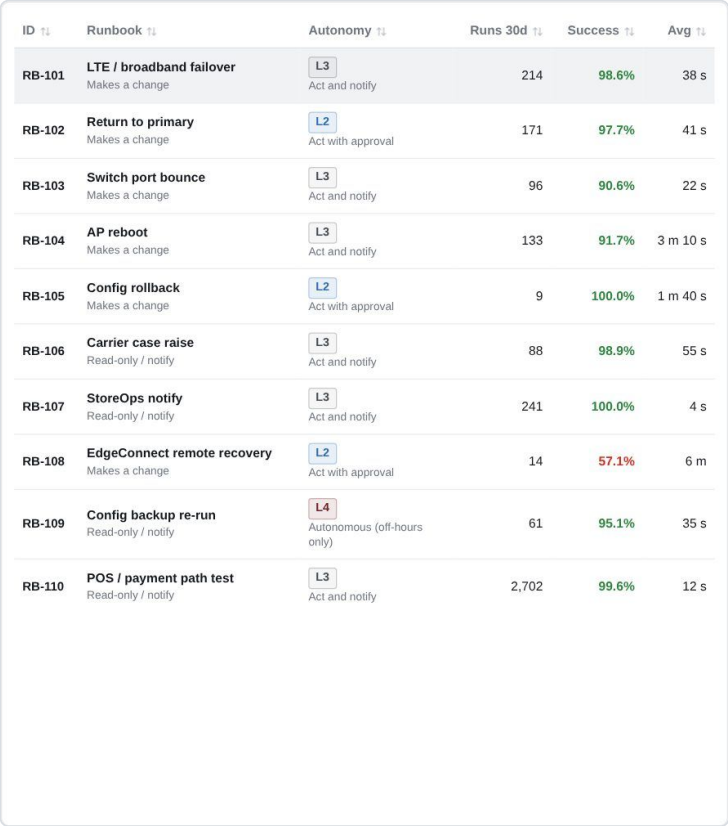


Figure 190. What to automate next

Repeated manual sequences found in work notes and command logs, with how often they happen, the hours they cost, their risk and the proposed autonomy level. **Draft runbook** starts the Toil Miner flow shown in the toil demo: a draft, generated checks, a replay against past cases and approval at the Stage tier.

The catalog



ID	Runbook	Autonomy	Runs 30d	Success	Avg
RB-101	LTE / broadband failover Makes a change	L3 Act and notify	214	98.6%	38 s
RB-102	Return to primary Makes a change	L2 Act with approval	171	97.7%	41 s
RB-103	Switch port bounce Makes a change	L3 Act and notify	96	90.6%	22 s
RB-104	AP reboot Makes a change	L3 Act and notify	133	91.7%	3 m 10 s
RB-105	Config rollback Makes a change	L2 Act with approval	9	100.0%	1 m 40 s
RB-106	Carrier case raise Read-only / notify	L3 Act and notify	88	98.9%	55 s
RB-107	StoreOps notify Read-only / notify	L3 Act and notify	241	100.0%	4 s
RB-108	EdgeConnect remote recovery Makes a change	L2 Act with approval	14	57.1%	6 m
RB-109	Config backup re-run Read-only / notify	L4 Autonomous (off-hours only)	61	95.1%	35 s
RB-110	POS / payment path test Read-only / notify	L3 Act and notify	2,702	99.6%	12 s

Figure 191. Runbook catalog

Column	Meaning
ID and runbook	RB number, name, and whether it makes a change or is read-only and notify
Autonomy	L1 to L4 and what it allows
Runs 30d, success, avg	Volume, success rate and average run time

Level	Name	Meaning
L1	Recommend only	Mova iO suggests it; a person runs it
L2	Act with approval	Stage tier: prepared, then waits for a named person
L3	Act and notify	Runs inside the gate rules and notifies at once
L4	Autonomous (off-hours only)	Runs on its own outside trading hours, for read-only or no-impact work

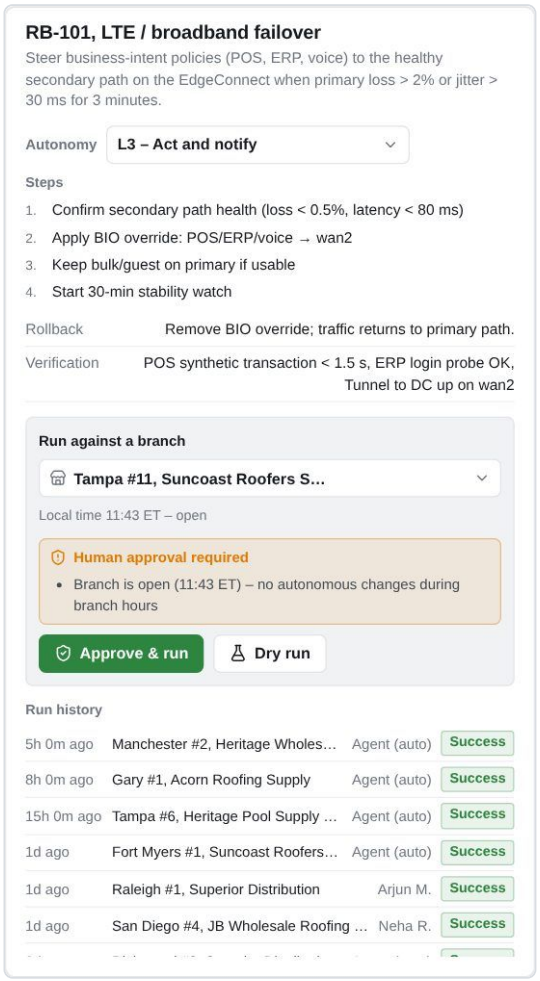


Figure 192. Runbook detail

The selected runbook shows its trigger, its autonomy level (with the levels it could move to), its steps, rollback, verification checks and recent runs. Changing the autonomy level is itself a change that SRS approves.

22. SLA & KPI Tracker

The SLA & KPI Tracker measures every service level and KPI agreed for the contract (Annexure B) against its minimum service level (MSL) and expected service level (ESL), with trends and breach risk.

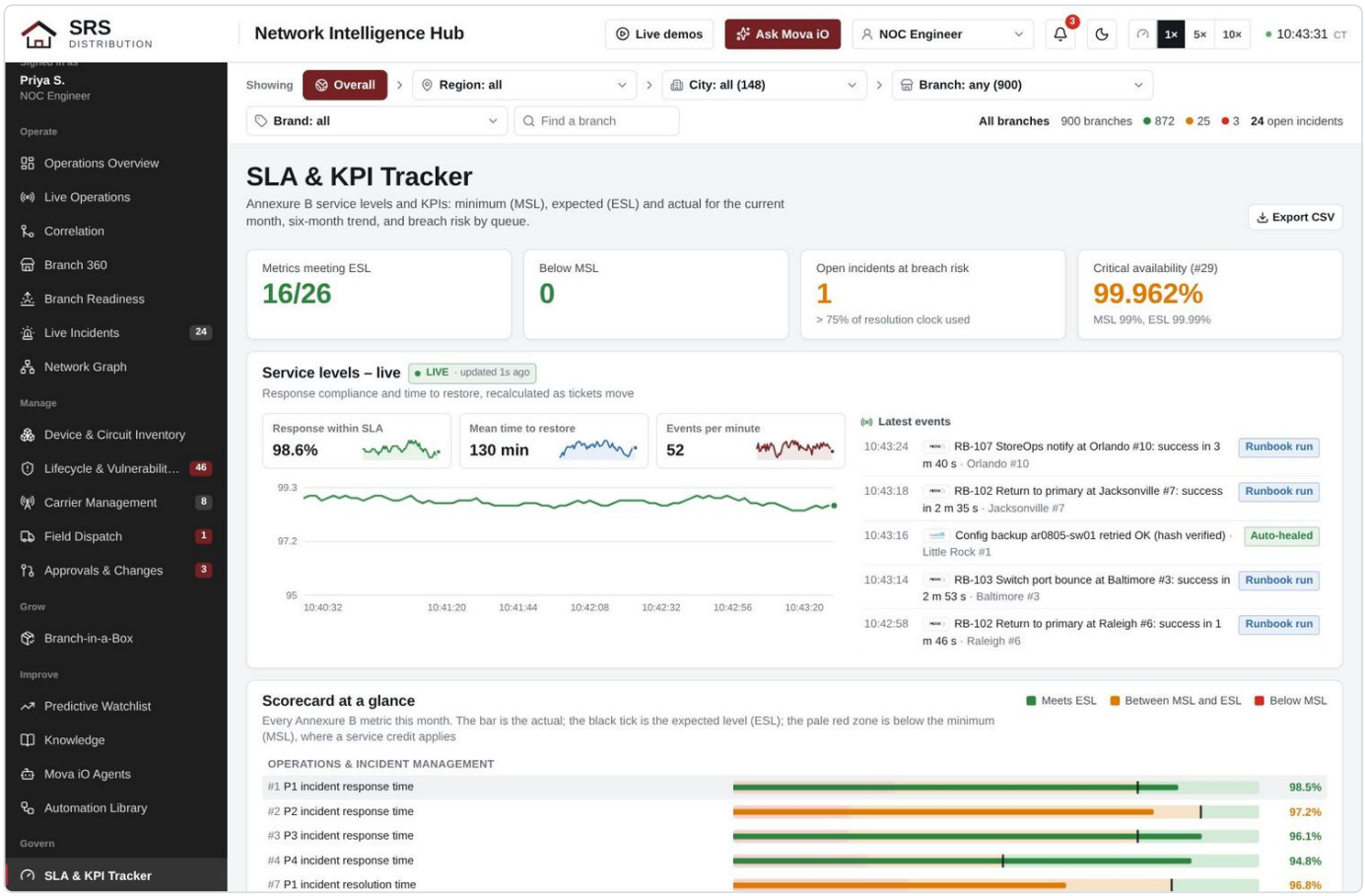


Figure 193. SLA & KPI Tracker

Tiles

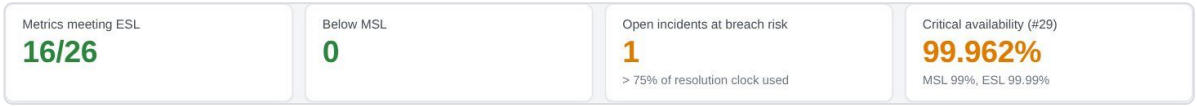


Figure 194. SLA tiles

Tile	Meaning
Metrics meeting ESL	Metrics at or above their expected level, out of all measured
Below MSL	Metrics under their minimum; below MSL in a month triggers a service credit
Open incidents at breach risk	Open incidents with more than 75% of their resolution clock used
Critical availability (#29)	Availability of in-scope critical network devices and services, against MSL 99% and ESL 99.99%

Service levels – live



Figure 195. Service levels live

Response within SLA, mean time to restore and events per minute, recalculated as tickets move, with the latest ticket events.

Scorecard at a glance

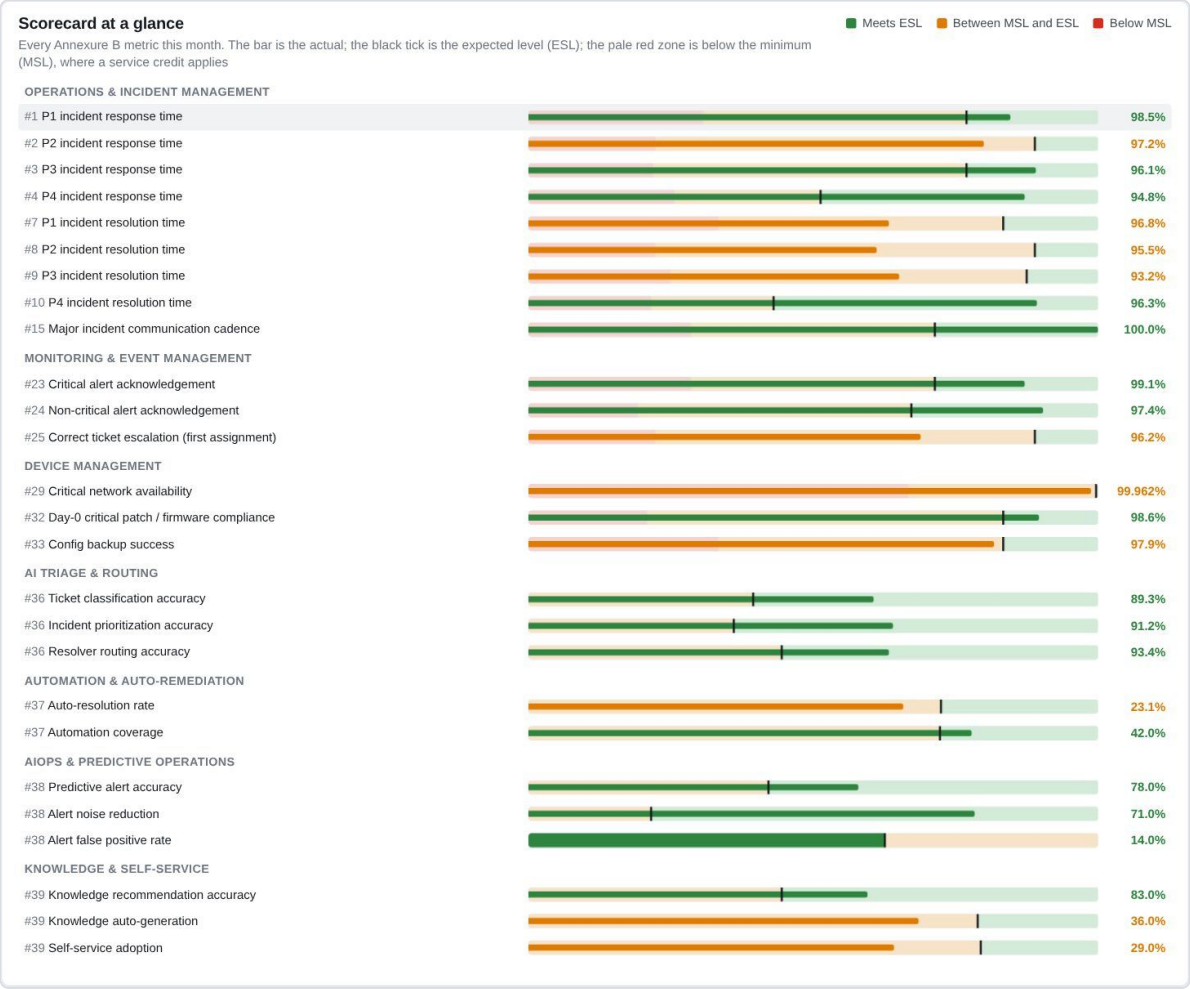


Figure 196. Scorecard at a glance

Every Annexure B metric this month on one card, grouped by service line. Each row is a bullet bar:

Part	Meaning
Coloured bar	This month's actual: green meets the expected level, amber is between the minimum and the expected level, red is below the minimum
Black tick	The expected service level (ESL), the target
Pale red zone	Below the minimum service level (MSL), where a service credit applies
Pale amber and green zones	Between MSL and ESL, and above ESL
Figure on the right	The actual, in the colour of its status

Click a row to show that metric's projection at the bottom of the screen. For a "lower is better" metric, such as the alert false positive rate, the bar is green while it stays under the tick.

Where the service stands



Figure 197. Where the service stands

A donut of all 26 metrics by status (meets ESL, between MSL and ESL, below target for KPIs, below MSL), and a bar per service line showing how many of its metrics meet the expected level.

Respond and resolve on time, by priority

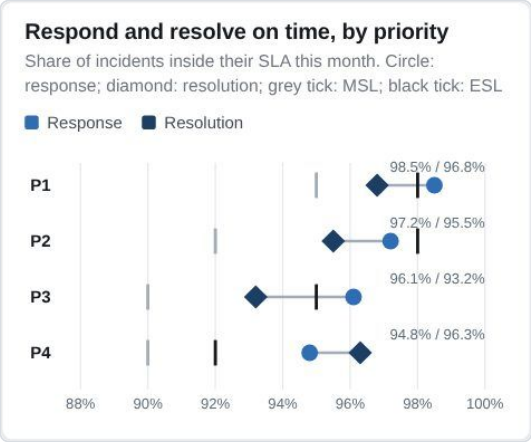


Figure 198. Respond and resolve by priority

One line per priority on a scale from 88% to 100%: the circle is the share of incidents answered within the response time, the diamond the share fixed within the resolution time. The grey tick is the MSL and the black tick the ESL. A circle or diamond left of its black tick is the place to focus.

Service credit exposure

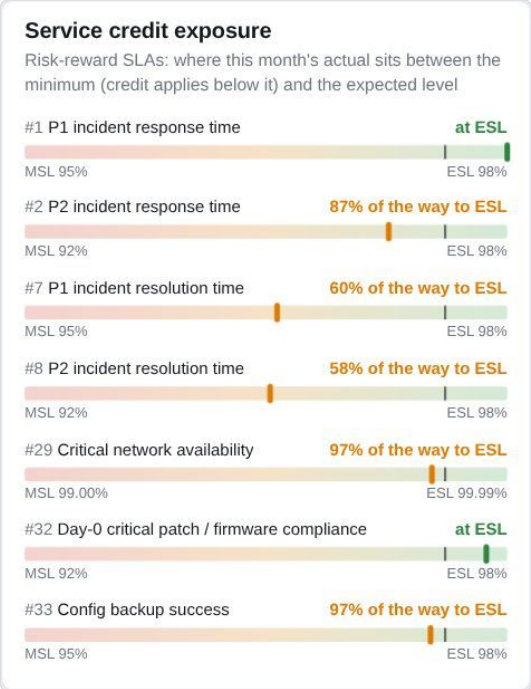


Figure 199. Service credit exposure

The risk-reward SLAs, which carry a service credit. Each bar runs from the minimum (left, red: a credit applies below it) to the expected level (the black line); the marker shows where this month's actual sits, labelled "at ESL" or how far of the way to ESL it is. A marker in the red end is a credit risk.

Service levels and error budgets

The same four error budgets as on the Operations Overview (chapter 5): branches online, jitter under 30 ms, loss under 1%, and P1/P2 answered in 15 minutes.

Mean time to restore, and tickets per month

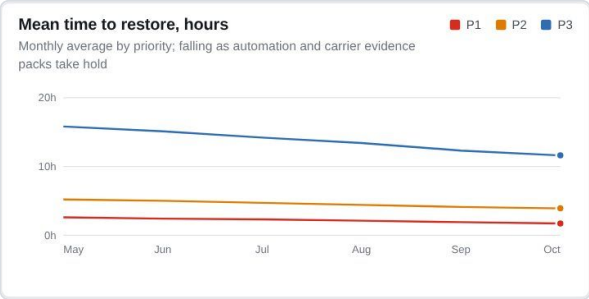


Figure 200. Mean time to restore

Average hours to restore service each month, for P1, P2 and P3. Falling lines show automation and carrier evidence packs shortening repairs.

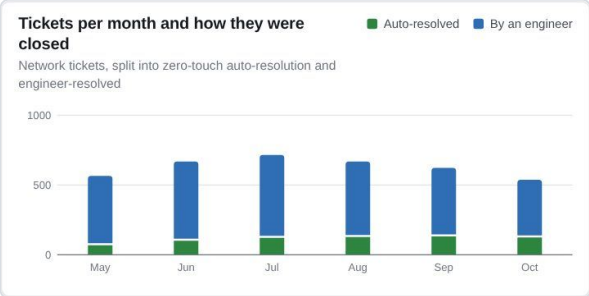


Figure 201. Tickets per month

Network tickets per month, split into those closed by zero-touch automation (green) and those resolved by an engineer (blue). The green share is the auto-resolution rate (KPI 37).

Six months at a glance



Figure 202. Six months at a glance

One row per metric and one column per month, each cell coloured by that month's status. Read across a row to see a metric's history and down a column to see which month slipped; an exclamation mark is a month below MSL. Click a row to show its projection.

Incidents per day, last 12 weeks

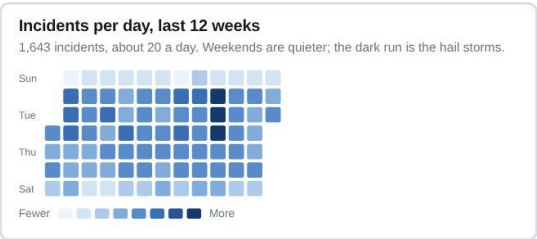


Figure 203. Incidents per day

A calendar of incidents per day for 12 weeks, one square per day, rows Sunday to Saturday; darker means more. Weekends are quieter; a dark run usually matches a storm.

Breach risk by queue

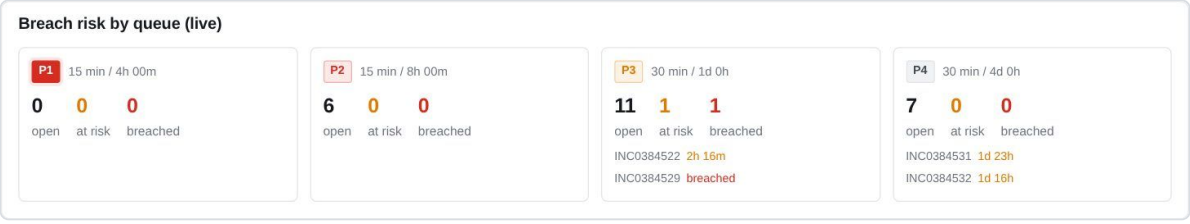
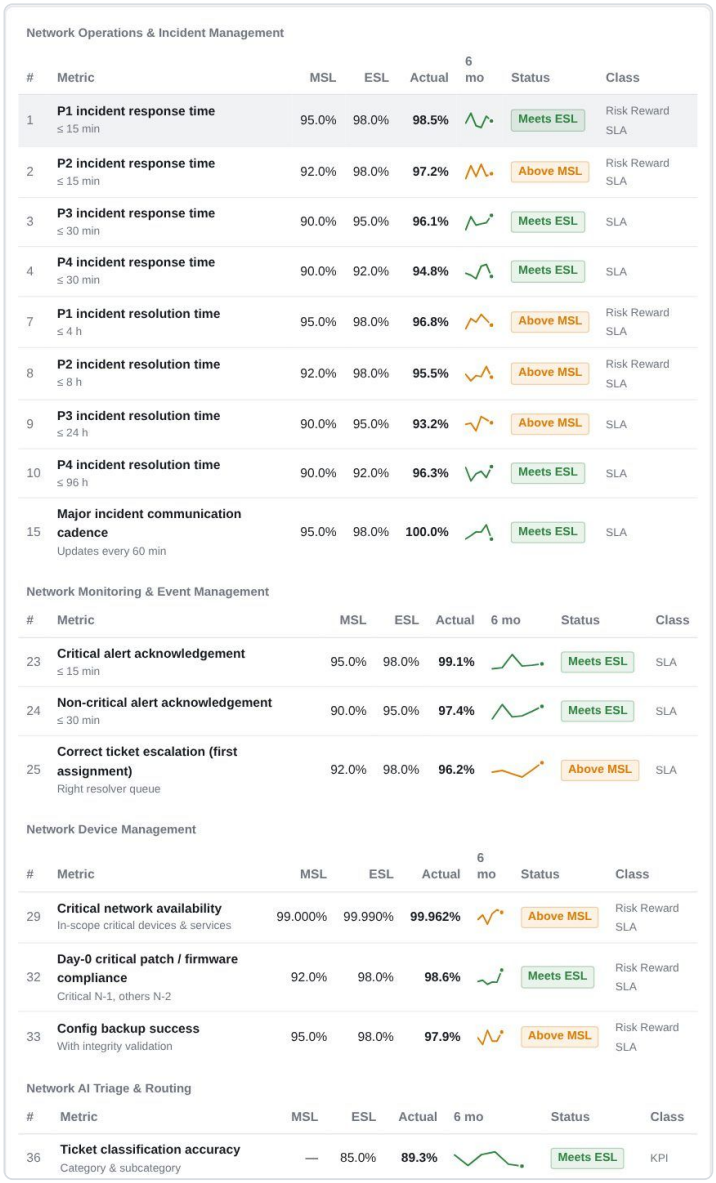


Figure 204. Breach risk by queue

One block per priority with its response and resolution targets, the number open, at risk and breached, and the incidents closest to their limit.

The metrics table



Network Operations & Incident Management

#	Metric	MSL	ESL	Actual	6 mo	Status	Class
1	**P1 incident response time** ≤ 15 min	95.0%	98.0%	**98.5%**		Meets ESL	Risk Reward SLA
2	**P2 incident response time** ≤ 15 min	92.0%	98.0%	**97.2%**		Above MSL	Risk Reward SLA
3	**P3 incident response time** ≤ 30 min	90.0%	95.0%	**96.1%**		Meets ESL	SLA
4	**P4 incident response time** ≤ 30 min	90.0%	92.0%	**94.8%**		Meets ESL	SLA
7	**P1 incident resolution time** ≤ 4 h	95.0%	98.0%	**96.8%**		Above MSL	Risk Reward SLA
8	**P2 incident resolution time** ≤ 8 h	92.0%	98.0%	**95.5%**		Above MSL	Risk Reward SLA
9	**P3 incident resolution time** ≤ 24 h	90.0%	95.0%	**93.2%**		Above MSL	SLA
10	**P4 incident resolution time** ≤ 96 h	90.0%	92.0%	**96.3%**		Meets ESL	SLA
15	**Major incident communication cadence** Updates every 60 min	95.0%	98.0%	**100.0%**		Meets ESL	SLA

Network Monitoring & Event Management

#	Metric	MSL	ESL	Actual	6 mo	Status	Class
23	**Critical alert acknowledgement** ≤ 15 min	95.0%	98.0%	**99.1%**		Meets ESL	SLA
24	**Non-critical alert acknowledgement** ≤ 30 min	90.0%	95.0%	**97.4%**		Meets ESL	SLA
25	**Correct ticket escalation (first assignment)** Right resolver queue	92.0%	98.0%	**96.2%**		Above MSL	SLA

Network Device Management

#	Metric	MSL	ESL	Actual	6 mo	Status	Class
29	**Critical network availability** In-scope critical devices & services	99.000%	99.990%	**99.962%**		Above MSL	Risk Reward SLA
32	**Day-0 critical patch / firmware compliance** Critical N-1, others N-2	92.0%	98.0%	**98.6%**		Meets ESL	Risk Reward SLA
33	**Config backup success** With integrity validation	95.0%	98.0%	**97.9%**		Above MSL	Risk Reward SLA

Network AI Triage & Routing

#	Metric	MSL	ESL	Actual	6 mo	Status	Class
36	**Ticket classification accuracy** Category & subcategory	—	85.0%	**89.3%**		Meets ESL	KPI

Figure 205. Metrics table

Metrics are grouped by area, for example Network Operations & Incident Management.

Column	Meaning
# and metric	The Annexure B number and name, with its definition (for example P1 response within 15 minutes)
MSL	Minimum service level; below it a credit applies
ESL	Expected service level, the target
Actual	This month's result
6 mo	The trend over six months

Column	Meaning
Status	Meets ESL (green), Above MSL (amber), Below MSL (red)
Class	**Risk Reward SLA** (carries a service credit and earn-back), **SLA** or **KPI**

The SLAs that carry credits are P1 and P2 response, P1 and P2 resolution, critical network availability (#29), Day-0 critical patch and firmware compliance (#32) and config backup success (#33). AI and automation KPIs, such as auto-resolution, automation coverage, predictive accuracy and alert noise reduction, have expected levels only.

Metric projection

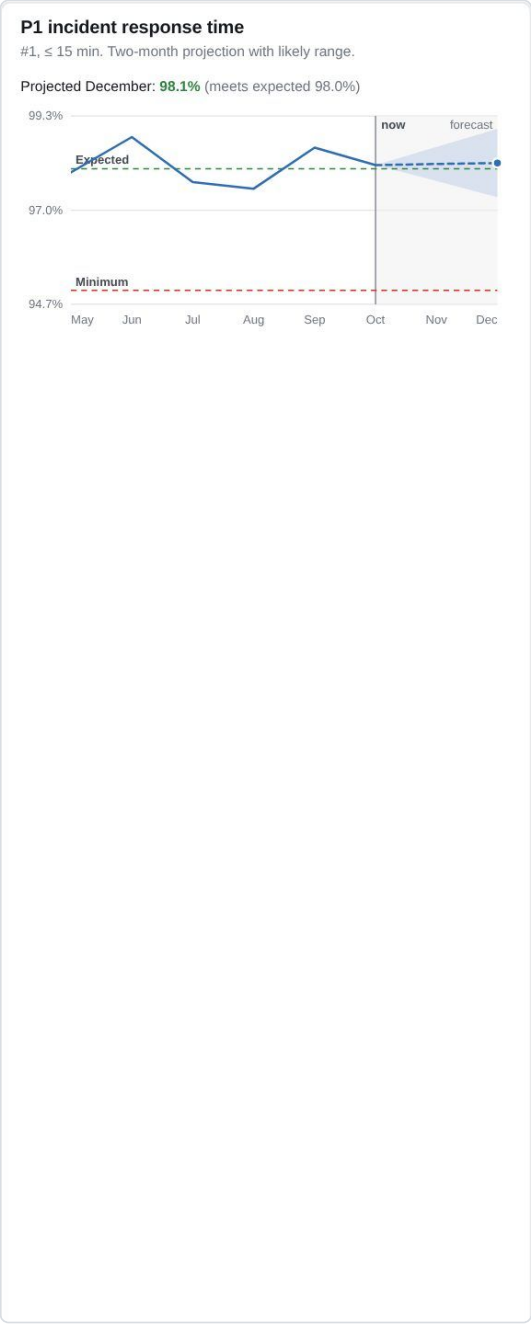


Figure 206. Metric projection

The selected metric's monthly results and a two-month projection with its likely range, against its minimum and expected lines, with a sentence on whether the projection meets the expected level. Click any row in the table to see its projection.

23. Reports

Every report is built from live console data at the moment it is opened, with who generated it and when. **Download HTML** saves it with its charts (it opens in Word or a browser); **Print / PDF** prints it. Each report opens with a row of charts, then the tables:

Report	Charts at the top
Shift handover	Open incidents by priority; SLA clock used for each open P1 and P2, with the 75% at-risk line; incidents opened per hour over the last 12 hours, by priority
Daily operations summary	Branch health now; incidents opened per hour over 24 hours, resolved or still open; incidents by category; WAN lines down or degraded, by carrier
Weekly service review pack	Annexure B metrics by status; the risk-reward SLAs this month; network tickets per month by close code; each carrier's restore commitment met, against 90%; each carrier's mean time to restore
Branch-in-a-Box onboarding report	Projects by stage and track; project health (RAG); cutover gates passed for each project, out of 13
Audit log	Who acted (Mova iO agents, people, system); actions per hour over the last 12 hours

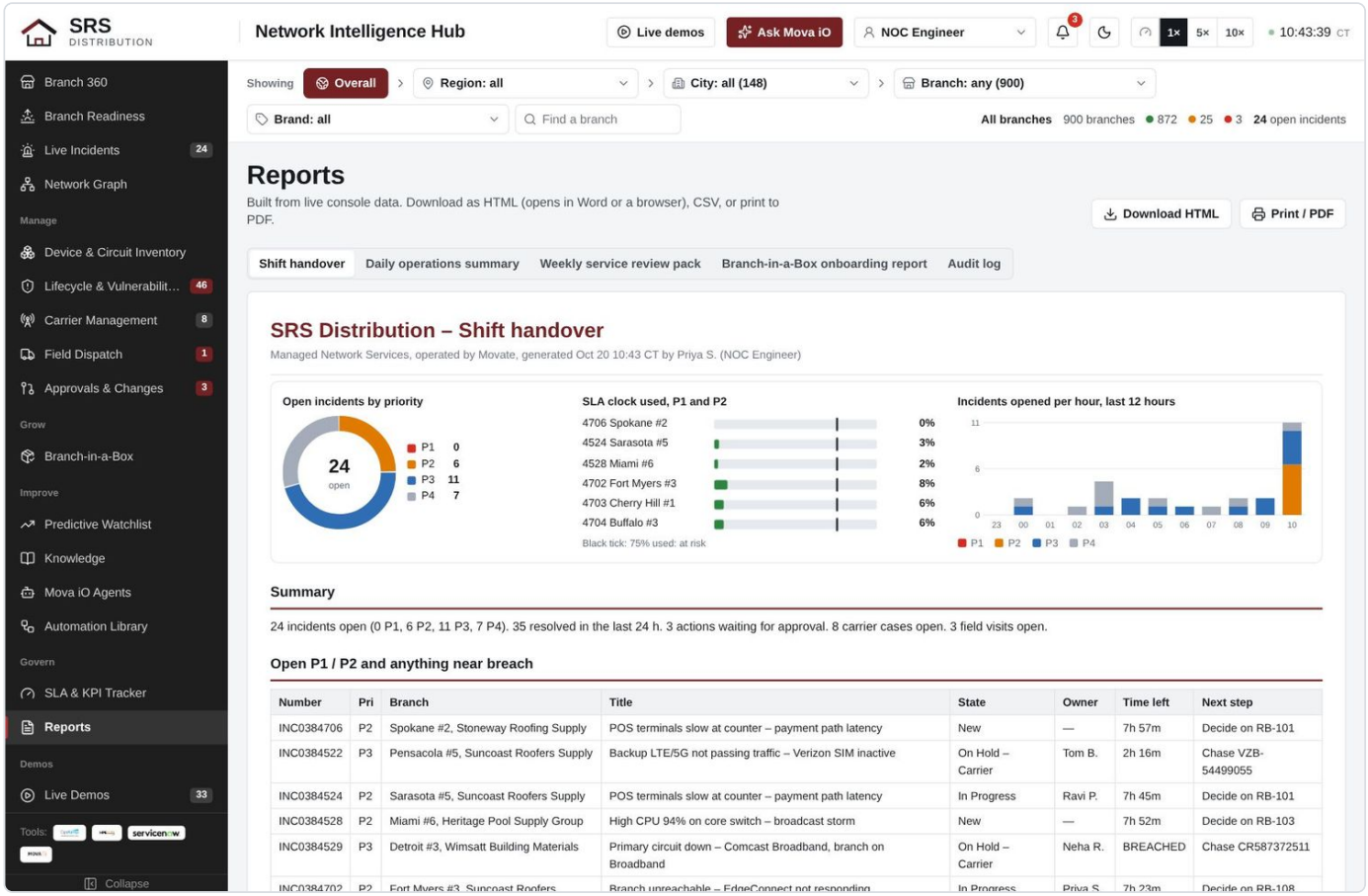


Figure 207. Reports: shift handover

Report	For	What it contains
Shift handover	The incoming NOC shift	Summary line; open P1 and P2 and anything near breach, with owner, time left and next step; actions waiting on a human; carrier cases
Daily operations summary	StoreOps and IT leadership	Network health; incidents in the last 24 hours; readiness this morning; carriers
Weekly service review pack	The weekly service review	Service levels against Annexure B; ticket mix for six months; problems and improvements; carriers
Branch-in-a-Box onboarding report	The M&A integration lead	The project portfolio and this weekend's cutover gates
Audit log	Security and audit	Every human and agent action this session: approvals, runbook runs, changes, demos and settings, with who and when

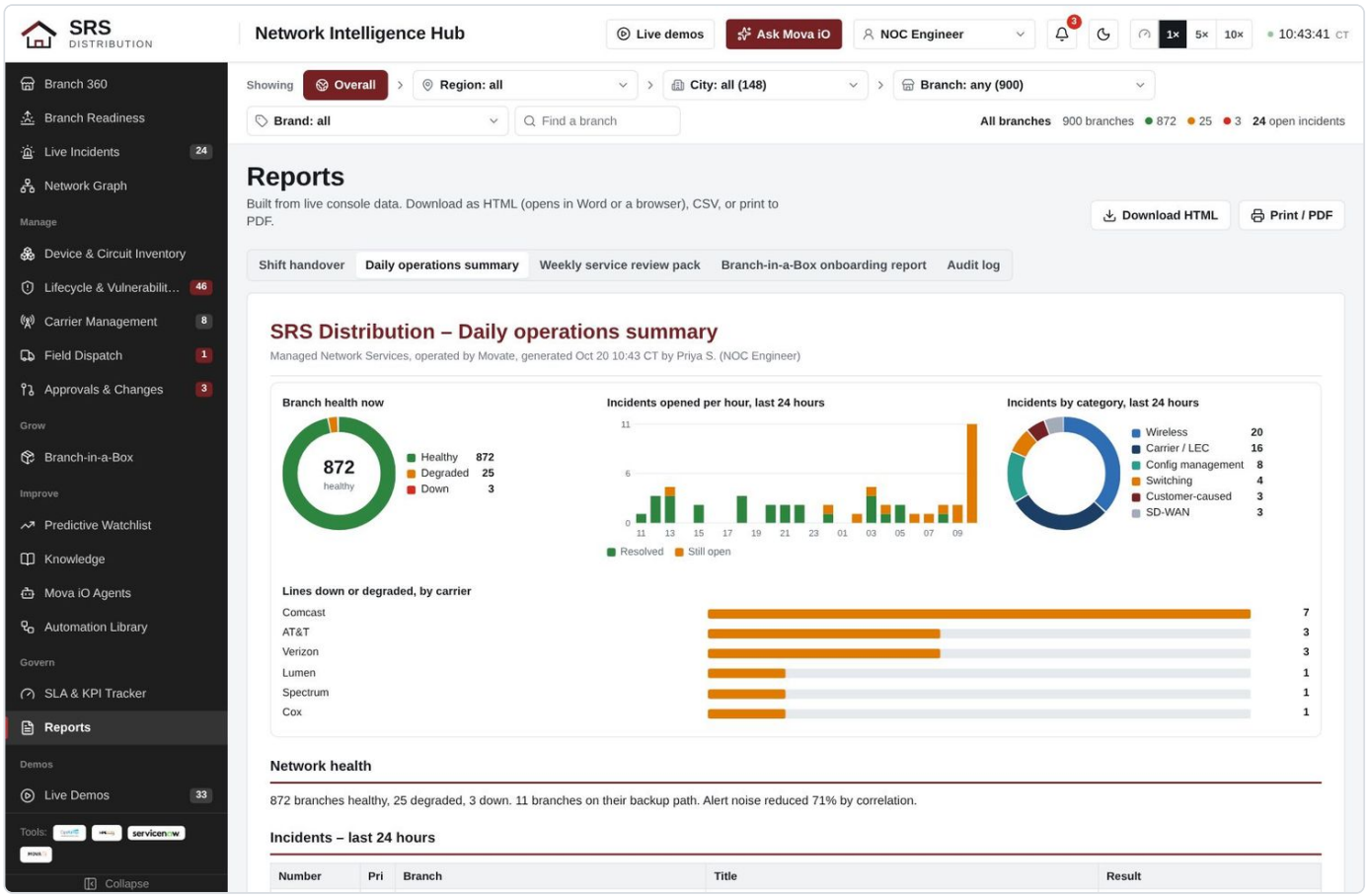


Figure 208. Daily operations summary

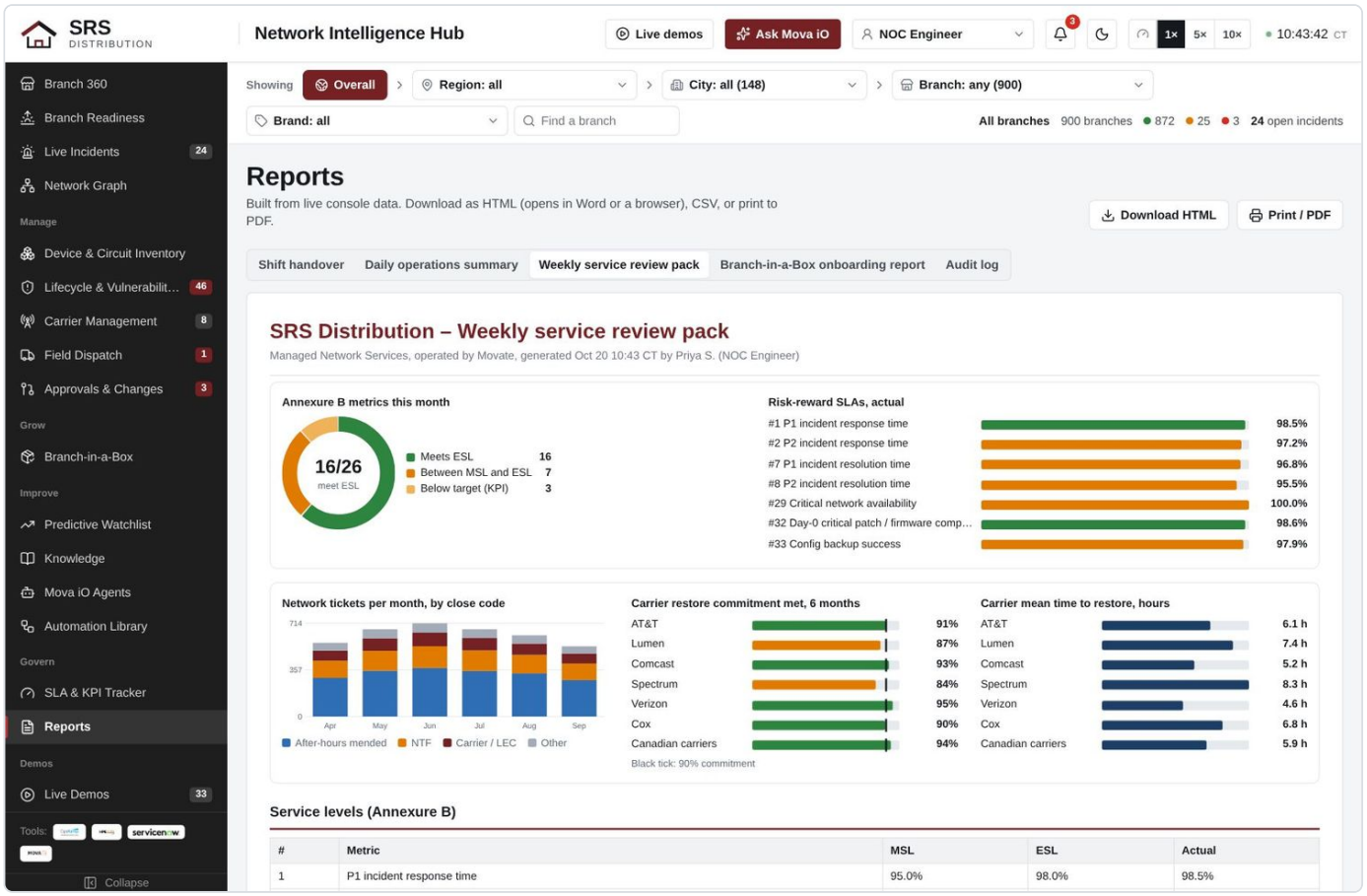


Figure 209. Weekly service review pack

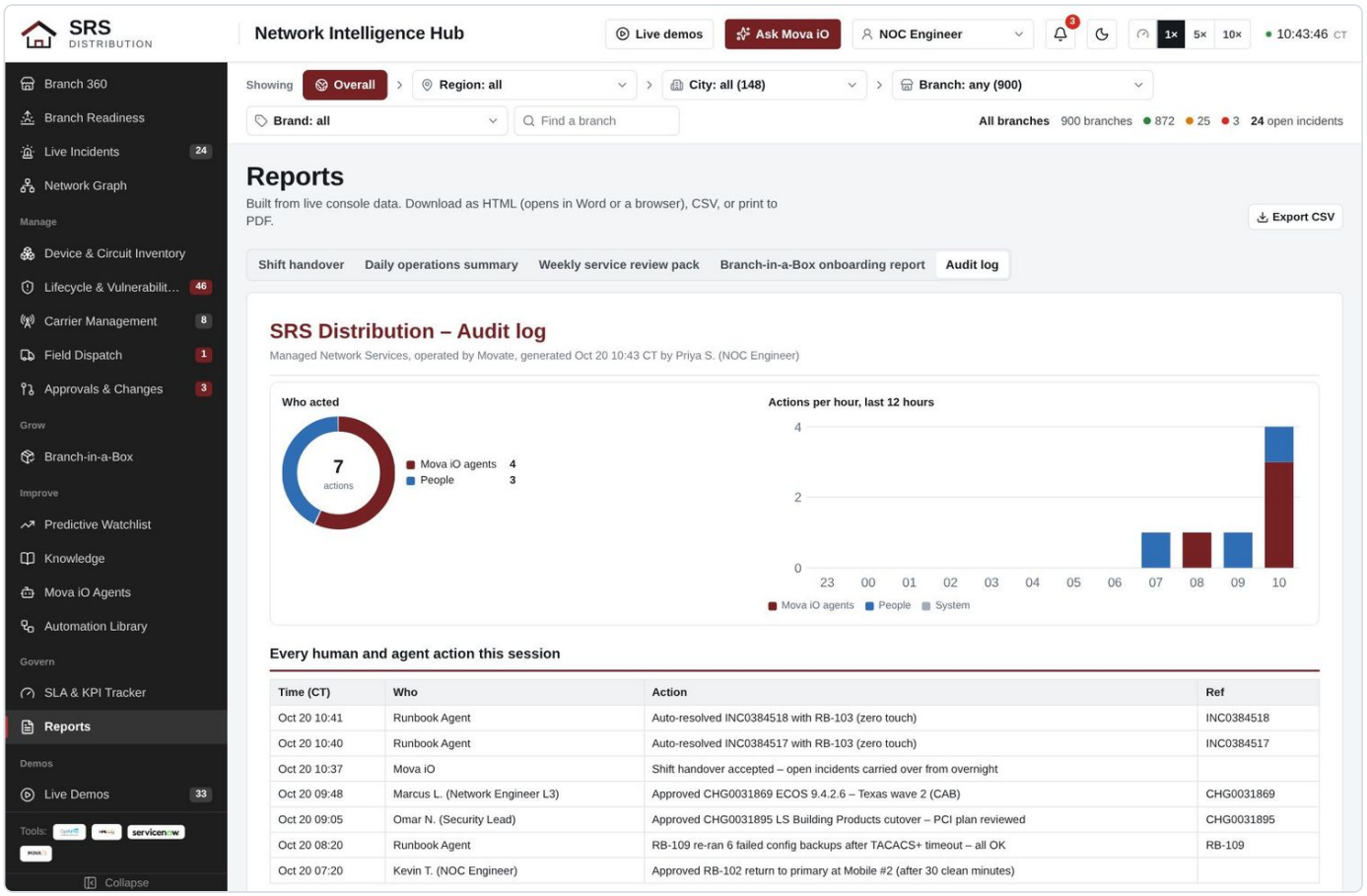


Figure 210. Audit log

24. Live Demos

Live Demos is the library of 33 scripted SRS branch situations, opened from the bottom of the left menu or the **Live demos** button in the header. Each demo plays on a full-screen stage that shows every system involved at once, then writes its incident, approvals, carrier cases and work orders into the live console where an engineer would find them. Part III describes each demo.

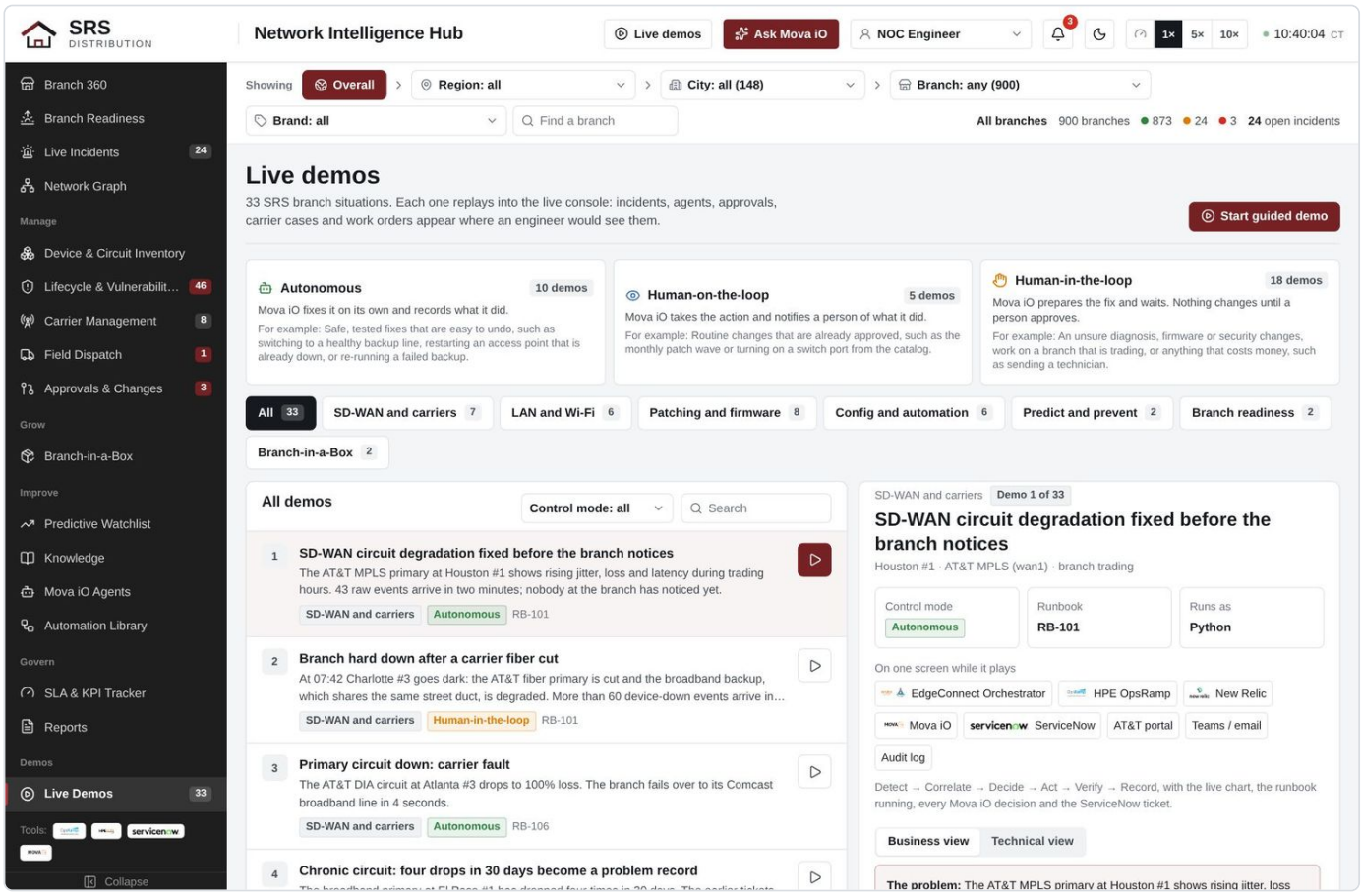


Figure 211. Live Demos screen

The demo library

Part of the screen	What it shows	How to use it
Start guided demo	Plays demo 1 and then moves through the list	Use it for a non-stop walkthrough
Control mode cards	Autonomous, Human-on-the-loop and Human-in-the-loop, each with its rule and how many demos use it	Click a card to show only that mode; click again to clear
Area chips	SD-WAN and carriers, LAN and Wi-Fi, Patching and firmware, Config and automation, Predict and prevent, Branch readiness, Branch-in-a-Box, with counts	Click to filter the list by area
Search and control-mode list	Text search over titles and problems	Find a demo by branch, device or keyword

Part of the screen	What it shows	How to use it
Demo list	Number, title, the problem in two lines, area, control mode and runbook	Click a row to see its details; click the play button to run it
Detail card	Control mode, runbook, whether it runs as Ansible or Python, the systems shown on stage, and the business or technical explanation	Read it before you play, to know what the audience will see
Open (screen)	Goes to the console screen that owns the demo's area	Use it after a demo to show the same record in the console

A demo adds to the console; it never resets anyone's work. The note under the detail card shows how many incidents are already open.

The demo stage

The stage is one screen with every system involved, in the order a real event moves through them.

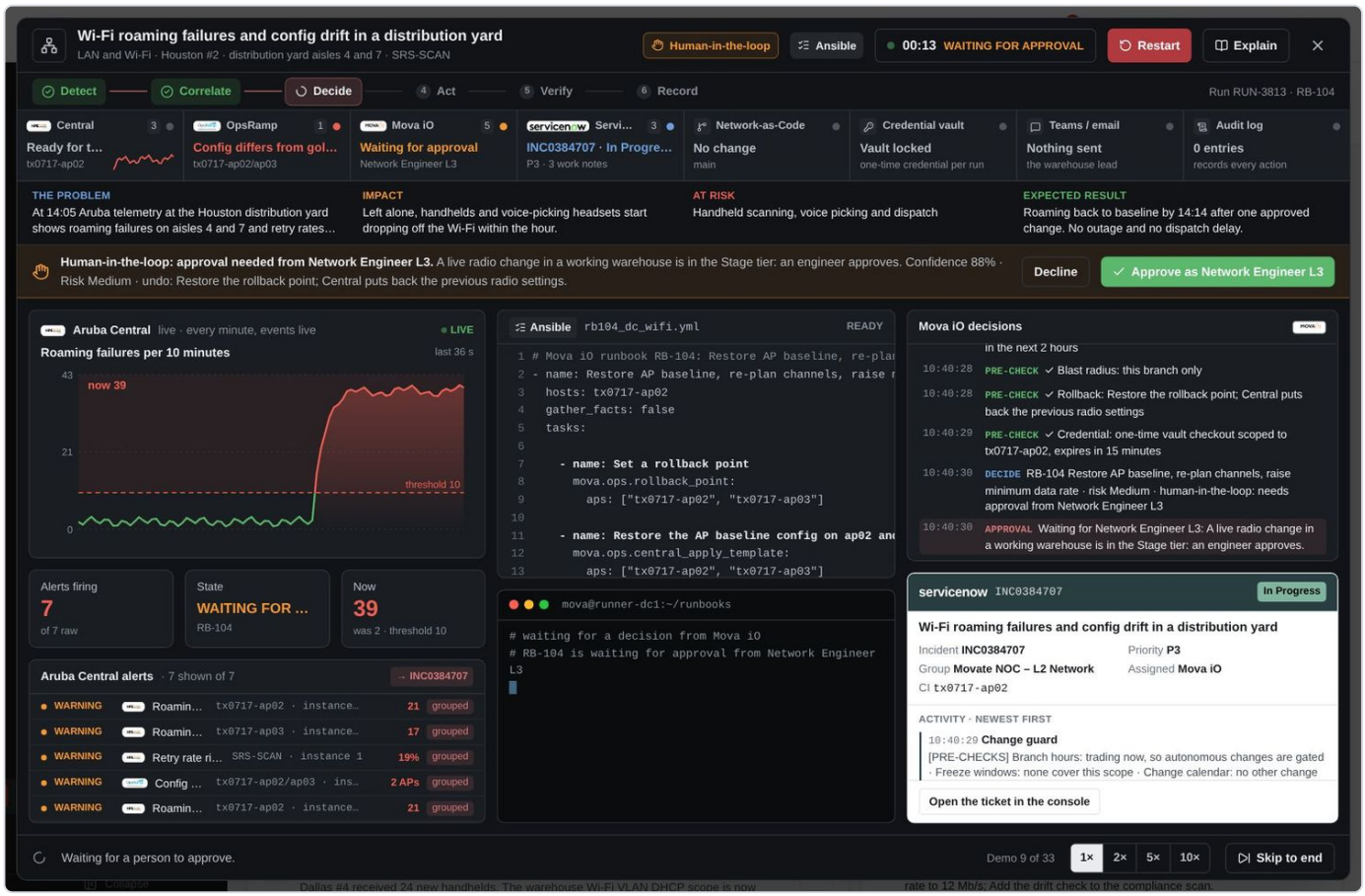


Figure 212. Stage waiting for an approval (Human-in-the-loop)

Area of the stage	What it shows
Title bar	Demo title, area, branch and device; the control mode badge; whether the runbook is Ansible or Python; the run clock and state; Restart (or Replay), Explain and close

Area of the stage	What it shows
Phase bar	Detect, Correlate, Decide, Act, Verify, Record. Each phase turns green when it is done; the current one is highlighted
System strip	One tile per system on stage (for example Central, OpsRamp, Mova iO, ServiceNow, Network-as-Code, credential vault, Teams, audit log). Each shows its state now, a count of updates and a status dot
Story strip	The problem, the impact, what is at risk (later "Protected") and the expected result (later "Result") in business words
Approval or stop bar	Appears at the decision. Human-in-the-loop shows **Approve as (role)** and **Decline**; Human-on-the-loop shows a countdown and **Stop**
Live chart	The key metric from the source system, with its threshold line, a red zone above it and a "Mova iO acted" marker when the runbook changes something
Three numbers	Alerts firing (raw count, then grouped, then cleared), state of the run, and the metric now against its normal value and threshold
Source alerts	The raw alerts from the source system, then "grouped" into the ticket, then "cleared"
Runbook and terminal	The Ansible playbook or Python job that runs, step by step, and the runner's terminal output including verification lines
Mova iO decisions	The decision log: intake, correlation, evidence, ruled-out causes, diagnosis with confidence, pre-checks, decision, approval, actions, verification, ticket, notice and learning
ServiceNow ticket	The real incident record: number, priority, group, assignee, CI and activity, newest first, with a button to open it in the console
Bottom bar	Status line, demo number, speed (1×, 2×, 5× or 10×) and **Skip to end**

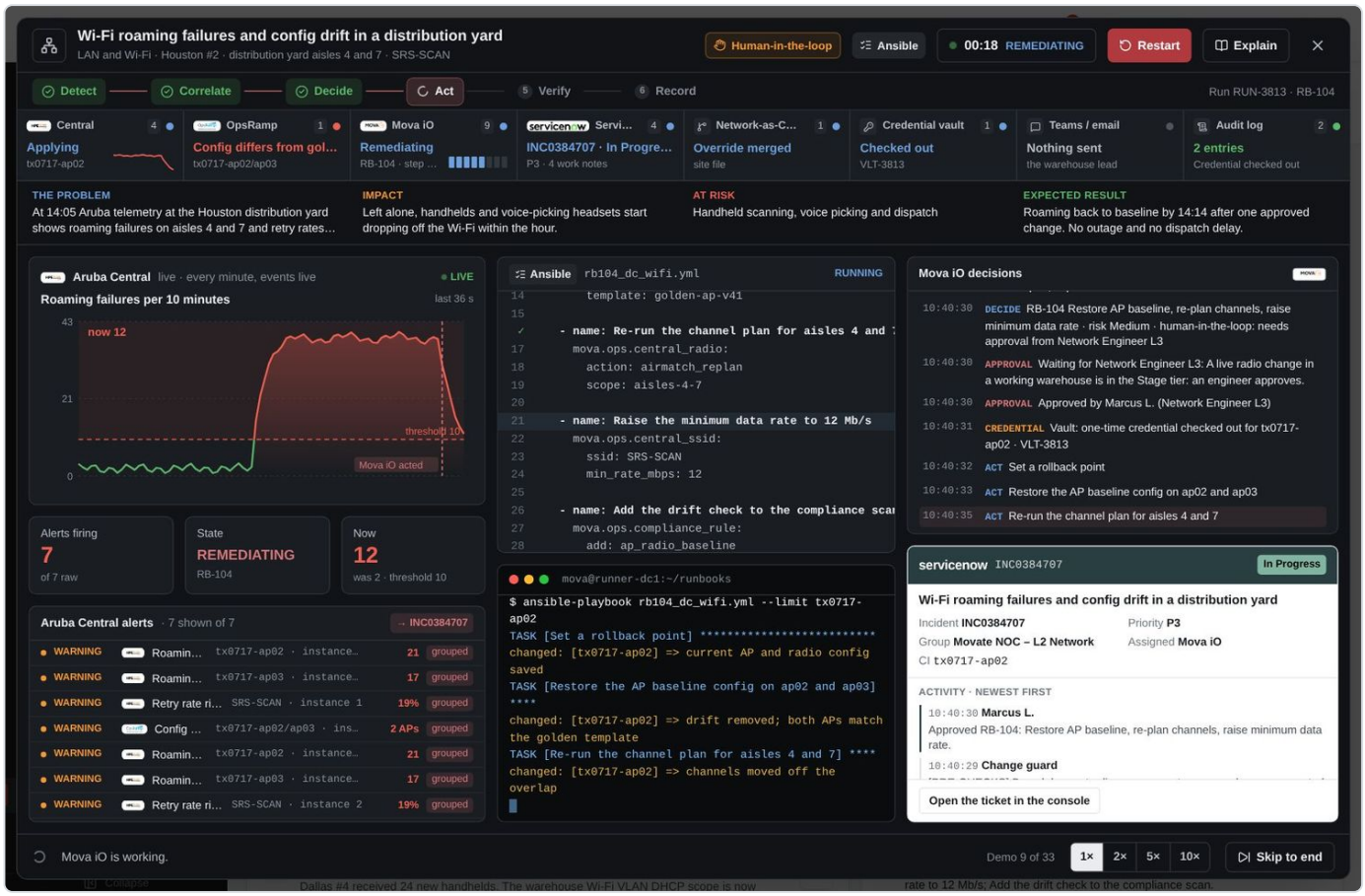


Figure 213. Stage running after approval

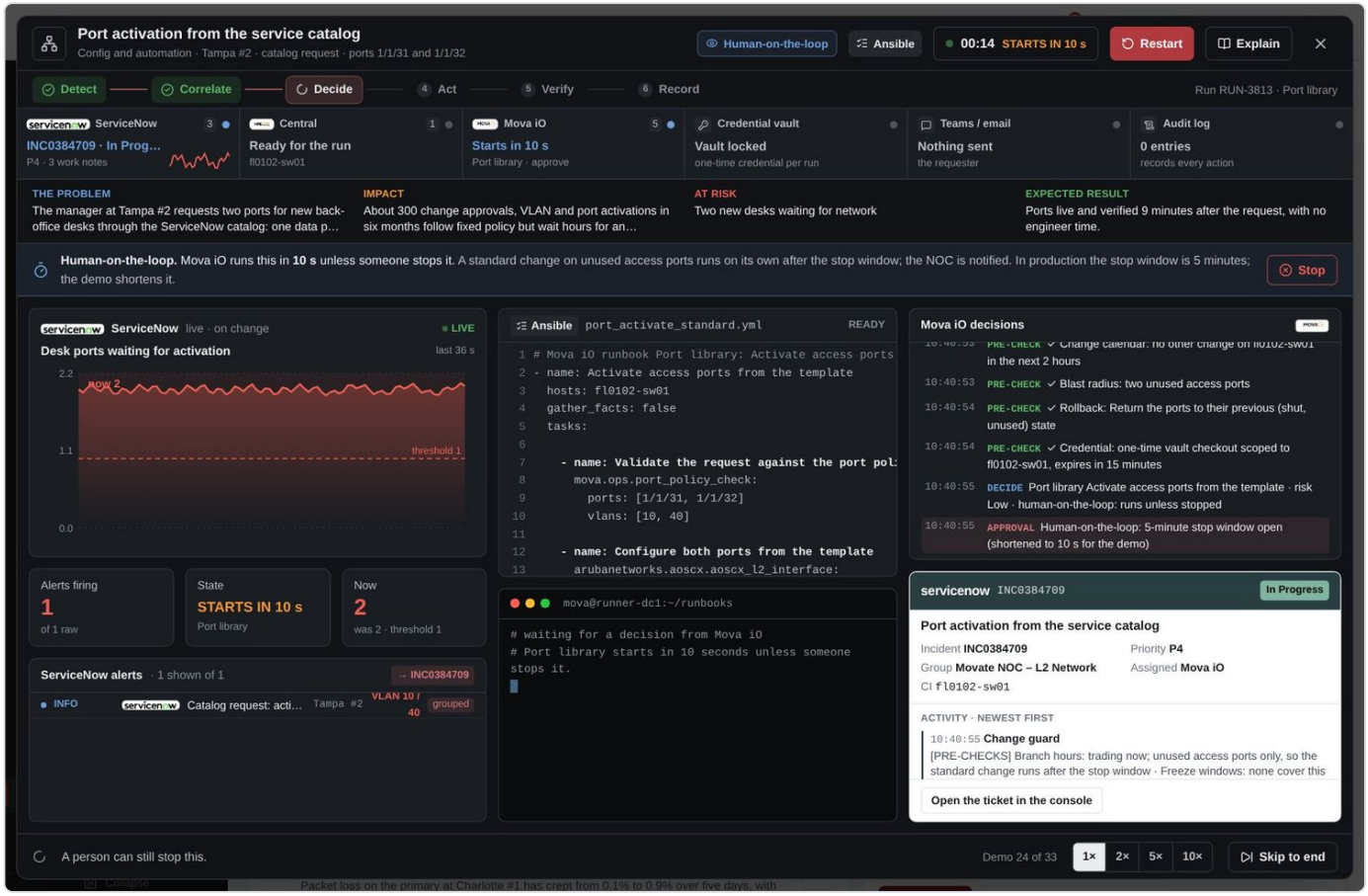


Figure 214. Stage on Human-on-the-loop: the stop window is counting down

When the run finishes, the bottom bar shows the outcome: the metric before and after, how many alerts became one ticket, how many people were needed, and how long the demo ran. **Branch 360** opens the branch, **Open in console** goes to the record, and **Next demo** plays the next one.

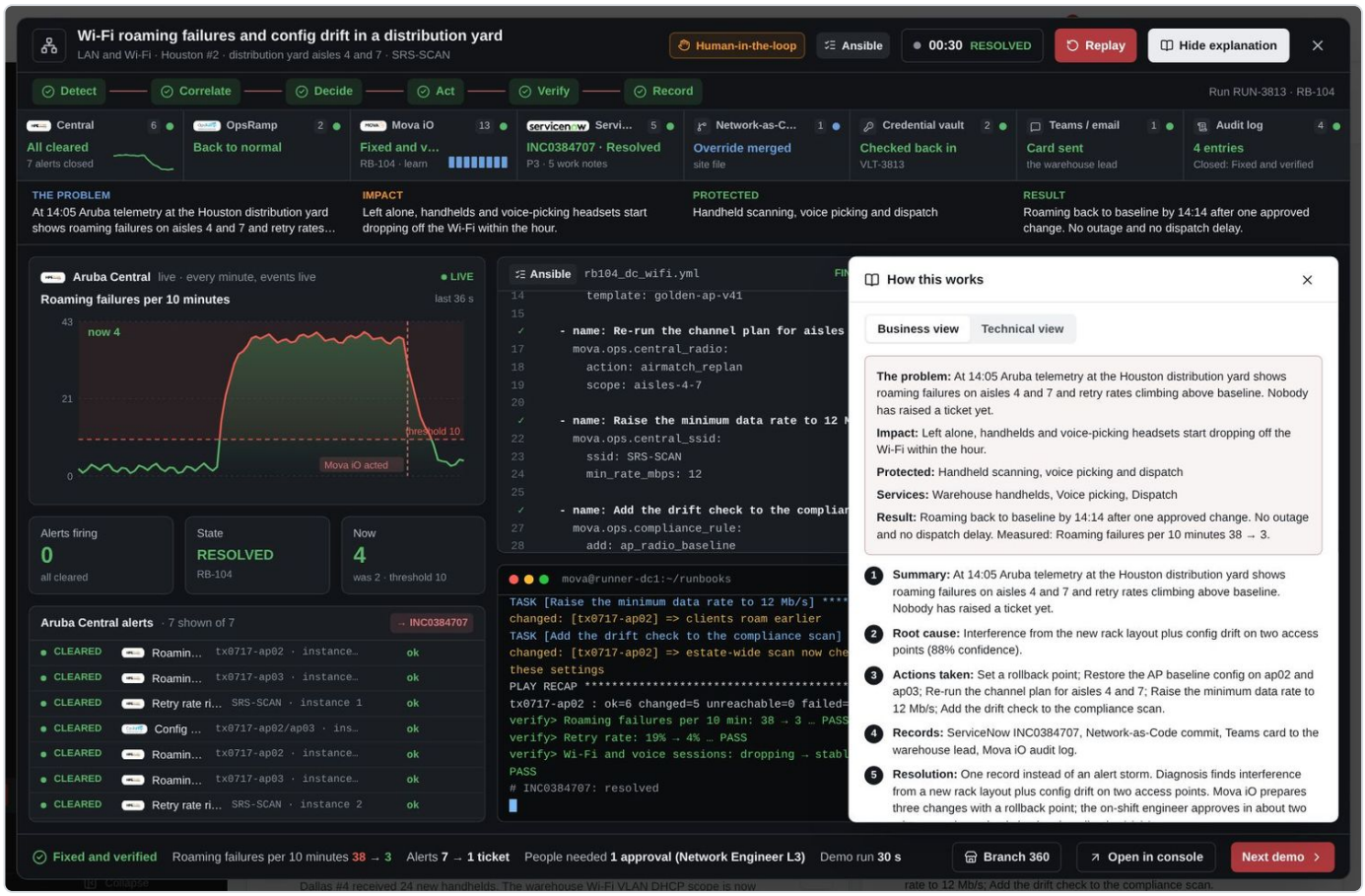


Figure 215. Finished run with the explanation open

The explanation panel

Explain opens a panel with two views. The **Business view** gives the problem, impact, what was protected, services, the result with the measured change, then five numbered points: summary, root cause with confidence, actions taken, records written and resolution. Two boxes list how Mova iO knew and what it ruled out.

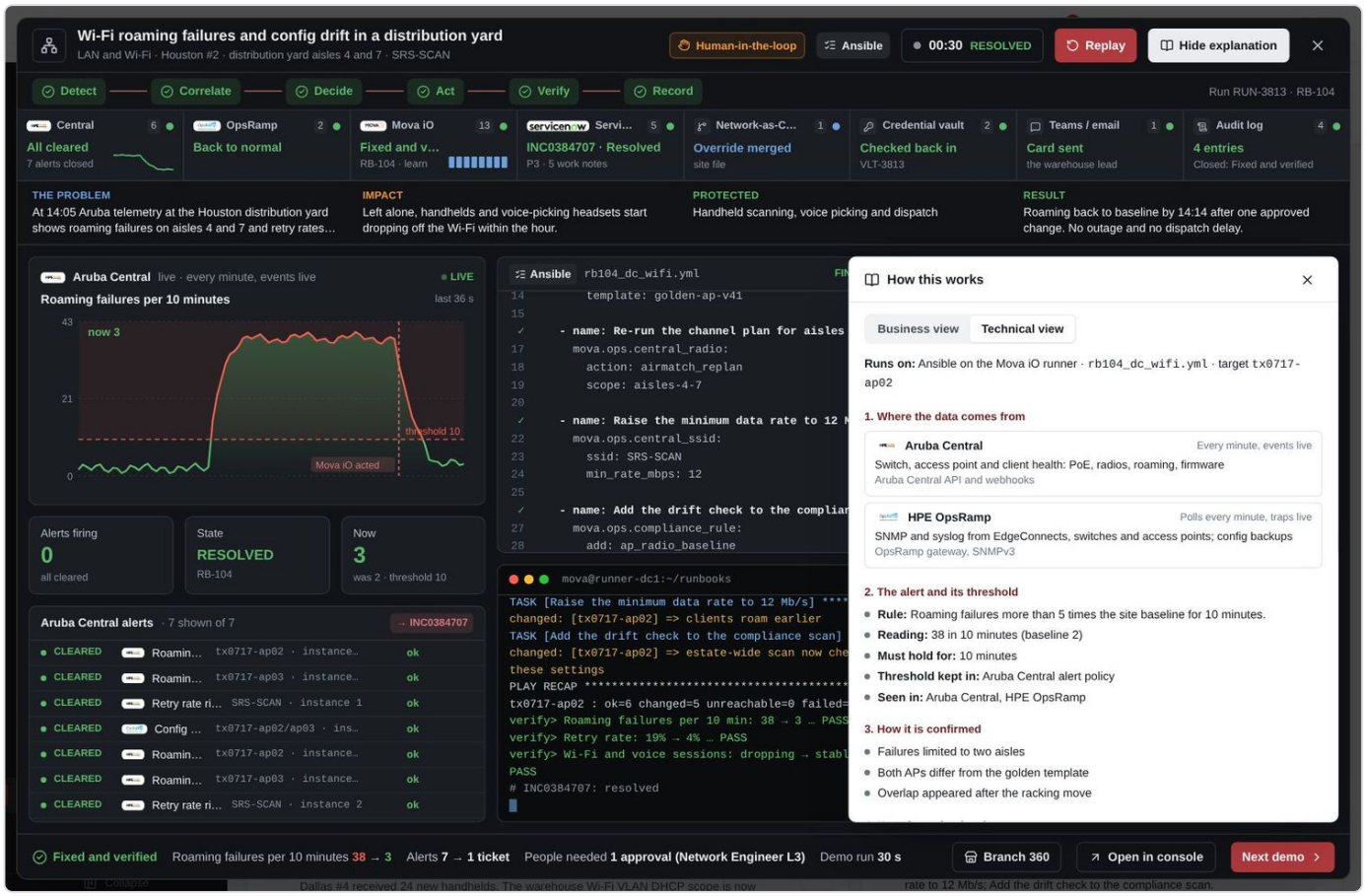


Figure 216. Technical view of the explanation

The **Technical view** answers the questions an engineer asks, in six sections:

1. **Where the data comes from:** each source system, how often it reports, what it measures and how it connects.
2. **The alert and its threshold:** the rule, the reading, how long it must hold and where the threshold is kept.
3. **How it is confirmed:** the cross-checks that rule out a false alarm.
4. **How the action is taken:** the engine, file, credential and each step, then the undo.
5. **How it is checked:** the verification tests.
6. **Control mode:** why this run is autonomous, on the loop or in the loop, with the rules.

Presenting a demo

- On a Human-in-the-loop demo, pause at the approval bar and read the evidence aloud before clicking **Approve**. **Decline** shows the other path: no change is made and the incident goes to the engineer.
- On a Human-on-the-loop demo, point at the countdown: the run will go ahead unless someone presses **Stop**.
- Use **Skip to end** for a long demo and then open the explanation.
- After the demo, click **Open in console** to show the same incident, carrier case or change in its own screen.

Part III. The live demos

25. The 33 demos

Every demo is an SRS branch situation built from the proposal documents: the Movate solution response (including its three "day in the life" scenarios), the RFP response, Annexure B (SLA and KPIs) and the console brief. Each one plays on the stage described in chapter 24 and leaves its records in the console.

Where each demo comes from

#	Demo	Mode	Source in the proposal
1	SD-WAN circuit degradation fixed before the branch notices	Autonomous	Solution response, day in the life #1
2	Branch hard down after a carrier fiber cut	Human-in-the-loop	Solution response, day in the life #3
3	Primary circuit down: carrier fault	Autonomous	RFP response: carrier isolation sequence and end-to-end carrier ownership
4	Chronic circuit: four drops in 30 days become a problem record	Autonomous	RFP response: problem rule (3 in 30 days); ticket data: 58% from repeat circuits
5	Return to primary circuit after carrier repair	Human-on-the-loop	Day in the life #3 (controlled failback); runbook catalog
6	EdgeConnect hardware failure	Human-in-the-loop	Console brief; RFP response: break-fix and field dispatch
7	Severe weather across Dallas–Fort Worth	Human-in-the-loop	Solution response: peak-season and storm war room
8	Yard access point offline	Autonomous	RFP response: closed-loop example (AP unresponsive, Act tier)
9	Wi-Fi roaming failures and config drift in a distribution yard	Human-in-the-loop	Solution response, day in the life #2
10	CRC errors climbing on a switch uplink	Human-in-the-loop	RFP response: Stage example (switch CRC errors)
11	Automation does not fix it: engineer takes over with full context	Human-in-the-loop	RFP response: handoff when automated remediation fails
12	Broadcast storm from an unmanaged desk switch	Human-on-the-loop	Solution response: blameless review ends in an engineering action
13	DHCP scope full after a handheld rollout	Human-in-the-loop	RFP response: deterministic workflows (DHCP scope check)
14	Day-0 critical patch for all EdgeConnects	Human-in-the-loop	Annexure B #32; RFP response: emergency patch cadence
15	Monthly switch patch wave with automatic checks	Human-on-the-loop	RFP response: monthly minor patches; Firmware Upgrade Library
16	Critical access point vulnerability (HPESBNW04712)	Human-in-the-loop	RFP response: vulnerability management; CVE-to-device mapping
17	Access point firmware wave stopped at canary	Human-on-the-loop	Solution response: progressive rollout with automatic halt; console brief
18	Annual major switch upgrade with early life support	Human-in-the-loop	RFP response: major upgrades up to once a year, pilot, waves, early life support
19	Firmware compliance catch-up after hardware swaps	Human-in-the-loop	Annexure B #32; firmware policy (N-1, N-2)

#	Demo	Mode	Source in the proposal
20	Error budget burning: tonight's patch wave deferred	Autonomous	Solution response: error budget as a control on change
21	Aruba 2530 end of support: refresh wave 1	Human-in-the-loop	RFP response: 728 Aruba 2530 switches reach end of support in early 2027 (31 Jan for 48-port, 31 Mar for 24-port)
22	Nightly config backups fail and are re-run	Autonomous	Annexure B #33; RFP response: configuration management
23	Unapproved local admin account on a switch	Human-in-the-loop	RFP response: drift monitoring and access management
24	Port activation from the service catalog	Human-on-the-loop	Solution response: about 300 catalog changes follow fixed policy
25	Standalone access points brought under Aruba Central	Human-in-the-loop	RFP response: 2,080 devices (51%) outside Aruba Central
26	A repeated manual fix becomes a tested runbook	Human-in-the-loop	Solution response: toil budget and Toil Miner; RFP response: Eval Runner
27	Backup LTE fails the weekly failover test	Autonomous	Solution response: dual-ISP failover test library
28	Circuit degradation predicted: proactive carrier case	Autonomous	RFP response: predictive report example
29	WAN capacity forecast for hail season	Human-in-the-loop	RFP response: capacity management; peak-season planning
30	Branch fails its 05:30 readiness check	Autonomous	Console brief: Branch Readiness
31	After-hours link down held and checked at opening	Autonomous	Solution response: site-closed hold; single link down reclassified from P2 to P3
32	Acquired branches monitored from day one, before any kit changes	Human-in-the-loop	Solution response: Day-1 M&A playbook; RFP response: M&A transition service
33	Acquired branch cutover weekend	Human-in-the-loop	Console brief: Branch-in-a-Box; solution response: weekend cutovers

Every demo is a core network situation: WAN circuits, SD-WAN, switching, Wi-Fi, firmware and configuration.

SD-WAN and carriers

1. SD-WAN circuit degradation fixed before the branch notices (Houston, RB-101, Autonomous). At 09:12:00 jitter, loss and latency rise on the AT&T MPLS primary while the branch is trading. Forty-three events become one incident at 09:12:15. Diagnosis runs path tests, checks the carrier portal and compares the config with the golden template, and points to the carrier (87%). At 09:13:00 the pre-approved runbook steers POS, ERP and voice to broadband and re-applies QoS; at 09:13:25 synthetic tests pass and the AT&T case opens with the evidence; at 09:13:40 the incident closes on its own. **Point at:** the chart falling back under the threshold at "Mova iO acted", and the closing time under two minutes with no human touch.

2. Branch hard down after a carrier fiber cut (Charlotte, RB-101, Human-in-the-loop). At 07:42 the fiber primary is cut and the broadband backup, in the same duct, degrades; 62 events become one P1. Diagnosis rules out the branch using last-gasp messages and the out-of-band router. The carrier case opens with evidence, the branch manager's call arrives with the incident already on screen, and the NOC engineer approves bringing up the LTE/5G path with

POS, ERP and voice first. The branch trades again about 13 minutes after going dark; after the repair, failback is verified, the RCA is published and branches with a shared duct are flagged. **Point at:** the approval, then the probes going from 0% to 100%.

3. Primary circuit down: carrier fault (Atlanta, RB-106, Autonomous). The AT&T primary drops and the branch fails over in 4 seconds. The Carrier Agent proves the fault is upstream of the demarc with a loopback test and opens the case with loss graphs, MTR and the loopback result, so the carrier dispatches on the first call instead of closing it as no trouble found.

4. Chronic circuit becomes a problem record (El Paso, Autonomous). A fourth drop in 30 days on the broadband primary. Mova iO links the four incidents into a problem record, sends a 30-day evidence pack to the carrier account team as a chronic-circuit review, and proposes a diverse second carrier for approval. **Point at:** the pattern finally has one owner.

5. Return to primary after carrier repair (Phoenix, RB-102, Human-on-the-loop). The primary has been clean for 40 minutes but the branch is still on LTE. The runbook proves the line is stable and the case closed, and runs unless the NOC engineer stops it in the stop window.

6. EdgeConnect hardware failure (Phoenix, RB-108, Human-in-the-loop). The EdgeConnect stops answering on both paths. Remote recovery is tried first: soft reboot, console read, reload from the backup partition. When it fails, a technician is dispatched with a pre-staged spare, zero-touch configures it and the CMDB updates itself.

7. Severe weather across Dallas–Fort Worth (Major incident, Human-in-the-loop). Hail hits seven branches in nine minutes. Sixty-four alerts become one major incident with one master carrier case, a regional change freeze, hourly leadership updates and an LTE plan uplift sent for approval.

LAN and Wi-Fi

8. Yard access point offline (Houston, RB-103, Autonomous). The yard AP is down with its PoE port drawing 0 W. Because the device is already down, the port bounce cannot make anything worse, so it runs at once during trading hours; the AP is back in three minutes.

9. Wi-Fi roaming failures and config drift in a distribution yard (Houston, RB-104, Human-in-the-loop). At 14:05 roaming failures rise on two aisles before anyone calls. Diagnosis finds interference from a new rack layout plus drift on two APs. Mova iO prepares three changes with a rollback point; the engineer approves in about two minutes; roaming is back to baseline by 14:14 and a drift check joins the estate-wide scan.

10. CRC errors climbing on a switch uplink (Memphis, RB-103, Human-in-the-loop). Errors rise over three days; diagnosis ranks a failing optic. The engineer approves taking that link out of the pair now and swapping the optic after close.

11. Automation does not fix it (San Antonio, Human-in-the-loop). An AP reboot runs but verification fails, so the runbook stops and the incident reaches the L2 engineer in the same minute with the SLA clock unchanged and the full handoff package. The package shows the switch PoE budget at 98% after new cameras; the engineer approves new PoE priorities and the AP returns. **Point at:** humans receive only the exceptions, and they receive them with context.

12. Broadcast storm from an unmanaged desk switch (Miami, RB-103, Human-on-the-loop). Loop-protect blocks the port, the branch is asked to unplug the desk switch, and a template change for BPDU guard on every branch follows.

13. DHCP scope full after a handheld rollout (Dallas, Human-in-the-loop). Twenty-four new handhelds fill the warehouse VLAN pool. The fix is proposed at 79% confidence, just under the gate, so an engineer approves it.

Patching and firmware

14. Day-0 critical patch for all EdgeConnects (Human-in-the-loop). A critical ECOS advisory is matched to all 933 EdgeConnects. Mova iO plans rings: a canary of 10 closed branches, then regions after close, HA pairs one unit at a time. The emergency change board approves, the canary passes its health gate and the waves are released with automatic rollback.

15. Monthly switch patch wave (Texas, Human-on-the-loop). An approved standard change patches 214 switches after close: backup, pre-checks, batches of 20 with stack members one at a time, and post-checks compared with pre-checks. One switch fails its post-check and rolls back on its own; 213 finish.

16. Critical access point vulnerability (Human-in-the-loop). Exposed APs are counted by branch, Security approves a mitigation for all access points, and patch waves run region by region after close with a canary first.

17. Access point firmware wave stopped at canary (Florida, Human-on-the-loop). The canary group shows older handhelds failing fast roaming; Change Validation stops the rollout on its own, the canary rolls back and a revised plan goes for approval.

18. Annual major switch upgrade with early life support (Human-in-the-loop). The once-a-year AOS-CX major upgrade is packaged with the lab result, 12 pilot branches, regional waves and the previous image for rollback. The change board approves; the pilot runs after close and enters 14 days of early life support, which spots a slow memory climb on one model. Those waves are held and a vendor case opened while the others continue.

19. Firmware compliance catch-up (Human-in-the-loop). The weekly scan finds 37 devices outside policy: replacement units on factory firmware and APs that were offline during the last wave. A catch-up wave is staged as a standard change, approved and run after close; the replacement process now upgrades units before shipping.

20. Error budget burning: tonight's patch wave deferred (Great Lakes, Autonomous). The region's availability budget is burning 3.1 times faster than planned. The change policy defers the non-essential AP wave by a week, keeps emergency security changes allowed and narrows the next wave to canary branches first.

21. Aruba 2530 end of support (Human-in-the-loop). The 728 switches reaching end of support on 1 February 2027 are ranked by failure risk and traffic, and the engineer approves the first refresh wave with the standard CX 6100 kit and spares pre-staged.

Config and automation

22. Nightly config backups fail and are re-run (Southeast, RB-109, Autonomous). Forty-six backups fail in the same minute; all used one TACACS+ server. Mova iO retries through the second server, verifies each file's hash and opens a problem record. Success is back above 99% before 03:00.

23. Unapproved local admin account on a switch (Charlotte, Human-in-the-loop). The nightly backup shows a local admin account outside the golden config. Security approves restoring the golden config after close.

24. Port activation from the service catalog (Tampa, Human-on-the-loop). A branch requests two desk ports. The request is checked against policy, built from the template after the stop window and verified with link, LLDP and DHCP checks; it closes in nine minutes.

25. Standalone access points brought under Aruba Central (Human-in-the-loop). Wave 1 of the 2,080 devices outside Central: 120 APs are checked, their configs captured and mapped to the golden group, then converted after close in batches. Telemetry and the AP runbooks now cover them.

26. A repeated manual fix becomes a tested runbook (Human-in-the-loop). The Toil Miner finds the same manual tunnel reset 23 times in 30 days. A runbook is drafted, checks are generated, and a replay against the 23 past cases passes 21 and correctly refuses 2. It joins the catalog at the Stage tier.

27. Backup LTE fails the weekly failover test (Tulsa, RB-106, Autonomous). The Sunday test finds the LTE backup passes no traffic because the SIM was suspended. An account case is opened, the SIM is reactivated and the re-test passes before Monday opening.

Predict and prevent

28. Circuit degradation predicted (Charlotte, Autonomous). Loss creeps from 0.1% to 0.9% over five days. A proactive carrier case opens with the trend, a failover test proves the backup, and the carrier replaces a failing line card before the circuit drops.

29. WAN capacity forecast for hail season (Dallas, Human-in-the-loop). A branch's WAN heads for 90% in about three weeks. An engineer approves a bandwidth upgrade and, until it lands, a policy that protects POS and ERP.

Branch readiness

30. Branch fails its 05:30 readiness check (Denver, Autonomous). Two checks fail before opening: a tunnel and a config backup. The branch is closed, so both are fixed at once and the 05:52 re-run passes all ten checks.

31. After-hours link down held and checked at opening (Lubbock, Autonomous). The backup line goes down at 22:15 at a closed branch with a clean primary. Mova iO sets P3 instead of P2, matches the carrier's maintenance window, holds the page and re-checks at opening; nobody is woken.

Branch-in-a-Box

32. Acquired branches monitored from day one (Mingledorff's wave 2, Human-in-the-loop). Ten acquired branches still on their own kit are discovered over a temporary VPN, monitored as they are, entered in the CMDB and given interim service levels and notify-only runbooks, weeks before their cutovers.

33. Acquired branch cutover weekend (LS Building Products, Human-in-the-loop). Thirteen timed gates from Friday close to Monday opening. Gate 9 fails because legacy handhelds are 2.4 GHz only; a site override is approved, the re-test passes and StoreOps signs off.

Part IV. Presenting and reference

26. Presenting the console

A 30-minute walkthrough that covers the story end to end:

1. **Operations Overview (3 min).** The estate on one screen: the tiles, the map, the error budgets. Pause updates with **LIVE** while you talk.
2. **Demo 1, SD-WAN degradation (4 min).** Autonomous: 43 alerts, one incident, fixed and closed in under two minutes.

3. **Demo 9, Wi-Fi drift (4 min).** Human-in-the-loop: read the evidence aloud, then approve.
4. **Demo 2, fiber cut (4 min).** A P1 handled end to end, including the branch manager's call and the RCA.
5. **Correlation (2 min).** The funnel and the flow: why engineers see only a small share of the alerts.
6. **Patching (5 min).** Demo 14 (Day-0 patch) or 18 (major upgrade), then Approvals & Changes on the firmware tab.
7. **Branch Readiness (2 min).** Every branch checked at 05:30 local, problems fixed before opening.
8. **Branch-in-a-Box (4 min).** Demo 32 (day-1 monitoring) and demo 33 (cutover weekend).
9. **SLA & KPI Tracker (2 min).** Every Annexure B metric, live.

Phrases that help

- "One incident per branch per event, not one per alert."
- "Mova iO prepares the evidence and the rollback; a person decides whenever the rules say so."
- "Firmware never changes without a person, but the waves run themselves once approved."
- "A failed check at 05:30 is fixed or owned before the doors open."
- "The branch keeps its name; the network becomes standard over one weekend."

27. Common questions

Question	Answer	Where to show it
Can the AI change things on its own?	Only actions in the Act tier, inside the gate rules: confidence 80% or more, branch closed for changes, no freeze. Firmware, routing, security policy and QoS changes always need a person.	Chapter 3; Approvals & Changes, Rules
What happens when automation fails?	Verification catches it, the runbook stops and the incident goes to an engineer in the same minute with the full context and the SLA clock unchanged.	Demo 11
How do you stop alert storms?	Correlation by topology, time, change, carrier and similarity; transients are cleared after a hold-down and warnings go to a digest.	Correlation
How are carriers held to account?	Every case carries evidence, the scorecard compares carriers fairly and repeat circuits become problem records.	Carrier Management; demo 4
How is patching kept safe?	Rings with a canary, pre and post checks, HA pairs one unit at a time, automatic rollback, early life support for major releases and an error budget that can hold non-essential changes.	Demos 14, 15, 18, 19, 20
How do acquisitions join?	Day-1 monitoring as they are, then the standard kit over one weekend with 13 gates, then 14 days of hypercare.	Branch-in-a-Box; demos 32, 33
How is the service measured?	Every Annexure B SLA and KPI with its MSL and ESL, updated live, with service credits on the risk-reward SLAs.	SLA & KPI Tracker

28. Glossary

Short definitions; chapter 3 explains each network term in more detail, with normal values and what it means for the business.

Term	Meaning
Act / Act and notify / Stage	The three catalog tiers: acts on its own; acts and tells a person at once; prepares and waits for approval
Advisory	A vendor security bulletin, such as HPESBNW04712
Alert	An event worth acting on, left after noise removal
Annexure B	The contract schedule of SLAs and KPIs, with MSL and ESL for each
AP	Access point: the Wi-Fi radio
Aruba Central	HPE Aruba's cloud management for switches and APs; 1,062 SRS devices are in it
Auto-cleared	An alert that recovered by itself and was checked by ping and SNMP
Availability	The share of time a service works
Backup line	The second WAN line a branch uses when its primary fails
Bandwidth	The capacity of a line, in megabits per second (Mb/s)
Blast radius	Everything affected if one element fails or changes
BPDU guard	A switch setting that shuts a port when another switch is plugged into it, preventing loops
Broadband	A cable or fibre internet line shared with other customers
Broadcast storm	Traffic copied endlessly round a cabling loop, slowing or stopping the branch network
Canary	A small first group in a rollout, checked against a control group before the rest proceed
Carrier	The company that sells a WAN line
CI / CMDB	Configuration item / the ServiceNow database of all devices, circuits and their relationships
Circuit	One WAN line from a carrier
Confidence	Mova iO's certainty in a diagnosis or action, 0 to 100%; below 80% a person decides
Config backup	A nightly copy of each device's configuration, taken at 02:00 CT
Correlation	Grouping related alerts into one incident
CRC errors	Corrupted frames on a link, often a failing optic or cable
CVE / CVSS	A published vulnerability / its severity score from 0 to 10; 9.0 and above is critical
Degraded line	A line that works but with high latency, jitter or loss
Demarc / NTE	The point where the carrier's equipment ends and the branch's begins
DHCP / scope	The service that gives devices addresses / the pool of addresses for a VLAN
DIA	Dedicated internet access: a business internet line with guaranteed speed

Term	Meaning
Digest	A daily summary of warnings that do not need a ticket
Drift	A difference between a device's running config and the approved golden config
Early life support	A period of closer monitoring after a major upgrade, before further waves
EdgeConnect / ECOS	The HPE Aruba SD-WAN appliance at each branch / its operating system. The EC-XS is replaced by the EdgeConnect 10104
End of sale	The last day a vendor sells a model; it is still supported
End of support	The last day of vendor support and security fixes; after it the device should be replaced
Error budget / burn rate	The failure a target allows over a period / how fast it is used compared with an even spend
ESL / MSL	Expected service level (target) / minimum service level (below it a service credit applies)
Event	Any single message or threshold crossing from a monitoring tool
Evidence pack	Graphs, traces, test results and timestamps attached to a carrier case
Failover / failback	Moving traffic to the other line automatically when one fails / moving it back once the main line is stable
Firmware / N, N-1, N-2	Device software / the target release, one behind and two behind
Flap / hold-down	A line or port that goes down and up quickly / the wait before acting, in case it recovers
Freeze window	A period when only emergency changes are allowed
Golden config	The approved configuration in the Network-as-Code repository
HA pair	Two EdgeConnects at one branch, one taking over if the other fails
Hypercare	14 days of tighter thresholds after a branch goes live
Incident	A ServiceNow record of a problem, with priority and SLA clocks
Jitter	How much the delay varies from packet to packet; above 30 ms calls become choppy
LAN	The branch network inside the building: switches, APs and cabling
Latency	Time for a request to reach the data centre and come back, in milliseconds
LEC	Local exchange carrier, which owns the last mile to the branch
Limited lifetime warranty	HPE's warranty for Aruba switches and APs, running until five years after the model's end of sale
LTE/5G backup	A cellular backup line; slower and metered, but independent of cables
MOS	Voice call quality on a 1 to 4.4 scale; good above 4.0
Mova iO	Movate's AI layer: agents that correlate, diagnose, act, verify and record
MPLS	A private line inside a carrier's network, steady and expensive
MTTR	Mean time to restore
Network-as-Code	Configurations kept as files in a repository, changed by reviewed merge requests

Term	Meaning
Noise removal	Dropping duplicates, flaps and information-only events before they become alerts
NTF	No trouble found: a carrier closing a ticket without finding a fault
OOB	Out-of-band: a cellular console path to devices when the WAN is down
Orchestrator	The central system that configures every EdgeConnect and collects its statistics
Overlay	An SD-WAN rule that says which line each kind of traffic uses
p95	The 95th percentile: 95% of measurements are at or below this value
Packet loss	The share of data that never arrives; above 1% payments retry and calls break up
Payment path round trip	A synthetic test payment from the counter VLAN to the payment processor and back; normally 100 to 200 ms, slow above 400 ms
PCI segment	VLAN 20, the isolated network for card terminals
PoE	Power over Ethernet, which powers APs, phones and cameras from the switch
Primary line	The main WAN line a branch normally uses
Problem record	A ServiceNow record for the cause behind repeat incidents
Readiness check	The ten network checks at 05:30 local before each branch opens
Repeat offender	A circuit with three or more tickets in 90 days
Roaming	A handheld moving from one AP to the next
RB-1xx	A runbook number in the Automation Library
Runbook	An approved, tested automation for one kind of fix
SD-WAN	Software that uses both lines together and picks the best one for each application
SLA clock	Time left to respond to or resolve an incident
Standalone	A switch or AP managed one by one, outside Aruba Central, and monitored by OpsRamp; 2,080 at SRS
Support contract	The yearly HPE support agreement covering the EdgeConnects
Synthetic probe	A scripted test run every minute from each EdgeConnect, such as a test payment or ERP login
Toil	Repeated manual work that should be automated
Tunnel	An encrypted path over one line from the EdgeConnect to the data centre; 1,866 in total
VLAN	A separate network segment, such as 20 for payments or 30 for warehouse Wi-Fi
WAN	The lines that connect a branch to the data centre, Azure and the internet
WAN use	The share of a line's bandwidth in use; above 85% traffic queues
Wi-Fi experience	Aruba Central's 0 to 100 score for Wi-Fi inside the branch; independent of the WAN
Zero-touch	An EdgeConnect or switch that configures itself from its template when plugged in